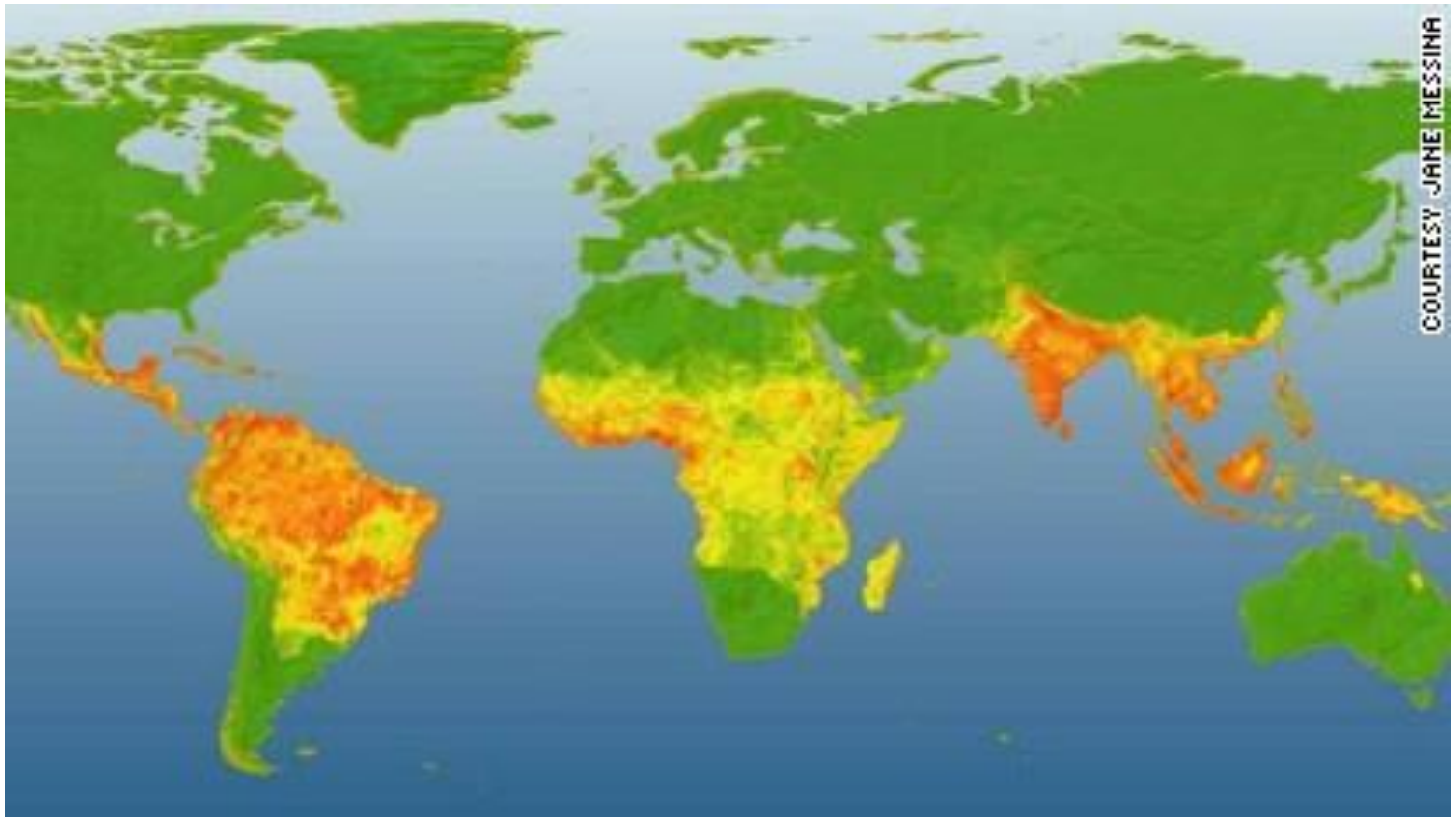
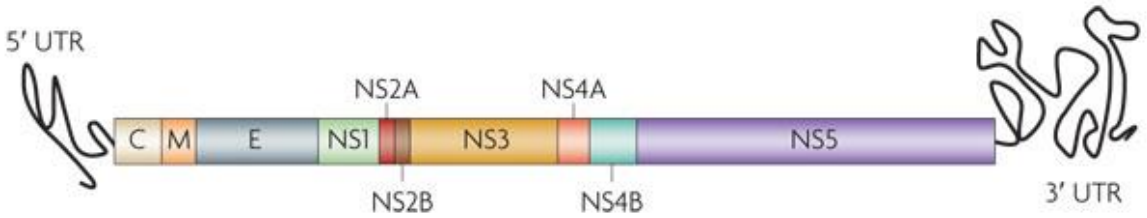
Dengue / Yellow Fever

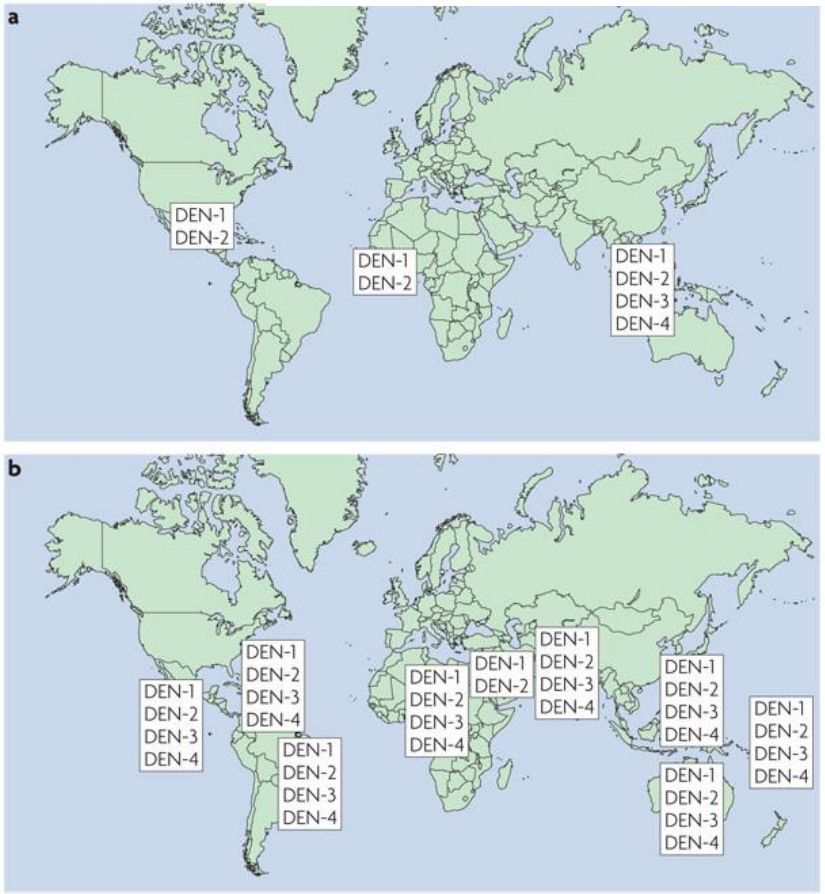
- 100 million + infections / Year
- 50 thousand + deaths
- RNA virus
- Aedes mosquitoes



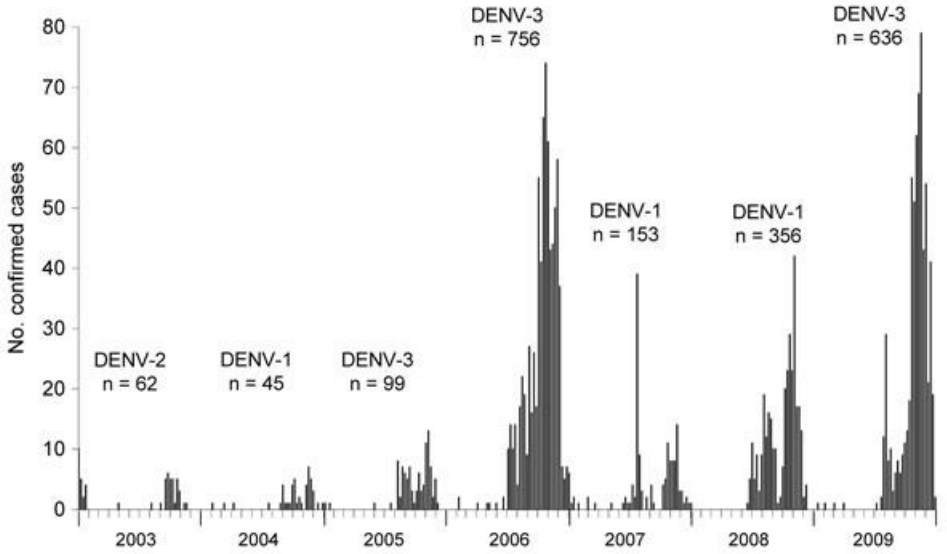


Dengue consists of four closely related but antigenically distinct viral serotypes (DEN1-4)
All four serotypes can cause the full spectrum of disease.





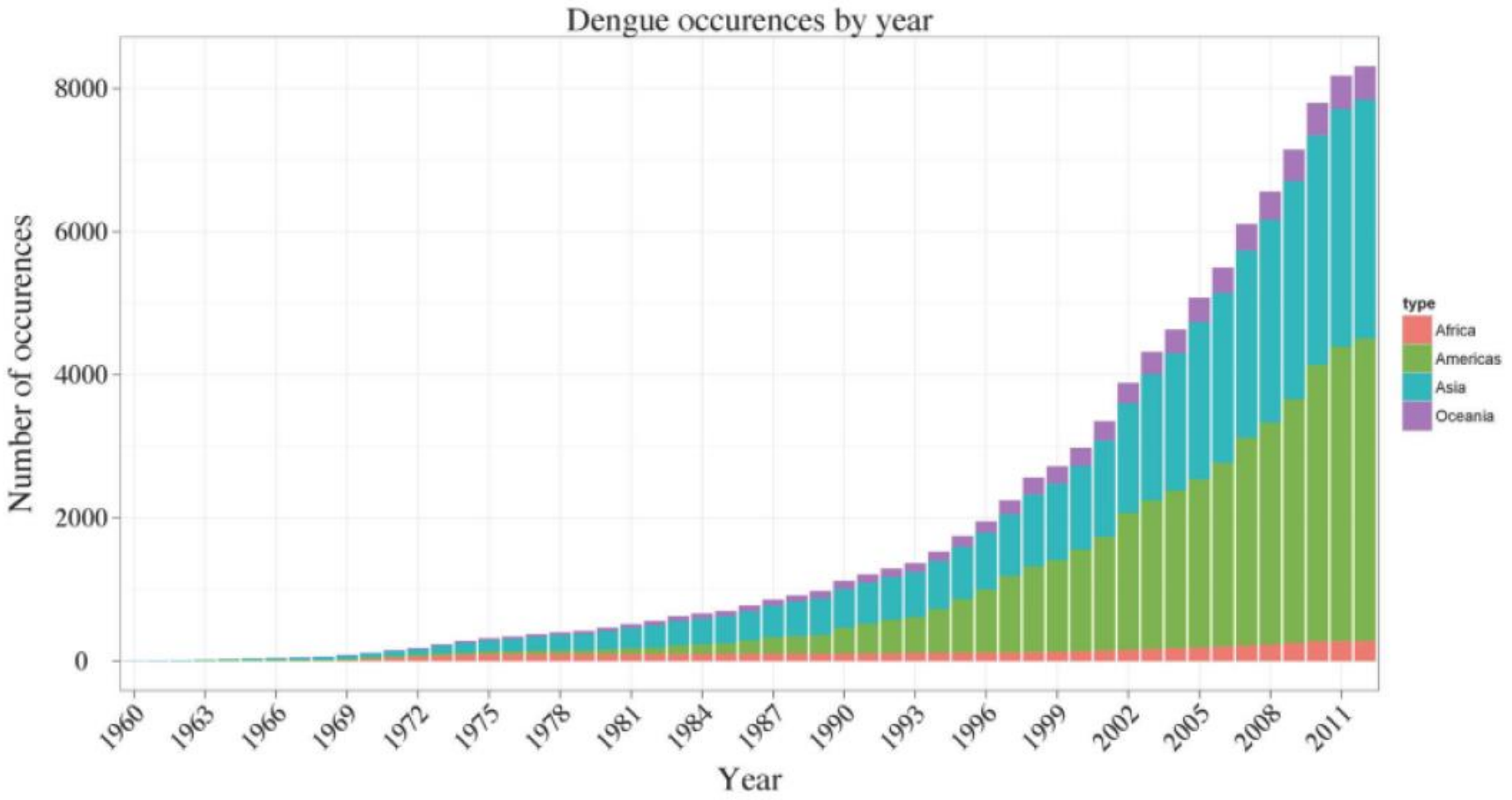
Nature review of Microbiology 2010



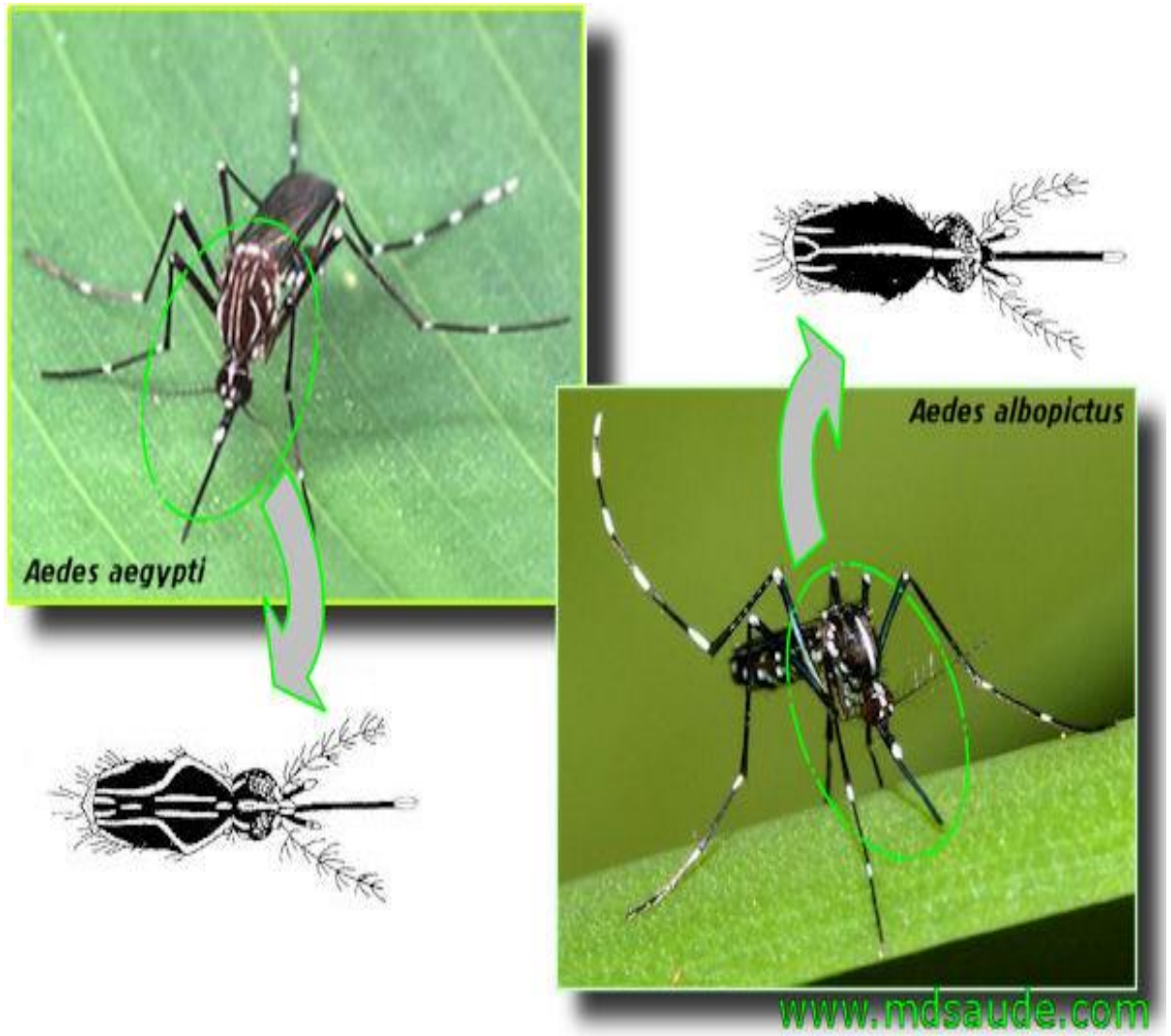
Dengue Outbreaks , Kaohsiung City, Taiwan, 2003–2009

Taiwan CDC

WHO: Failure of mosquito spraying to stop dengue raises questions for battle against Dengue



埃及斑蚊　　白線斑蚊



Remove the mosquitoes

Insecticides

Insects evolve resistance

Effect on non-target species





Mosquito source reduction

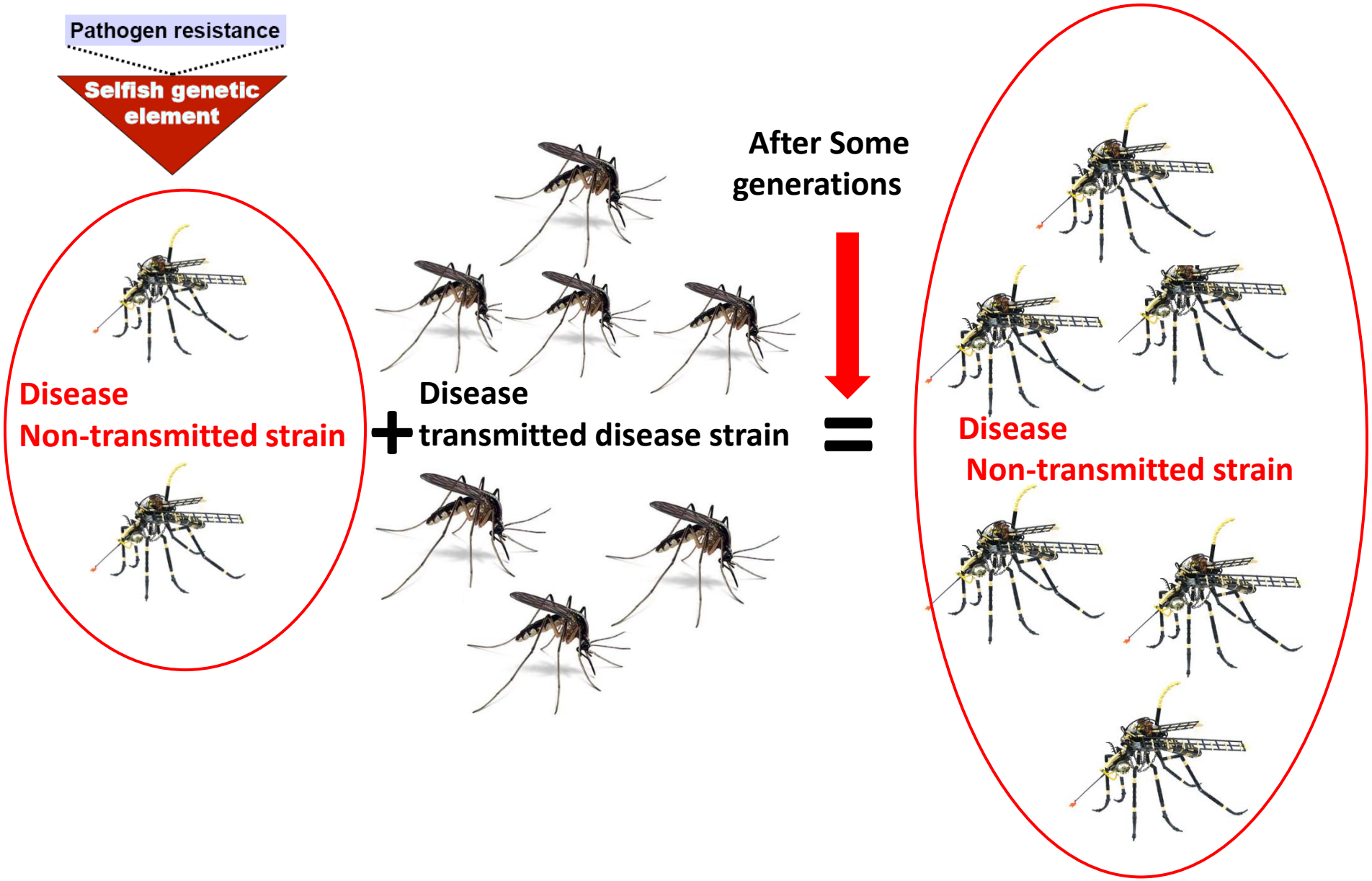


No Drug and Vaccine available yet





A New Strategy: A synthetic anti-pathogen population

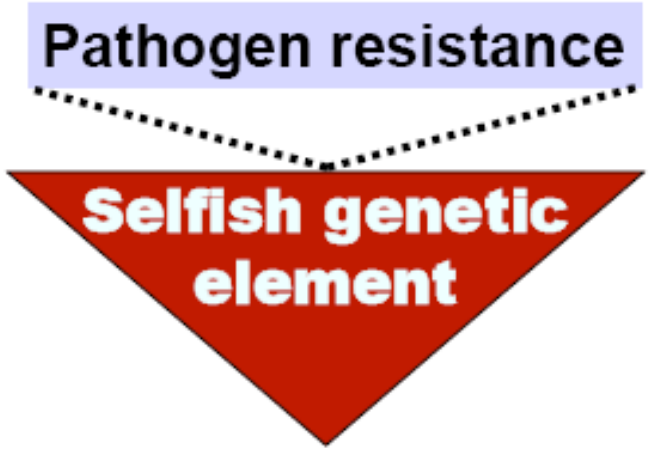


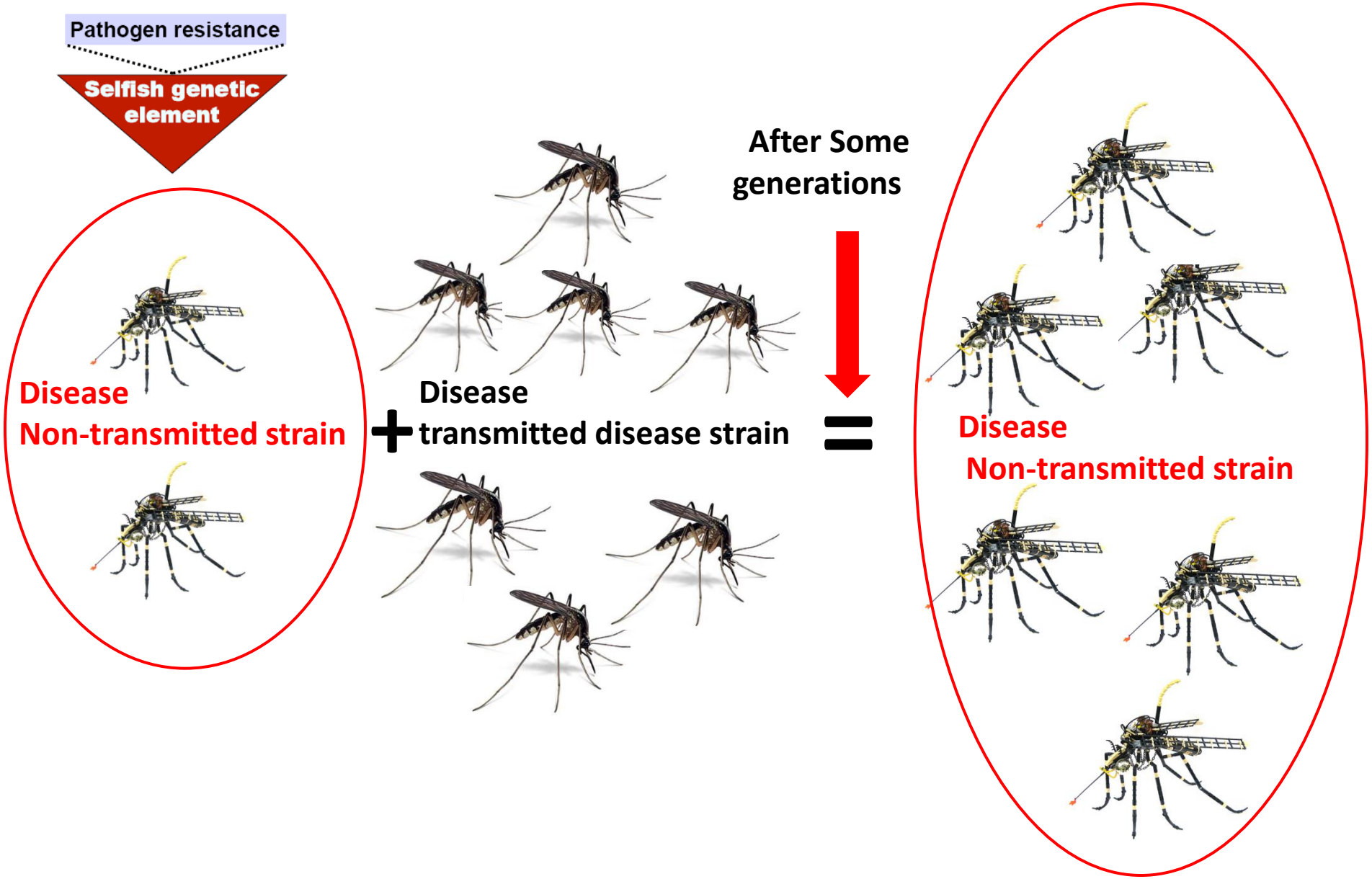
Two Major Goals:

Effector genes for Virus resistance

A driver gene for spreading the effector gene into population

The solution: Link genes for pathogen resistance with a selfish genetic element



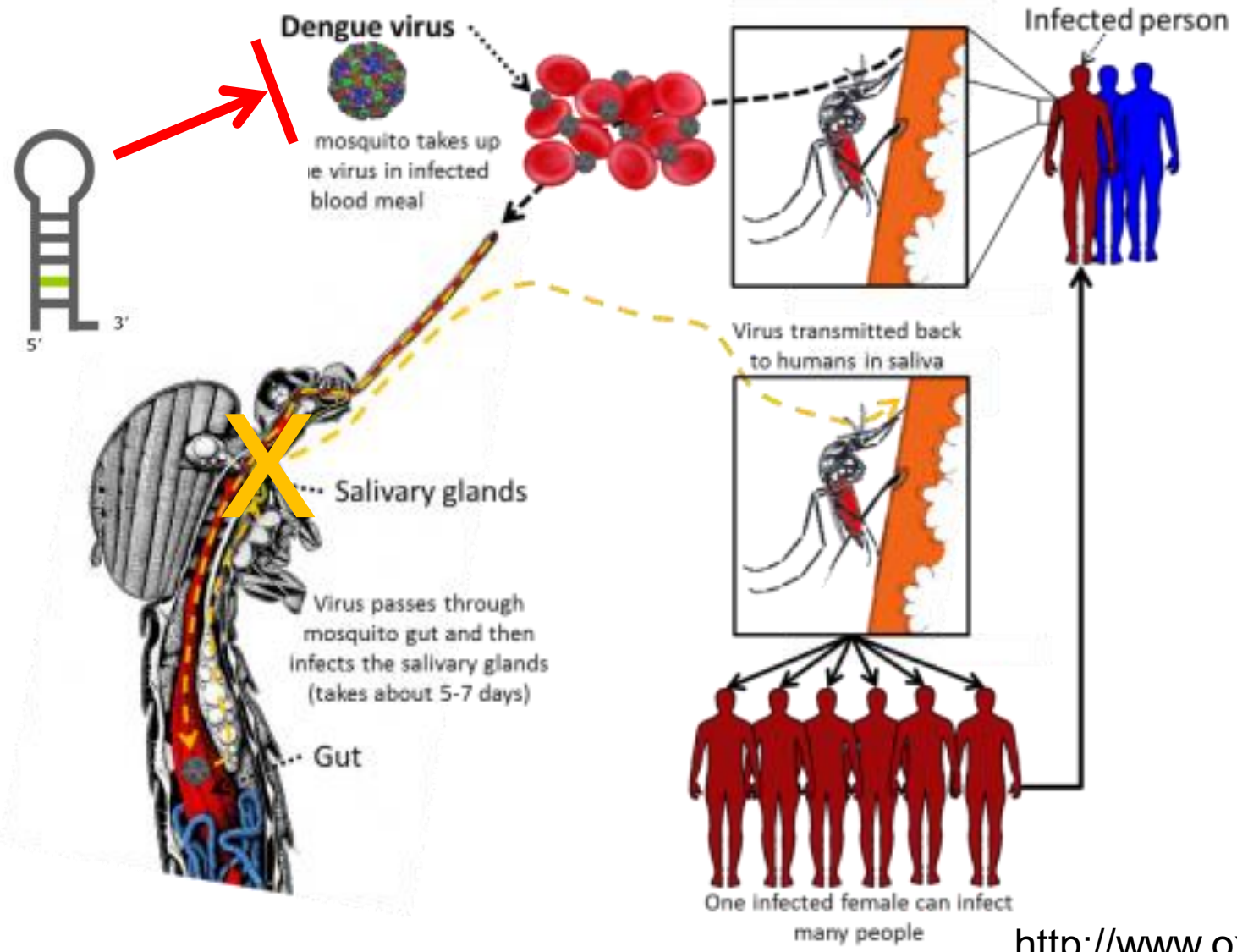

Pathogen resistance
Selfish genetic element
Disease
Non-transmitted strain
Disease
transmitted disease strain
After Some generations
Disease
Non-transmitted strain

Two effector genes

1 Targeting Dengue genomes or mosquito genes

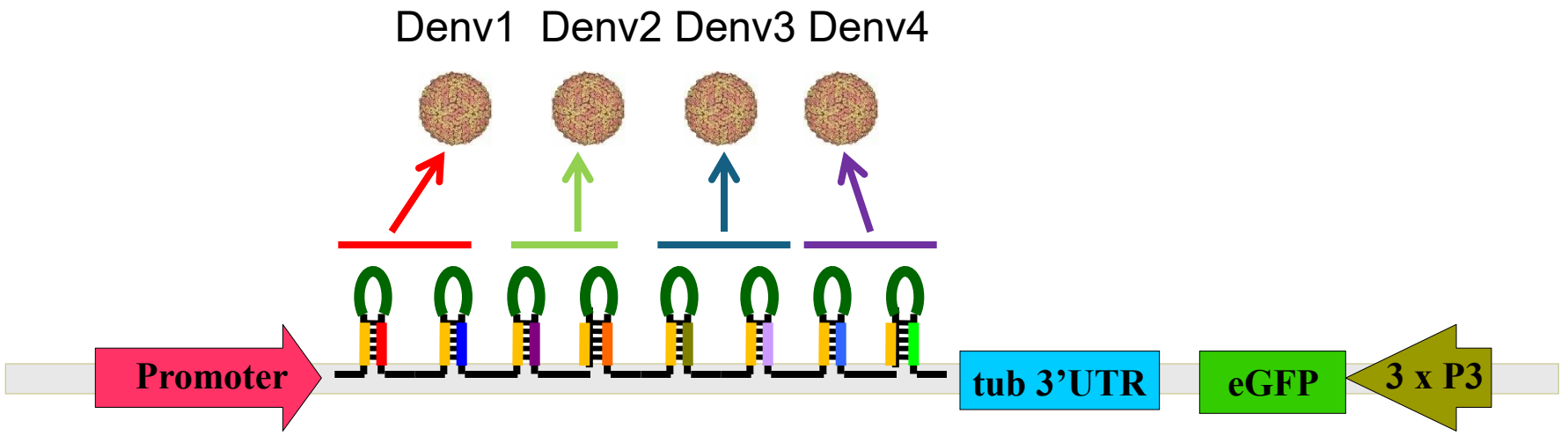
2 Altering Host seeking behavior

The life cycle of the dengue virus

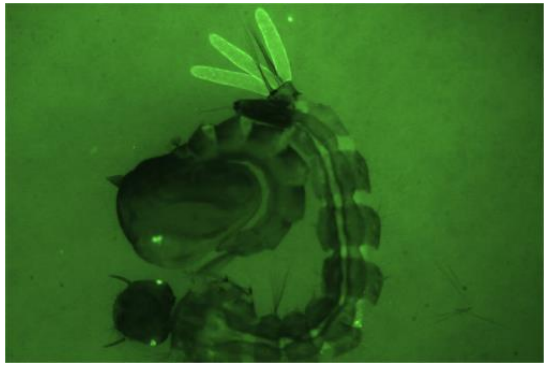


http://www.oxitec.com/

Using multiple microRNAs to KD single or multiple Dengue virus genomes







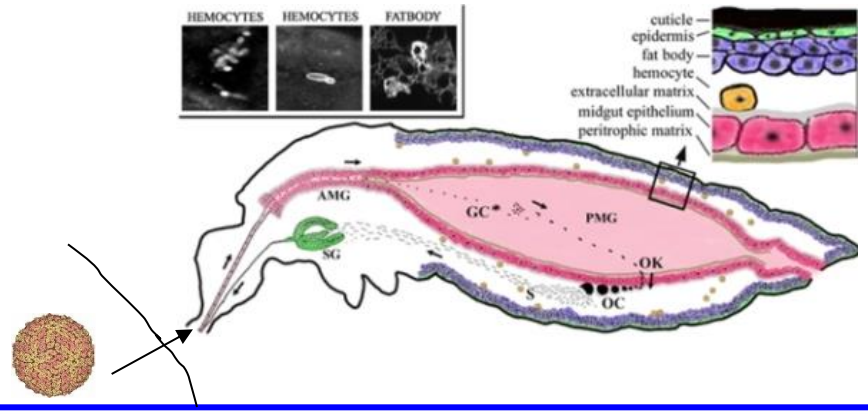


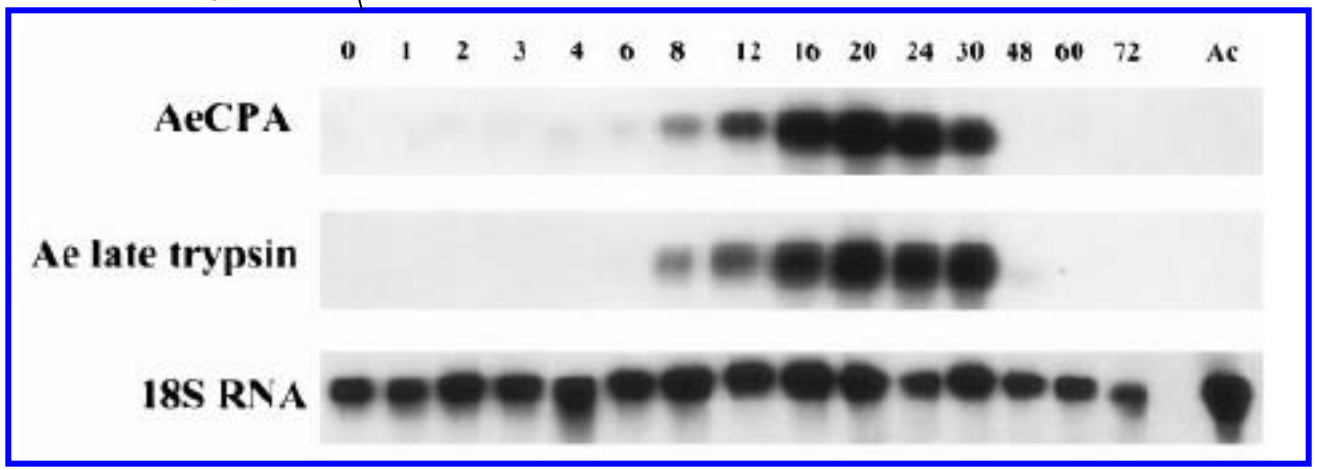
Polycistronic microRNAs were driven by a Blood meal induced promoter

Characterization of a carboxypeptidase A gene from the mosquito, *Aedes aegypti*

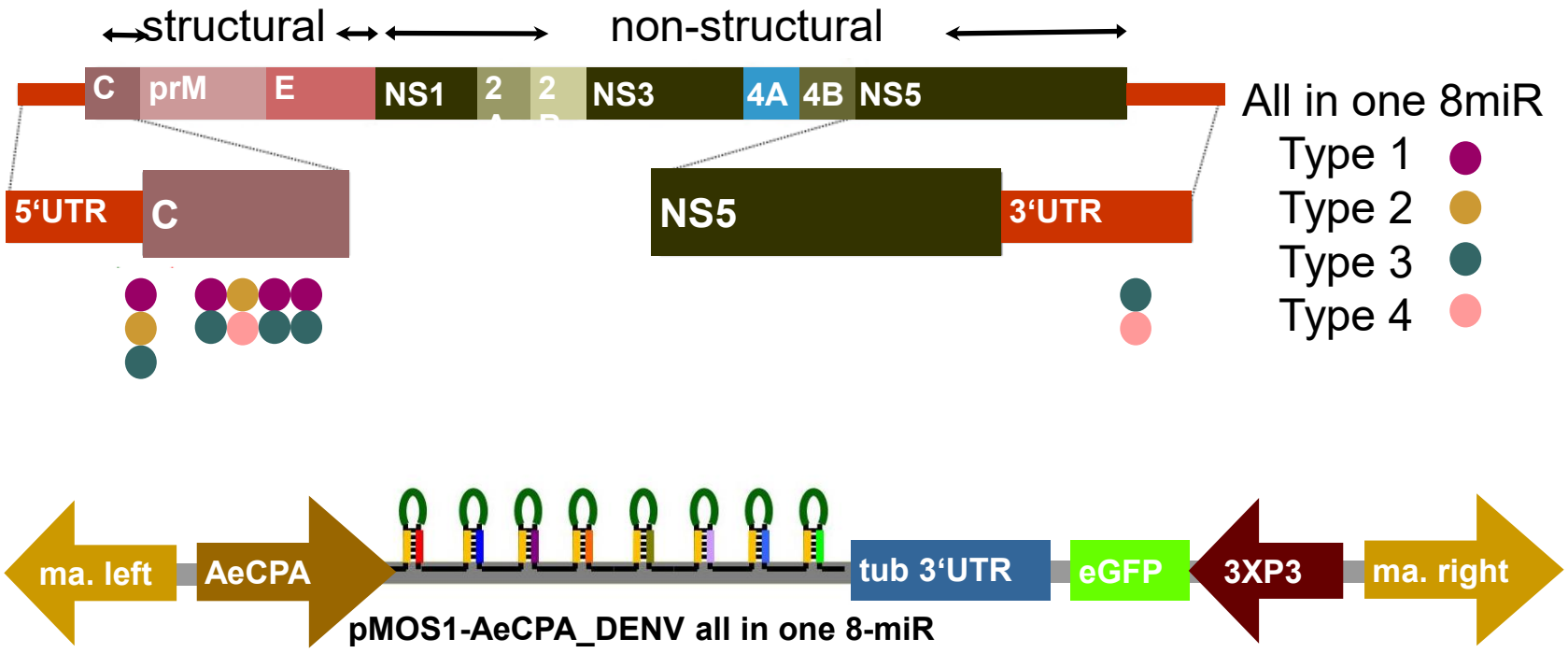
Insect Molecular Biology (2000) 9(1), 33–38

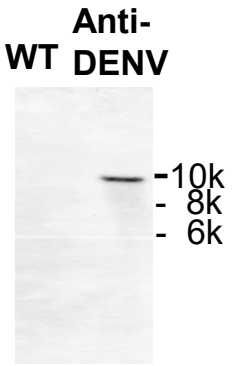
Time course of AeCPA and Aedes late trypsin mRNA accumulation after a blood meal.





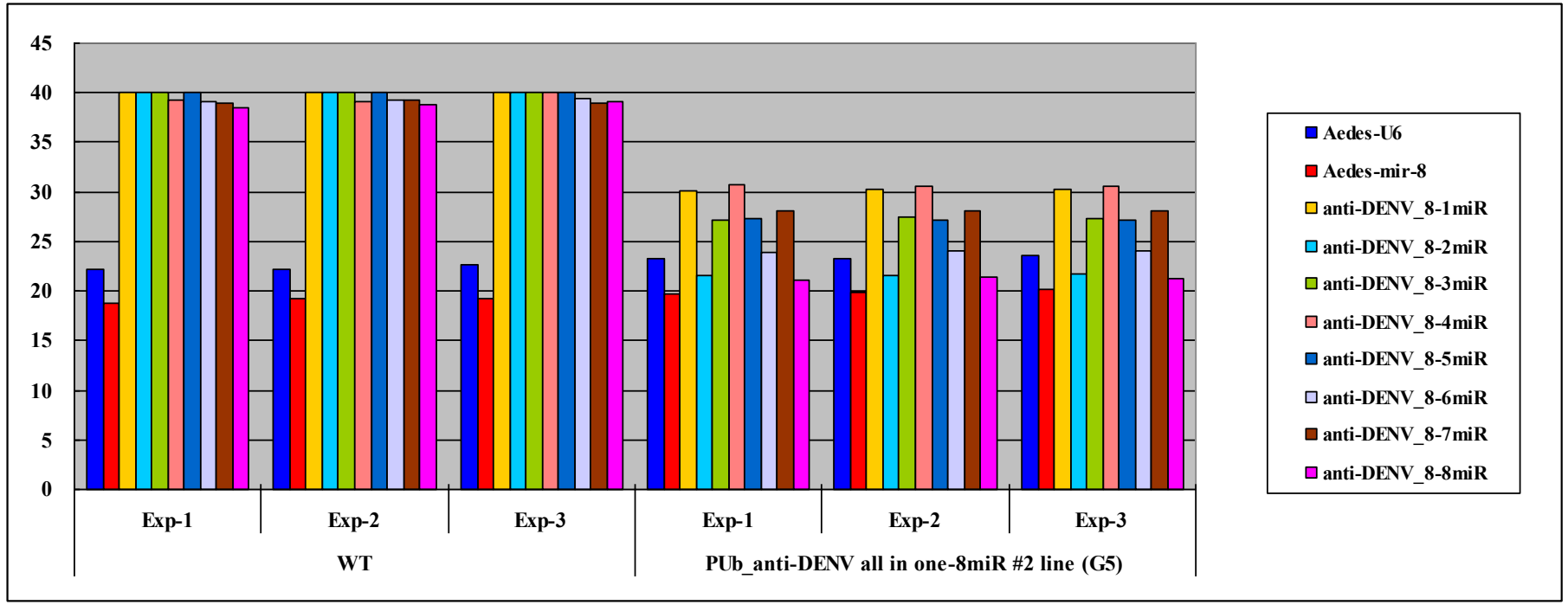
The anti-4-dengue transgenic mosquito





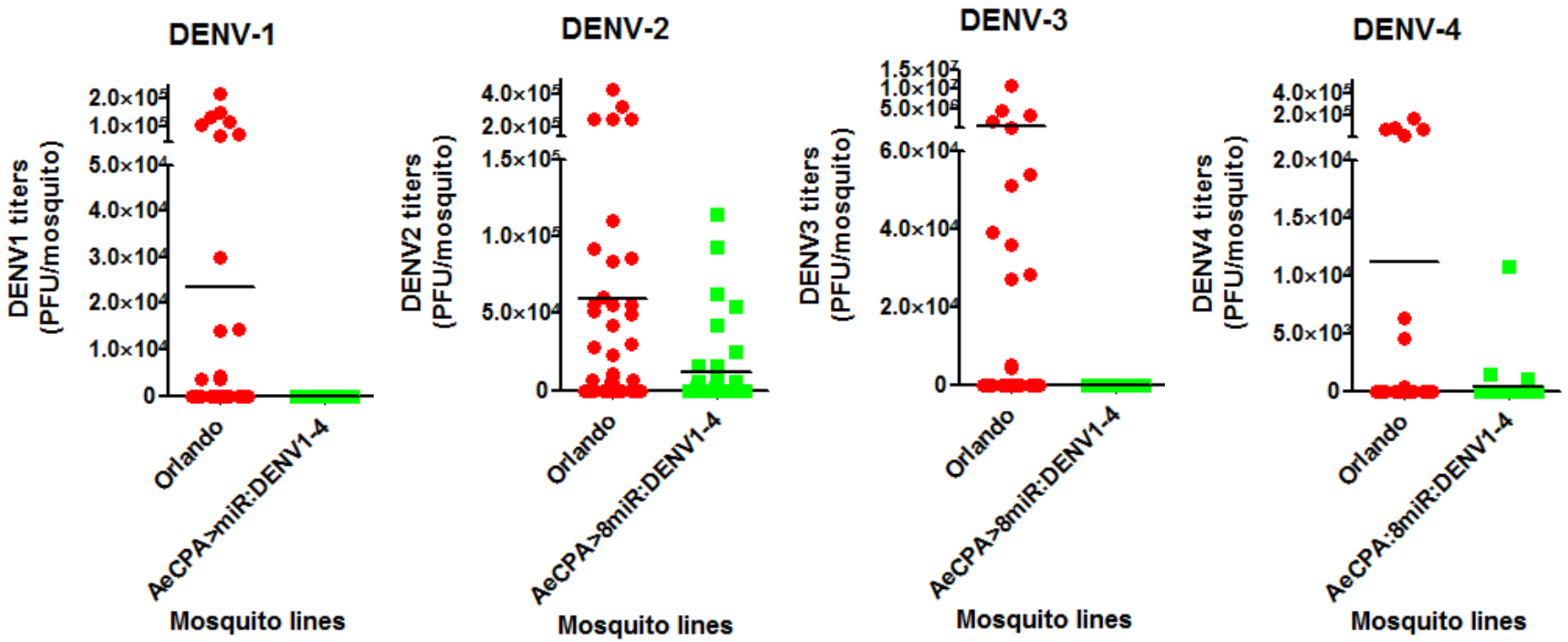
(Yen *et al.*, unpublished data)

miRNA expression profile of the anti-Dengue virus cassette

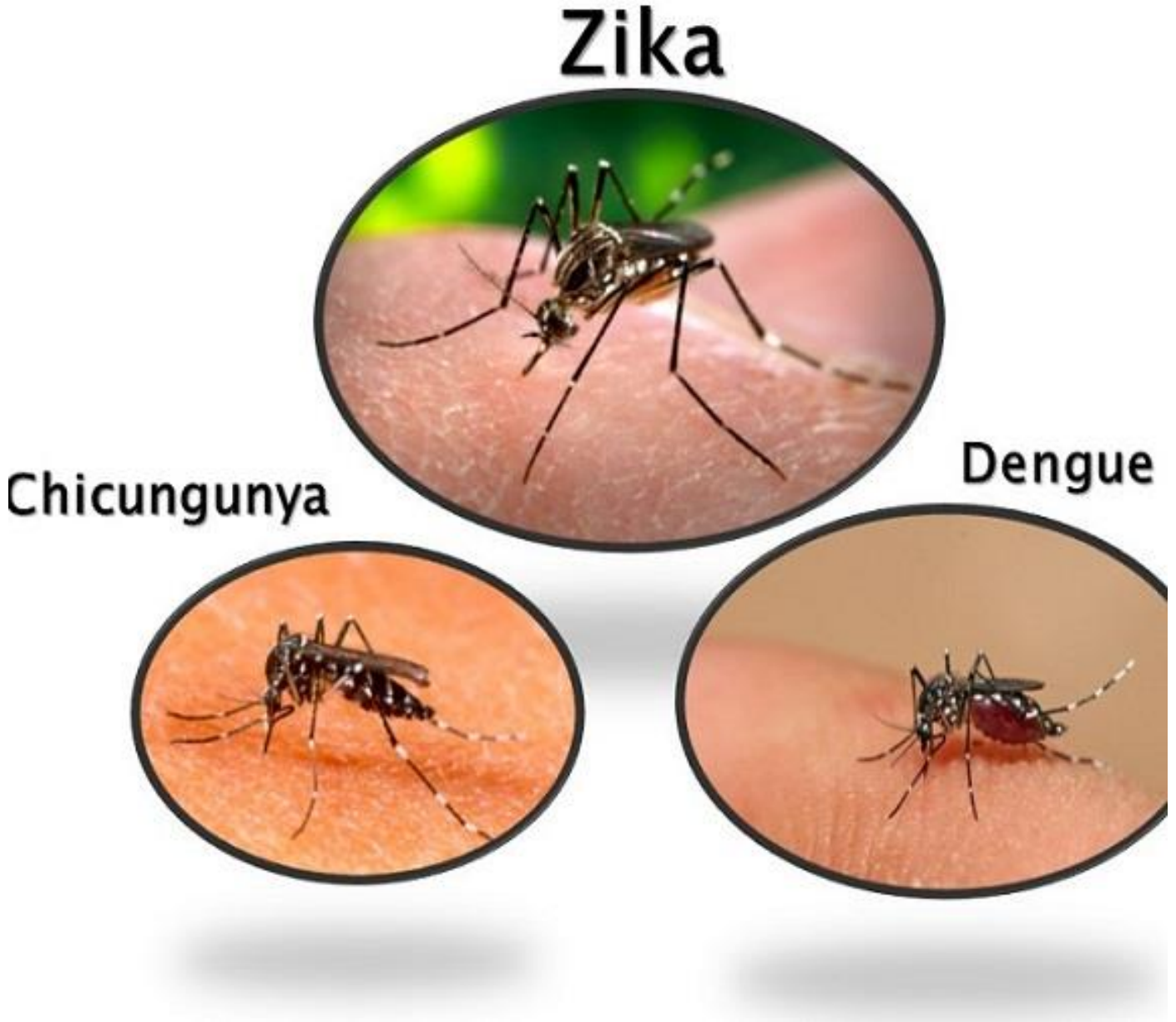


Generation	miR	Target Site	Reference Sequence Position			
			Den-1_Myanmar 3886201	Den-2_New Guinea-C	Den-3_98TW503	Den-4_H241
DENV-8miR	miR-4 (8-1miR)	TAGAGAGCAGATCTCTGATGAA	5'UTR	5'UTR	No	No
	miR-3 (8-2miR)	TATGCTGAAACGCGAGAGAAAC	No	Capsid	No	Capsid
	miR-1 (8-3miR)	CGTCTTTCAATATGCTGAAACG	Capsid	No	No	No
	miR-2 (8-1miR)	CGCGAGAAACCGCGTGTCAACT	Capsid	No	No	No
	miR-5 (8-4miR)	CGTCTATCAATATGCTGAAACG	No	No	Capsid	No
	miR-6 (8-5miR)	CGTGAGAAACCGTGTGTCAACT	No	No	Capsid	No
	miR-7 (8-6miR)	AGCAAACCGTGCTGCCTGTAGC	3'UTR	No	3'UTR	3'UTR
	miR-8 (8-7miR)	CACAGTTGGCGAAGAGATTCTC	Capsid	No	Capsid	No

The anti-DENV phenotype of the all in one transgenic mosquito



Aedes are the major vector of these three virus





Viral Bioinformatics Resource Center

Virus Selector: Togaviridae | Manage
User Name: Guest | Login

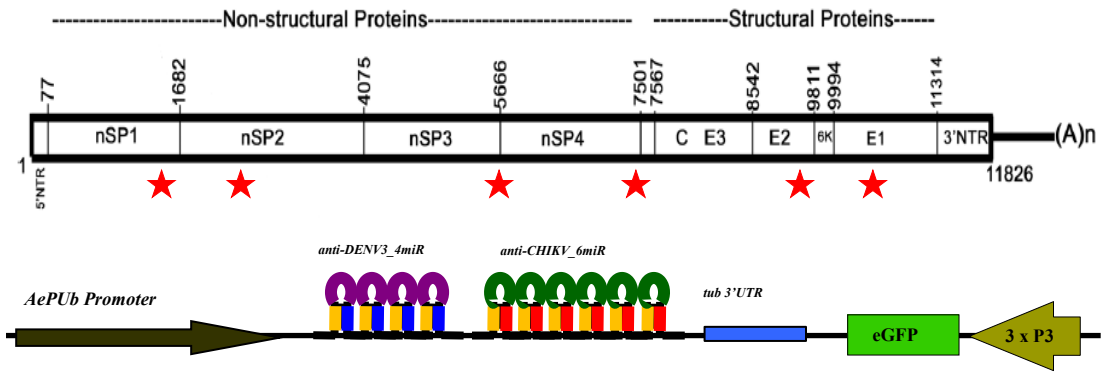
home | data | analysis | information | about | contact | ID Search | Search

Download genome(s) in FASTA as Nucleotide Sequence in new browser (Help)

Family:	*Togaviridae*	linkout	(1 genus)
Genus:	*Alphavirus*	linkout	(14 species)
Species:	*Aura virus*		(1 isolate)
Species:	*Barmah Forest virus*		(1 isolate)
Species:	*Chikungunya virus*		(32 isolates)
Species:	*Eastern equine encephalitis virus*		(11 isolates)
Species:	*Getah virus*		(6 isolates)
Species:	*Mayaro virus*		(2 isolates)
Species:	*Middelburg virus*		(1 isolate)
Species:	*O'nyong-nyong virus*		(3 isolates)
Species:	*Ross River virus*		(4 isolates)
Species:	*Salmon pancreas disease virus*		(3 isolates)
Species:	*Semliki forest virus*		(7 isolates)
Species:	*Sindbis virus*	linkout	(7 isolates)
Species:	Unclassified Alphavirus		(4 isolates)
Species:	*Venezuelan equine encephalitis virus*		(37 isolates)
Species:	*Western equine encephalitis virus*		(1 isolate)

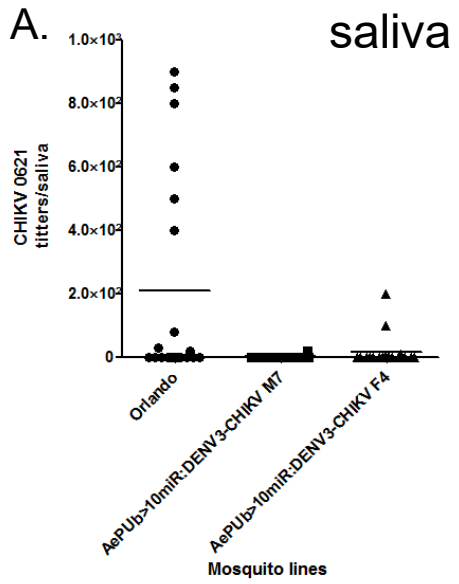
Download genome(s) in FASTA as Nucleotide Sequence in new browser (Help)

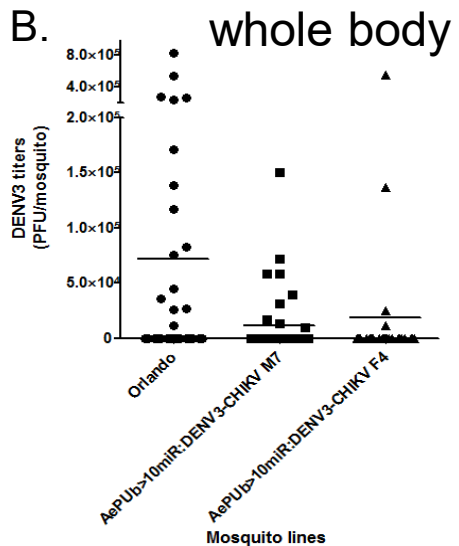
The efficacy of anti-DENV3-CHIKV mosquito



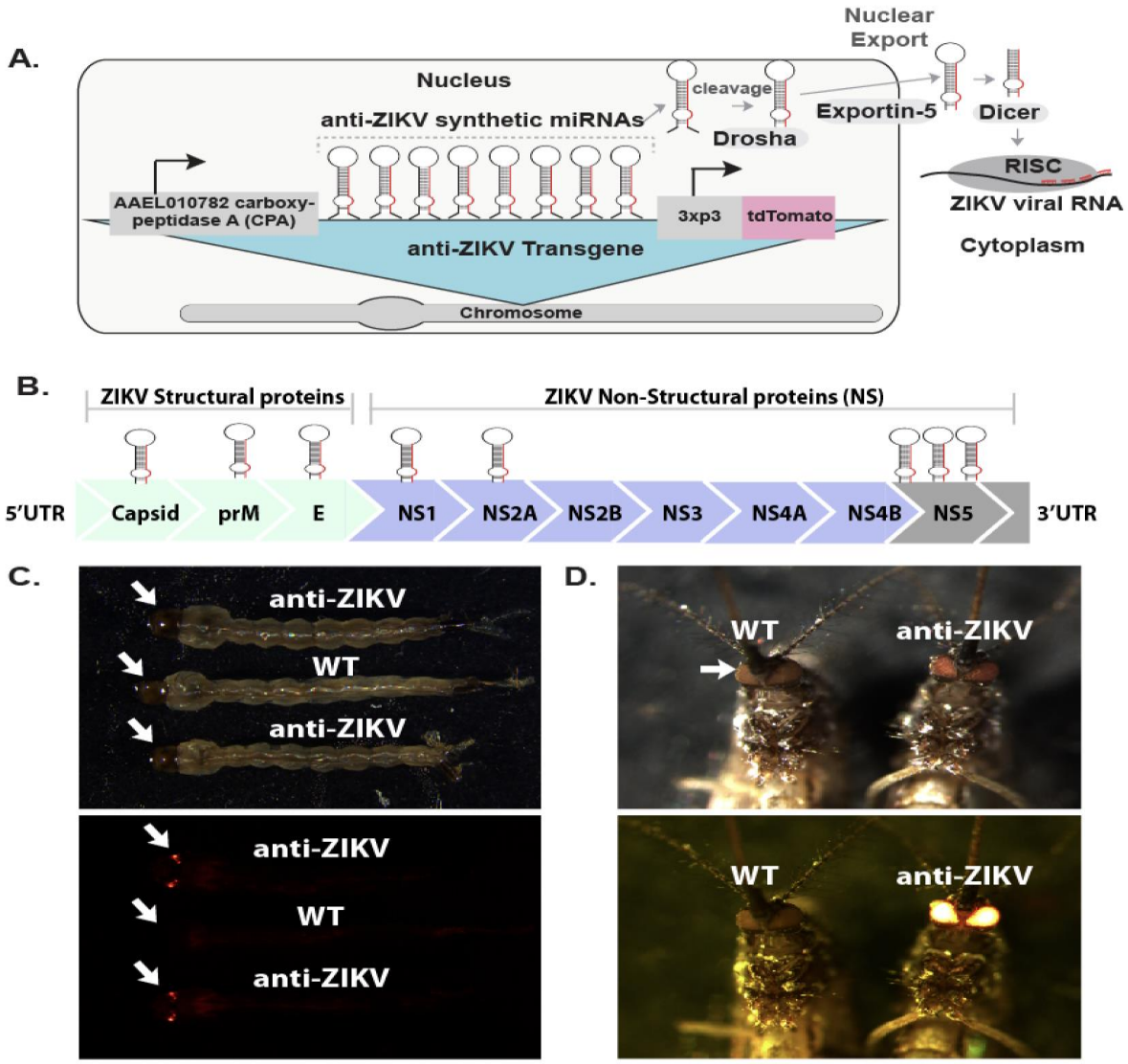
	miRNA	target sequence	coverage CHIKV (32)	targeting site
anti-CHIKV miRNAs	CHIKV-1★	AGAACGAGAAGCAGAACTGACT	31/32 (96.9%)	nSP1
	CHIKV-2★	AAAATGCGCACTACGAATGAGT	31/32 (96.9%)	nSP2
	CHIKV-3★	AGACAGGGCAGGTGGGTATATA	31/32 (96.9%)	nSP3-nSP4
	CHIKV-4★	ACATGGAAGTGAAGATCATAGA	32/32 (100%)	nSP4
	CHIKV-5★	GAACCAAACTATCAAGAAGAGT	31/32 (96.9%)	E2
	CHIKV-6★	AGAAGTCCGAATCATGCAAAAC	31/32 (96.9%)	E1

A pool of three biological replicates is shown, and p values were determined using the Mann-Whitney U test (*, $p<0.05$; **, $p<0.005$).

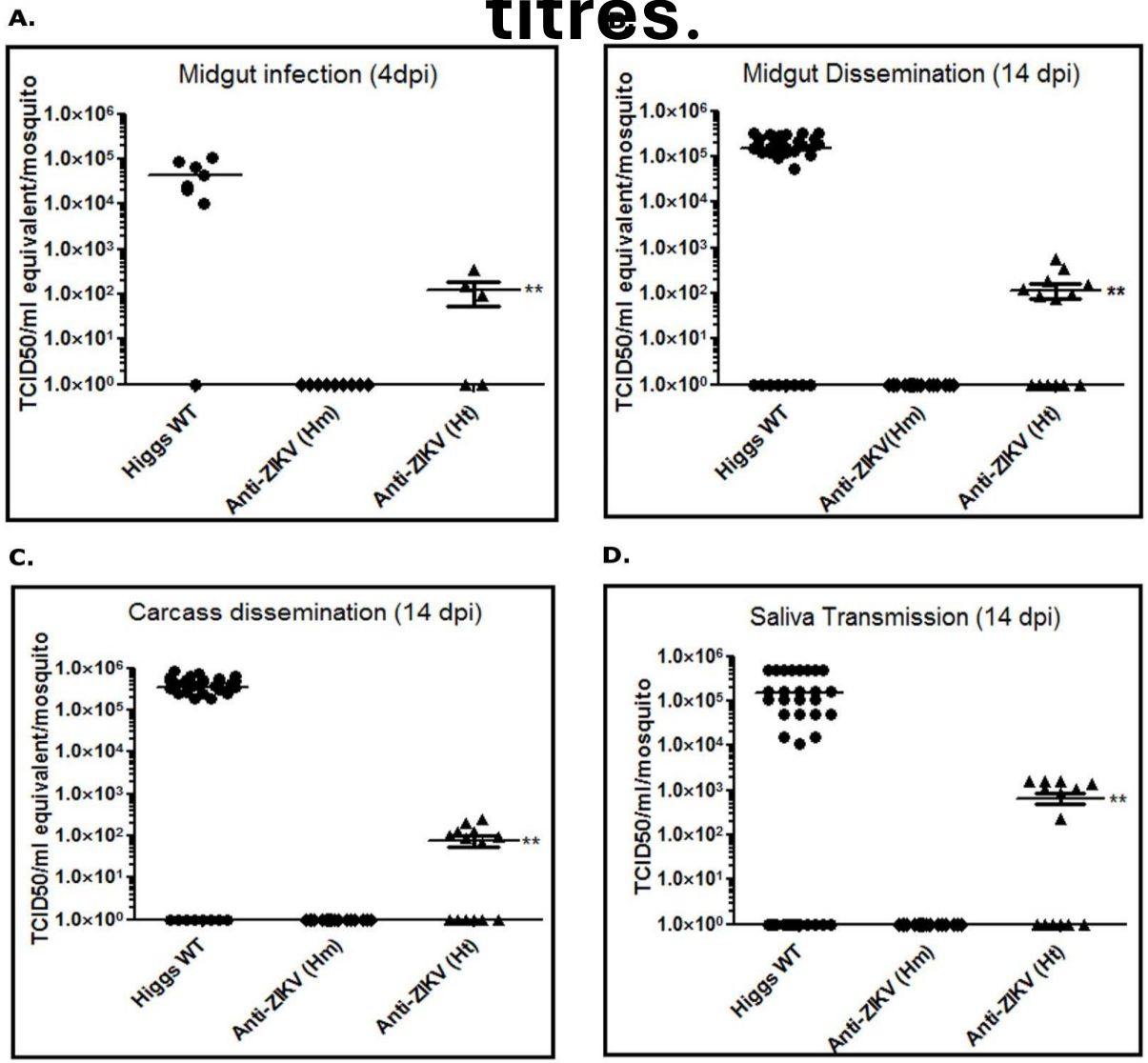




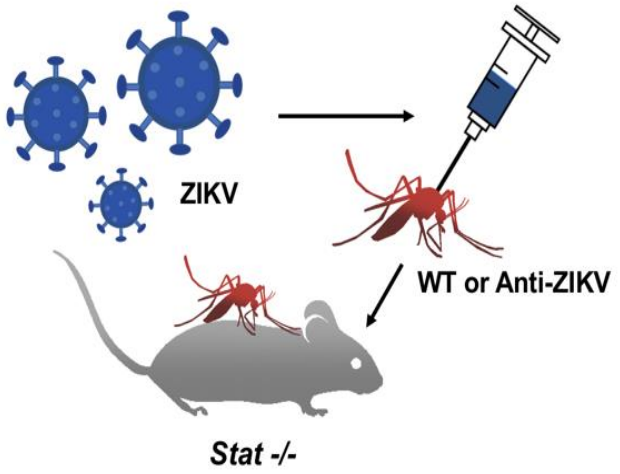
Schematic of anti-ZIKV effector cassette.

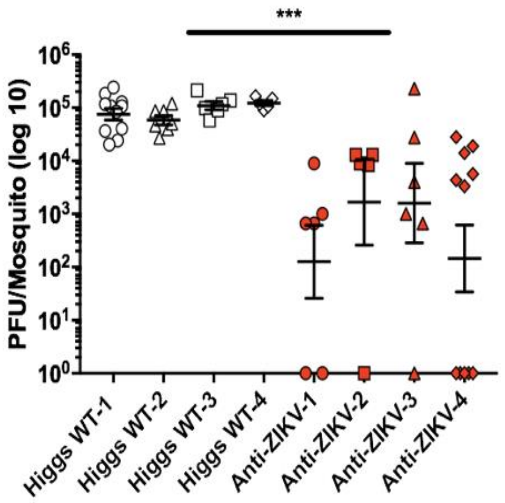


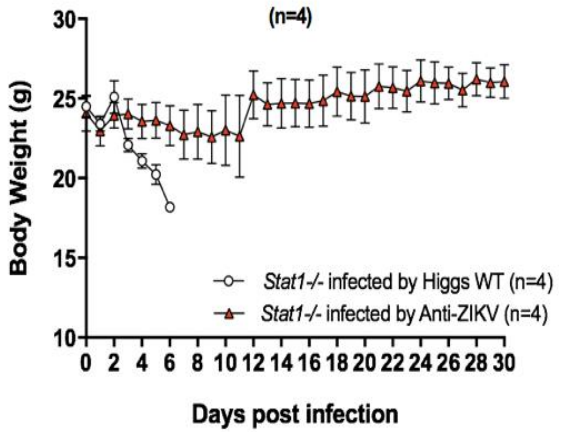
Effect of anti-ZIKV transgene on ZIKV titres.

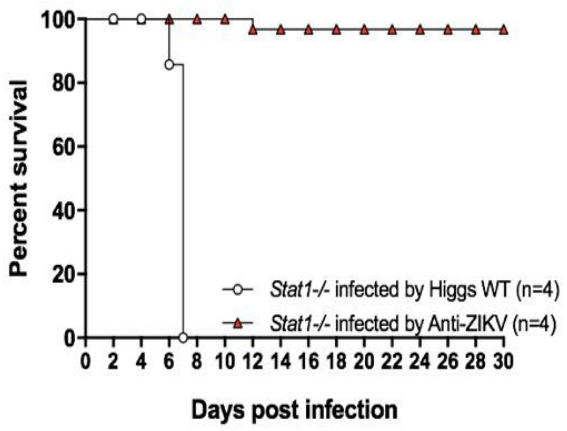


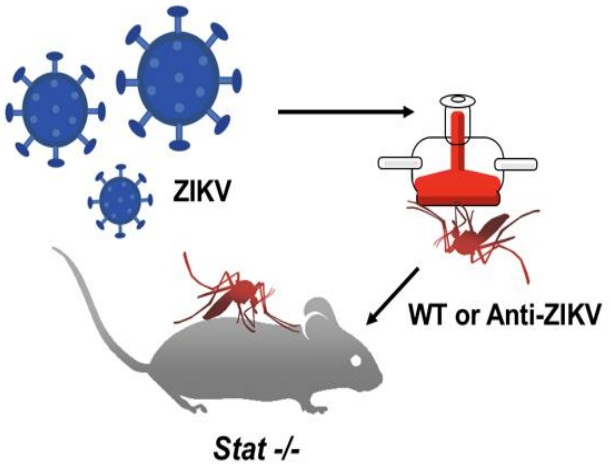
Anti-Zika mosquito prevent viral transmition

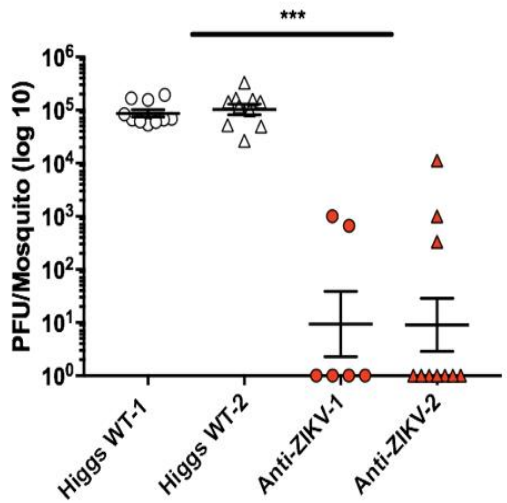


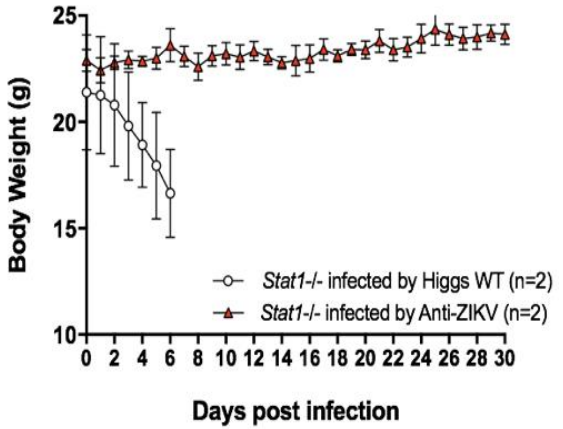


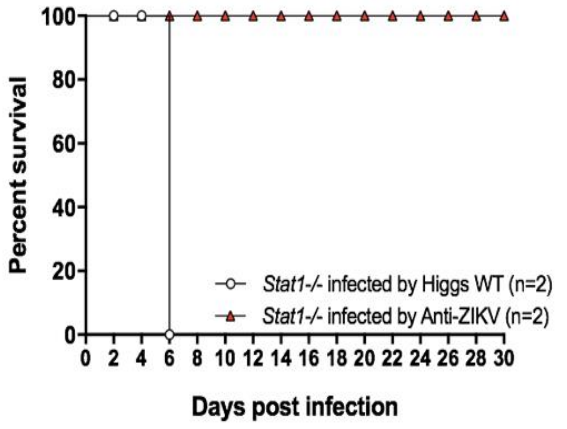










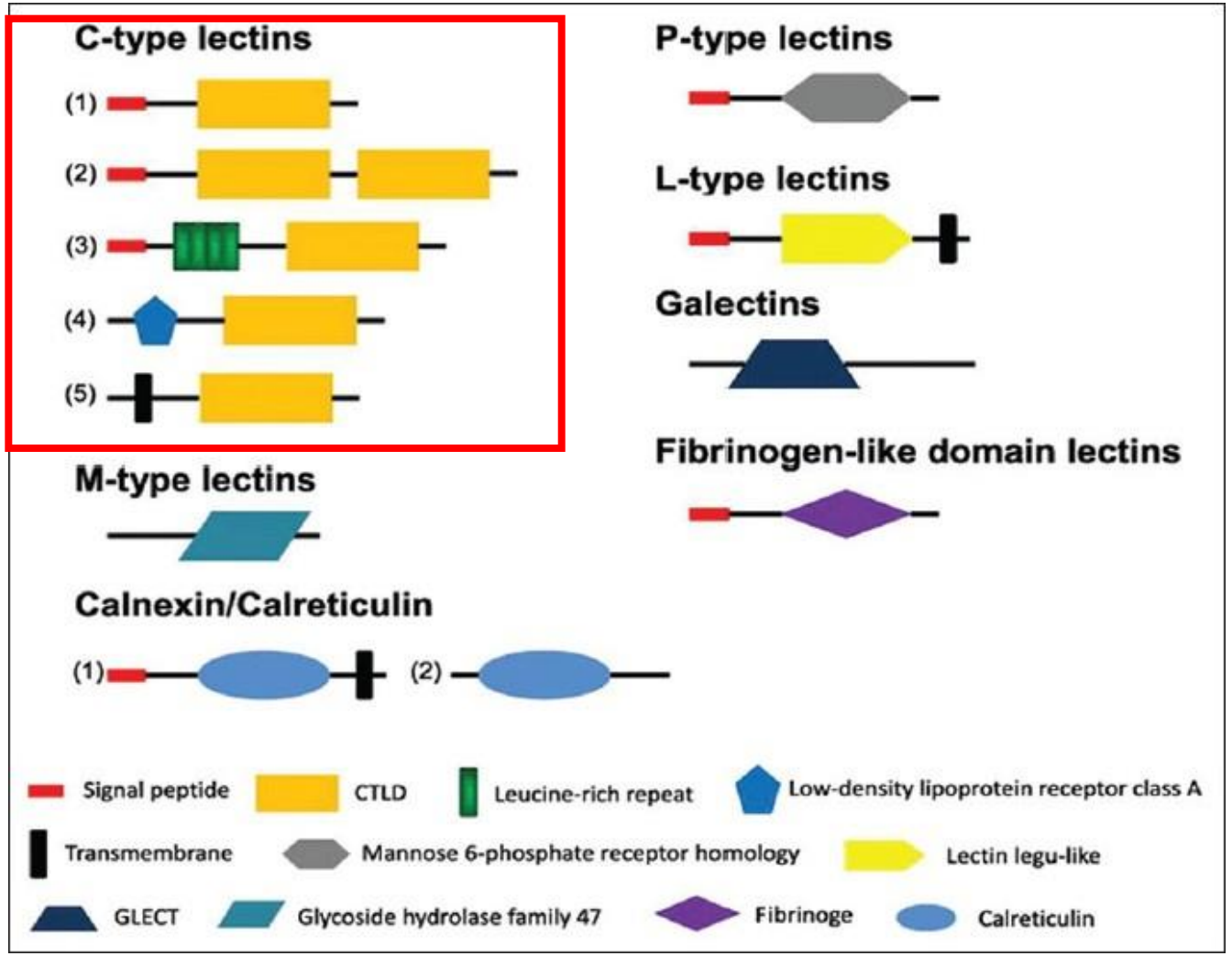


Two effector genes

1 Targeting Dengue genomes or mosquito genes

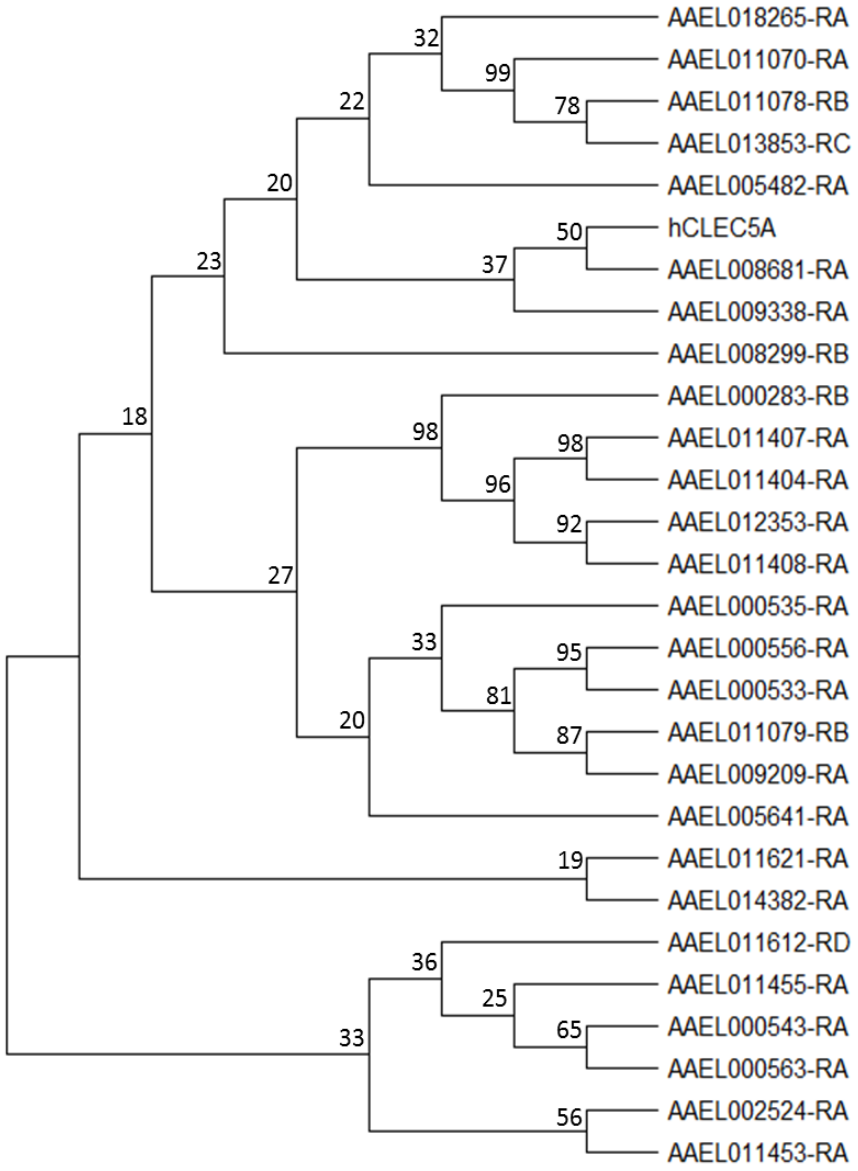
2 Altering Host seeking behavior

C Type Lectin involved in innate immunity



C-type lectins with carbohydrate binding activity
immune activation , viral pathogenesis

There are 27 *Aedes aegypti* C-type lectin proteins in the genome



OPEN ACCESS Freely available online

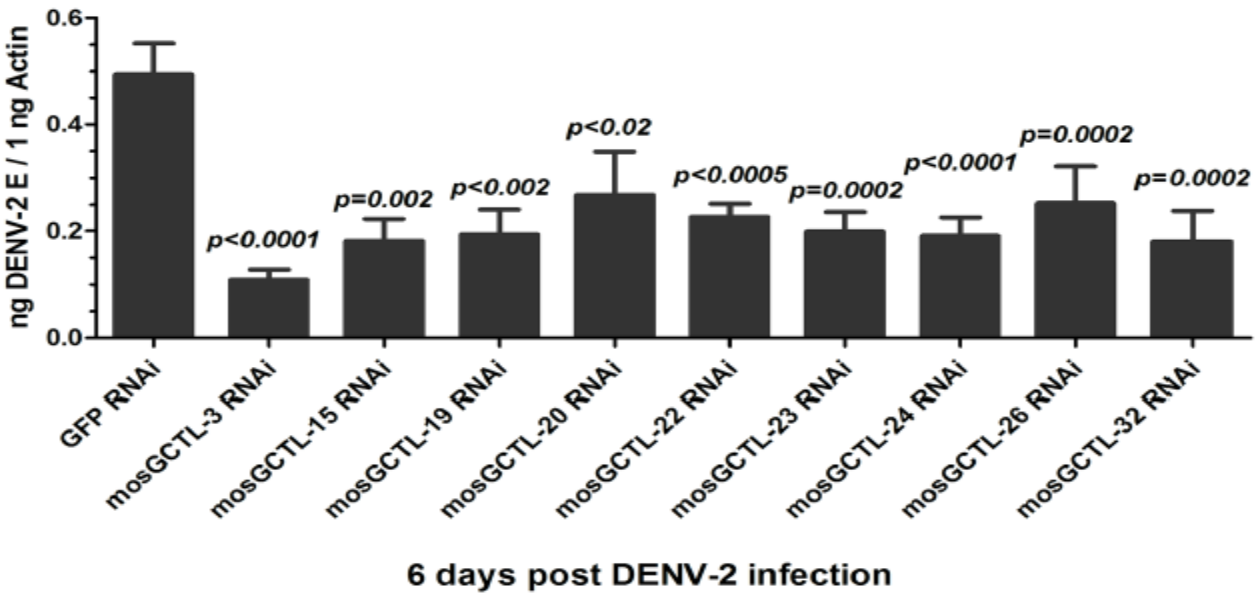
PLOS | PATHOGENS

Transmission-Blocking Antibodies against Mosquito C-Type Lectins for Dengue Prevention

Yang Liu[1], Fuchun Zhang[2], Jianying Liu[1], Xiaoping Xiao[1], Siyin Zhang[1], Chengfeng Qin[3], Ye Xiang[1], Penghua Wang[4], Gong Cheng[1,5]*

1 Department of Basic Medical Sciences, School of Medicine, Tsinghua University, Beijing, People's Republic of China, 2 Guangzhou 8th People's Hospital, Guangzhou, People's Republic of China, 3 State Key Laboratory of Pathogen and Biosecurity, Beijing Institute of Microbiology and Epidemiology, Beijing, People's Republic of China, 4 Section of Infectious Diseases, Yale University School of Medicine, New Haven, Connecticut, United States of America, 5 Collaborative Innovation Center for Diagnosis and Treatment of Infectious Diseases, Hangzhou, People's Republic of China

2014

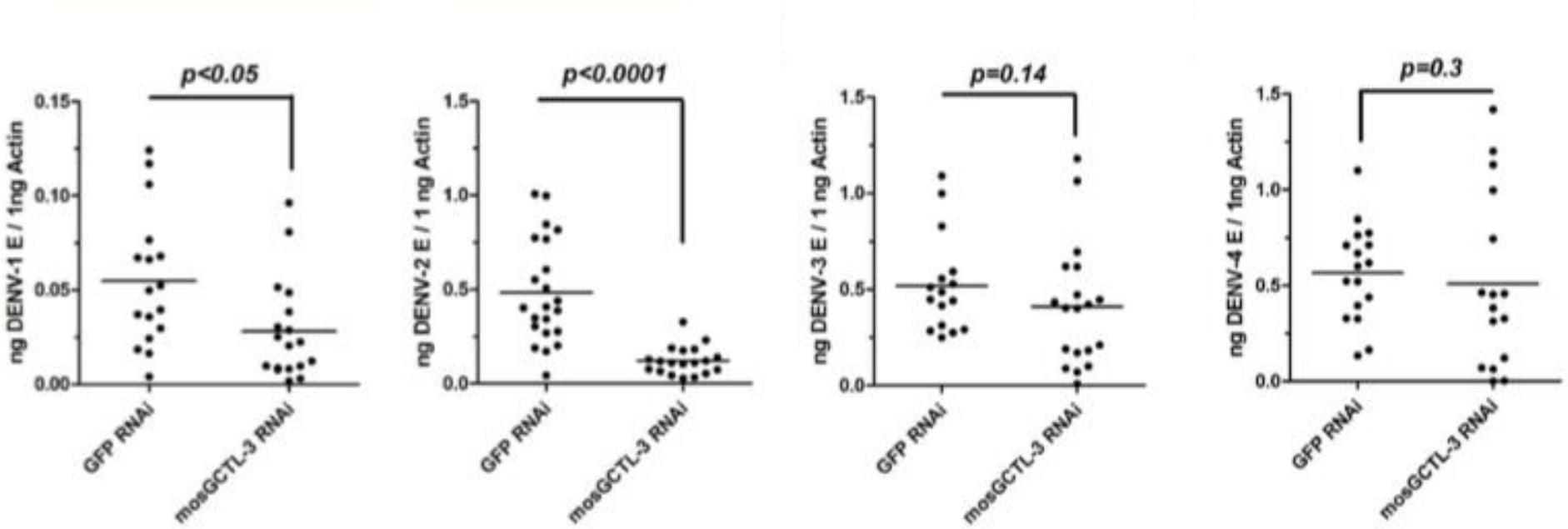




Gong Cheng
Tsinghua University

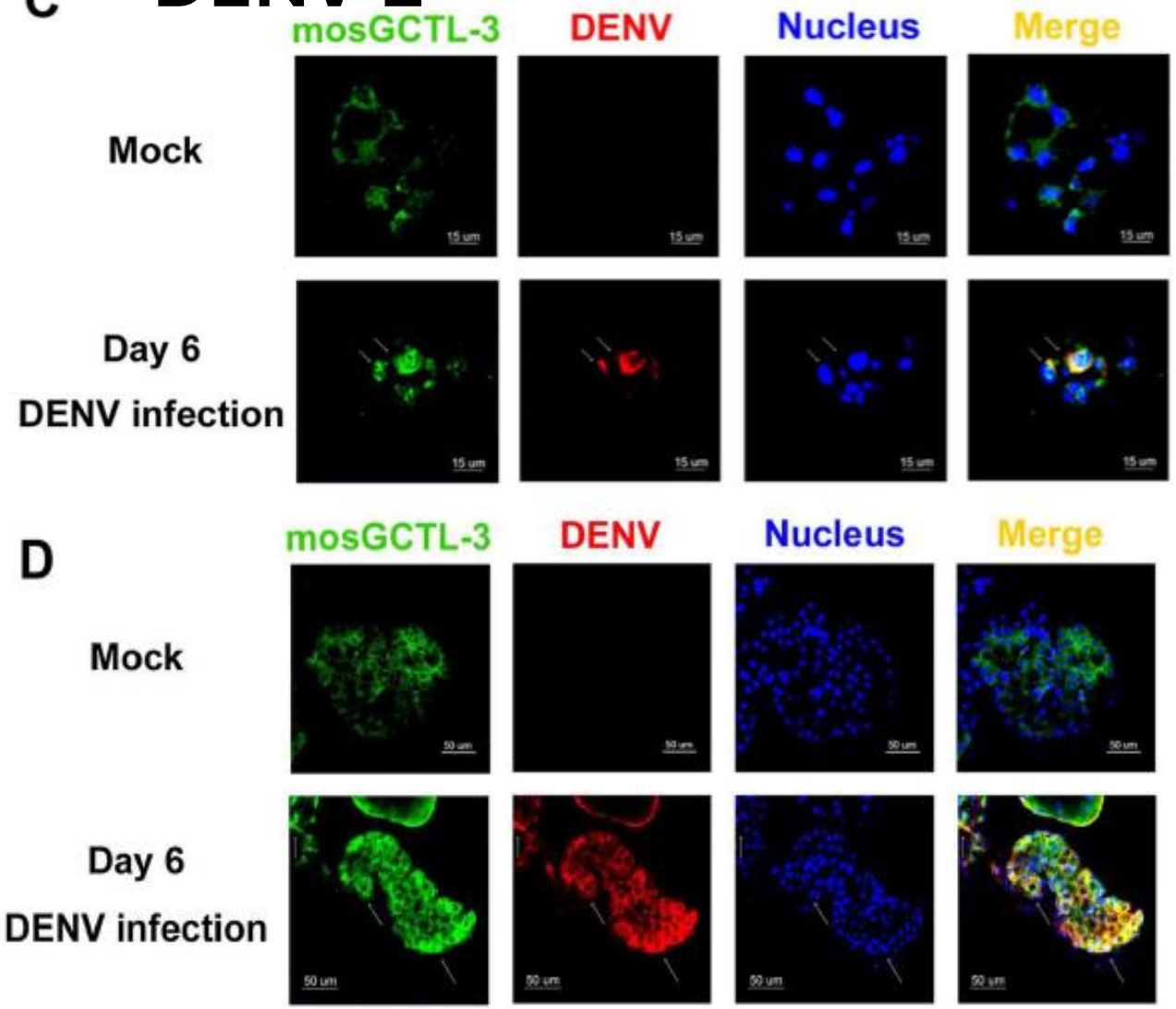
(Liu, Zhang et al. 2014)

Kock down mosGCTL-3 suppress the replication of four serotype Dengue virus



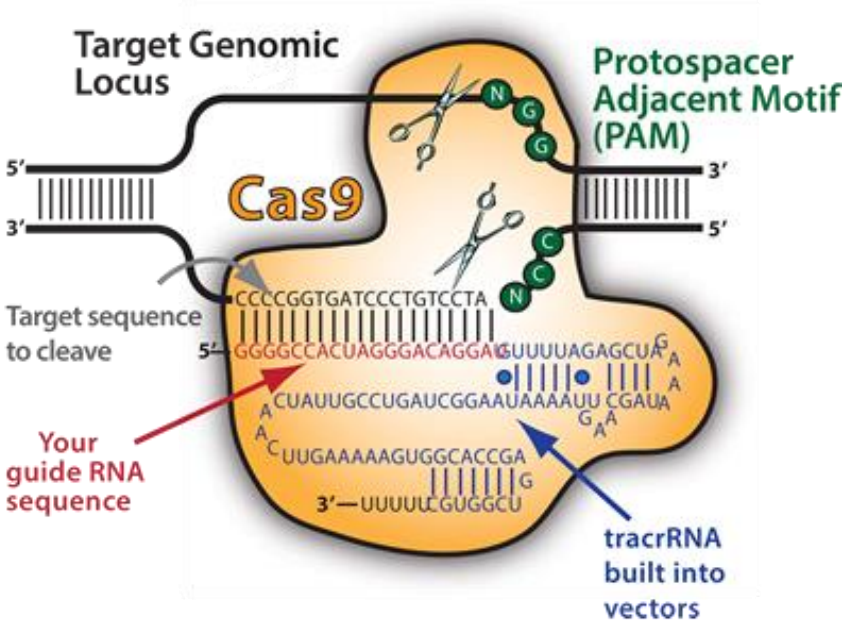
(Liu, Zhang et al. 2014)

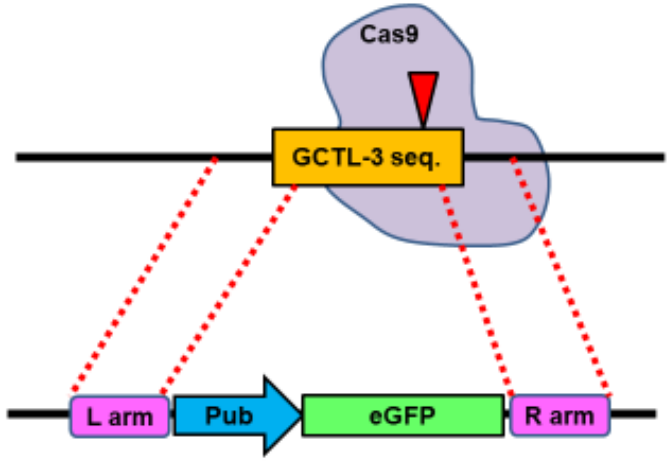
mosGCTL-3 could interact with DENV-2



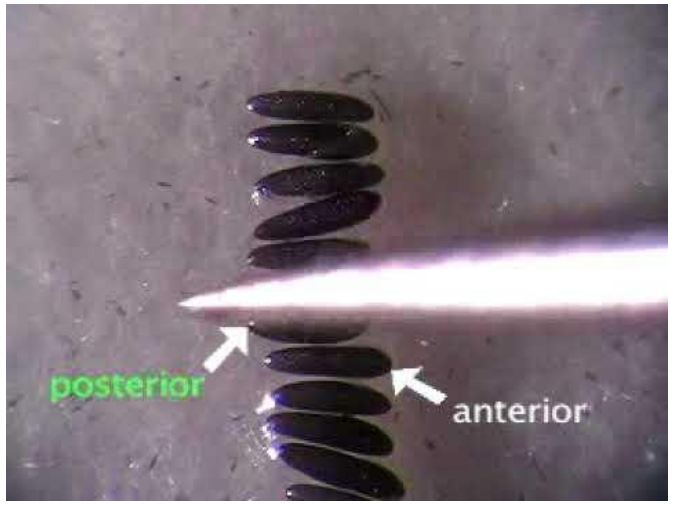
(Liu, Zhang et al. 2014)

Using CRISPR/cas9 method to create Aedes mutants

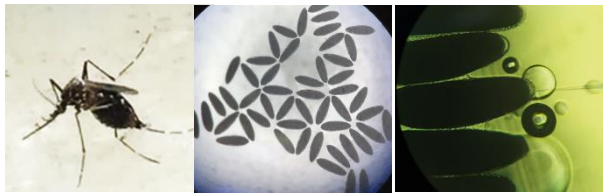




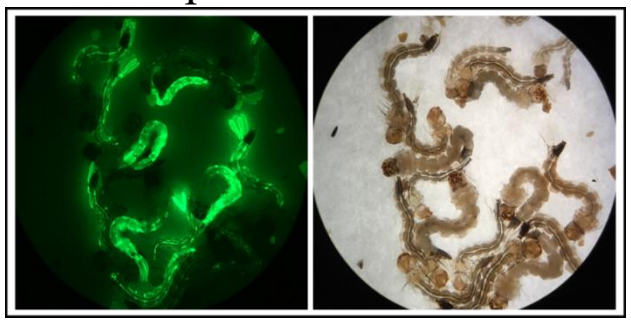




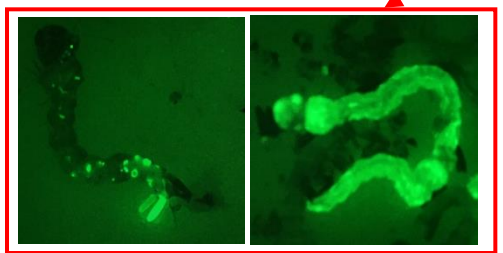
To generate the GCTL3 mutant by CRISPR/CAS9 method and screen by eGFP marker



Pub promoter>eGFP



Embryos	Larvae	% survival	Fluorescent	G0 adults	G1 adults	
					Visible fluorescent	Germline mutants
795	210	26%	168 (80%)	176	8 (4.5%)	2 (1.1%)
467	77	16.5%	49 (64%)	69	2 (2.9%)	1 (1.4%)

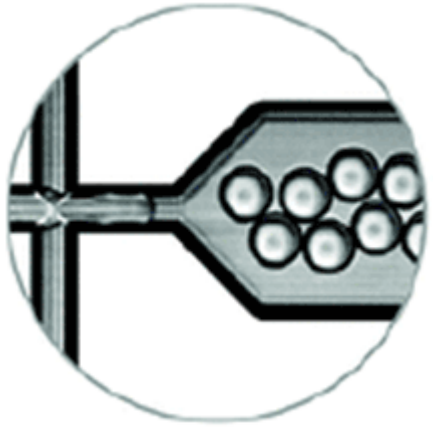


G2 without fluorescent

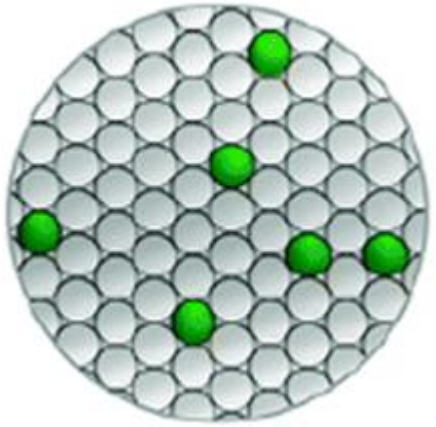
G2 with fluorescent

Droplet Digital PCR (ddPCR) Technology

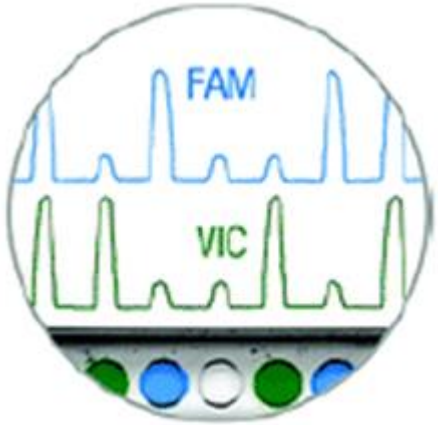
Partition

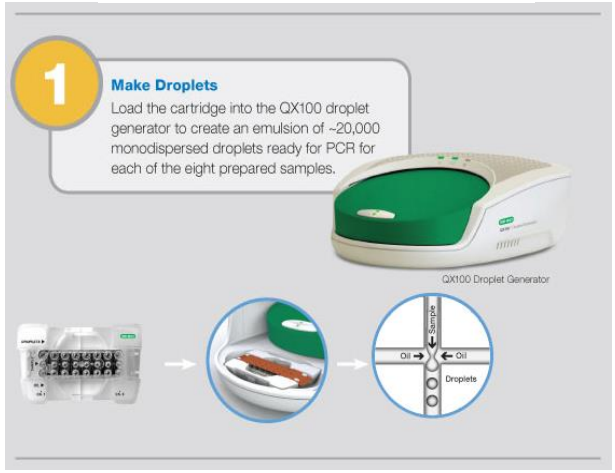


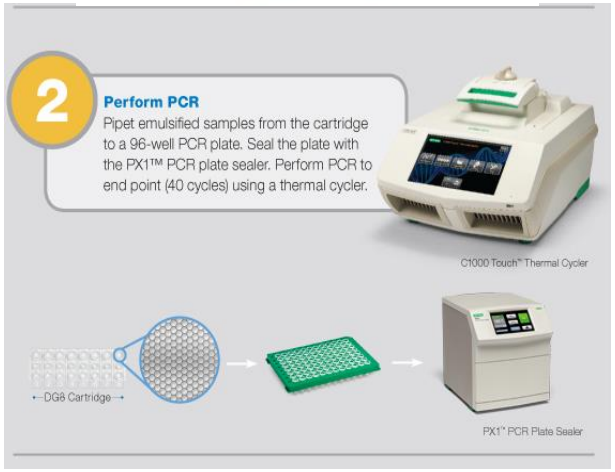
PCR amplification

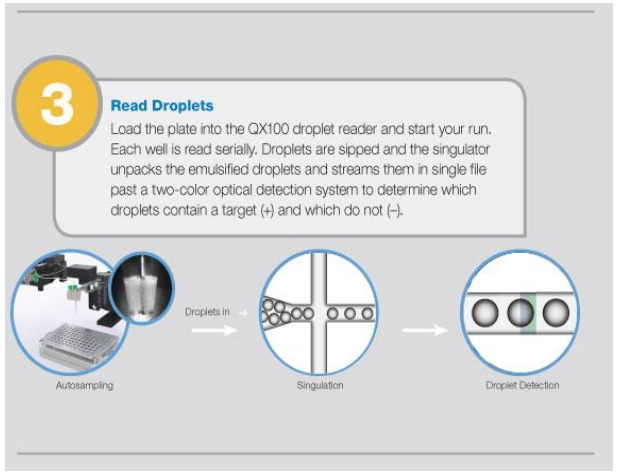


Detection

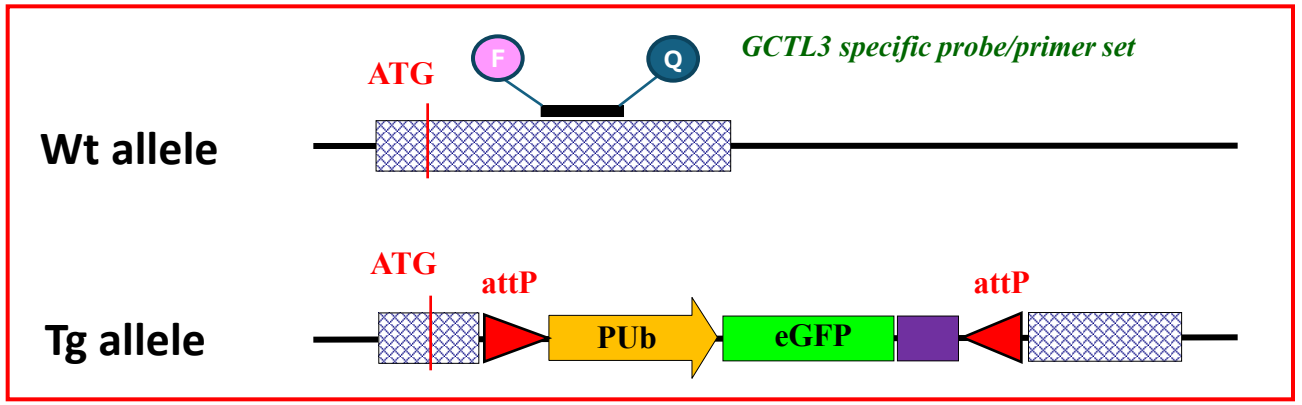


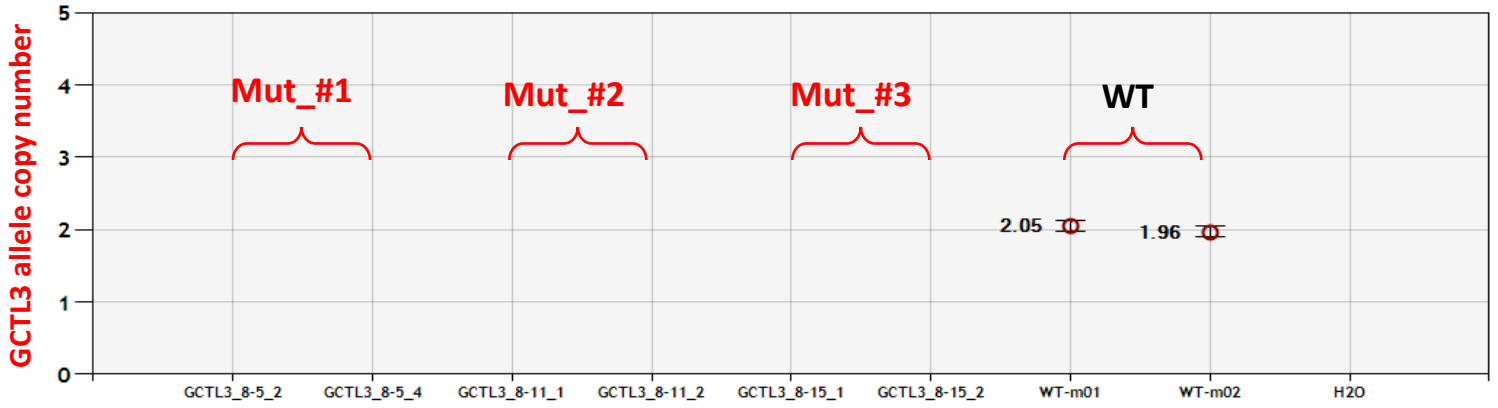


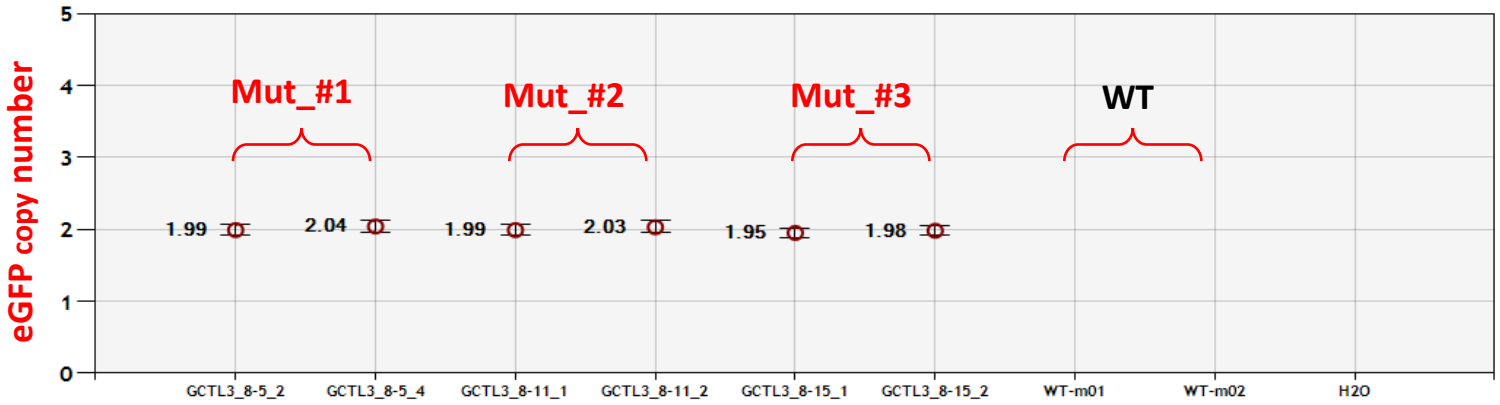




To confirm the GCTL-3 _ Homo mutant by ddPCR

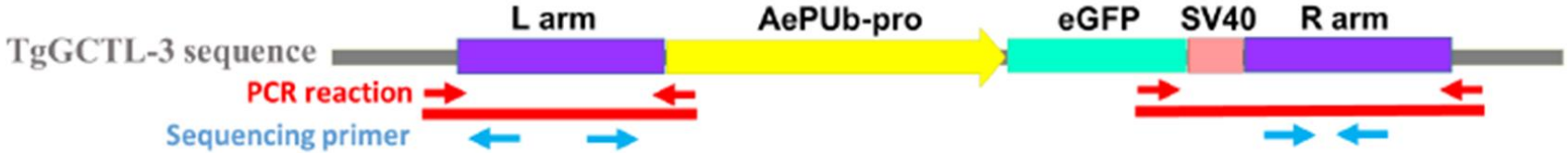




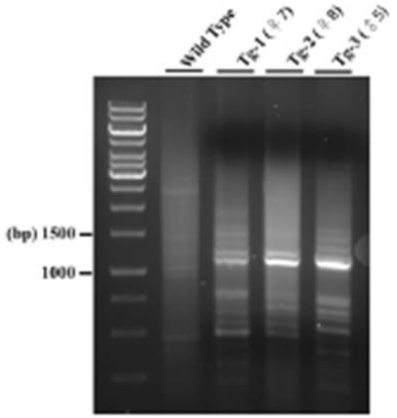


Genome mapping of GCTL-3 mutant mosquitoes

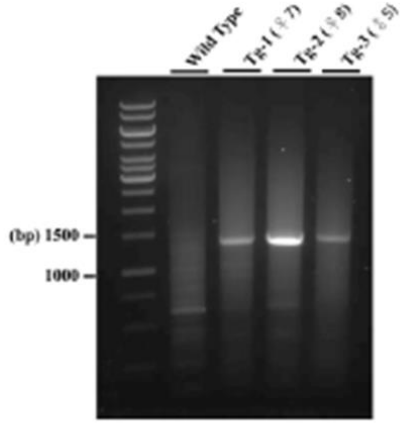
A



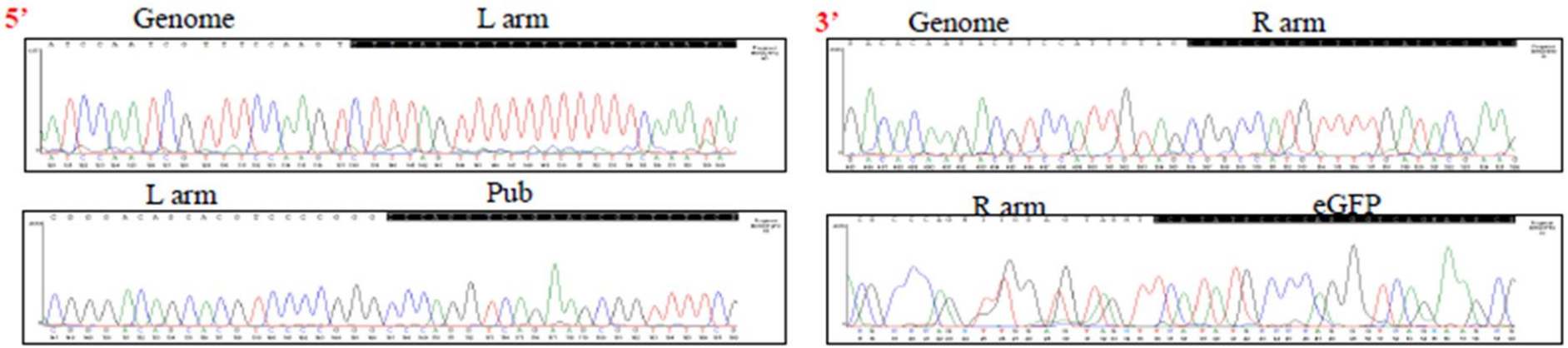
B



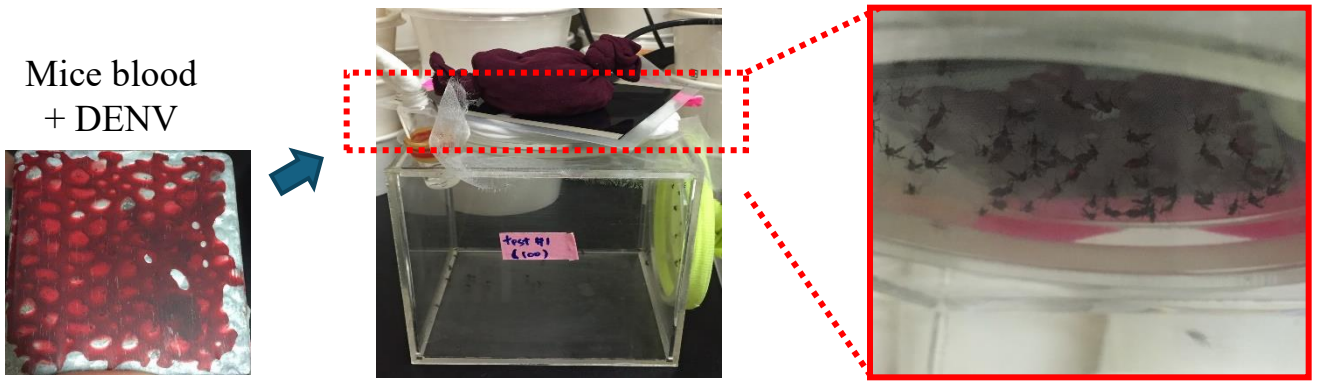
C

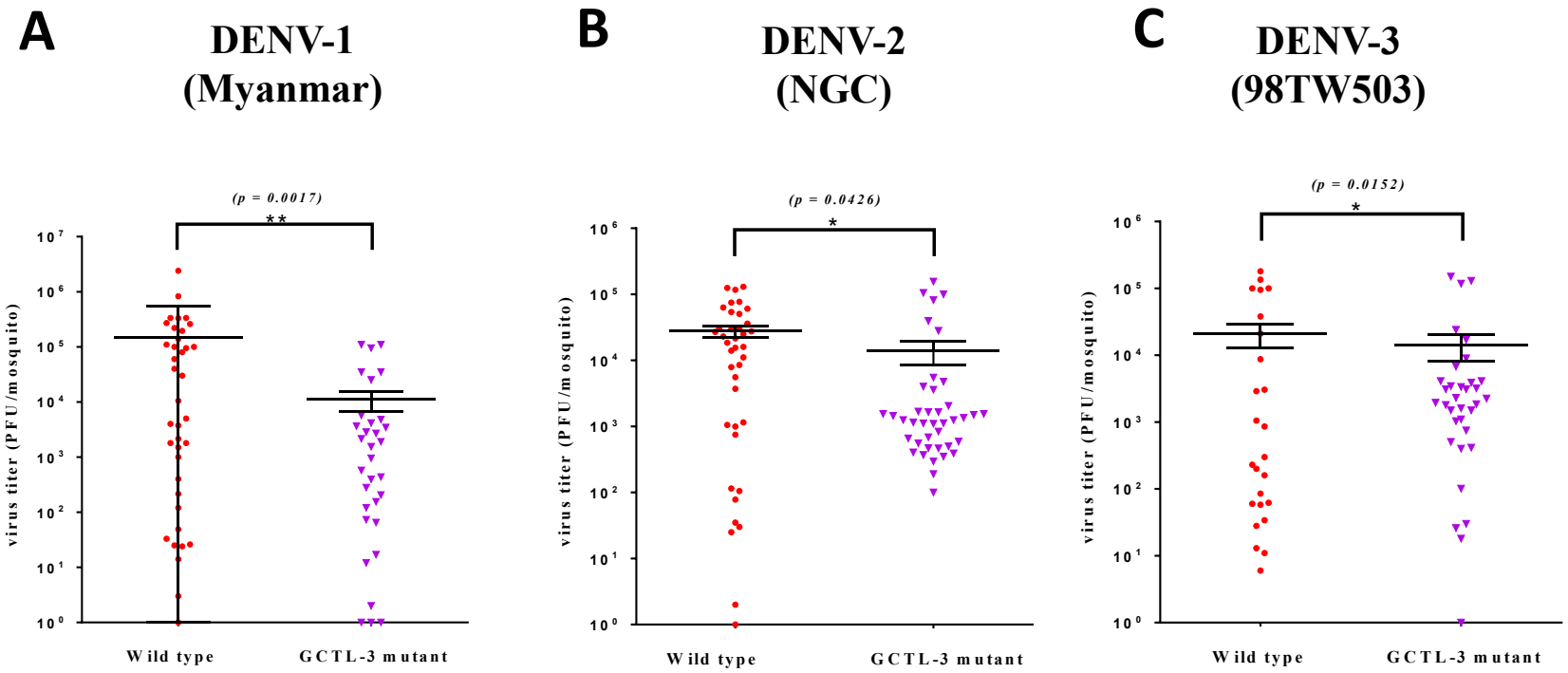


D



DENV viral load were suppressed in the GCTL-3 mutant mosquitoes at day 7 after post blood-meal



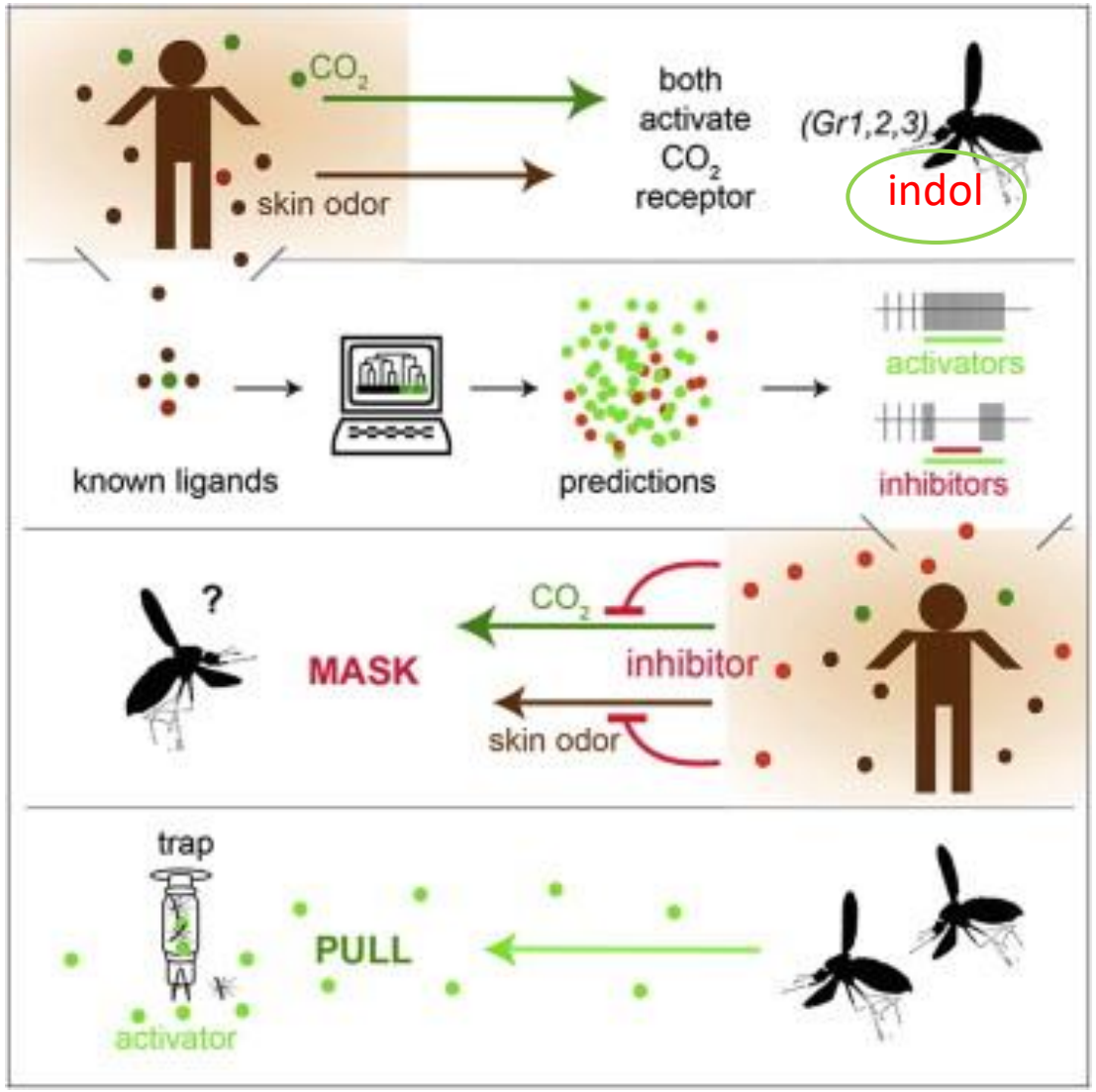


Two effector genes

1 Targeting Dengue genomes or mosquito genes

2 Altering Host seeking behavior

How mosquito detect people?



Indol occurs naturally in human feces and has an intense fecal odor.



Cell 2014

New Insights into the Mechanism of Odorant Detection by the Malaria-transmitting Mosquito *Anopheles gambiae*[*][S]

Received for publication, June 22, 2011, and in revised form, July 20, 2011 Published, JBC Papers in Press, August 4, 2011, DOI 10.1074/jbc.M111.274712

Foteini Davrazou[‡1], Emily Dong[‡1], Emma J. Murphy[‡], Hannah T. Johnson[‡], and David N. M. Jones[‡§2]

From the [‡]Department of Pharmacology and the [§]Program in Structural Biology and Biophysics, University of Colorado School of Medicine, Aurora, Colorado 80045

J Biol Chem. **2011**, 286(39):34175-83

(a). OBP4 has a higher affinity for indole than OBP1
(b). OBP1 and OBP4 co-localize within sensilla
(c). OBP4 can cooperatively increase the binding of a fluorescent dye (1-NPN) to OBP1 or OBP3

OBP1 and OBP4 are as important for perception of indole by the *Anopheles gambiae* mosquito

(Biessmann, H *et al*., PLoS ONE 5, **2010**; Qiao, H *et al*., *Cell. Mol. Life Sci.*, **2011**)

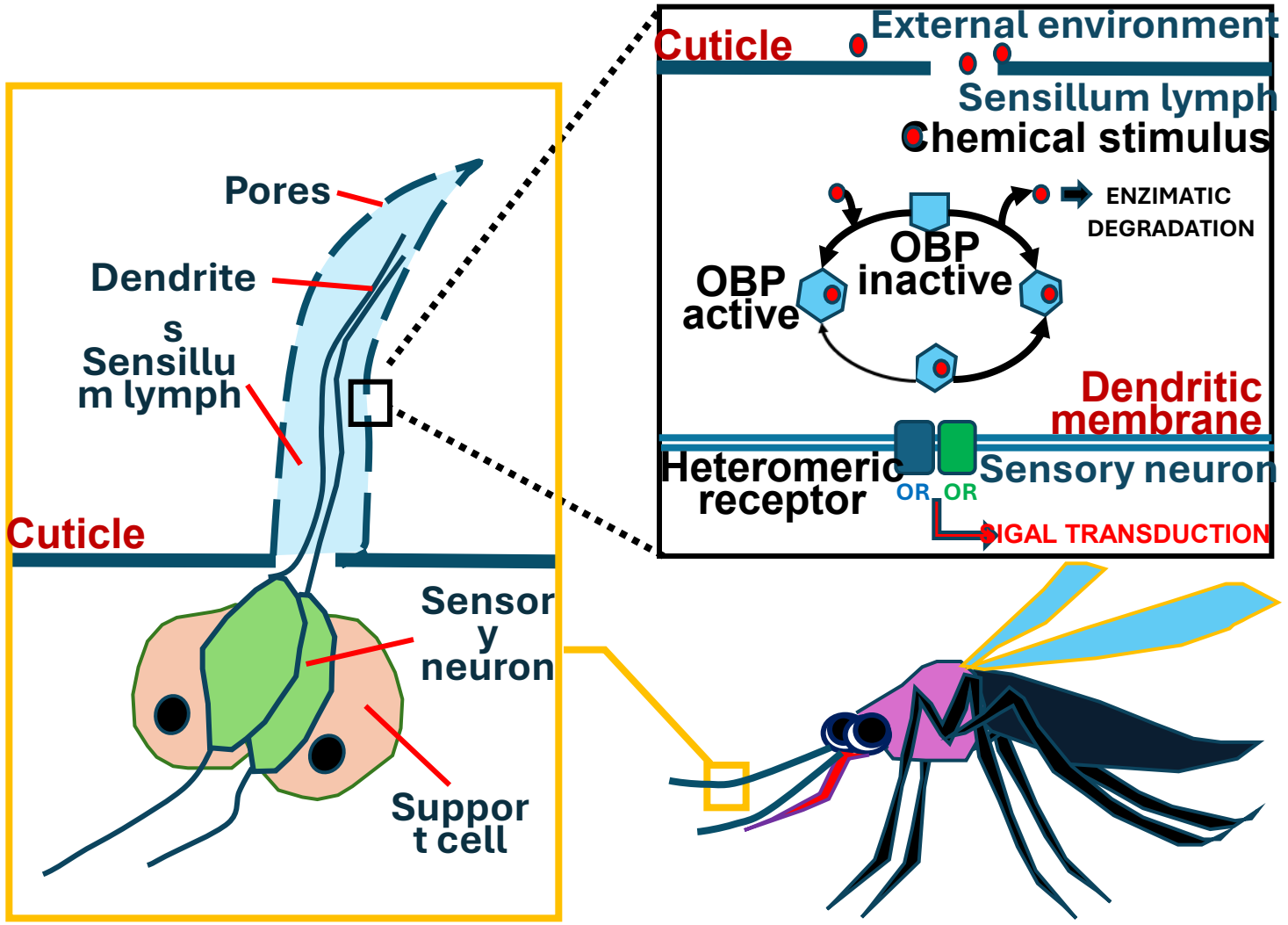
orthologous genes:

Agam_OBP1

- Aaeg_OBP39
- Aaeg_OBP56

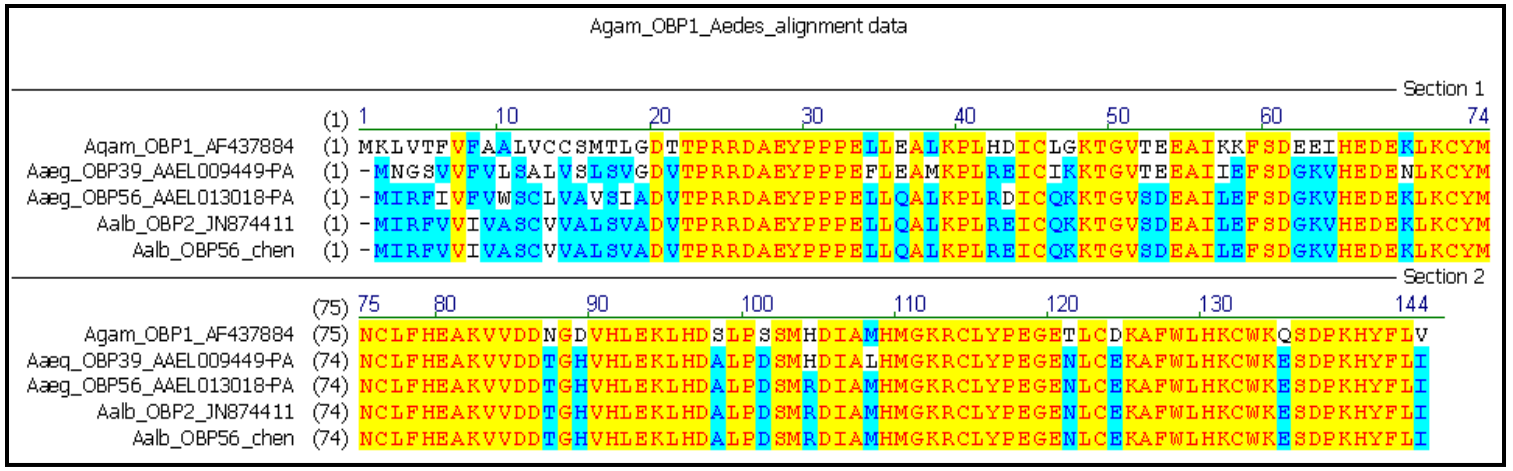
Agam_OBP4

- Aaeg_OBP1
- Aaeg_OBP34

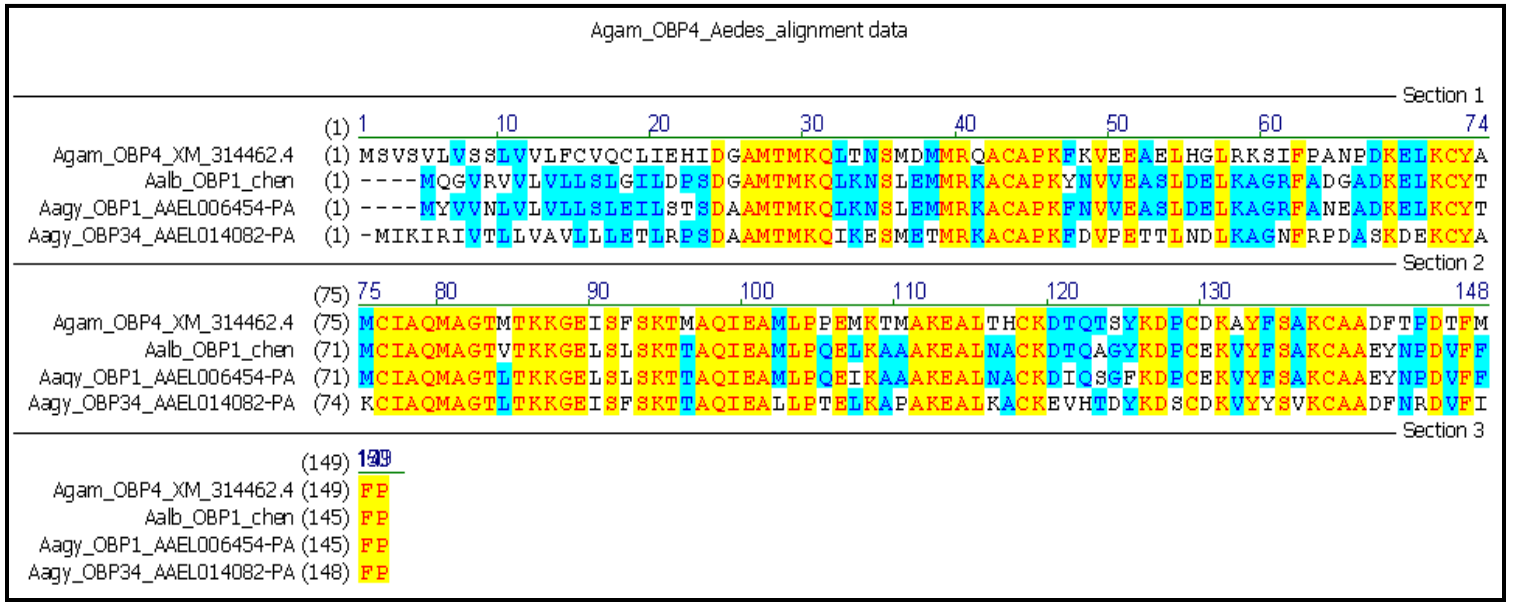


Alignment sequence between *Anopheles gambiae* and *Aedes aegypti*

A



B



(Done by Jian-Chiuan Li)

Screening of transgenic mosquitoes under the fluorescence microscopy

A

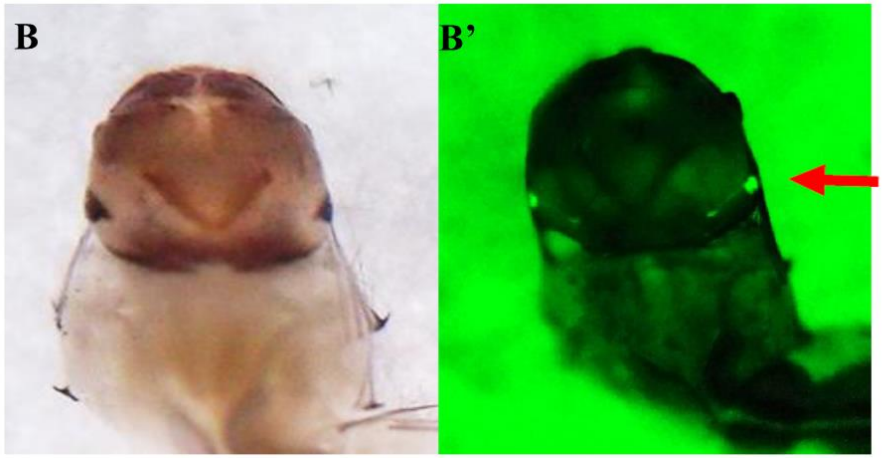
OBP34gRNA (vector)+HR-3xp3-eGFP+Cas9 protein

Construct marker	Strain	Hatch/ Injection Mount	Survival rate	Fluorescent (F0)	Remark
eGFP	Higgs	28/208	14 %	13/28 (46%)	♀1 (G1:N=2)

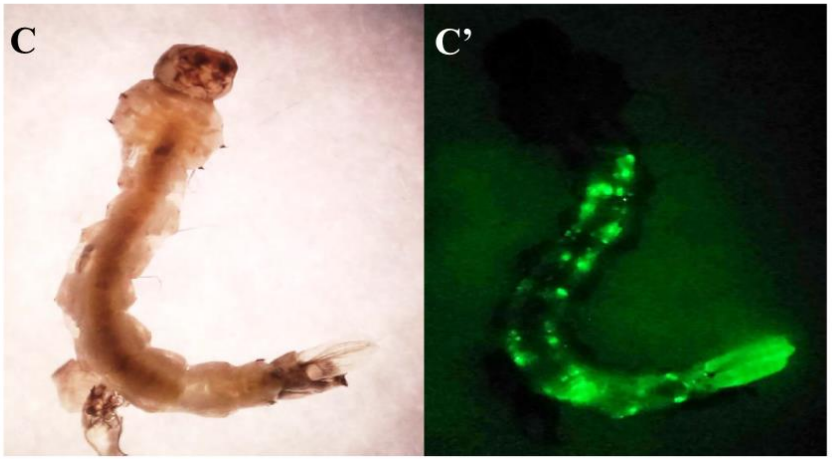
OBP39gRNA (RNA)+HR-Pub-eGFP+Cas9 protein

Construct marker	Strain	Hatch/ Injection Mount	Survival rate	Fluorescent (F0)	Remark
eGFP	Higgs	105/262	40 %	24/105 (23%)	♀3 (G1:N=41)

Cas9_OBP34_3xp3-eGFP

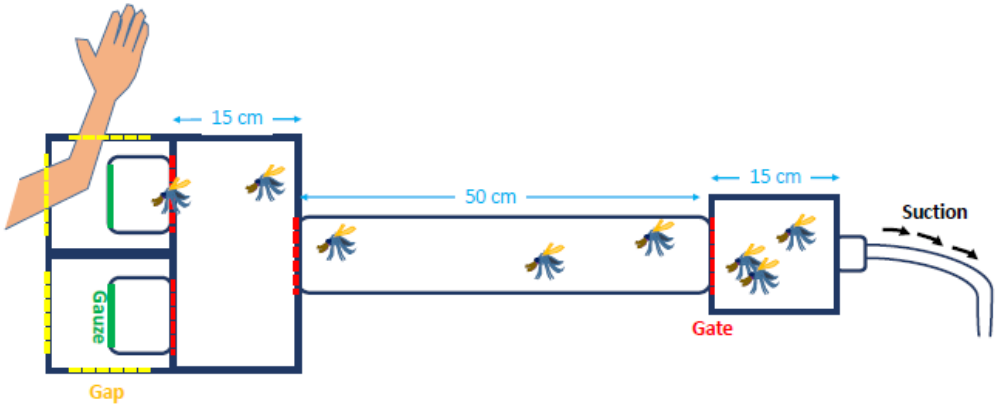


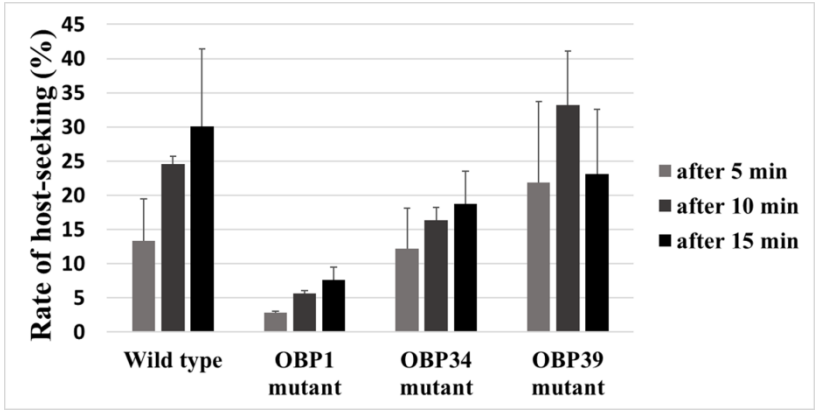
Cas9_OBP39_Pub-eGFP



Knockout Odorant-Binding Proteins affect host-seeking behavior in *Aedes* transgenic mosquitoes

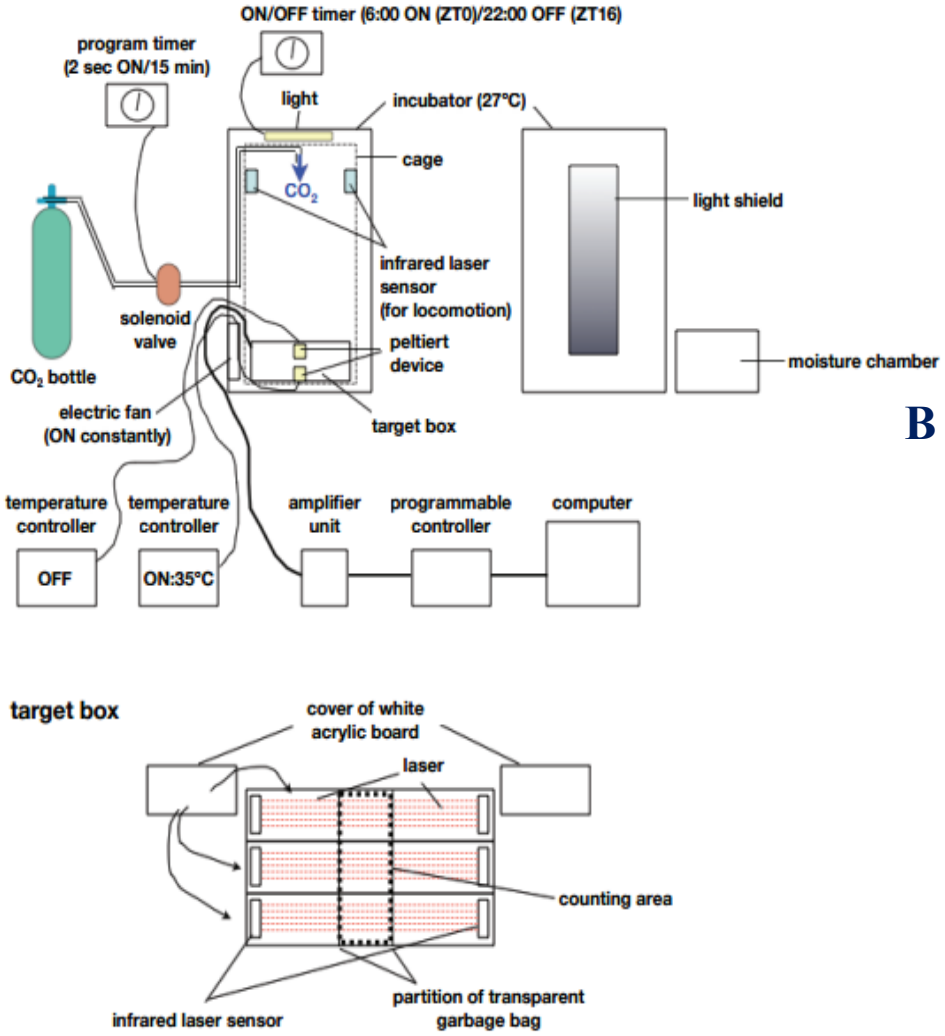




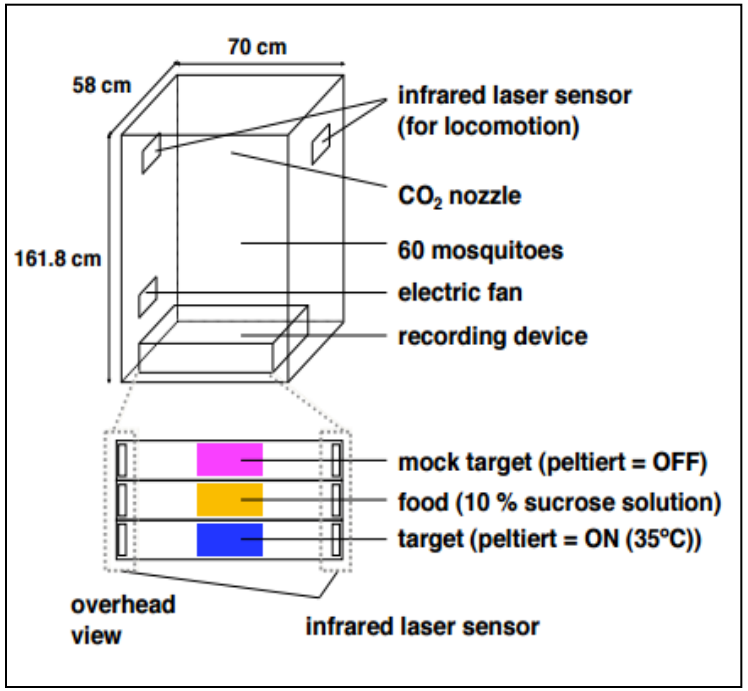


Automated-recording device for quantifying the selected host-seeking behavior

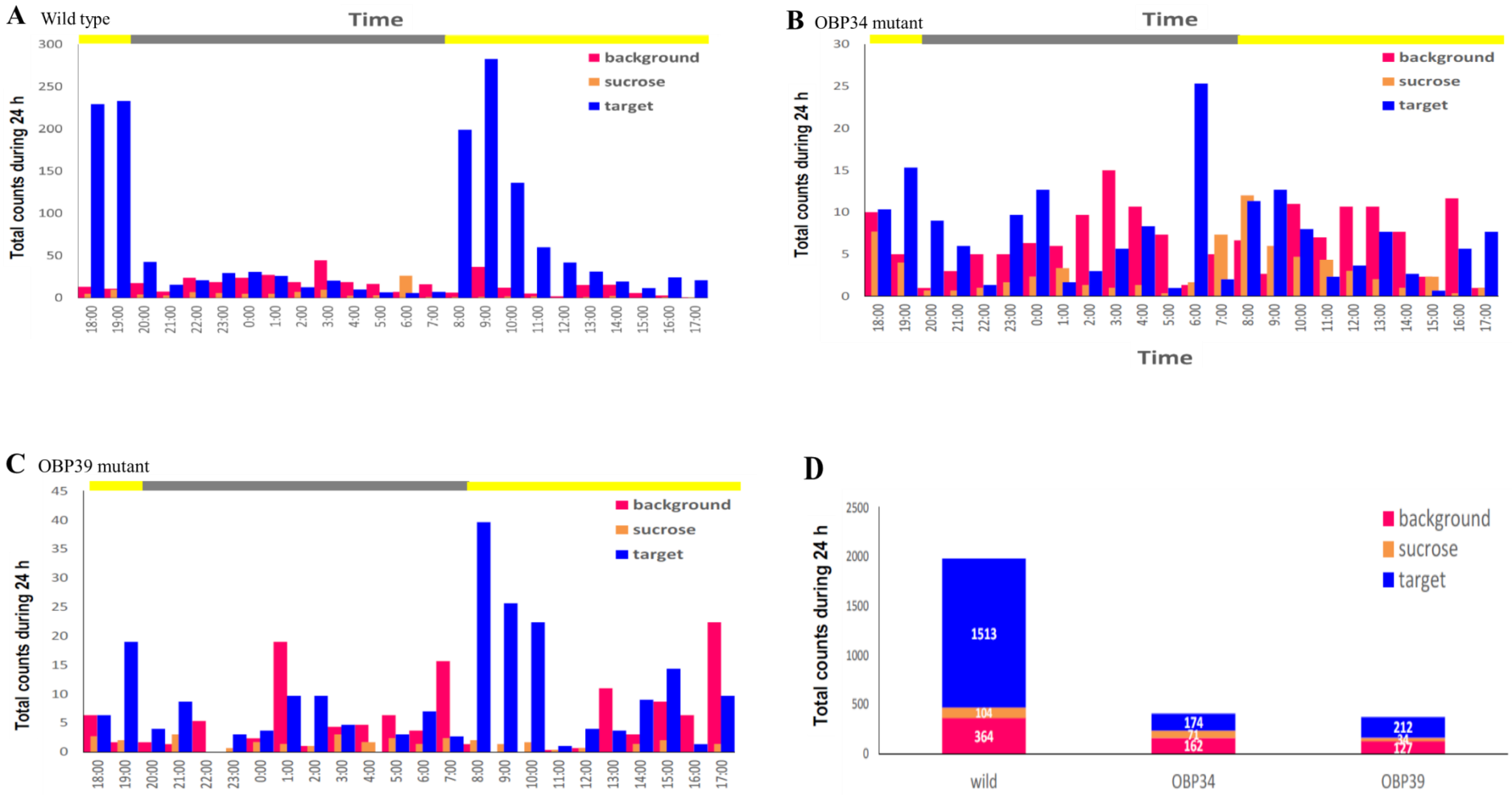
A

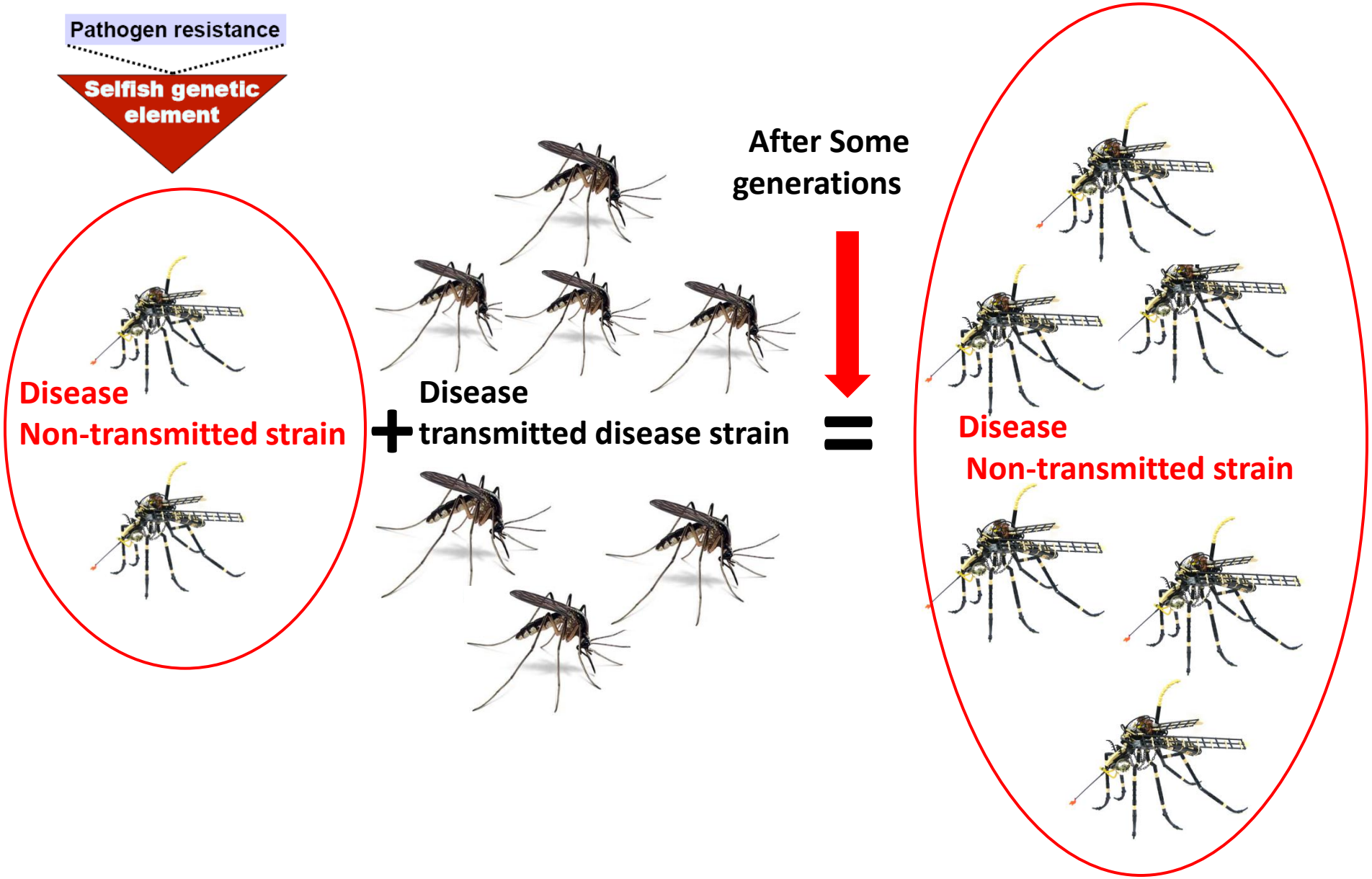


B



Mosquito host-seeking behavior is reproduced in the artificial environment (target: thermo)



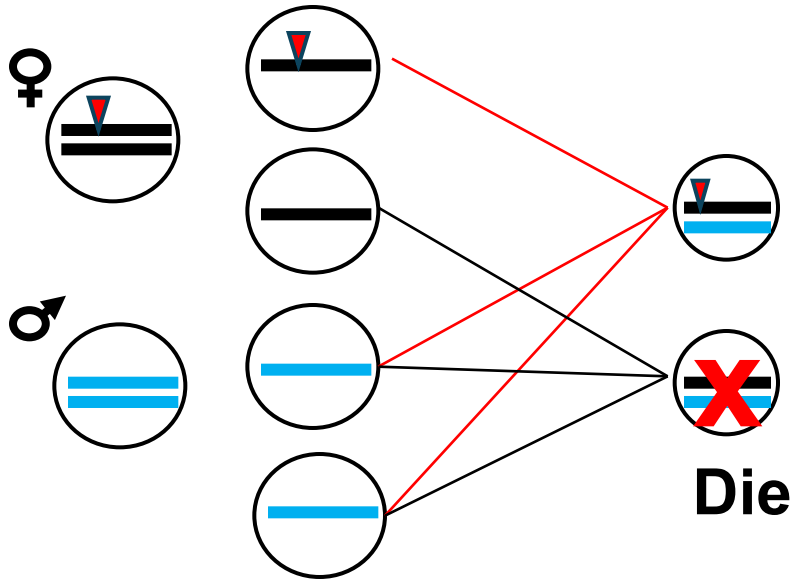

Pathogen resistance
Selfish genetic element
Disease
Non-transmitted strain
Disease
transmitted disease strain
After Some generations
Disease
Non-transmitted strain

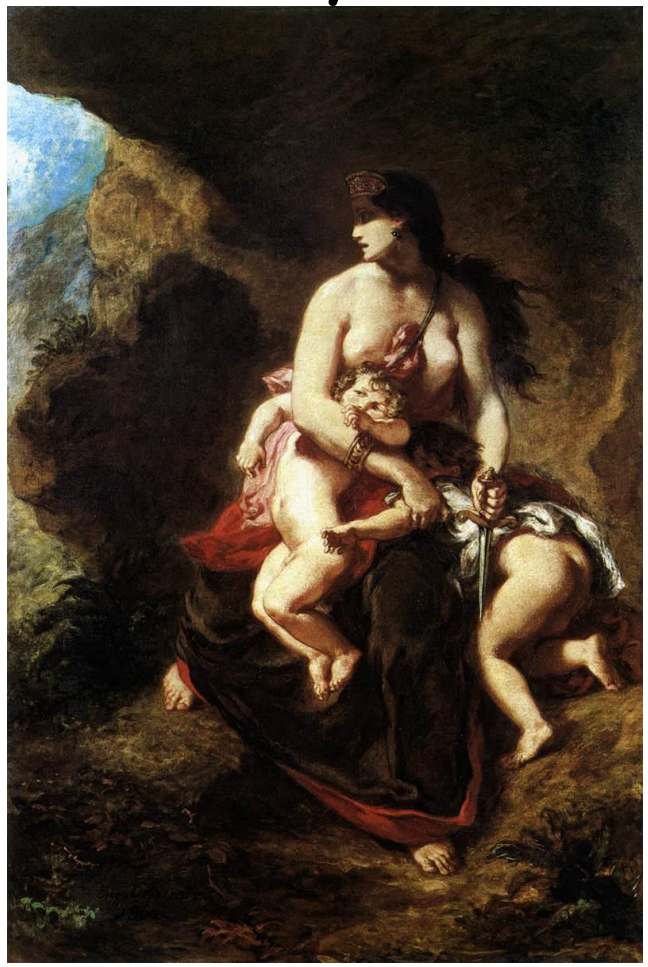
A Natural Maternal selfish element in *Tribolium castaneum*
Is a gene driver
Called: *Medea* Maternal-effect dominant embryonic arrest

Tribolium castaneum











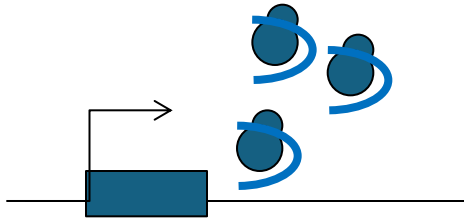
Mendelian inheritace
50%

Non-Mendelian inheritace
"Gene driver"
100%

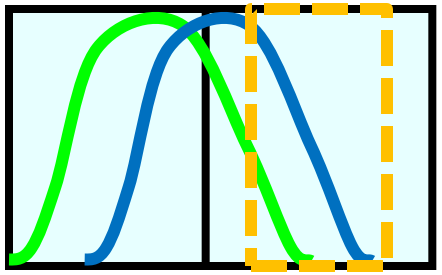
Beeman et al 1992 Science

The synthetic *Medea* can be made by rewiring a circuit of a maternal essential gene!

A maternally-expressed gene and essential gene whose product are not needed until embryogenesis

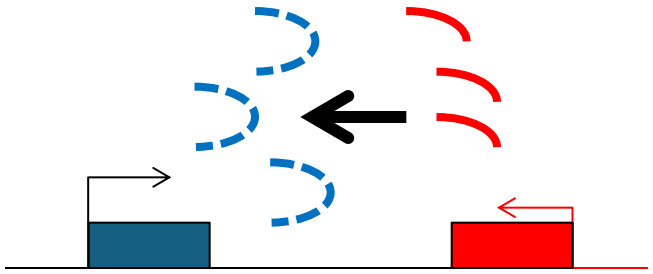


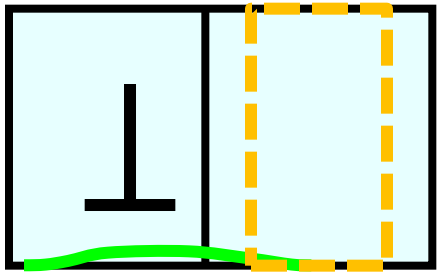
required



O Wild type

Maternal toxic microRNA as a maternal toxin to block this essential gene

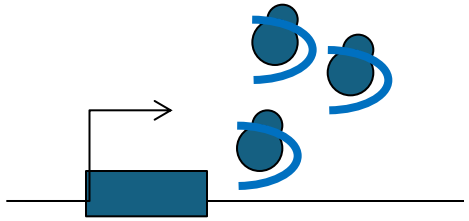


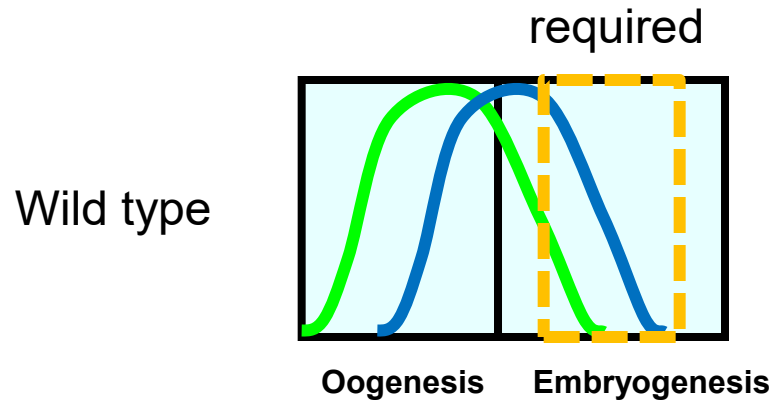


X Female sterile

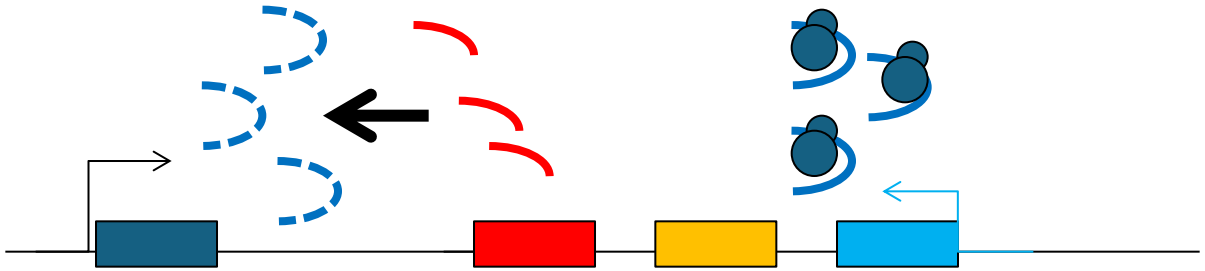
The synthetic *Medea* can be made by rewiring rewiring a circuit of a maternal essential gene !

A maternally-expressed and essential gene whose product are not needed until embryogenesis



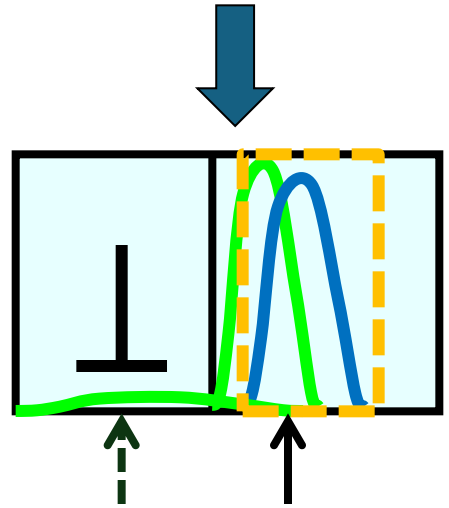


Altered the expressing timing from maternal to zygotic statue

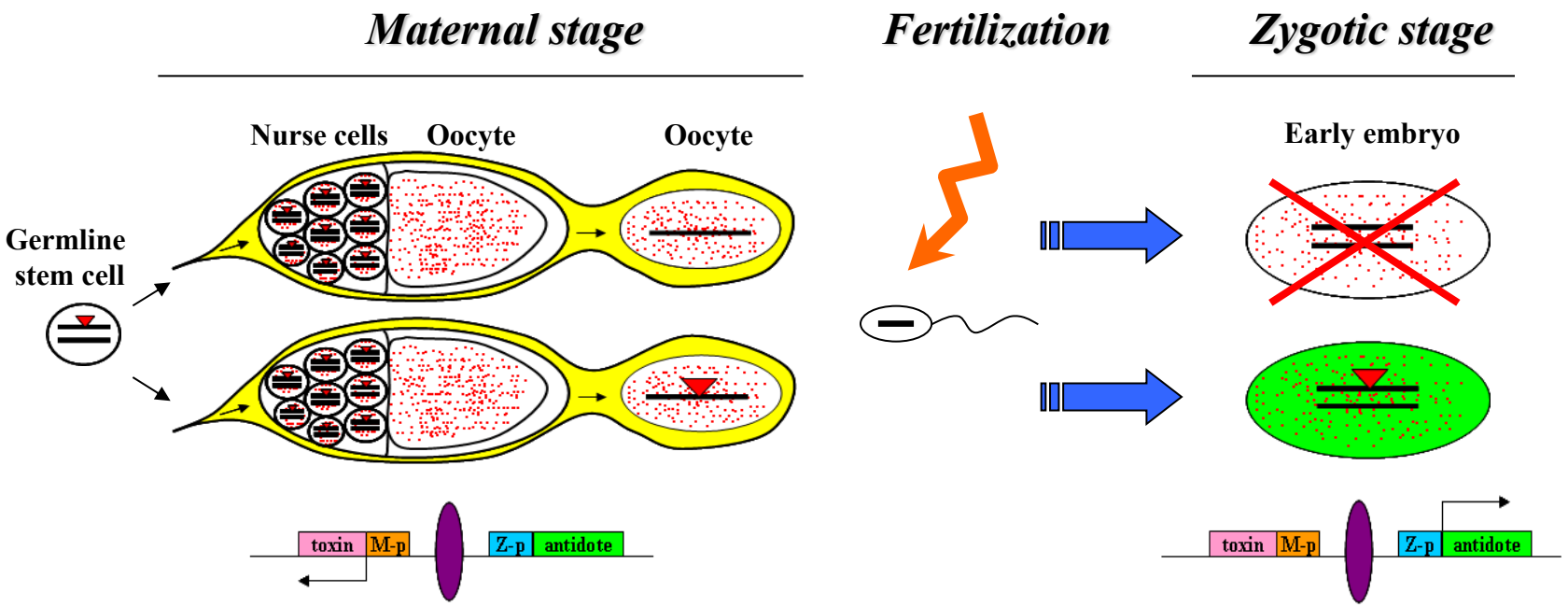


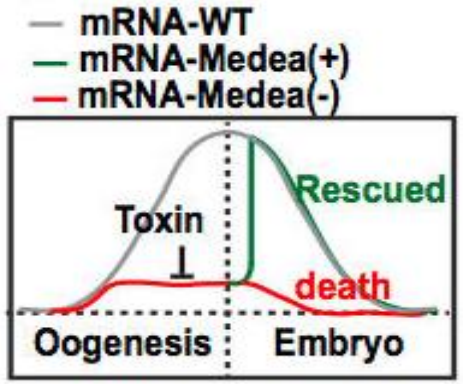
Rescue gene only transcribed in embryos which carrying this transgene.

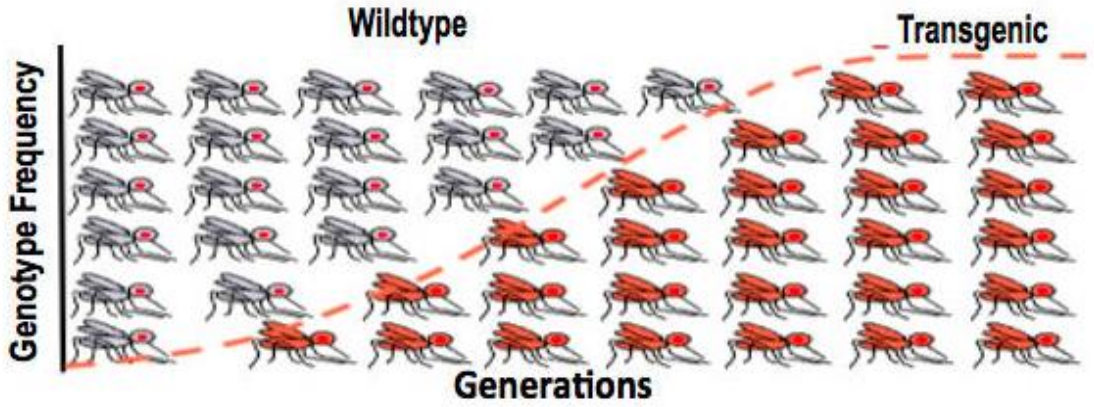
Rewiring this essential gene circuit



Synthetic population



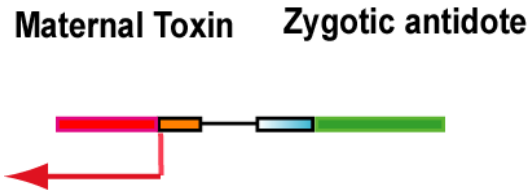




Akbari OS, Chen CH, Marshall JM, Huang H, Antoshechkin I, Hay BA. ACS Synth Biol. 2014

Three elements for Making a Synthetic population by *Medea* strategy

1. A Good Maternal Promoter



2. An Early Zygotic Promoter

3. A Good Target Gene for the Toxin and Antidote

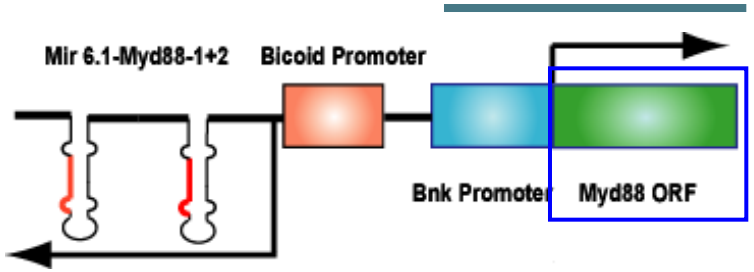


Making a mosquito Gene driver*Medea*

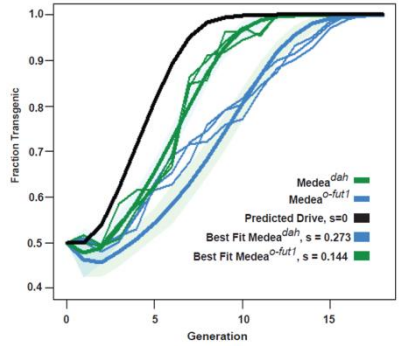




Zygotic Antidote



Chen et al. Science 2007



Akbari OS, Chen CH, Marshall JM, Huang H, Antoshechkin I, Hay BA. ACS Synth Biol. 2014

The zygote-specific promoter

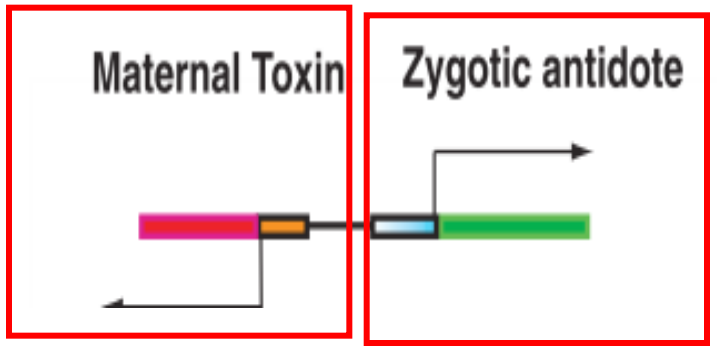
- **Identification of Early Zygotic Genes in the Yellow Fever Mosquito *Aedes aegypti* and Discovery of a Motif Involved in Early Zygotic Genome Activation**

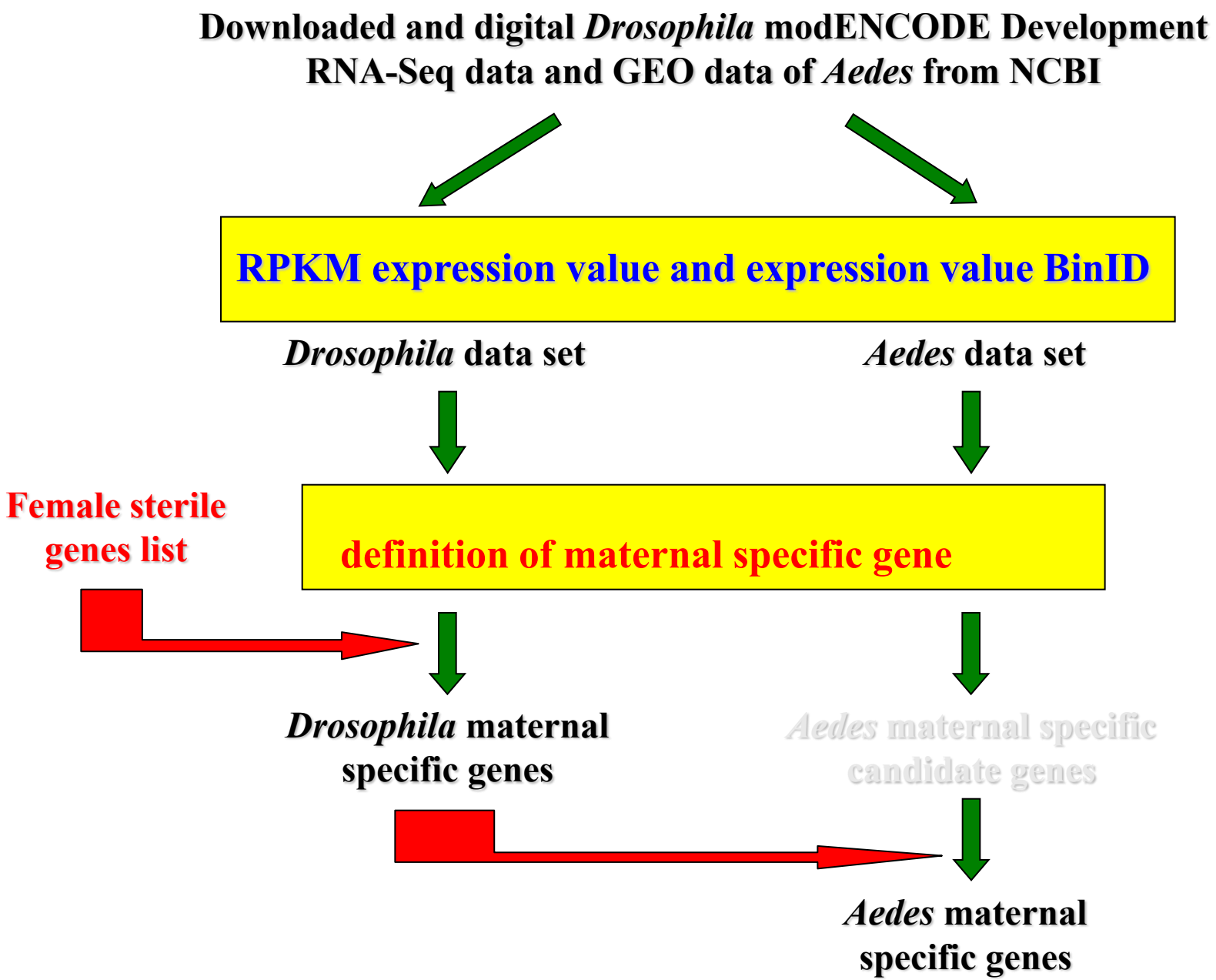
James K. Biedler,* Wanqi Hu, Hongseok Tae, and Zhijian Tu,* Plos One 2012

(James *et al*, 2012)



PWM	v	b	r	g	g	t	a
a	47	0	84	0	0	0	104
c	20	26	0	0	0	0	0
g	37	18	40	104	104	0	0
t	0	60	0	0	0	104	0

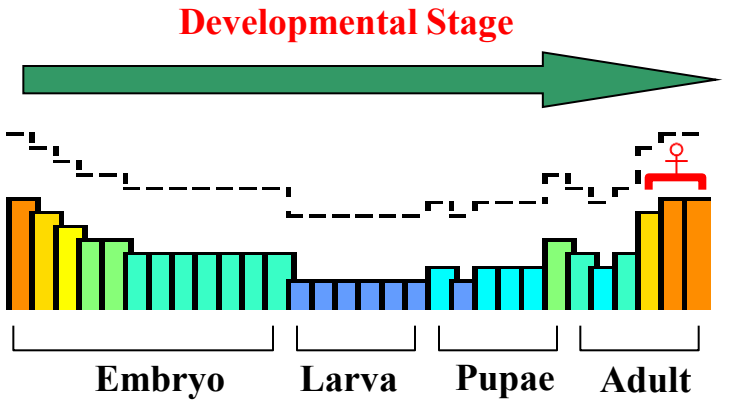



Downloaded and digital *Drosophila* modENCODE Development
RNA-Seq data and GEO data of *Aedes* from NCBI
RPKM expression value and expression value BinID
Drosophila data set
Aedes data set
Female sterile genes list
definition of maternal specific gene
Drosophila maternal specific genes
Aedes maternal specific candidate genes
Aedes maternal specific genes
(promoter analysis and maternal specific motif finding)

Definition of maternal specific gene

modENCODE Development RNA-Seq data of dhd gene





RPKM => binID
No Data => 0
0~10 => 1
11~100 =>2
101~400 =>3
401~1400 => 4
1401~4000=> 5
4001~10000=> 6
10000~100000=> 7
>100000 => 8

			embryo												larva						pupae						adult					
flyGeneID	flyGeneName	modENCOREstageSign	S01	S02	S03	S04	S05	S06	S07	S08	S09	S10	S11	S12	S13	S14	S15	S16	S17	S18	S19	S20	S21	S22	S23	S24	S25	S26	S27	S28	S29	S30
FBgn0011761	dhd	|||:::::::

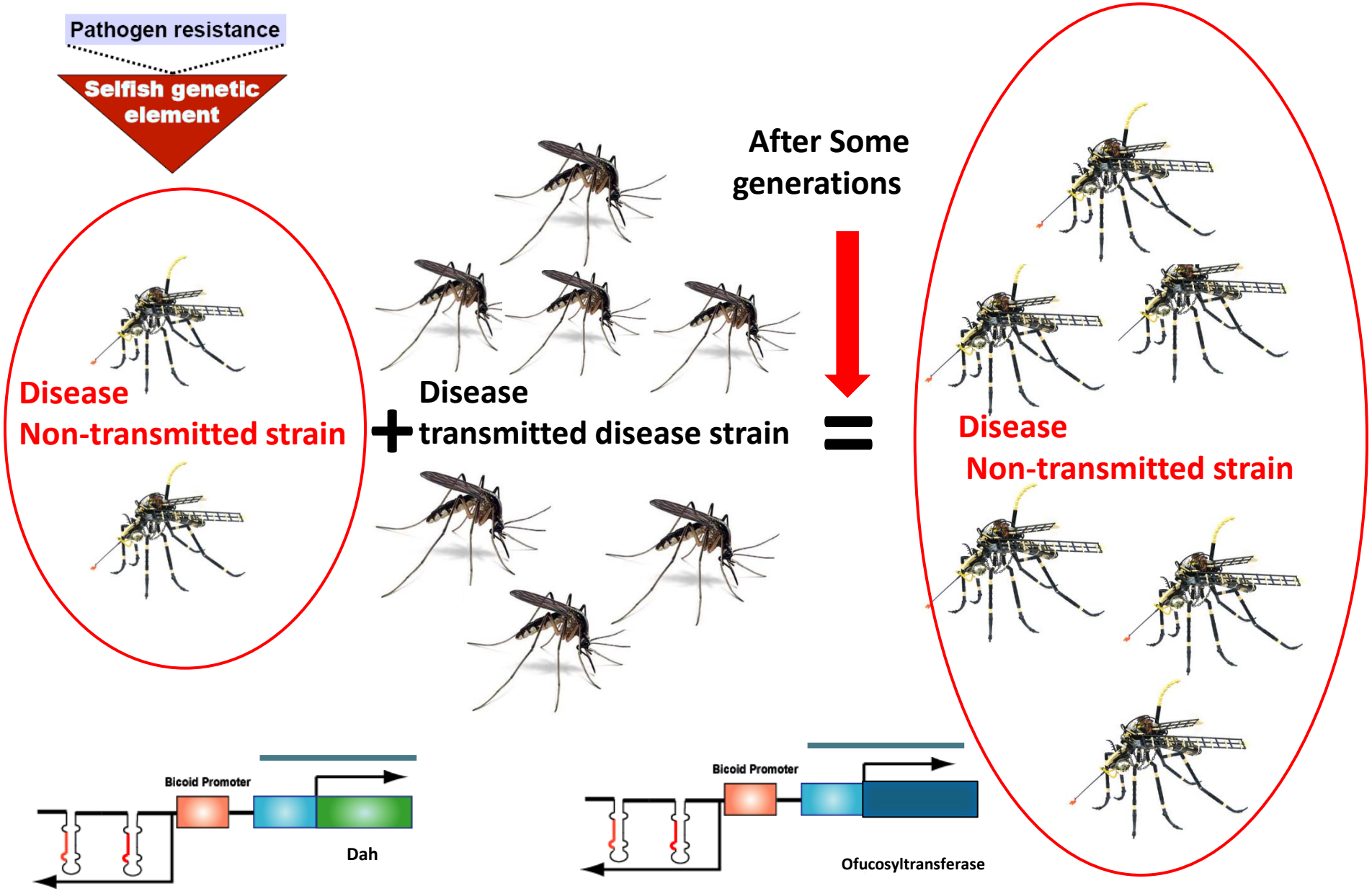
Endogenous Maternal Promoter drive shRNAi for Dah

Cas9_AAEL004112>4miR-AeDah_Pub-eGFP (Homo)
Cas9_AAEL007097>4miR-AeDah_Pub-eGFP (Homo)

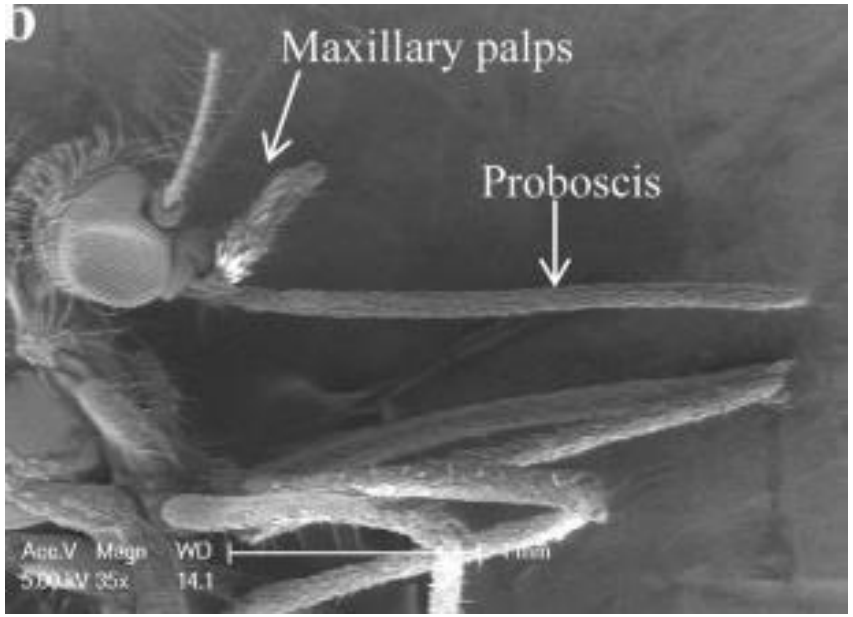


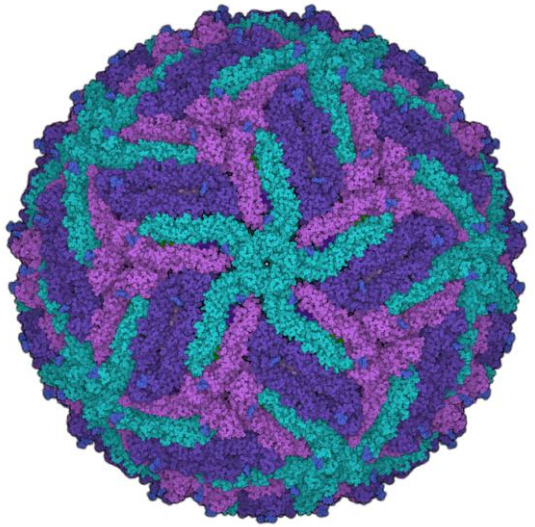


pMOS1-AeCPA_DENV all in one 8-miR



Mechanical transmission of dengue by *Aedes* mosquitoes contributes to disease proliferation







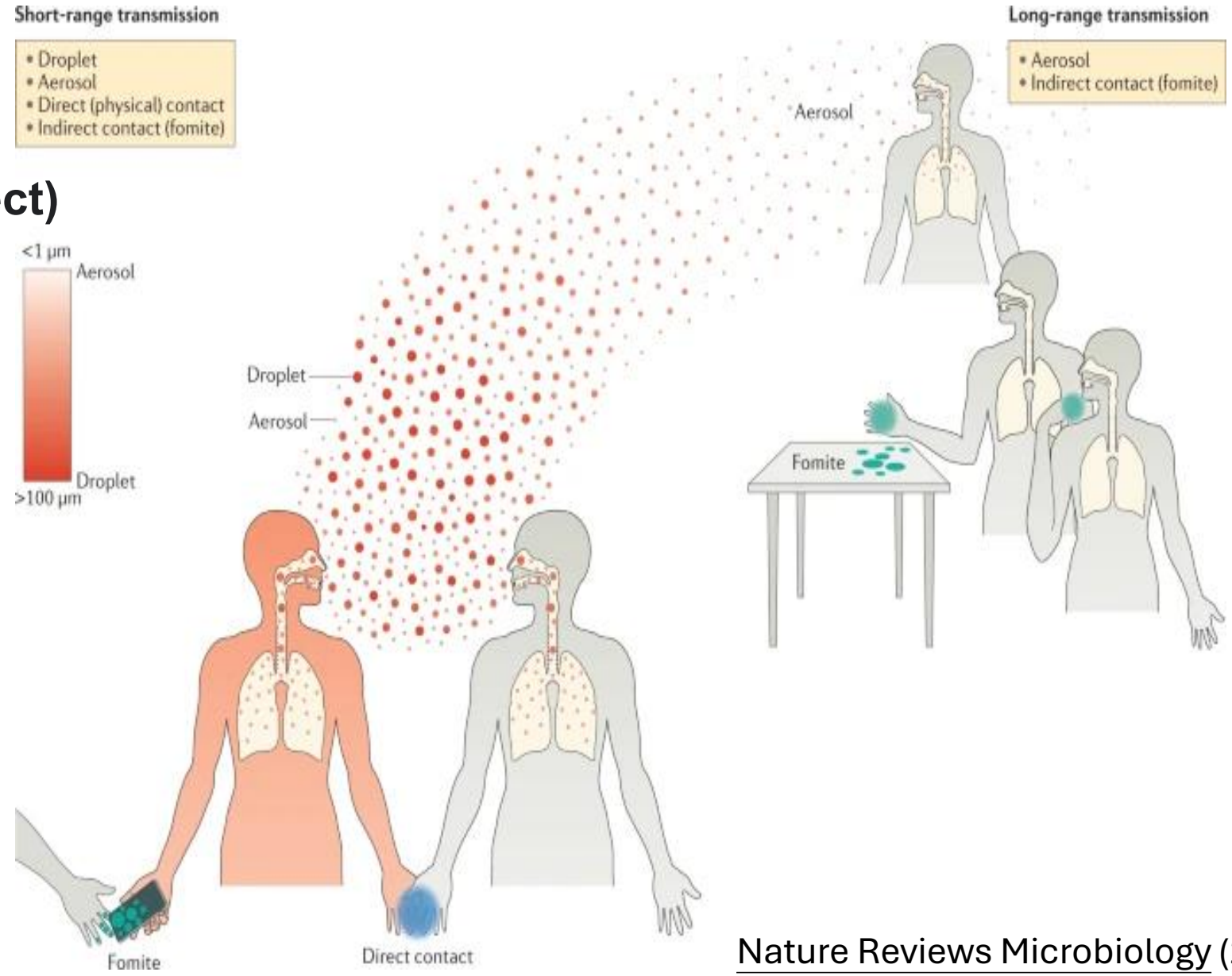
How can pathogens be transmitted?

1) Contact (direct and/or indirect)

2) Droplet

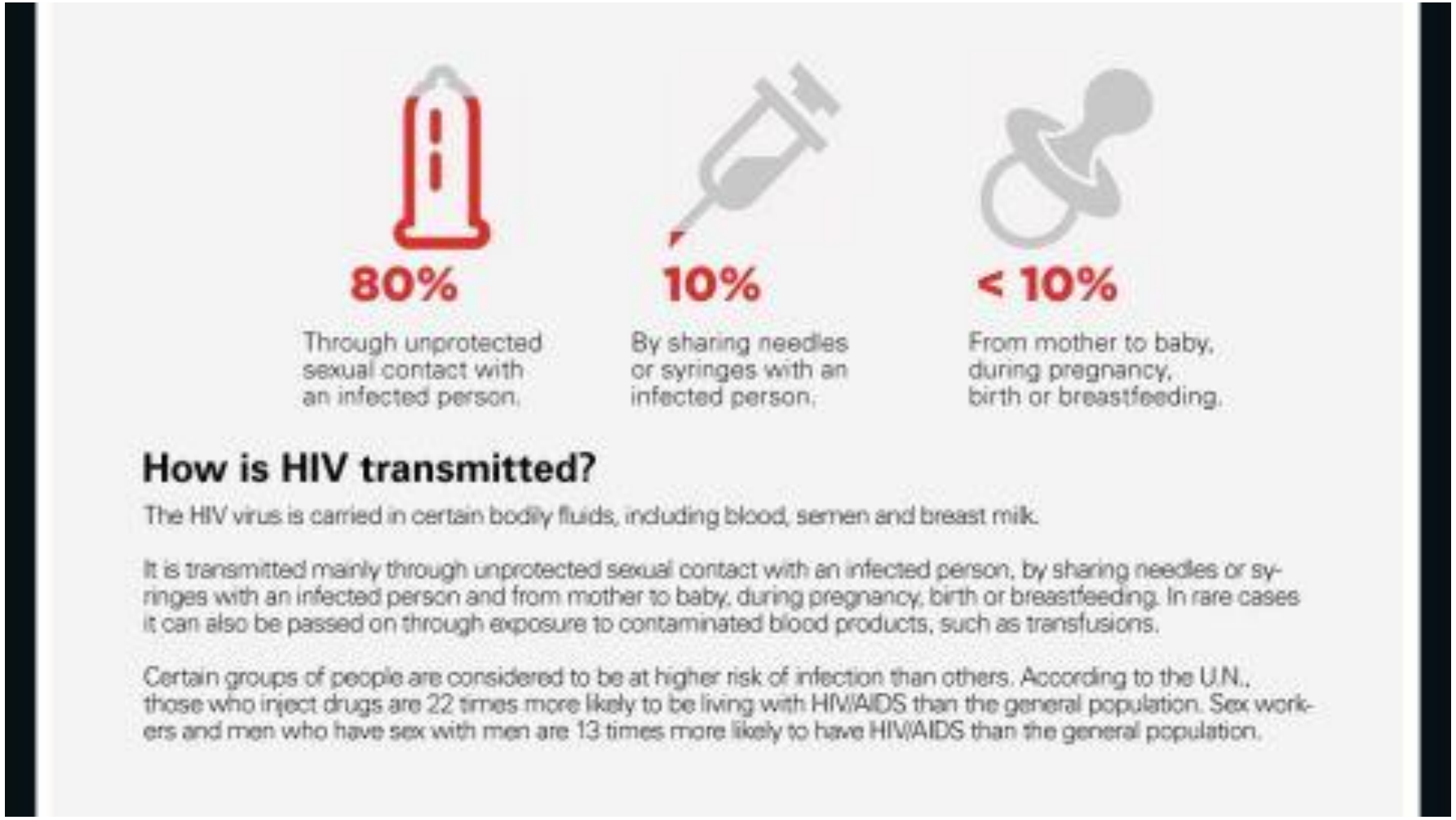
3) Airborne

4) Vector (insect)

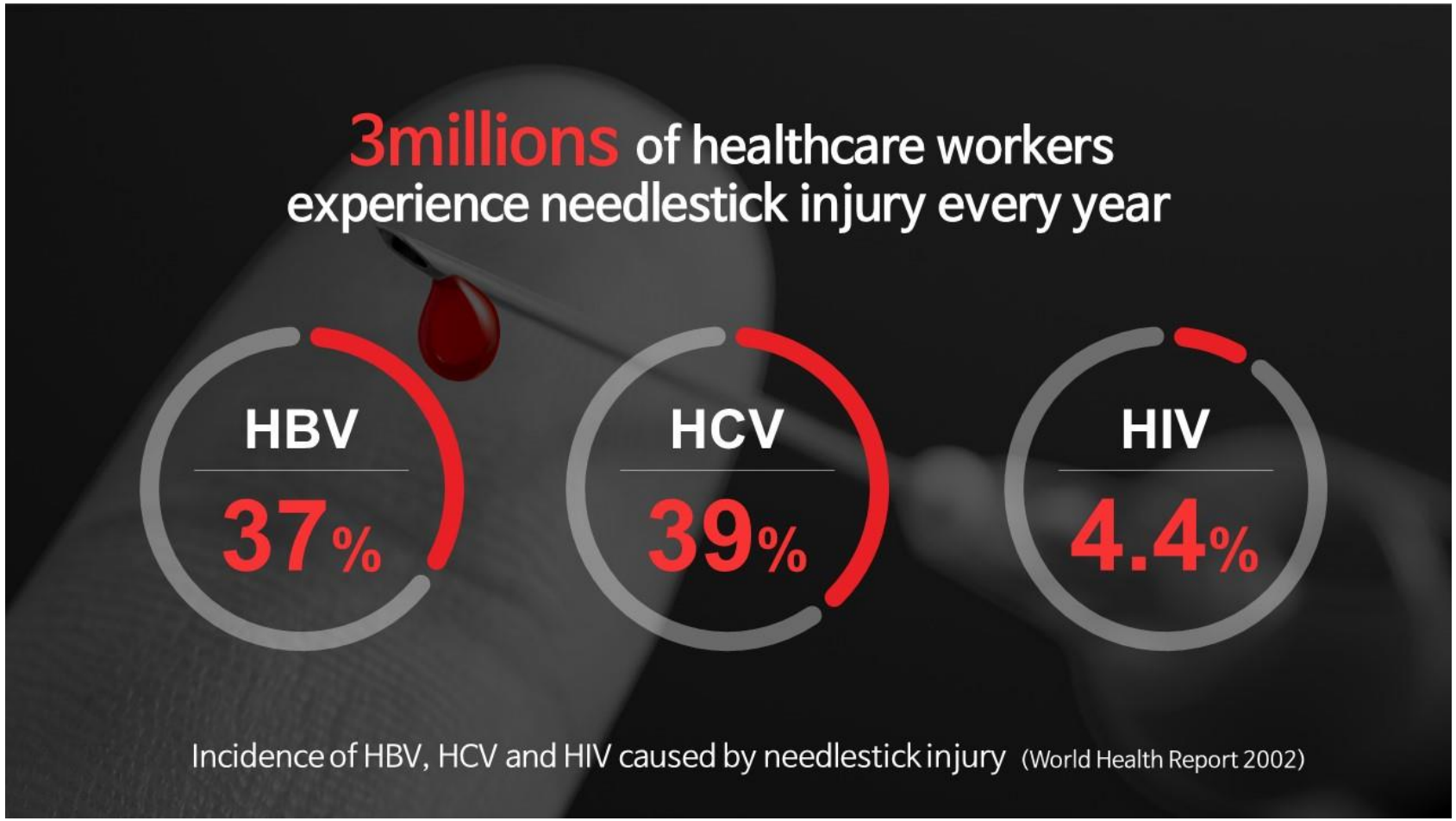


Nature Reviews Microbiology (2021)

HIV transmission routes



Three major viral diseases transmitted via needlestick injury



Two types of vector transmission

Disease Transmission: Source of Infection

- Biological vector
  - An organism which not only transports a pathogen but also plays a role in the life cycle of the pathogen (virus inside of mosquito, bacteria inside of tick)
- Mechanical vector
  - An organism which only transports a pathogen (fly)



(a) Biological vectors are infected.



(b) Mechanical vectors are not infected.

The blood sucking proboscis of mosquitoes

Two directions of fluid exchange

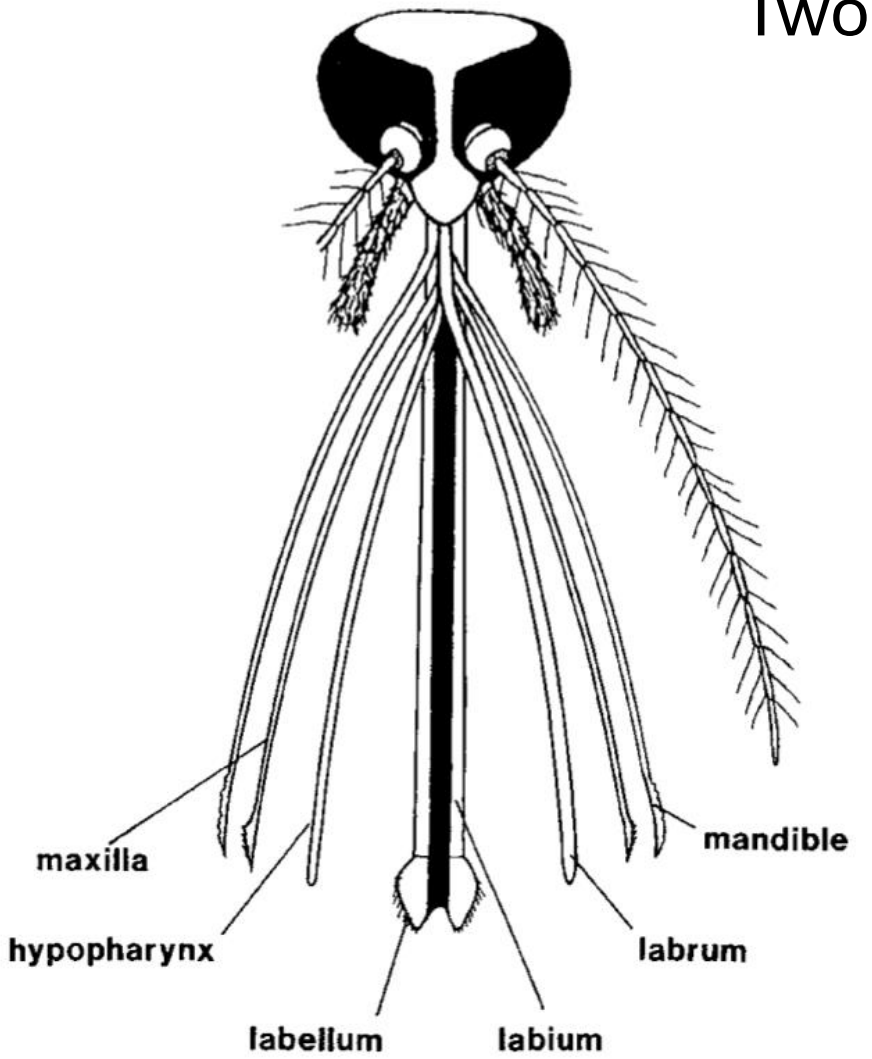


Figure 1.4 Diagram of the head of a female culicine mosquito, showing the components of the mouthparts spread out from the labium.

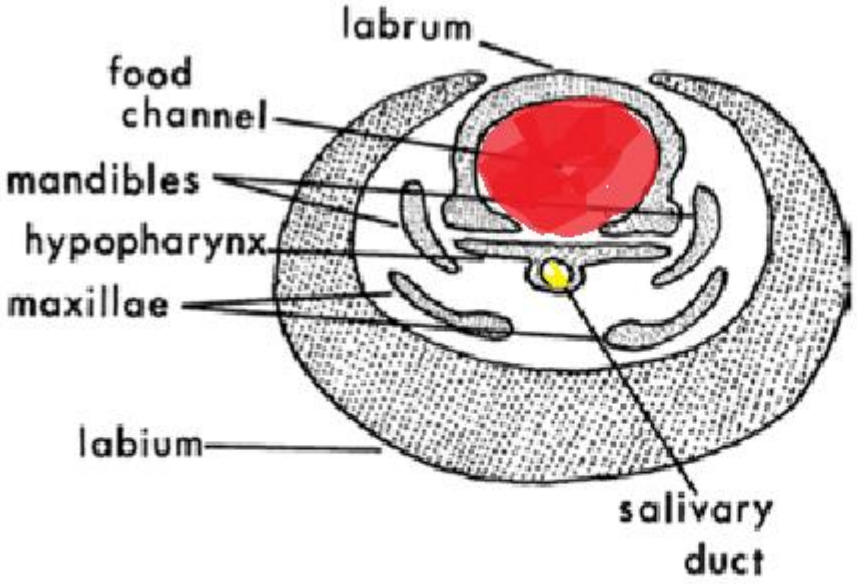
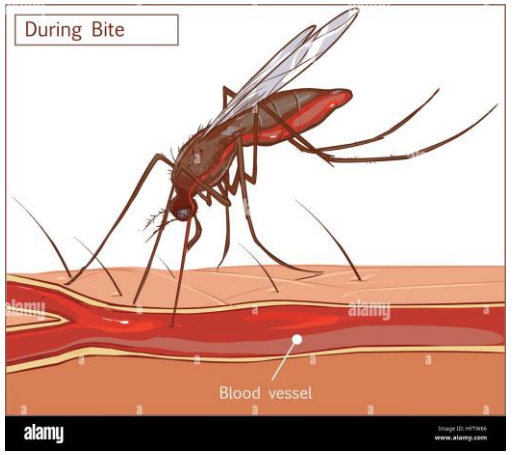


Figure 1.3 Diagram of a cross-section through the proboscis of a mosquito, showing components of the mouthparts and food channel.



From Biology of Mosquitoes, Volume 2 Sensory Reception and Behaviour Alan Clement

Which viruses are known to be transmitted by Mechanical Transmission?

Virus			
mosquitoes	fowl pox virus	fowl pox	birds
mosquitoes	myxomatosis virus	myxomatosis	rabbits

From Biology of Mosquitoes, Volume 2 Sensory Reception and Behaviour Alan Clement





wound contamination



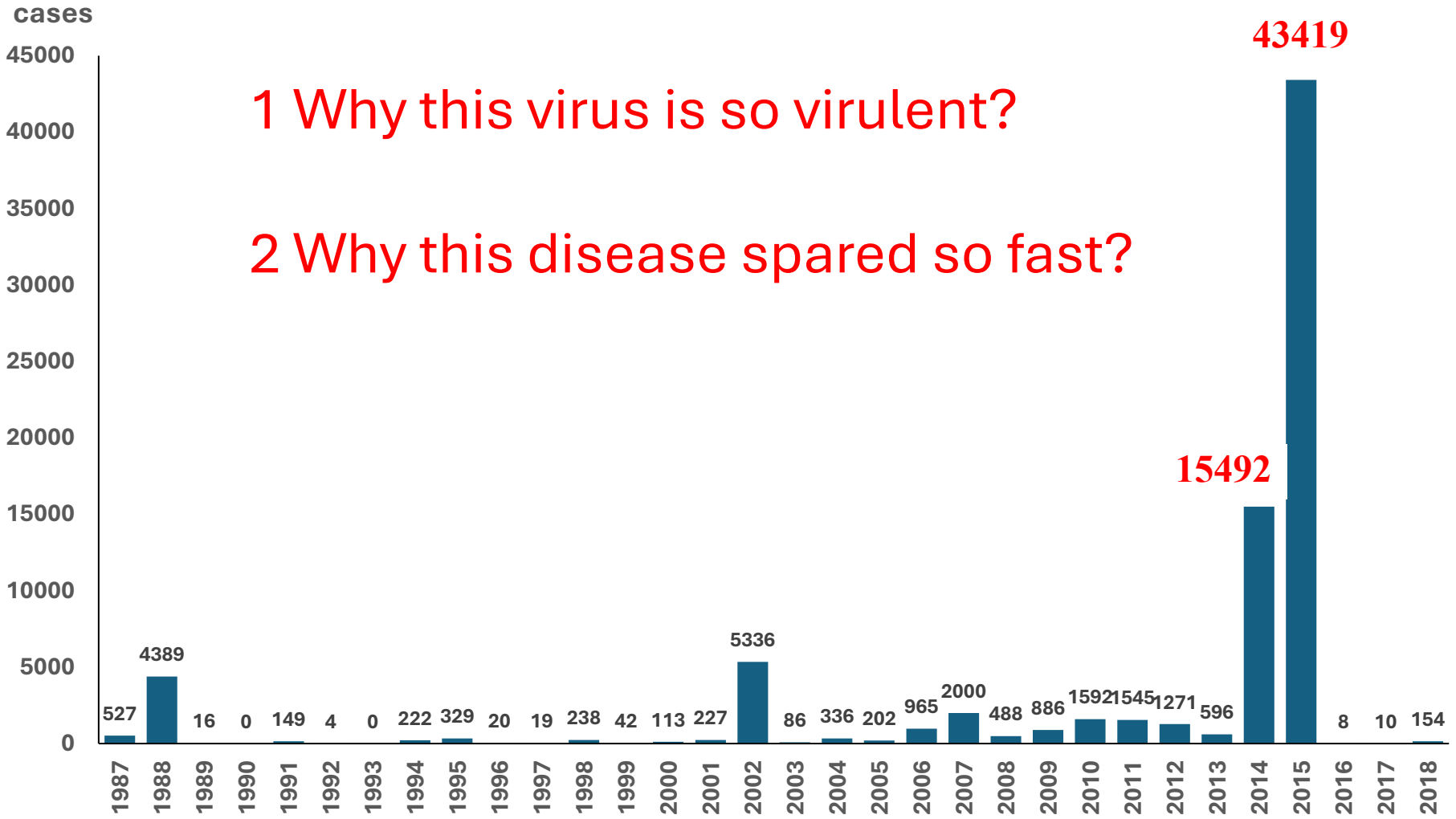
How about Dengue virus?

Two major dengue outbreaks in Taiwan (2014 and 2015)

What did we learn from these two big outbreaks?

1 Why this virus is so virulent?

2 Why this disease spared so fast?

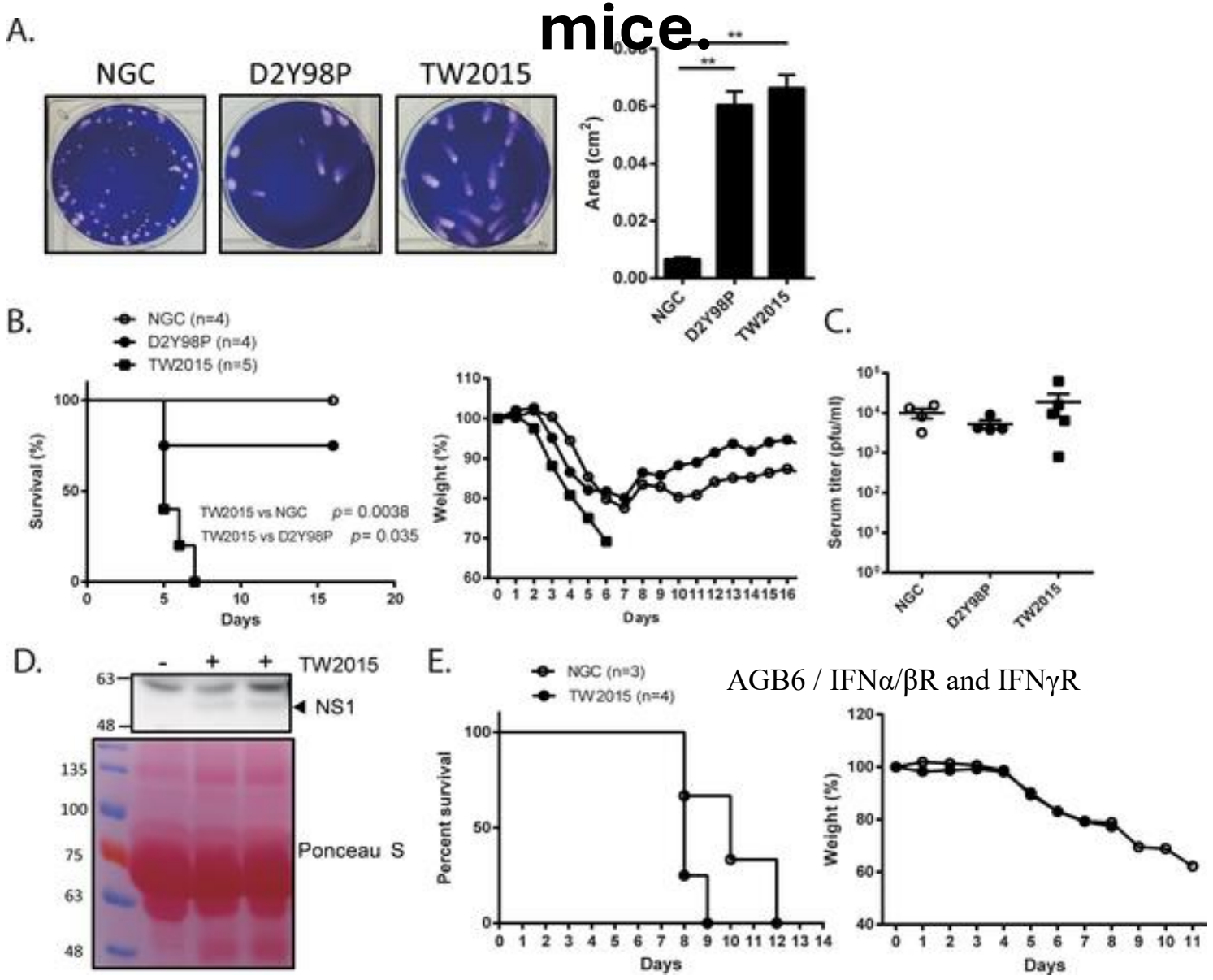






A clinical isolate of the dengue virus type 2 (DENV-2) TW2015 strain was highly virulent in Stat1$^{-/-}$ and AGB6 mice.

IFN-r stat1-/-



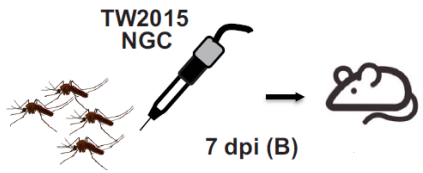
AGB6 / IFNα/βR and IFNγR

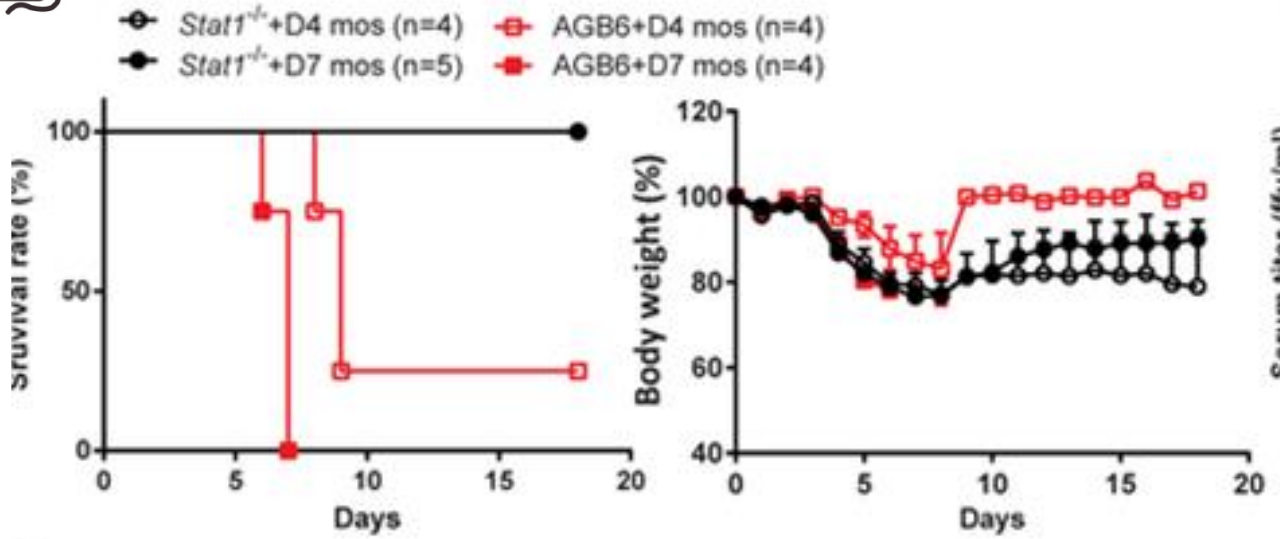


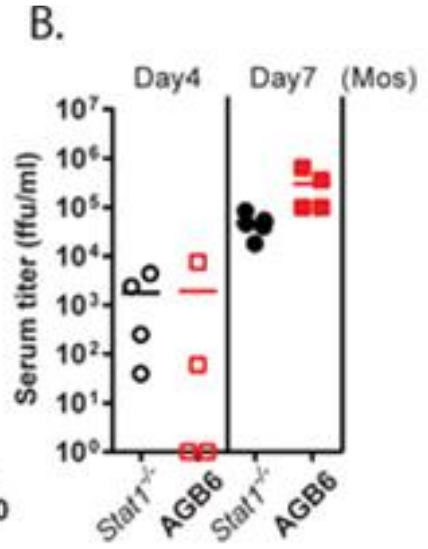
Guann-Yi Yu

PLOS Pathogen 2021

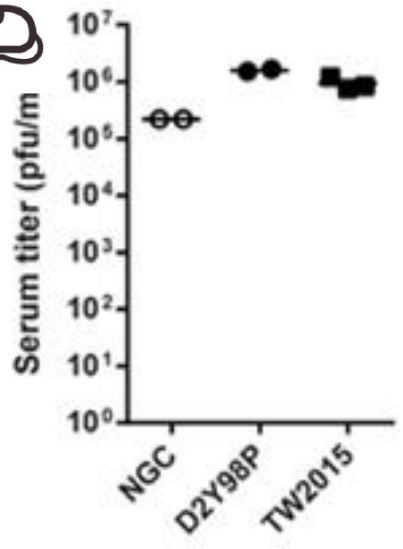
The TW2015 virus had high *Aedes aegypti* mosquito-mediated transmissibility to mice

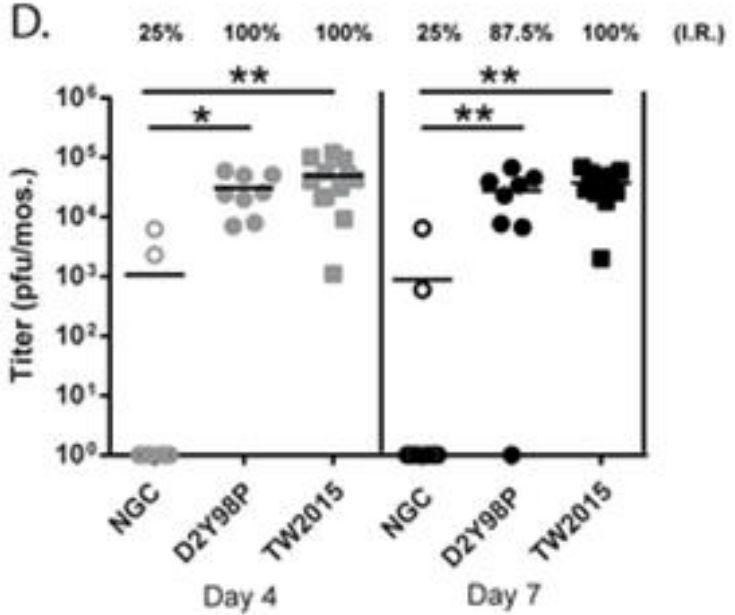






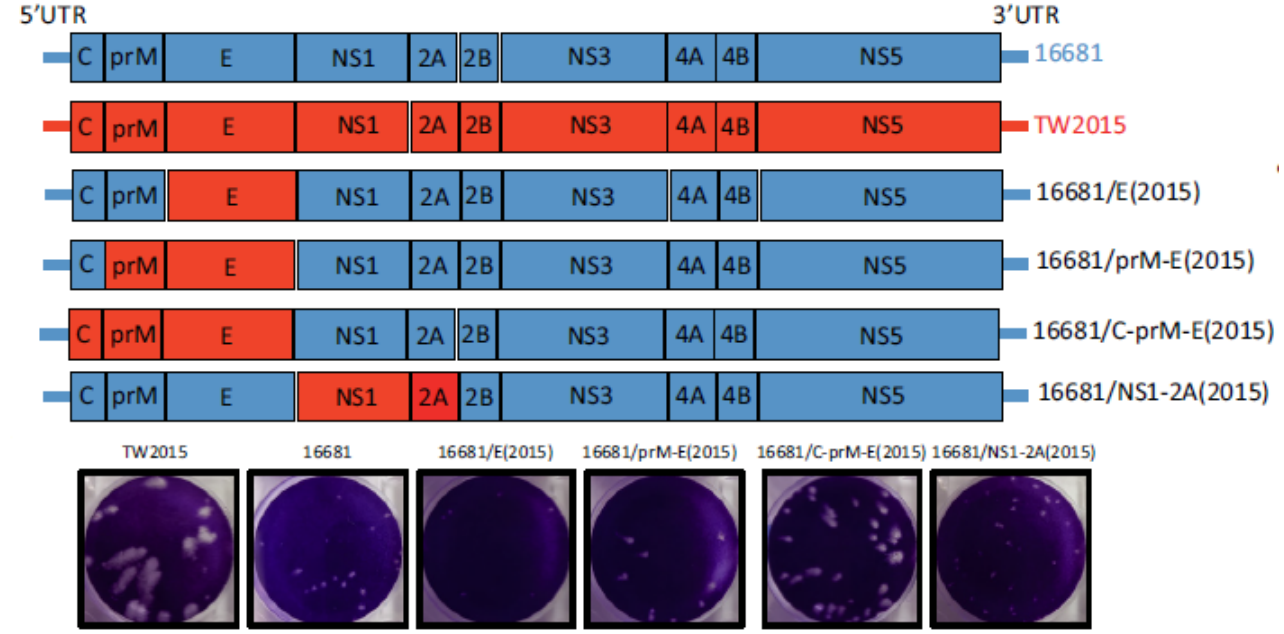


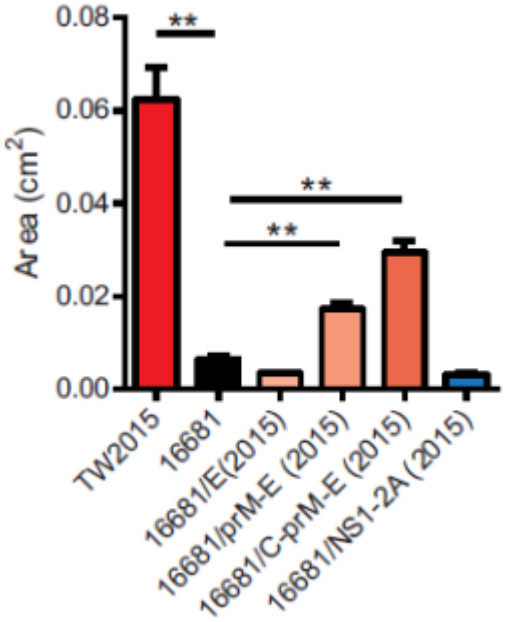


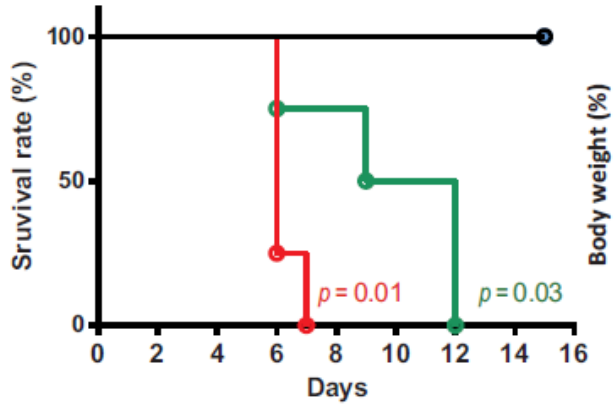


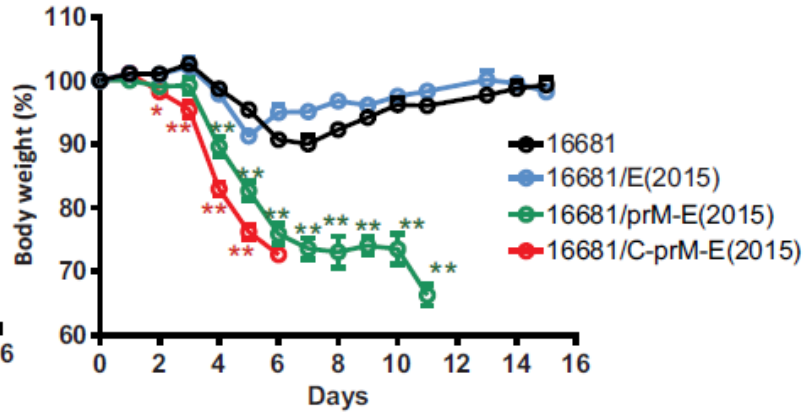
PLOS Pathogen 2021

The TW2015 PrM-E determined virulence in mice







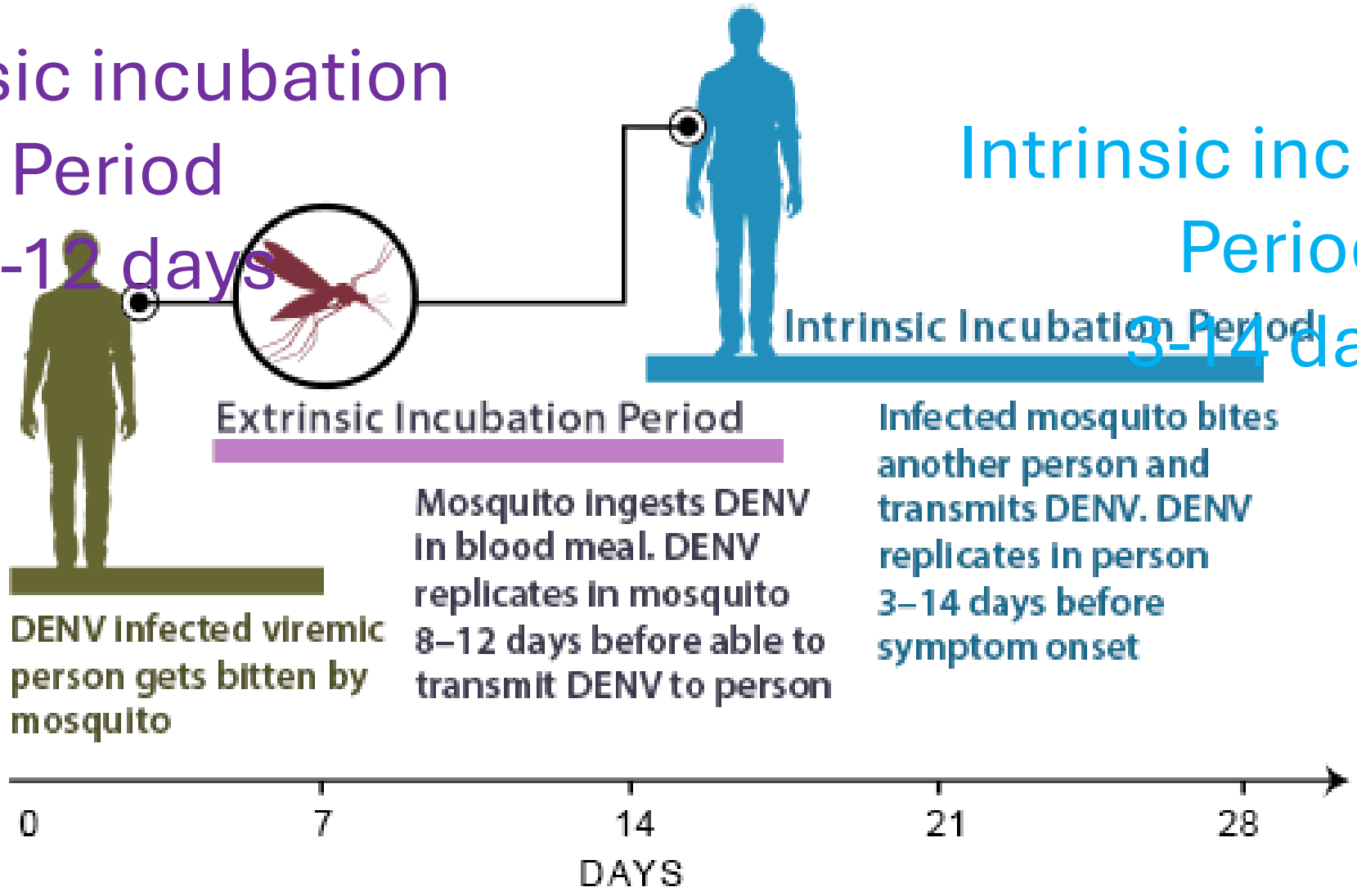


PLOS Pathogen 2021

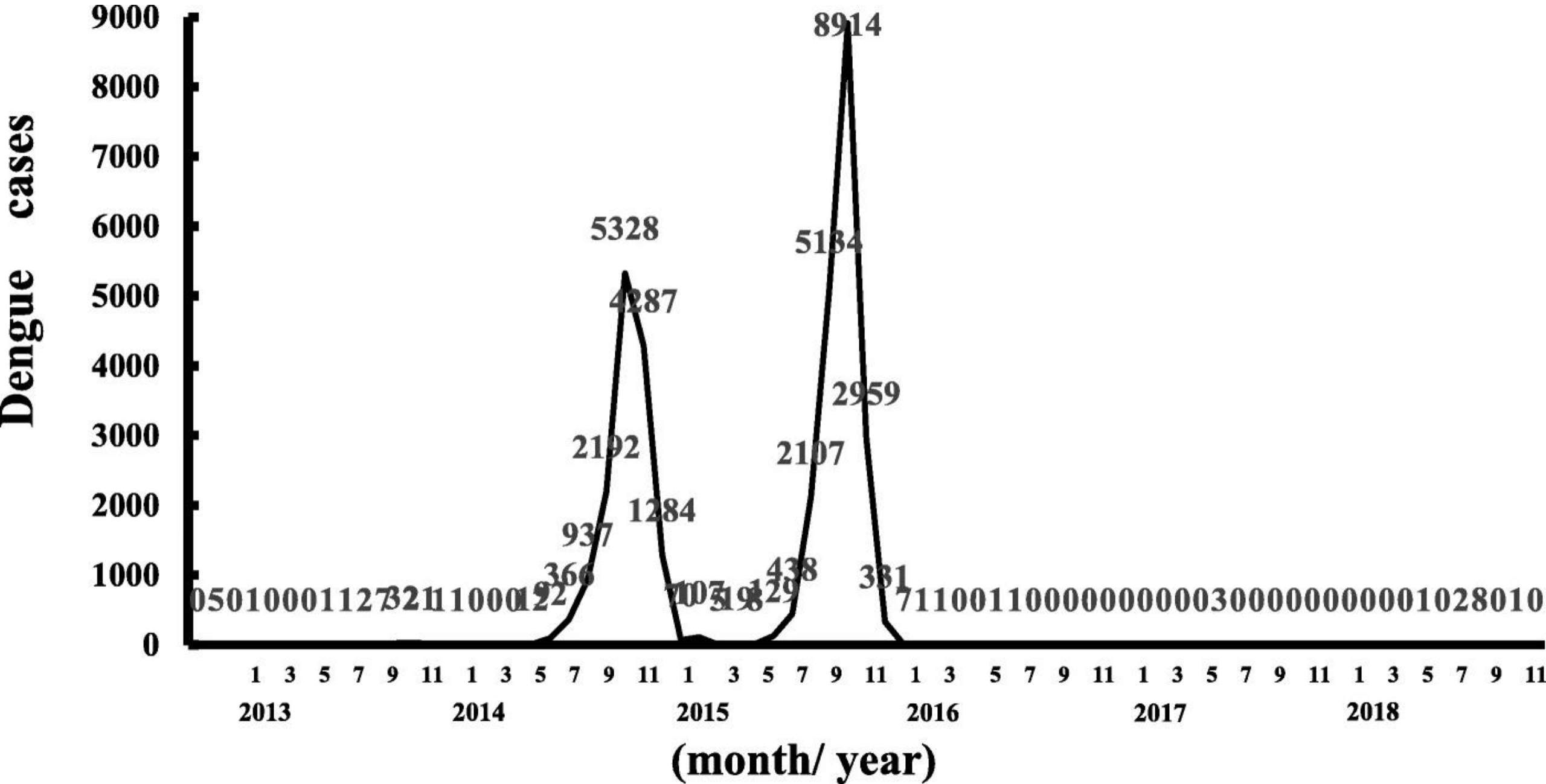
n theory, Dengue virus will take 11 to 26 days for transmissio from one infected person to the next

Extrinsic incubation Period 8-12 days

Intrinsic incubation Period 3-14 days



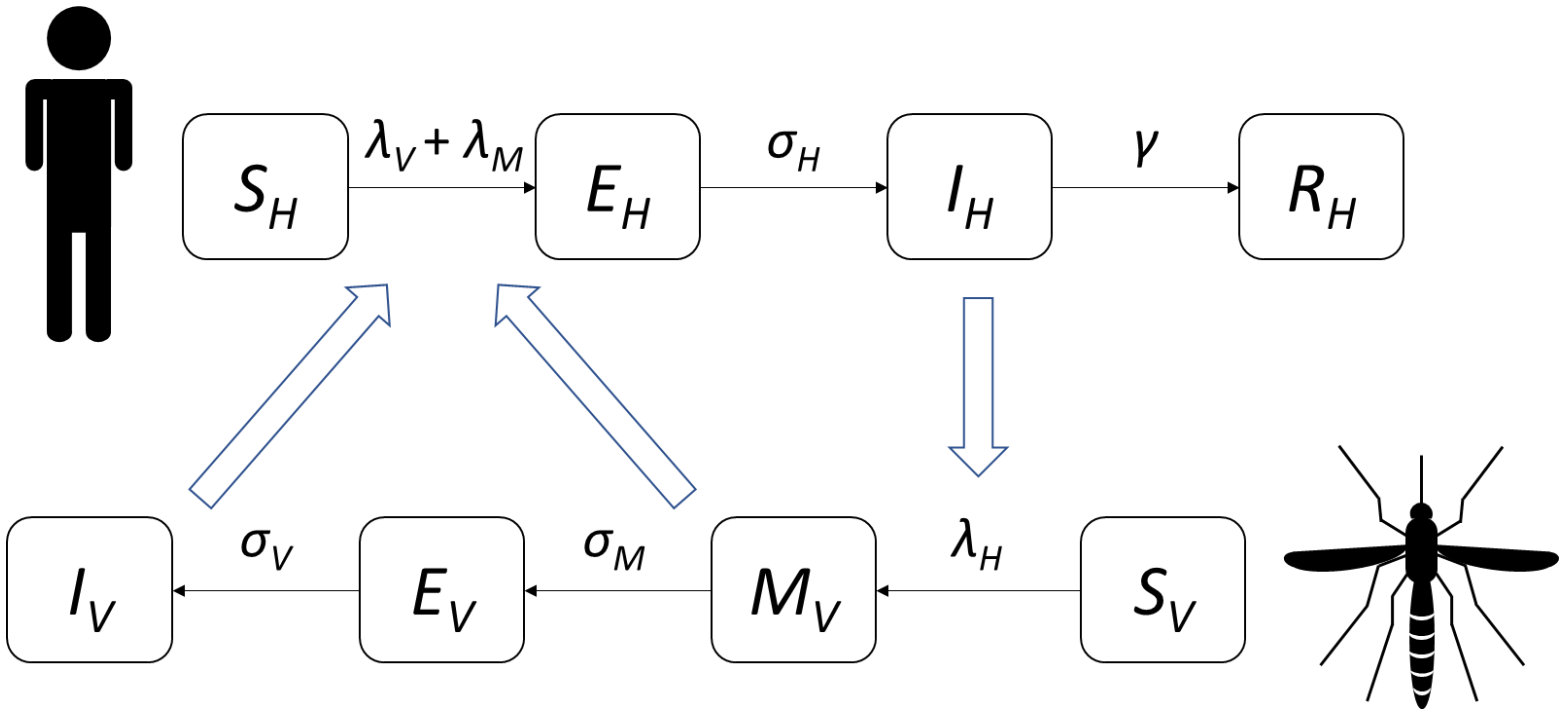
The number of indigenous cases of dengue in Kaohsiung City reported from 2013 to September 2018



Current model for Dengue clustering transmission

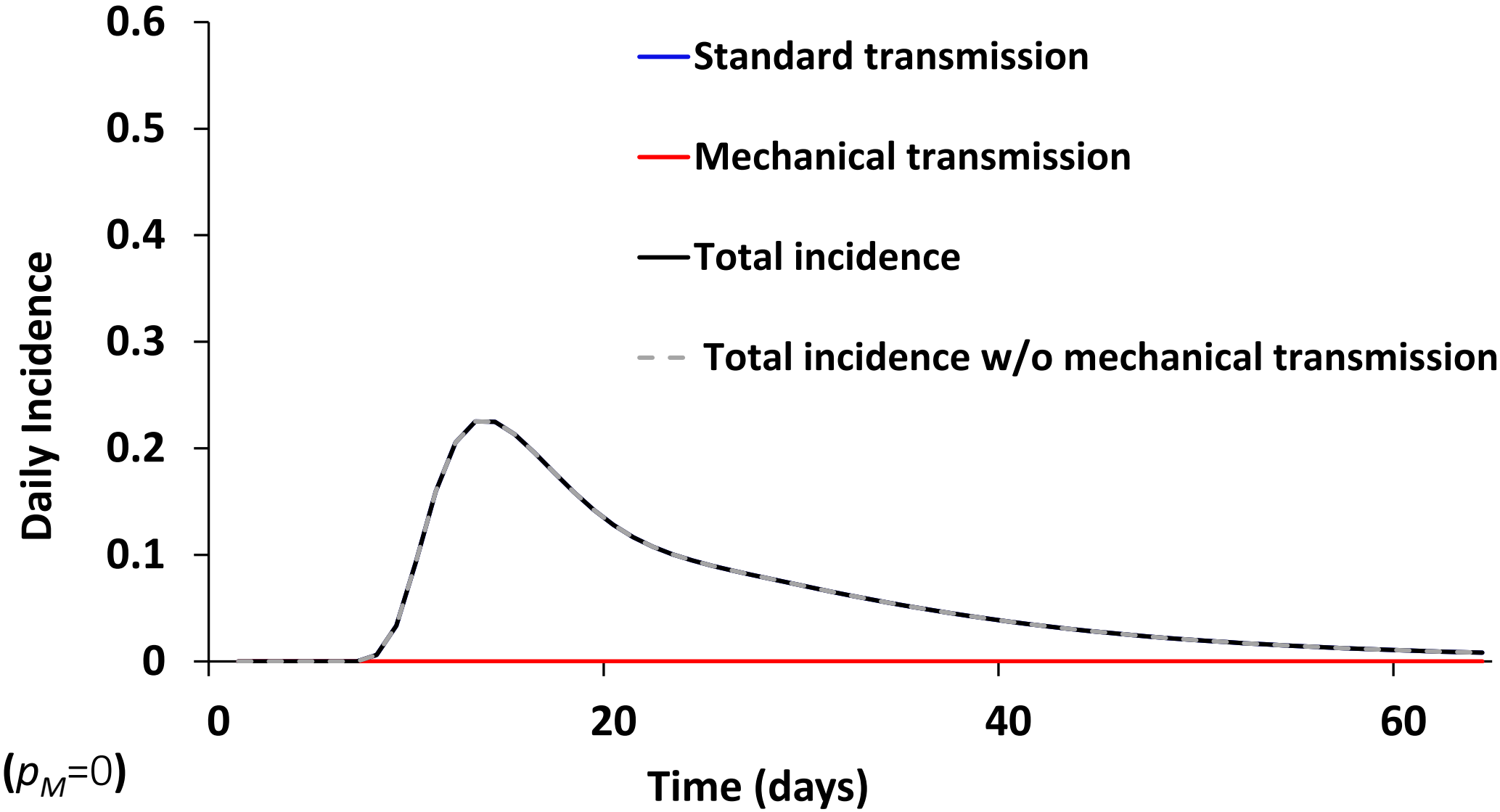


Mathematical model of dengue transmission dynamics including a mechanical transmission component.



Parameter:	Description:	Value:	Reference:
	Average adult human lifetime	70 years	
N_H	Human population size	10,000	
k	Average number of adult female mosquitoes per person	2, 2.5, 3, 5	(Newton and Reiter, 1992), (Focks et al., 2000)
	Average adult female mosquito lifetime	14 days	(Sheppard et al., 1969), (Trpis and Hausermann, 1986), (Trpis et al., 1995)
	Average latent period in human host	5 days	(Siler et al., 1926), (Sabin, 1952)
	Average latent period in mosquito host	10 days	(Siler et al., 1926), (Watts et al., 1987)
	Average infectious period in human host	6 days	(Gubler, 1988), (Vaughn et al., 2000)
b	Mosquito biting rate	0.3, 0.5 per day	(Trpis and Hausermann, 1986), (Putnam and Scott, 1995)
p	Mosquito-to-human transmission probability (following parasite incubation)	0.38	(Wearing and Rohani, 2006)
q	Human-to-mosquito transmission probability	0.38	(Wearing and Rohani, 2006)
a	Degree of seasonality in mosquito population size	0	
T	Period of seasonality	1 year	
	Average duration of mechanical transmission	1 hour	
n_M	Number of bites during mechanical transmission period	1, 2	
p_M	Mosquito-to-human transmission probability during mechanical transmission	0.1 (from this study), and 0.05, 0.025 (default), 0.0125	

In the current model, the first and next infection in the same cluster should be separated by at least 11 days

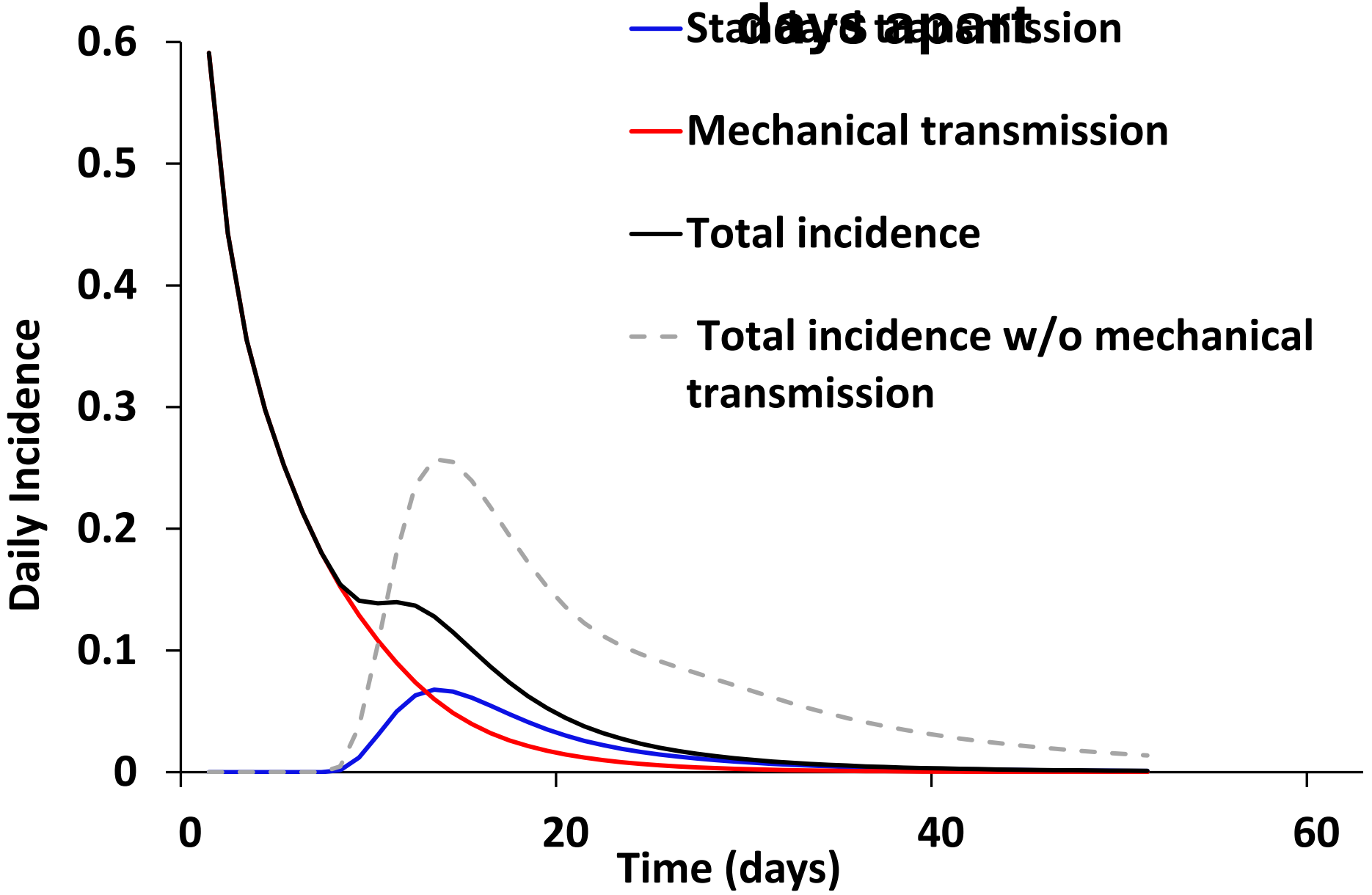


NH=5 (p_M=0)

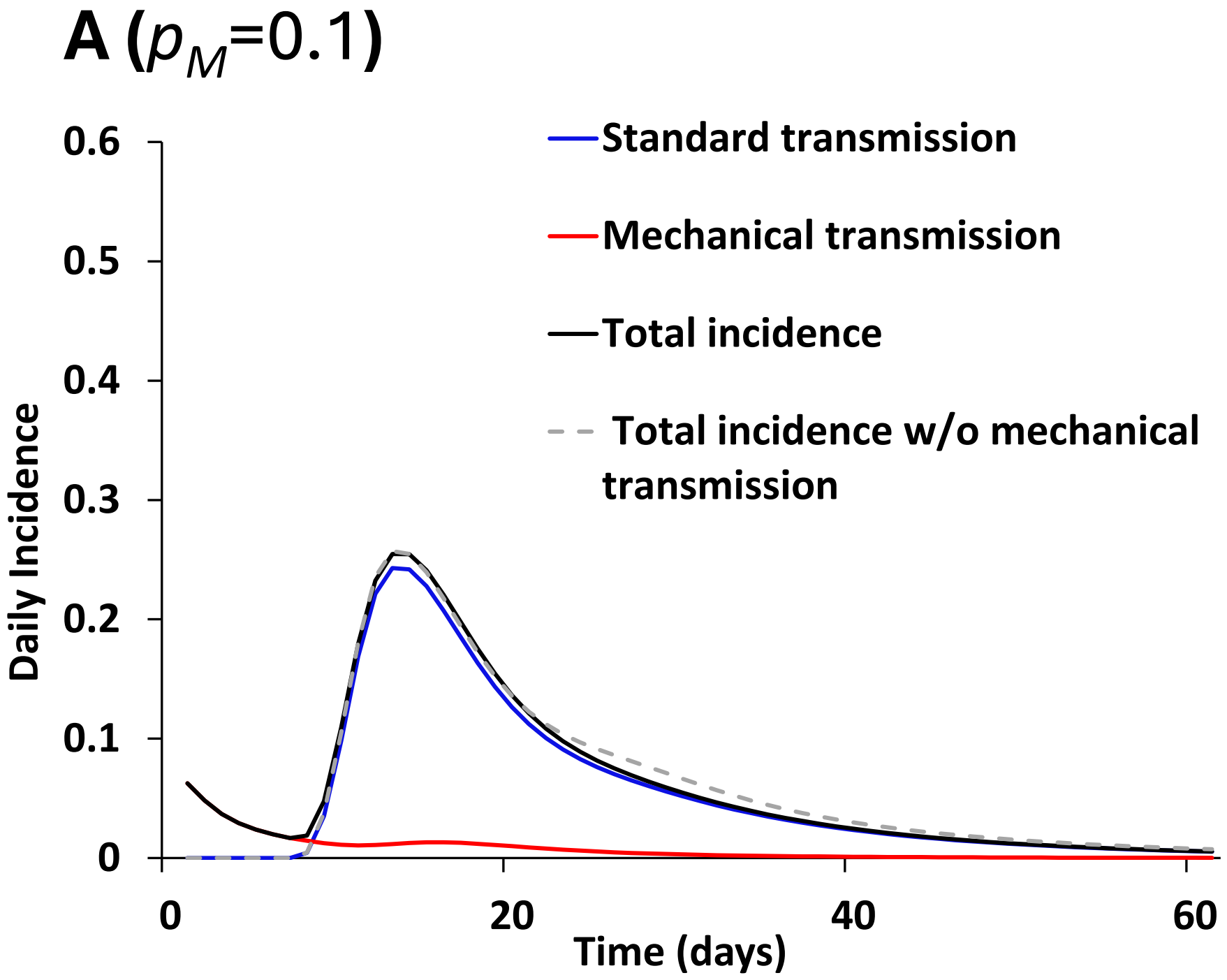
Potential scenario for Dengue clustering transmission without Extrinsic incubation Period



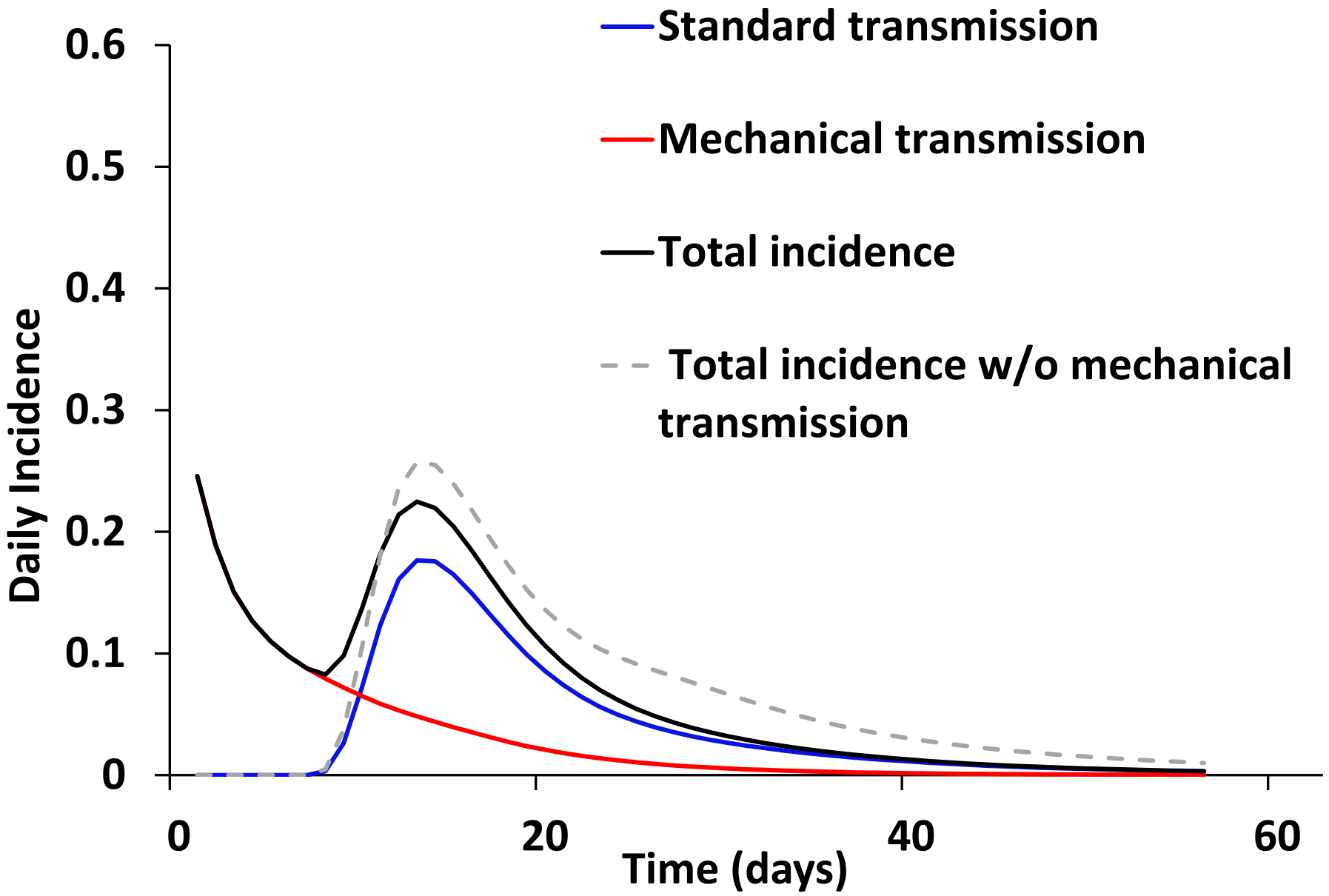
Without an Extrinsic incubation Period, the first and next infection in the same cluster should be less than 5 days apart

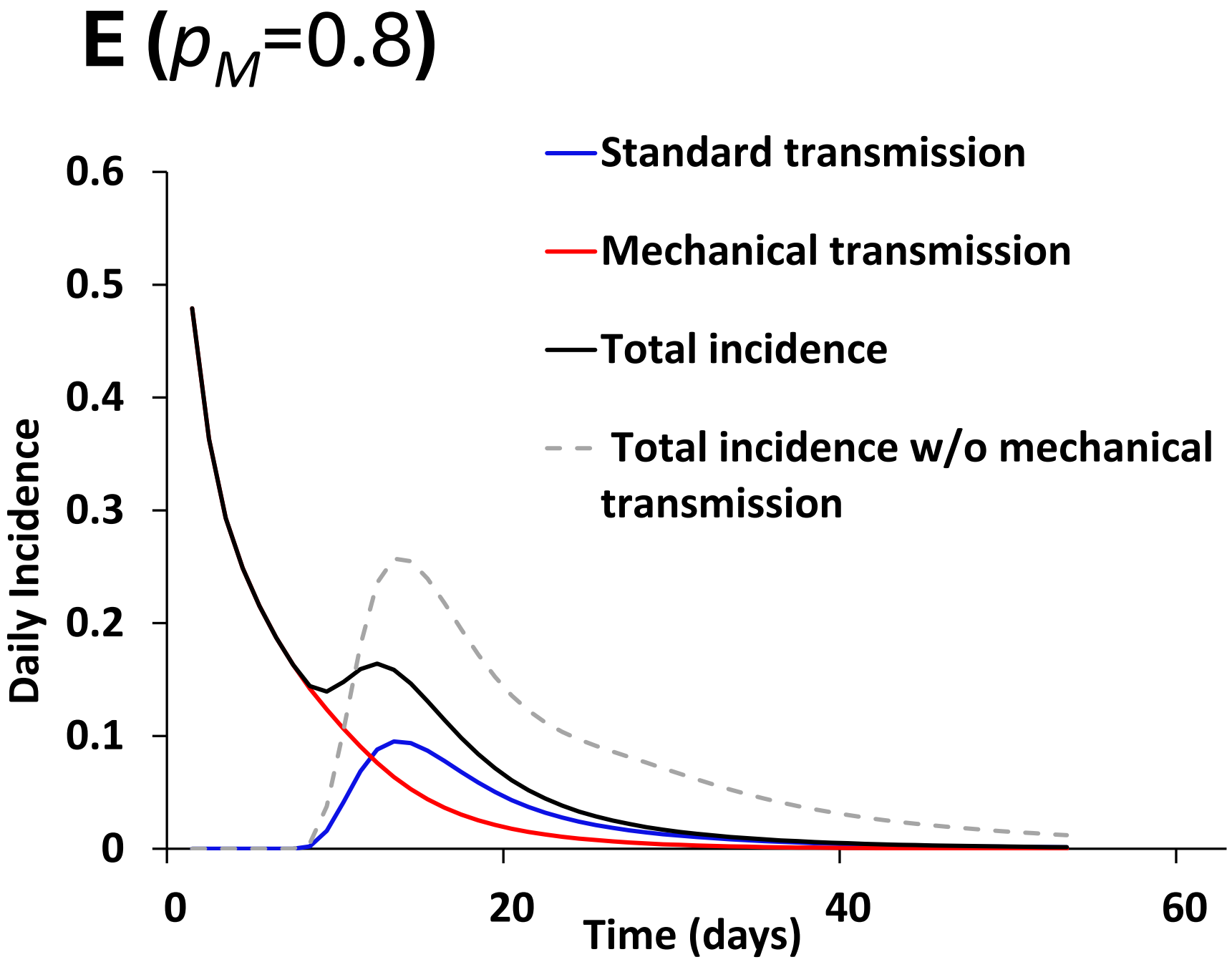


F (p_M=1)


A (p_M=0.1)
Standard transmission
Mechanical transmission
Total incidence
Total incidence w/o mechanical transmission
Daily Incidence
0.6
0.5
0.4
0.3
0.2
0.1
0
0
20
40
60
Time (days)

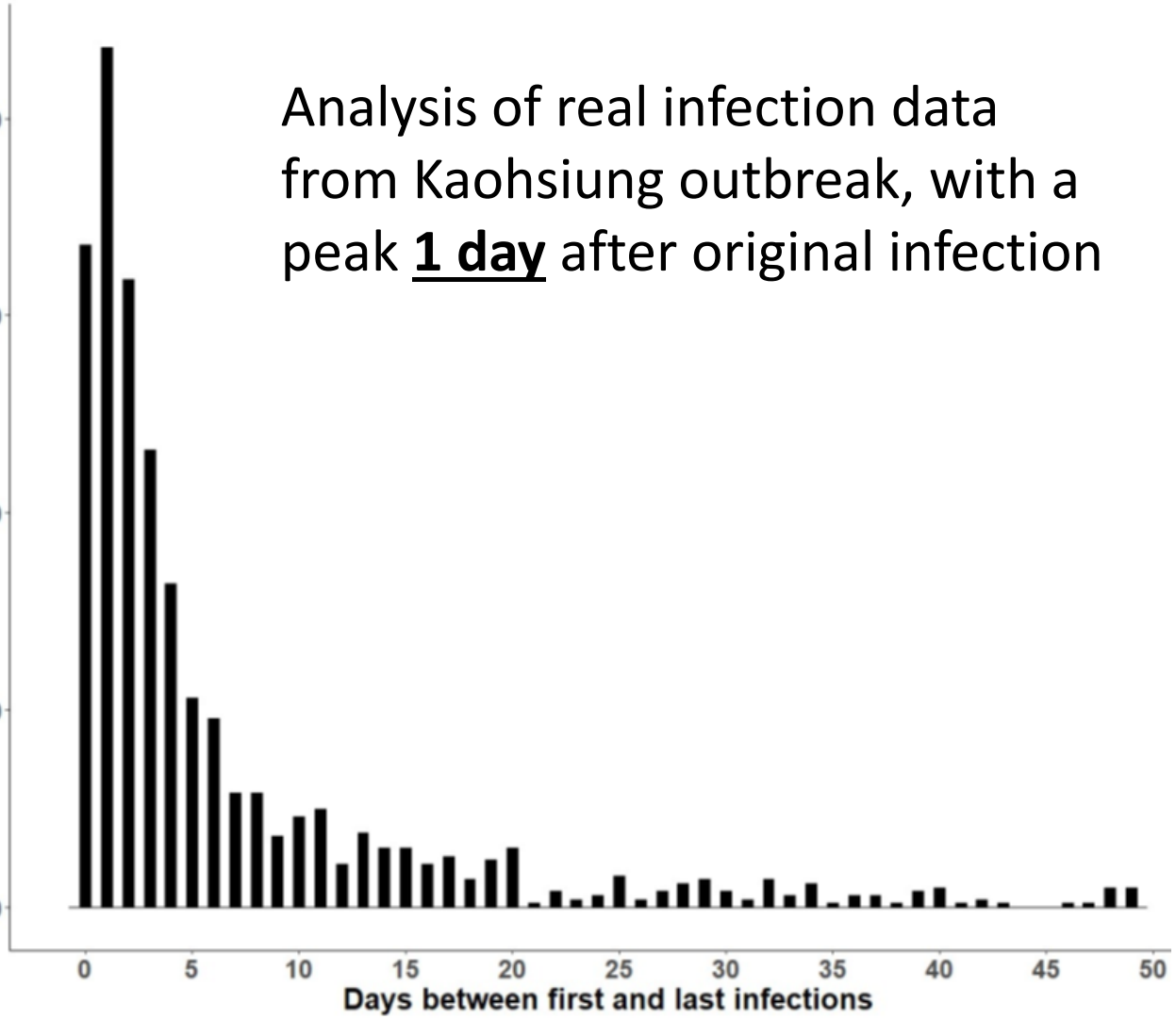
C (p_M=0.4)

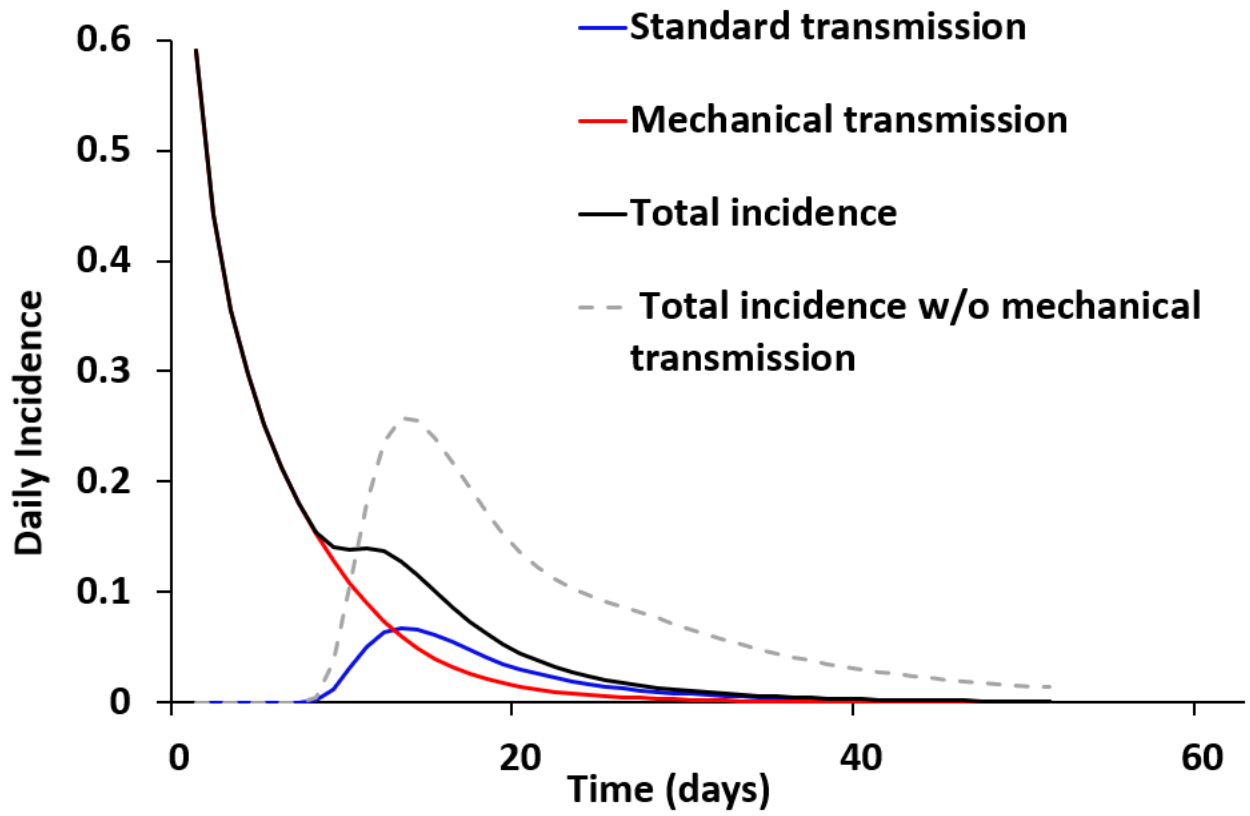

Standard transmission
Mechanical transmission
Total incidence
Total incidence w/o mechanical transmission
Daily Incidence
0.6
0.5
0.4
0.3
0.2
0.1
0
0
20
40
60
Time (days)


E (p_M=0.8)
Standard transmission
Mechanical transmission
Total incidence
Total incidence w/o mechanical transmission
Daily Incidence
0.6
0.5
0.4
0.3
0.2
0.1
0
0
20
40
60
Time (days)

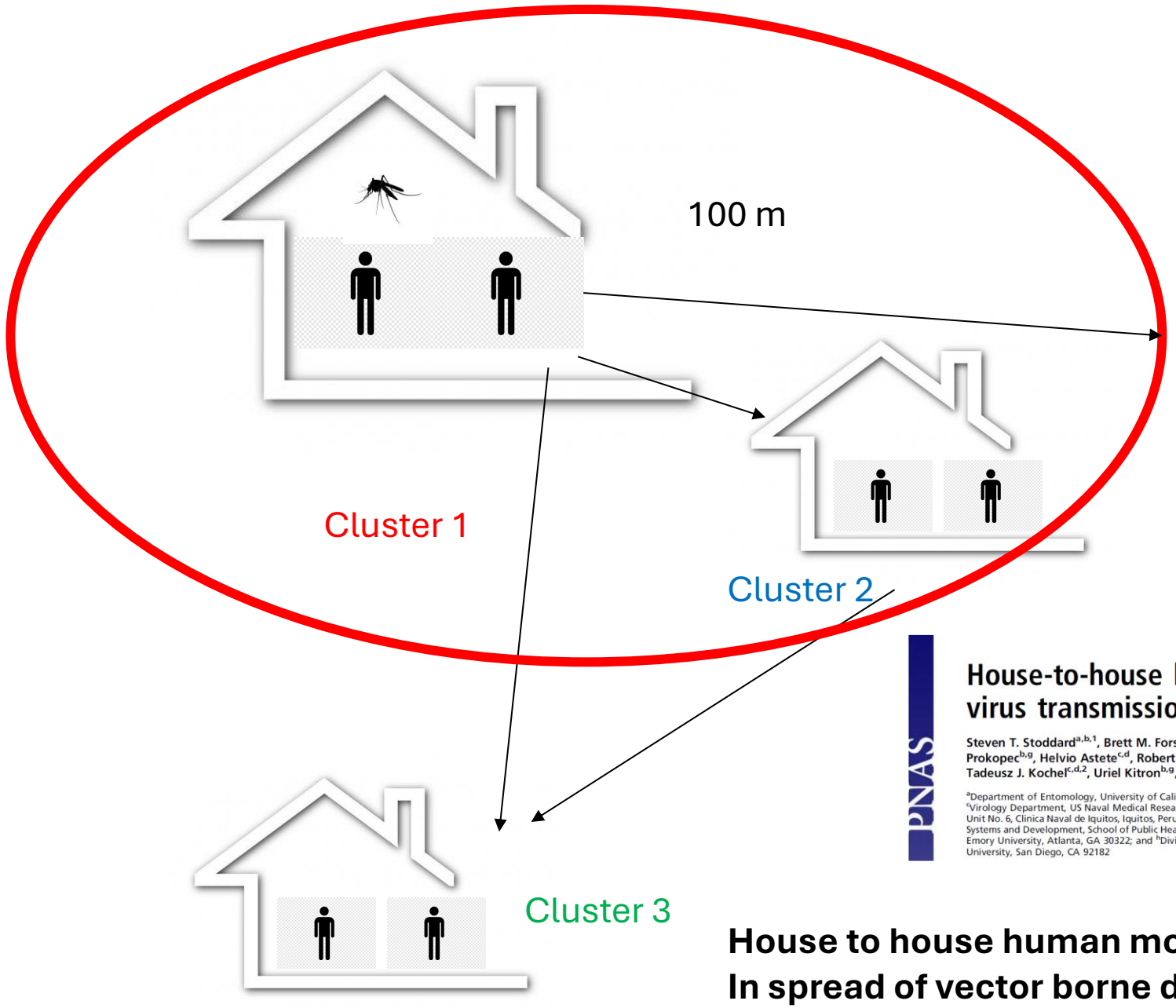
For 2015 KHC outbreak, in 85% of the clustered cases the time between first and next infection in the same cluster was less than 5days

Analysis of real infection data from Kaohsiung outbreak, with a peak **1 day** after original infection





In the current theory, cluster cases should be mostly within 100 m



PNAS

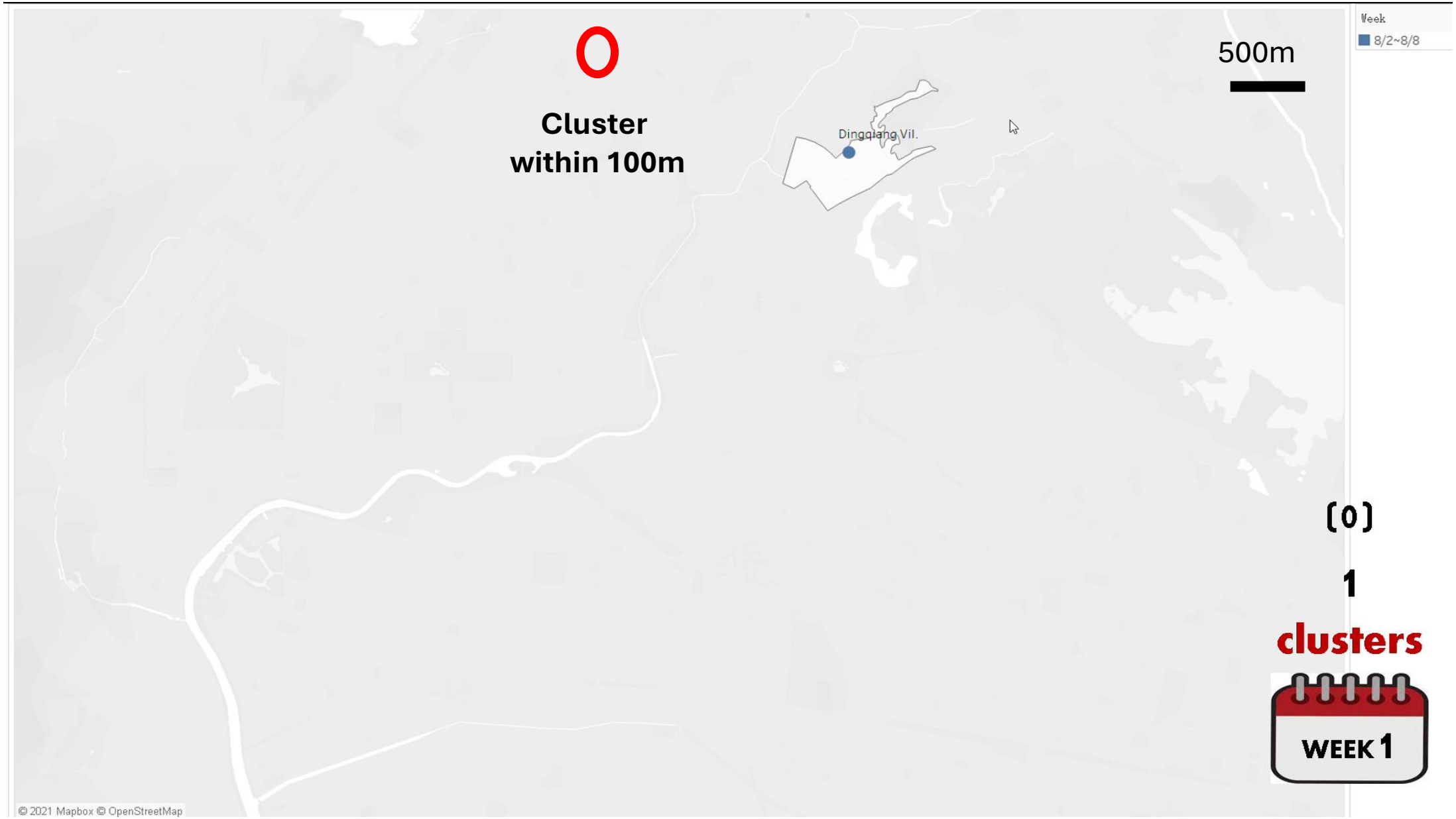
House-to-house human movement drives dengue virus transmission

Steven T. Stoddard[a,b,1], Brett M. Forshey[c,d,e], Amy C. Morrison[a,c,d], Valerie A. Paz-Soldan[f], Gonzalo M. Vazquez-Prokopec[b,g], Helvio Astete[c,d], Robert C. Reiner, Jr.[a,b], Stalin Vilcarromero[c,d], John P. Elder[h], Eric S. Halsey[c,d], Tadeusz J. Kochel[c,d,2], Uriel Kitron[b,g], and Thomas W. Scott[a,b]

[a]Department of Entomology, University of California, Davis, CA 95616; [b]Fogarty International Center, National Institutes of Health, Bethesda, MD 20892; [c]Virology Department, US Naval Medical Research Unit No. 6, Hospital Centro Medico Naval, Lima, Peru; [d]Virology Department, US Naval Medical Research Unit No. 6, Clinica Naval de Iquitos, Iquitos, Peru; [e]Department of Biostatistics, College of Public Health, University of Iowa, Iowa City, IA 52242; [f]Global Health Systems and Development, School of Public Health and Tropical Medicine, Tulane University, New Orleans, LA 70112; [g]Department of Environmental Studies, Emory University, Atlanta, GA 30322; and [h]Division of Health Promotion and Behavioral Sciences, Graduate School of Public Health, San Diego State University, San Diego, CA 92182

House to house human movement do play a key role In spread of vector borne disease

GIS location map for clustered cases from Kaohsiung 2015 outbreak (Aug to



Cluster within 100m: 67/397 16%

Very few Field-Caught *Aedes aegypti* were Dengue positive

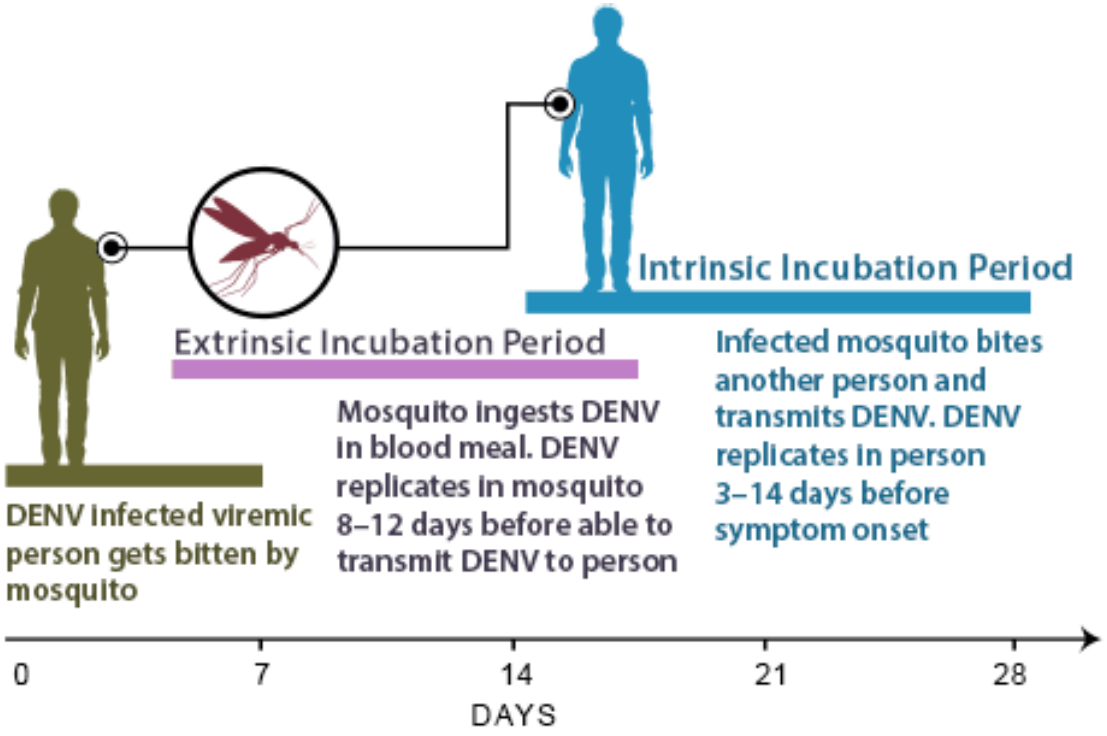
VECTOR-BORNE AND ZOONOTIC DISEASES
Volume 10, Number 10, 2010
© Mary Ann Liebert, Inc.
DOI: 10.1089/vbz.2008.0069

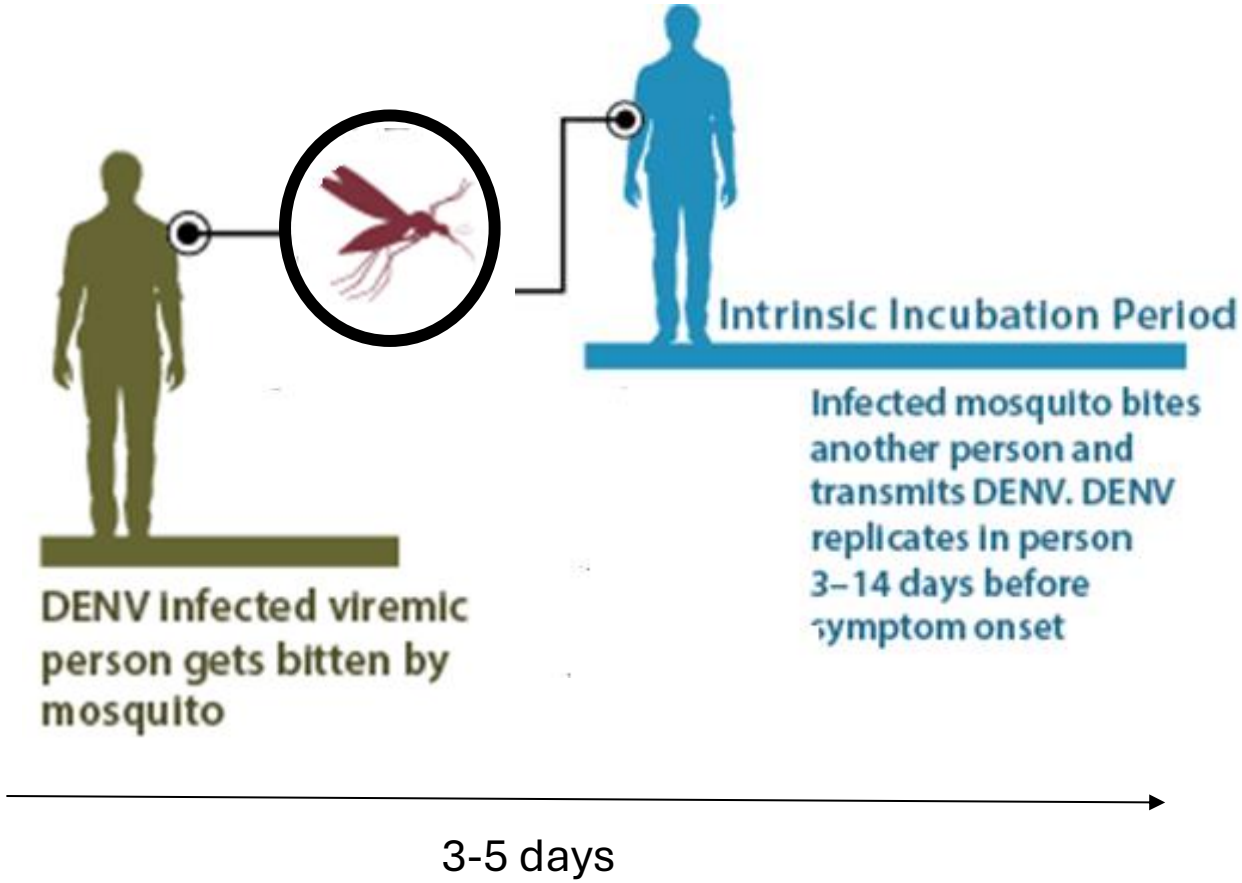
Screening of Dengue Virus in Field-Caught *Aedes aegypti* and *Aedes albopictus* (Diptera: Culicidae) by One-Step SYBR Green-Based Reverse Transcriptase-Polymerase Chain Reaction Assay During 2004–2007 in Southern Taiwan

Chien-Fu Chen,[1] Pei-Yun Shu,[1] Hwa-Jen Teng,[1] Chien-Ling Su,[1] Jhy-Wen Wu,[2] Jen-Hsin Wang,[2] Ting-Hsiang Lin,[1] Jyh-Hsiung Huang,[1] and Ho-Sheng Wu[1]

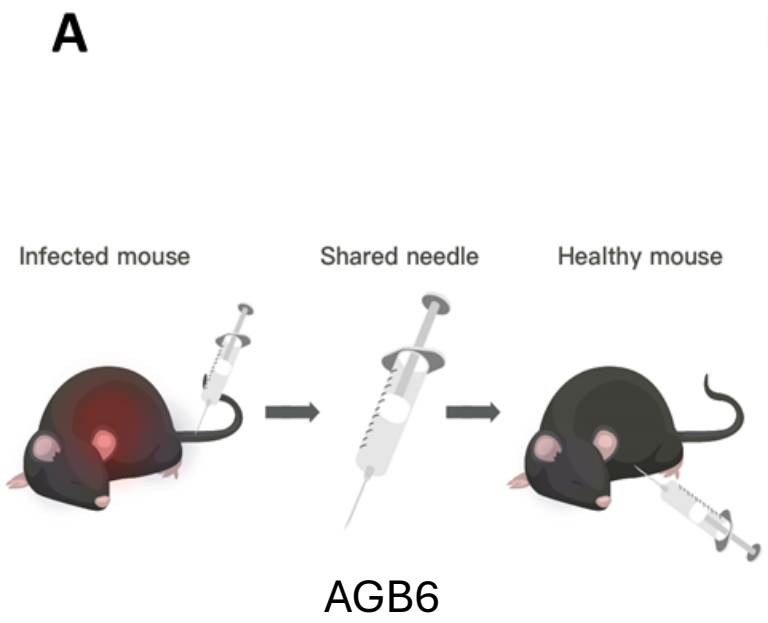
Infection rate 0.5-2%

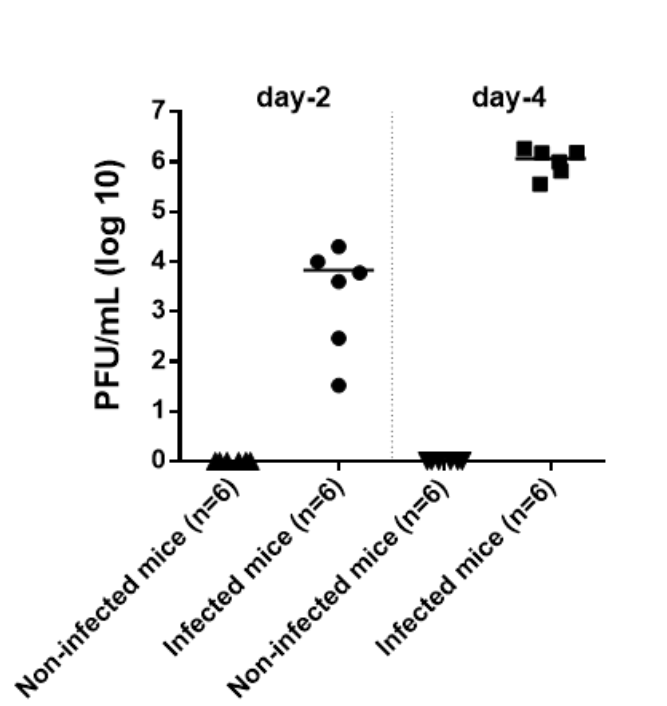
Do we have any biological evidence to prove EIP is not essential for Dengue transmission?

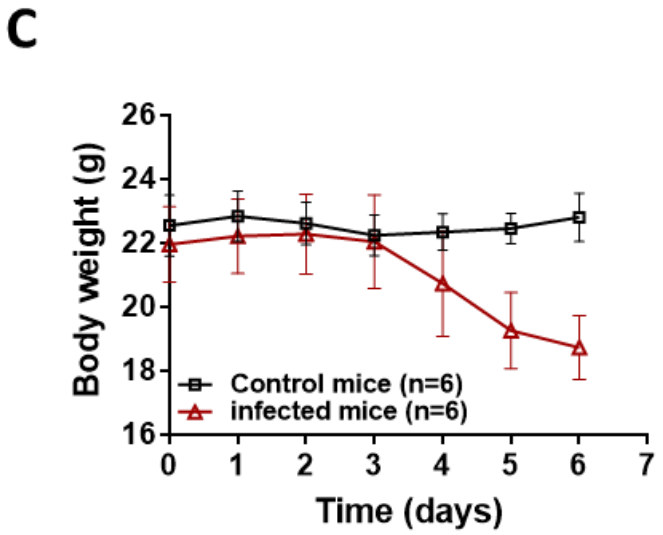


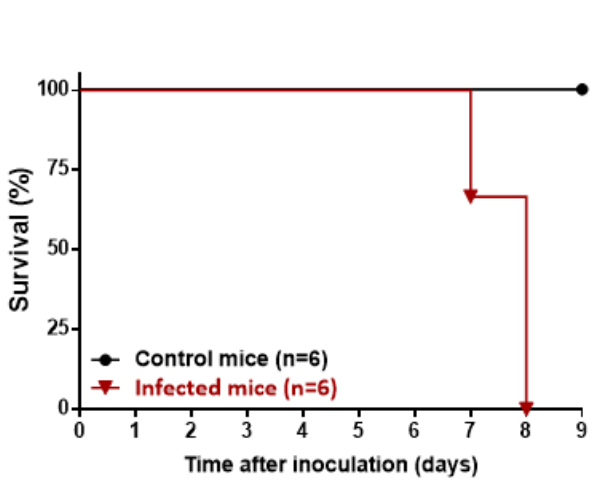


Dengue virus 2015 TW can be transmitted via needle sharing in mice



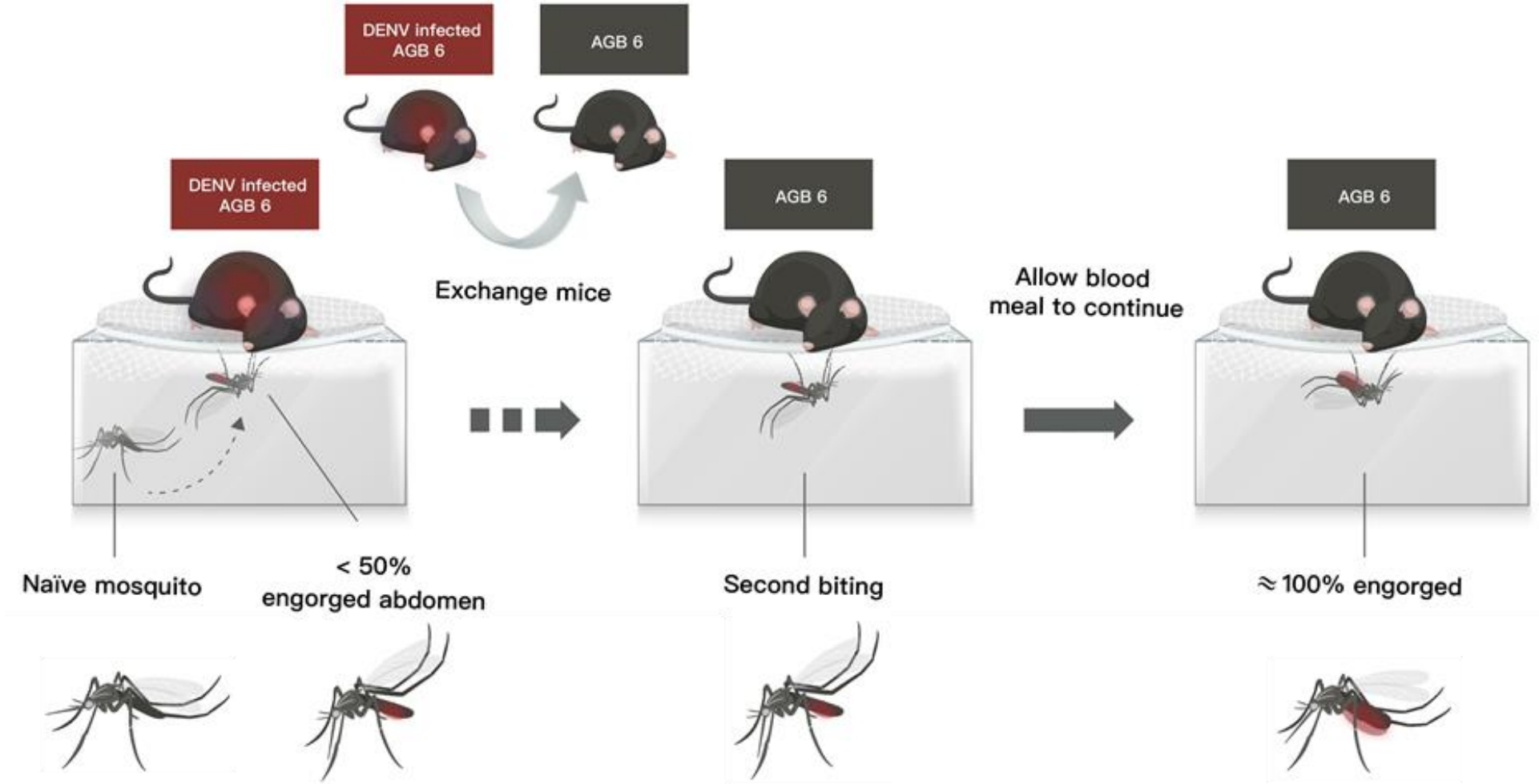






Schematic model to investigate mechanical transmission of DENV by *Aedes aegypti* mosquitoes





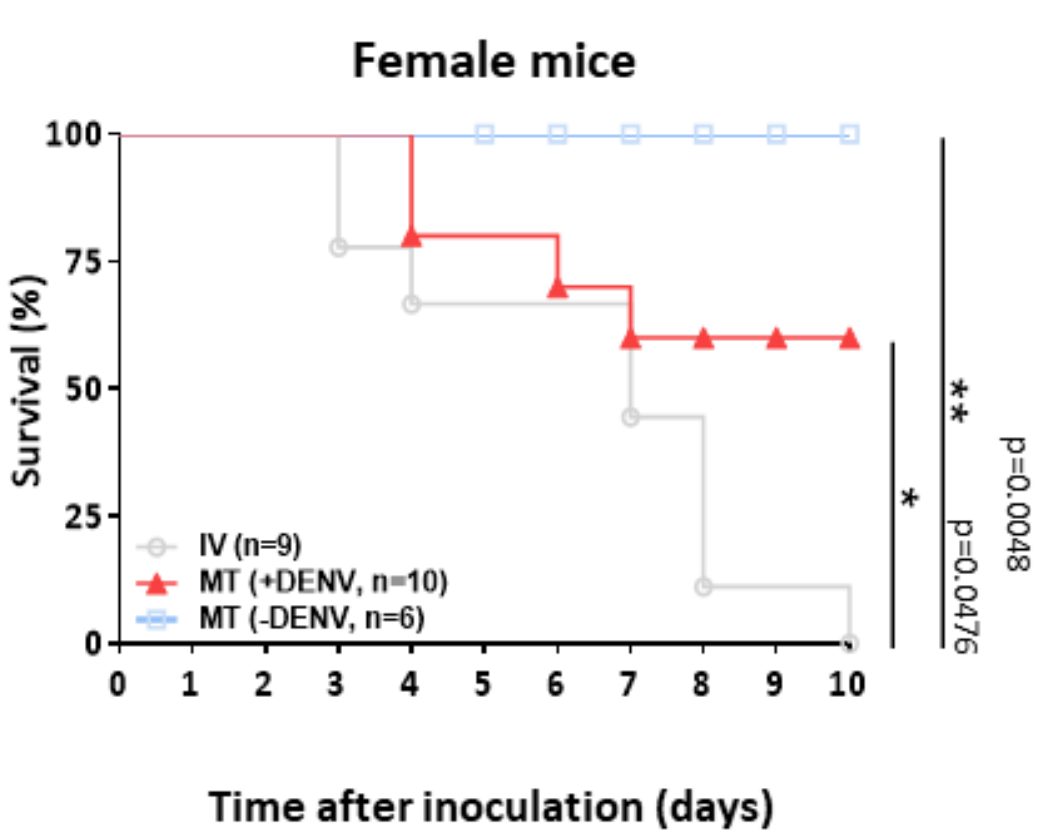
Mechanical transmission of DENV by *Aedes aegypti* mosquitoes

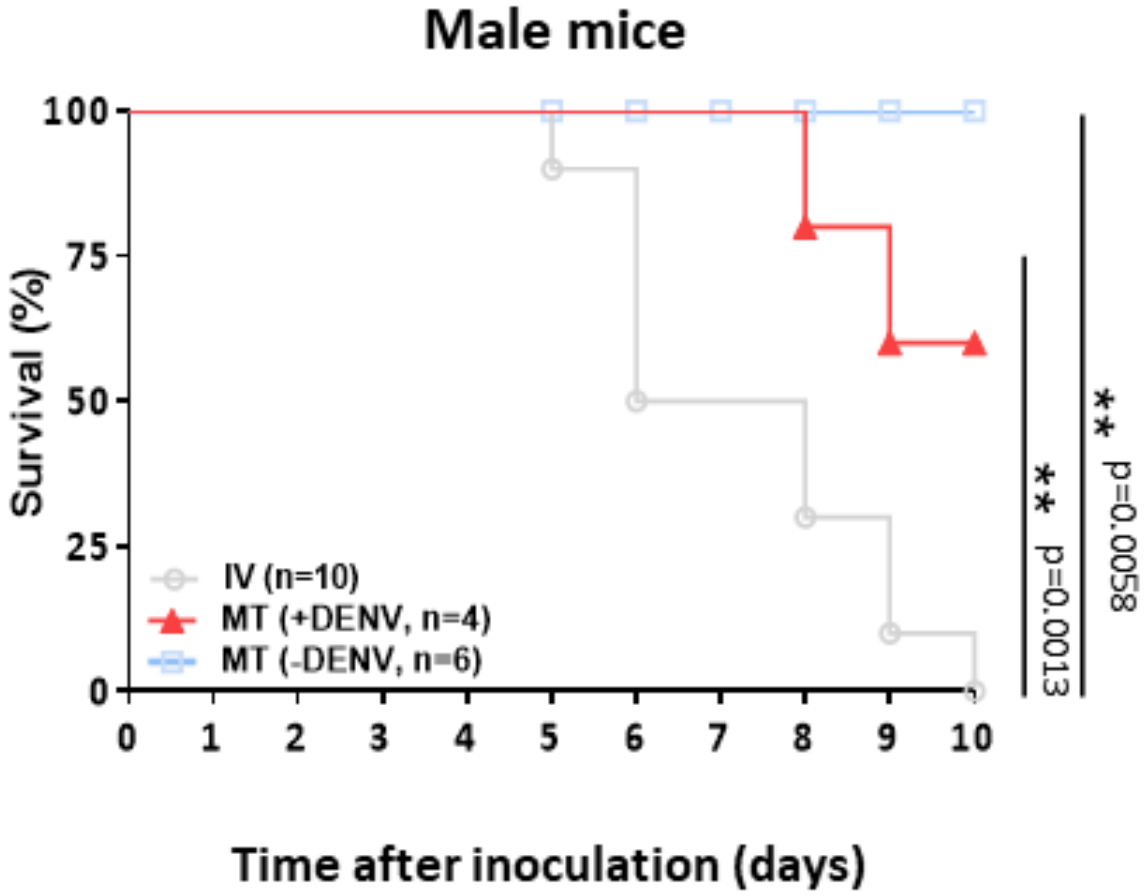






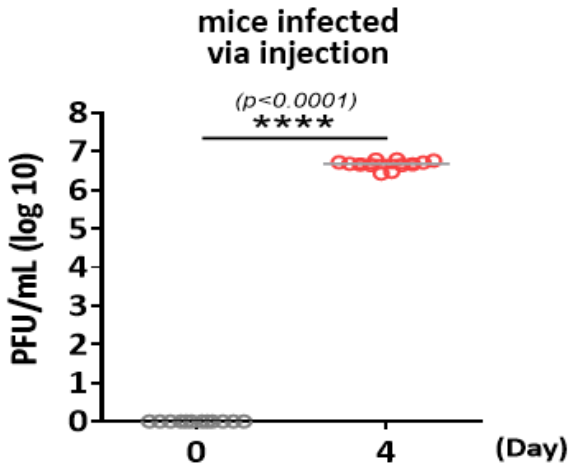
Mice were infected with DENV via mechanical transmission (MT) following 4 bites by DENV-exposed *A. aegypti*



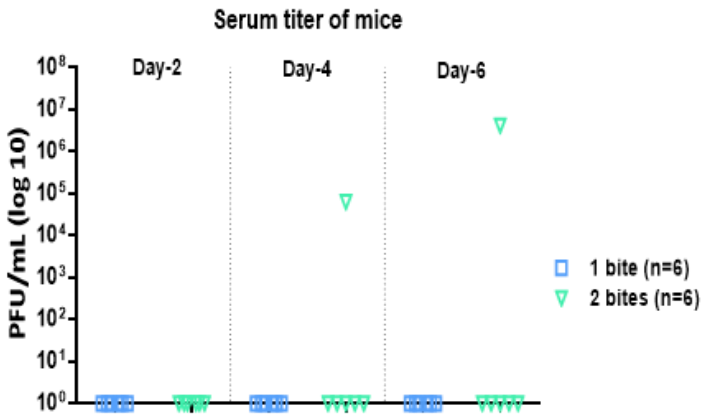


Mice can be infected via mechanical transmission (MT) by only two bites of DENV-exposed *A. aegypti*

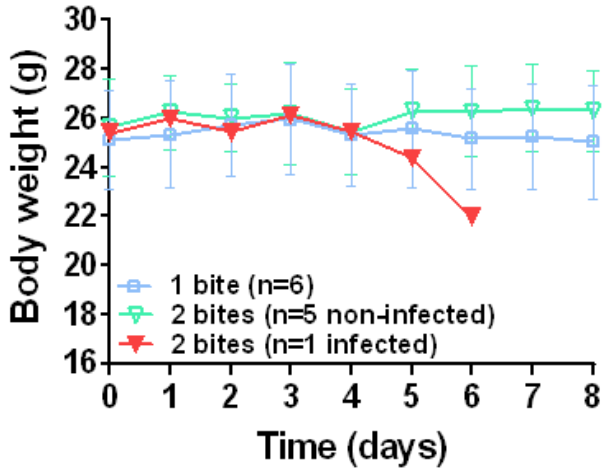
A



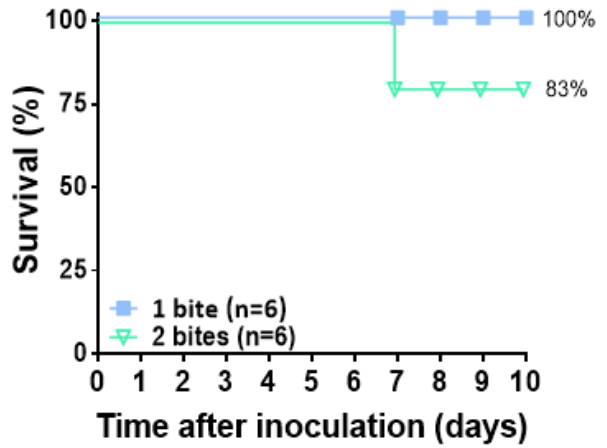
B



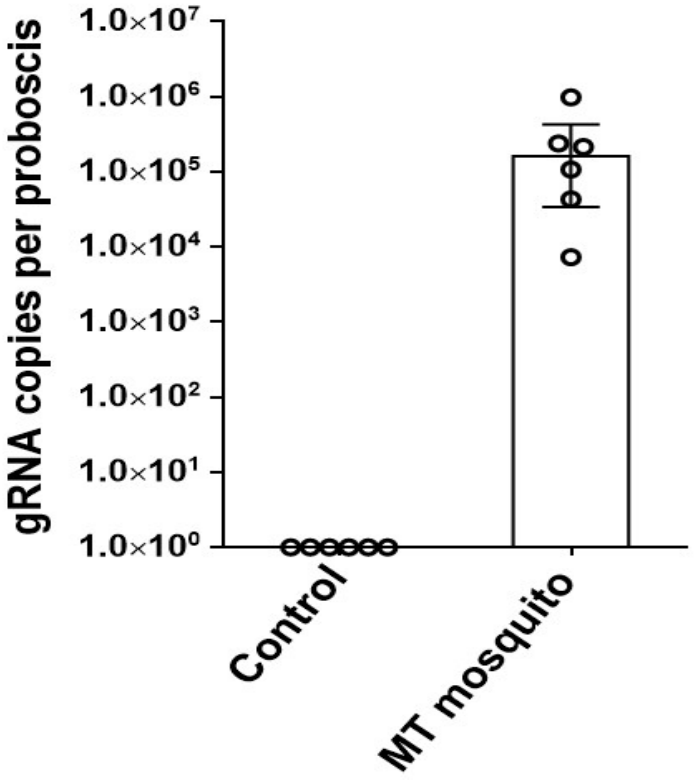
C

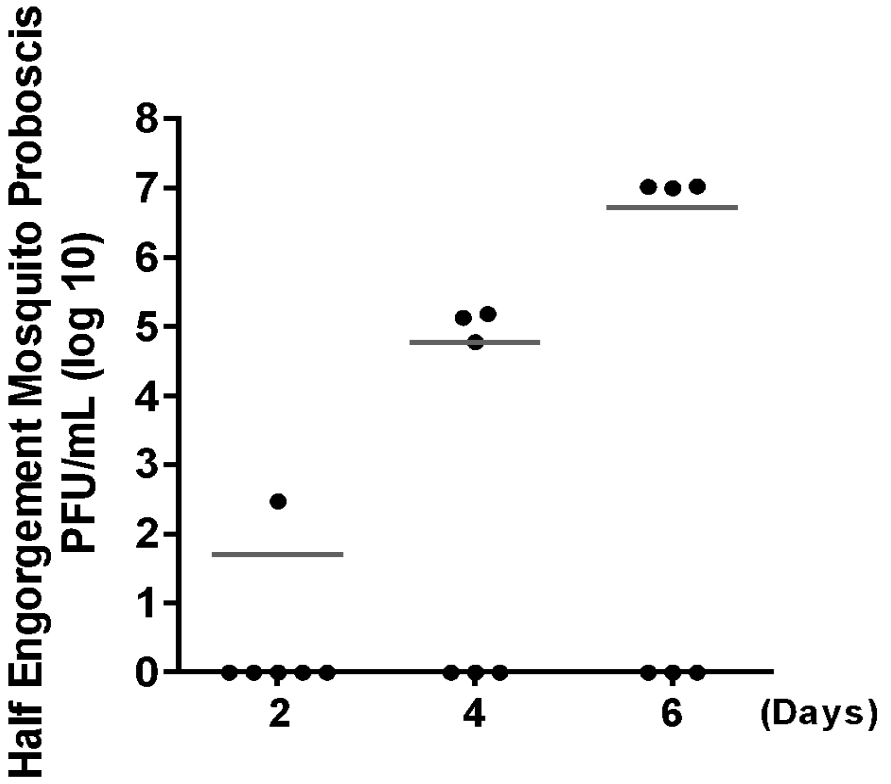


D

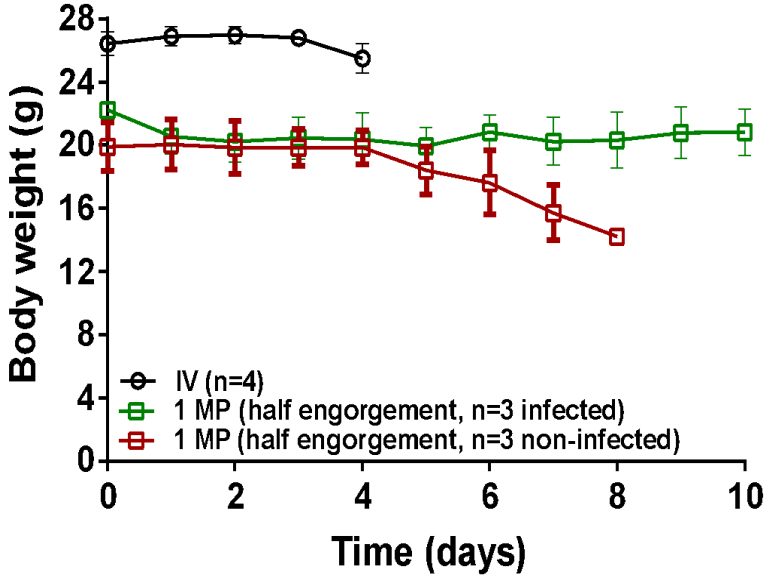


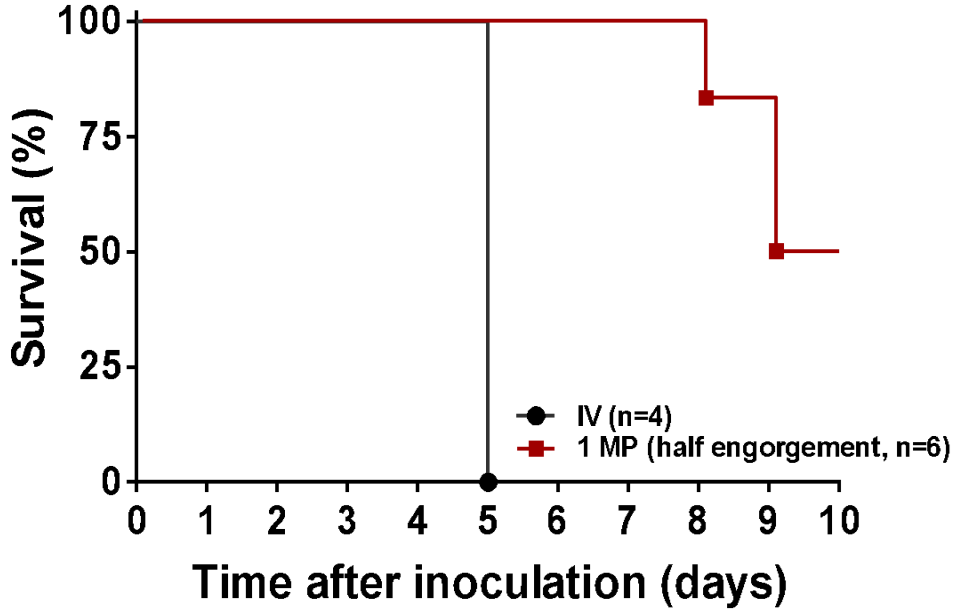
Residual DENV in proboscis of A. aegypti mediates dengue spread via mechanical transmission.





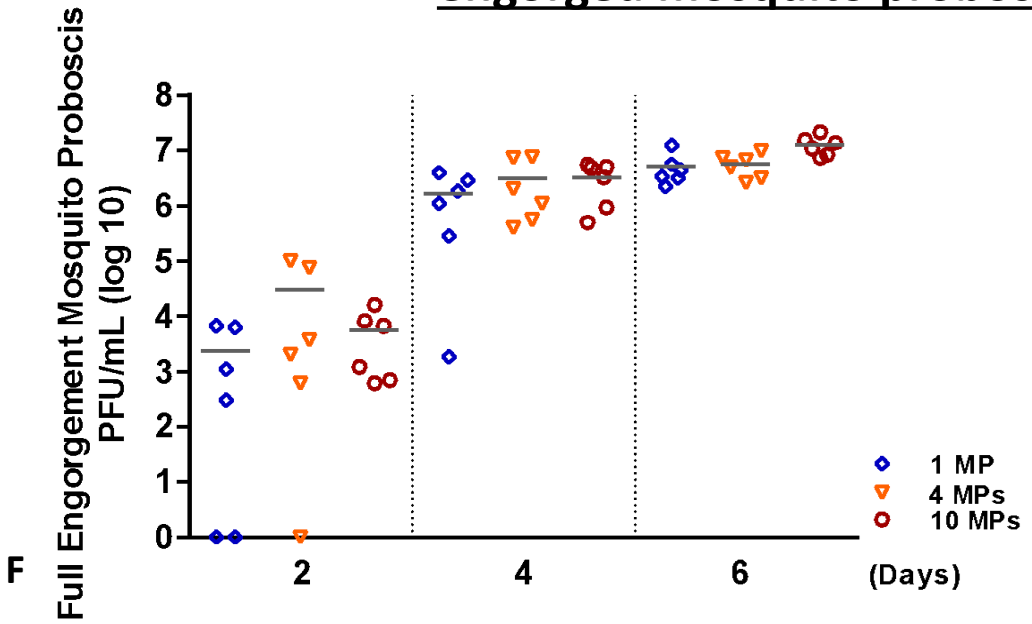
Survival curves of male mice intravenously injected with the extracts from the indicated number of half engorged mosquito proboscises (MPs) from DENV blood-exposed

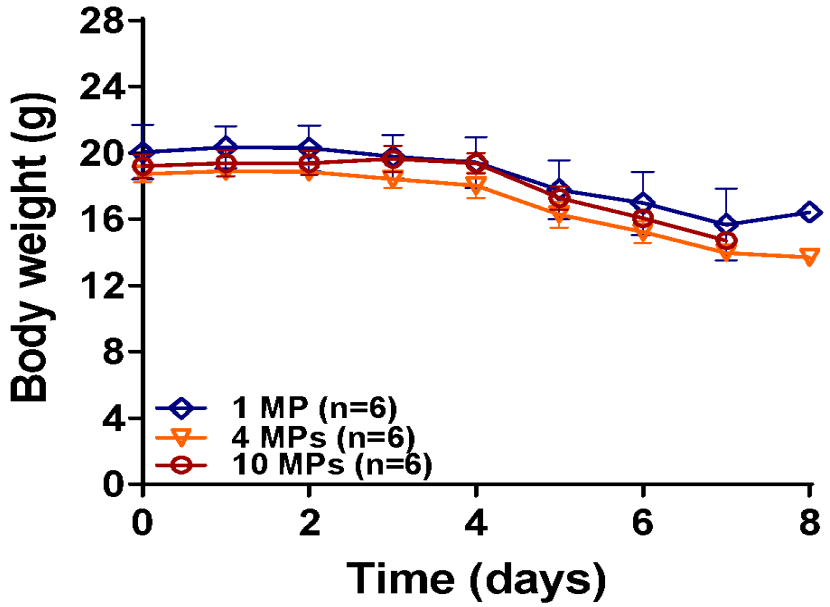


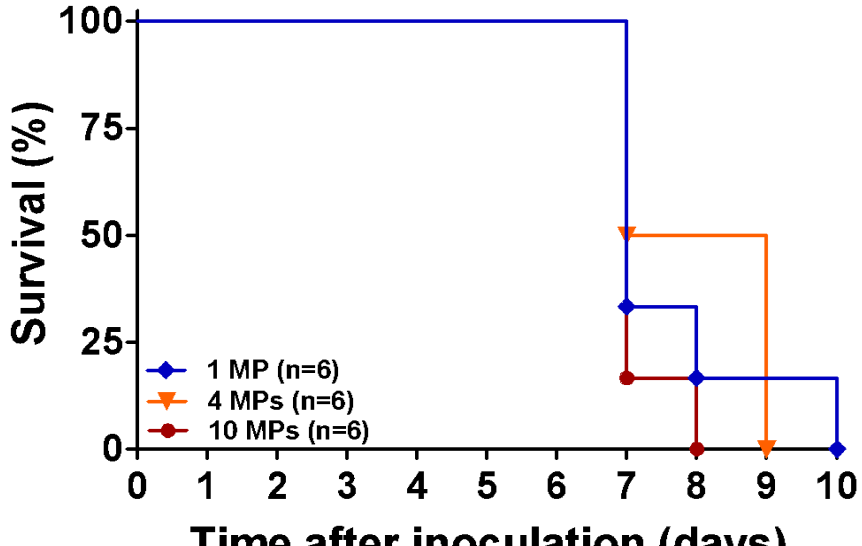


Survival curves of male mice intravenously injected with the extracts from the indicated number of full engorged mosquito proboscises (MPs) from DENV blood-exposed

F

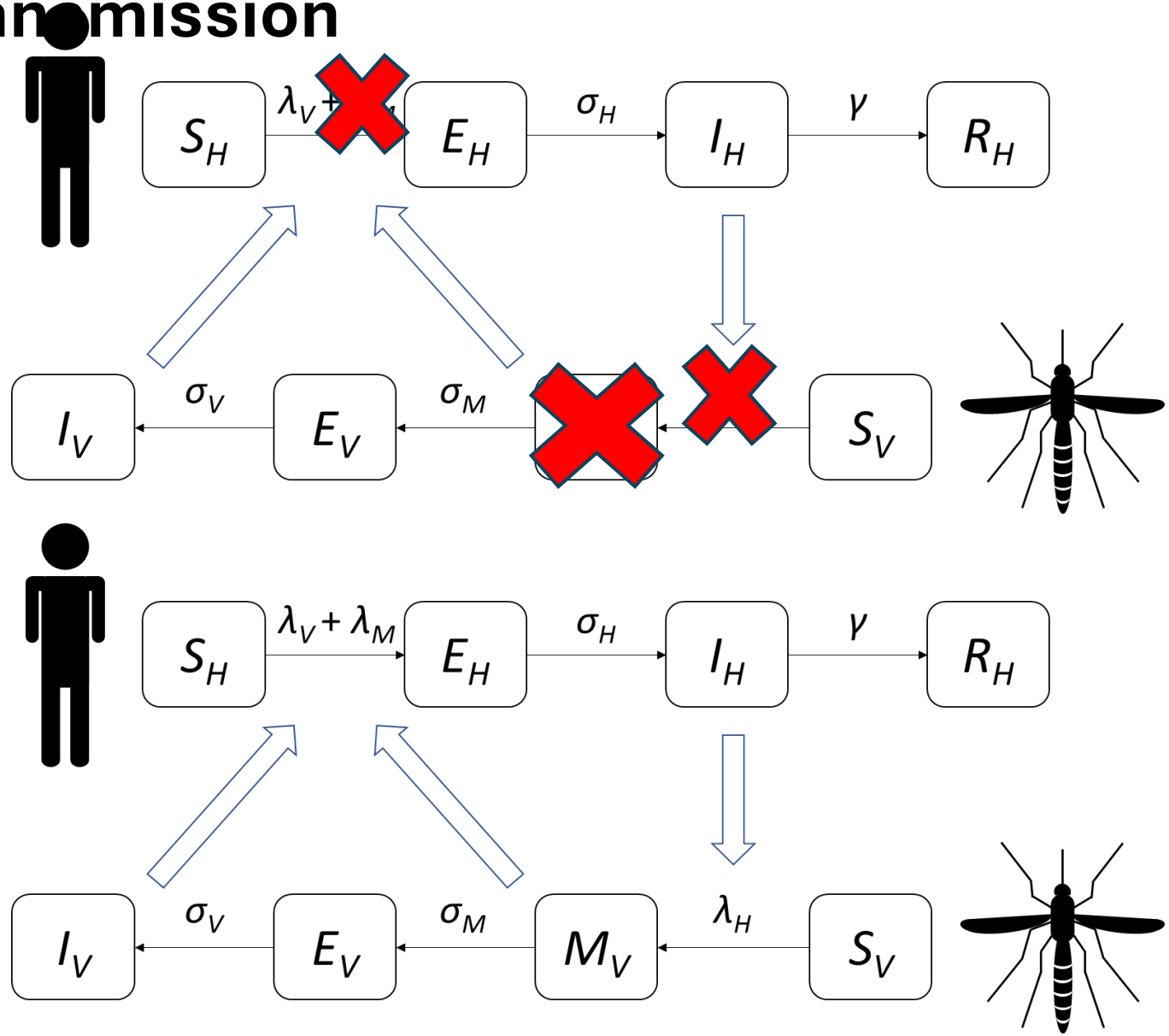




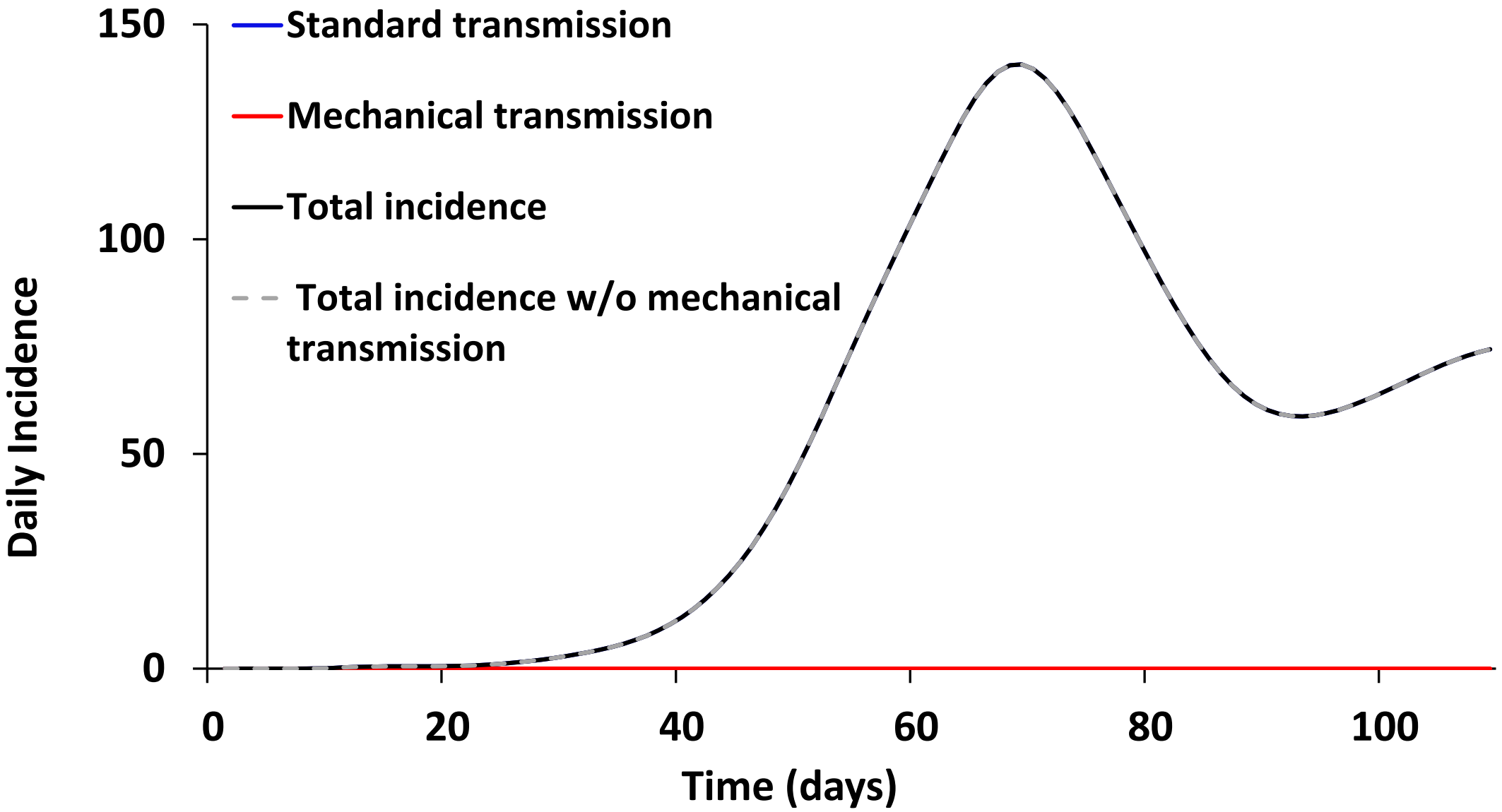


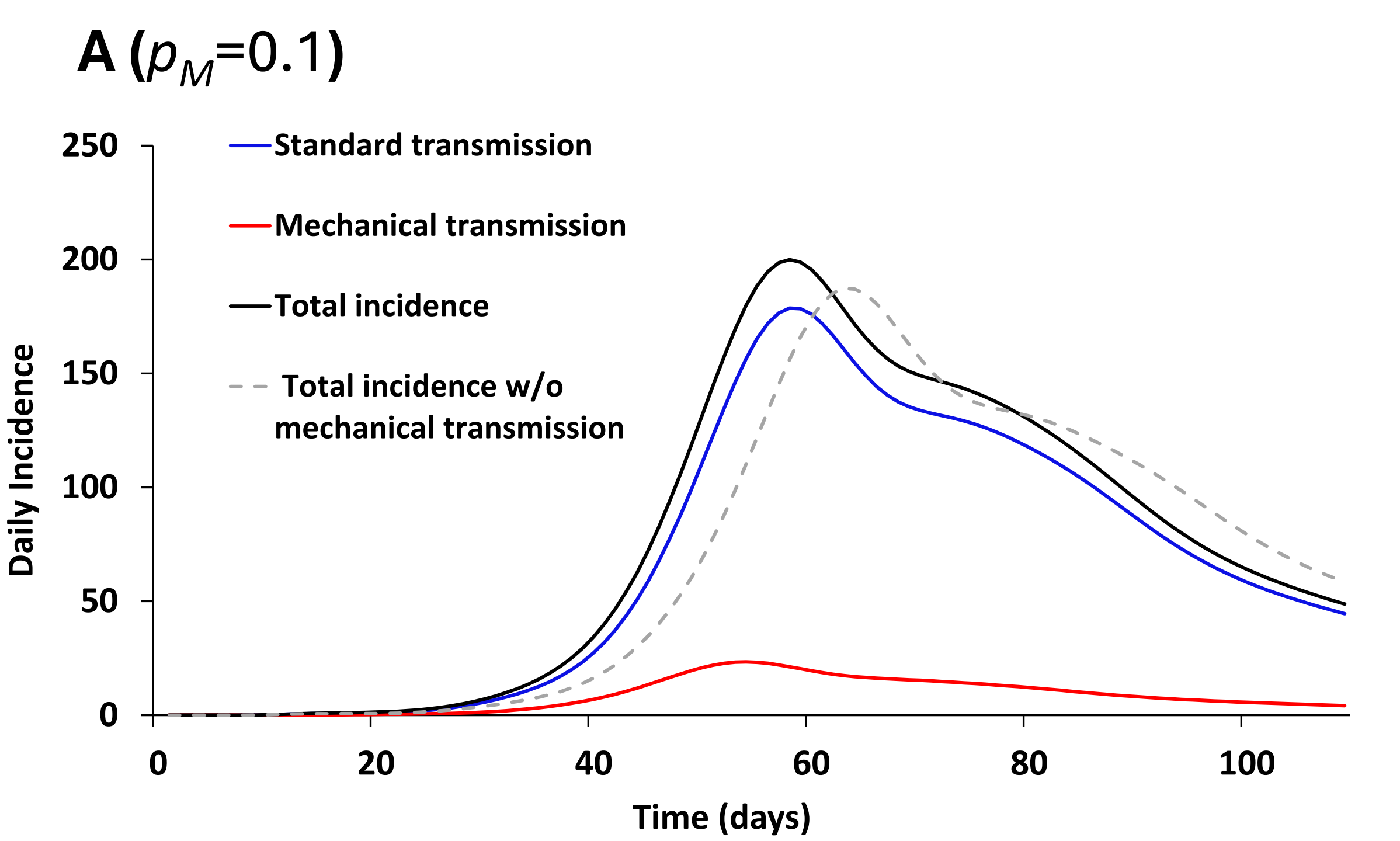
Mathematical model of dengue incorporating mechanical transmission

Two models created for comparison – one including only standard dengue transmission, the other incorporating both standard and mechanical transmission

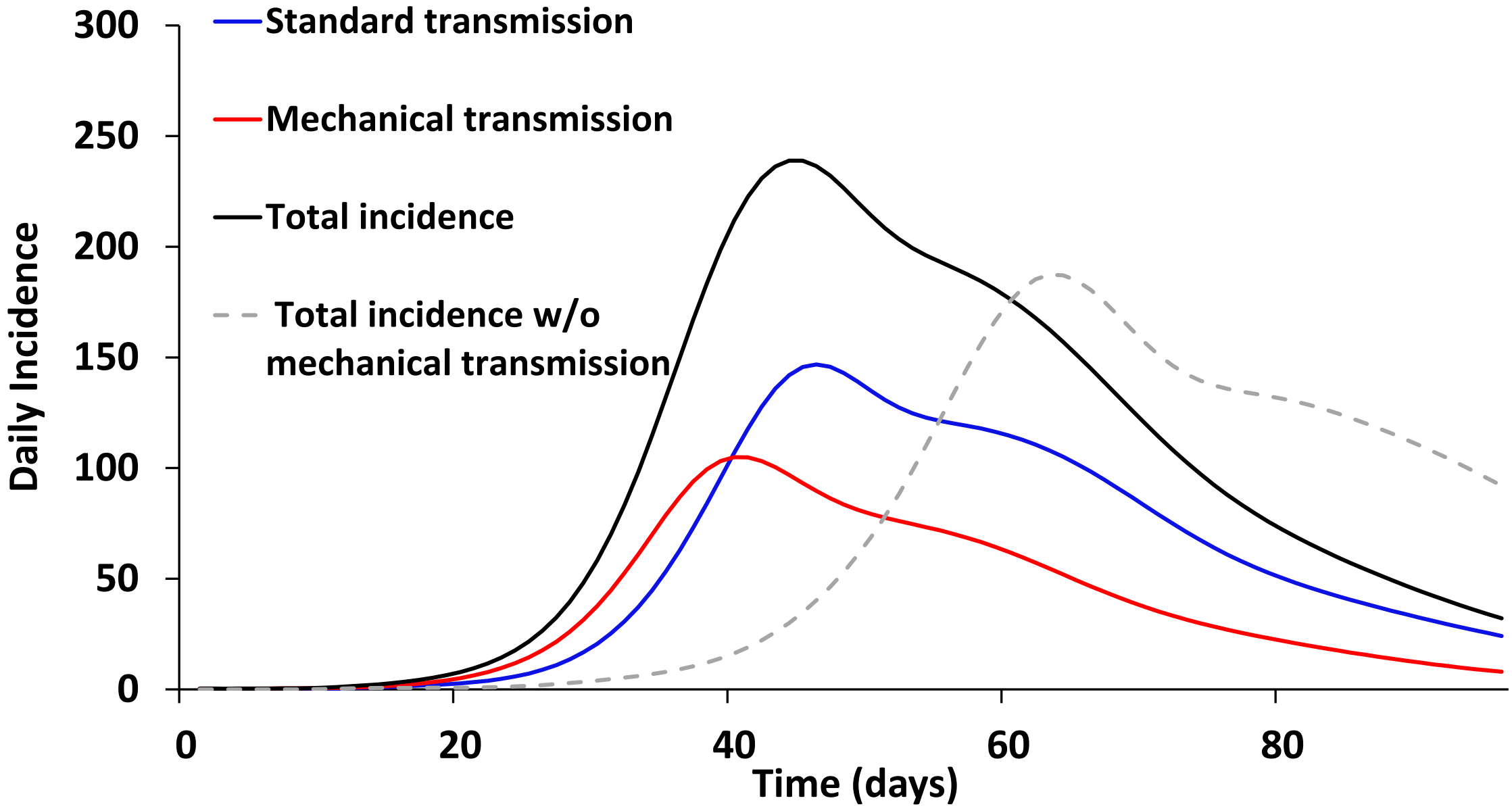


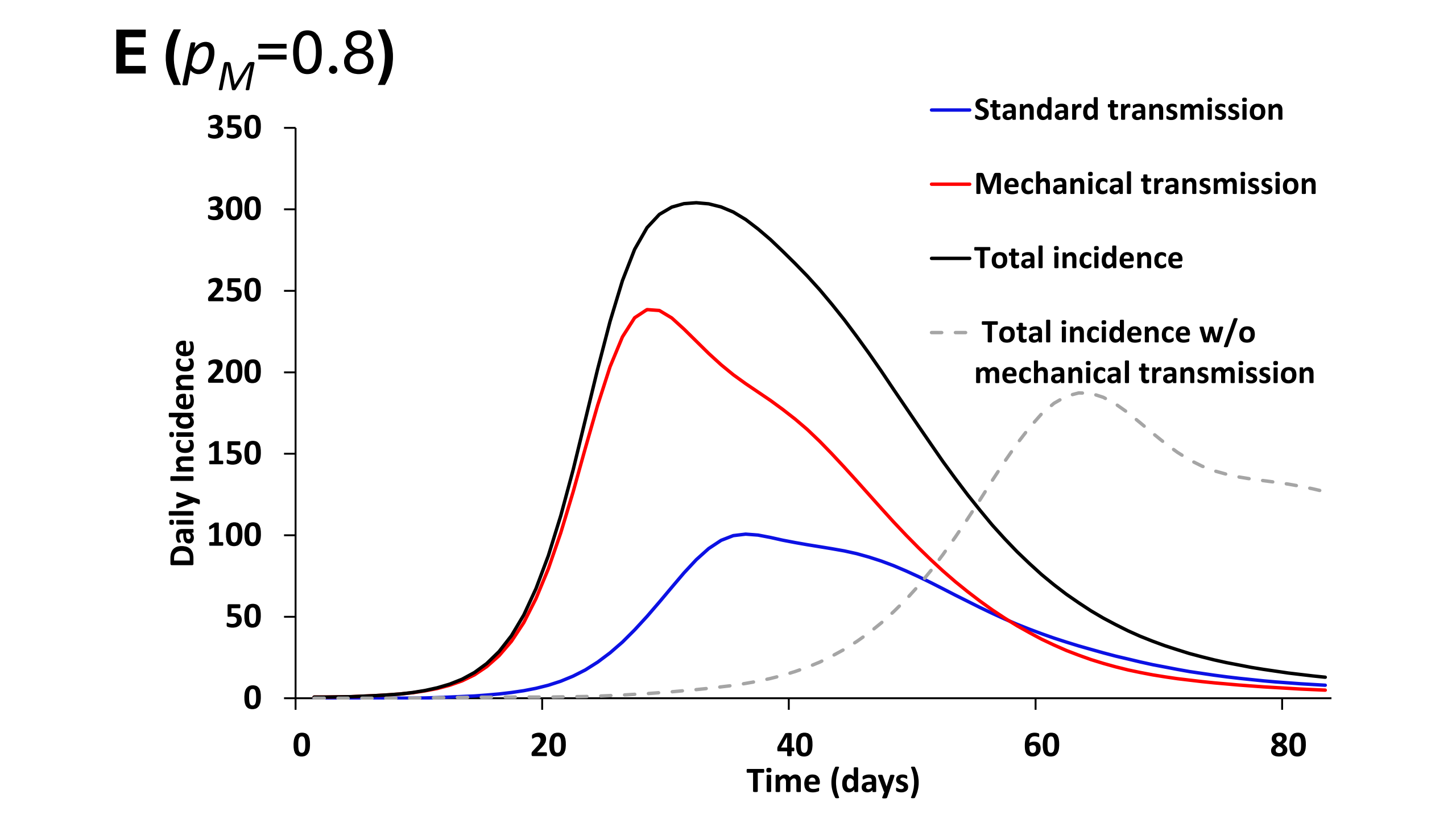
NH=10000 (p_M=0)


Standard transmission
Mechanical transmission
Total incidence
Total incidence w/o mechanical transmission
Daily Incidence
150
100
50
0
0
20
40
60
80
100
Time (days)

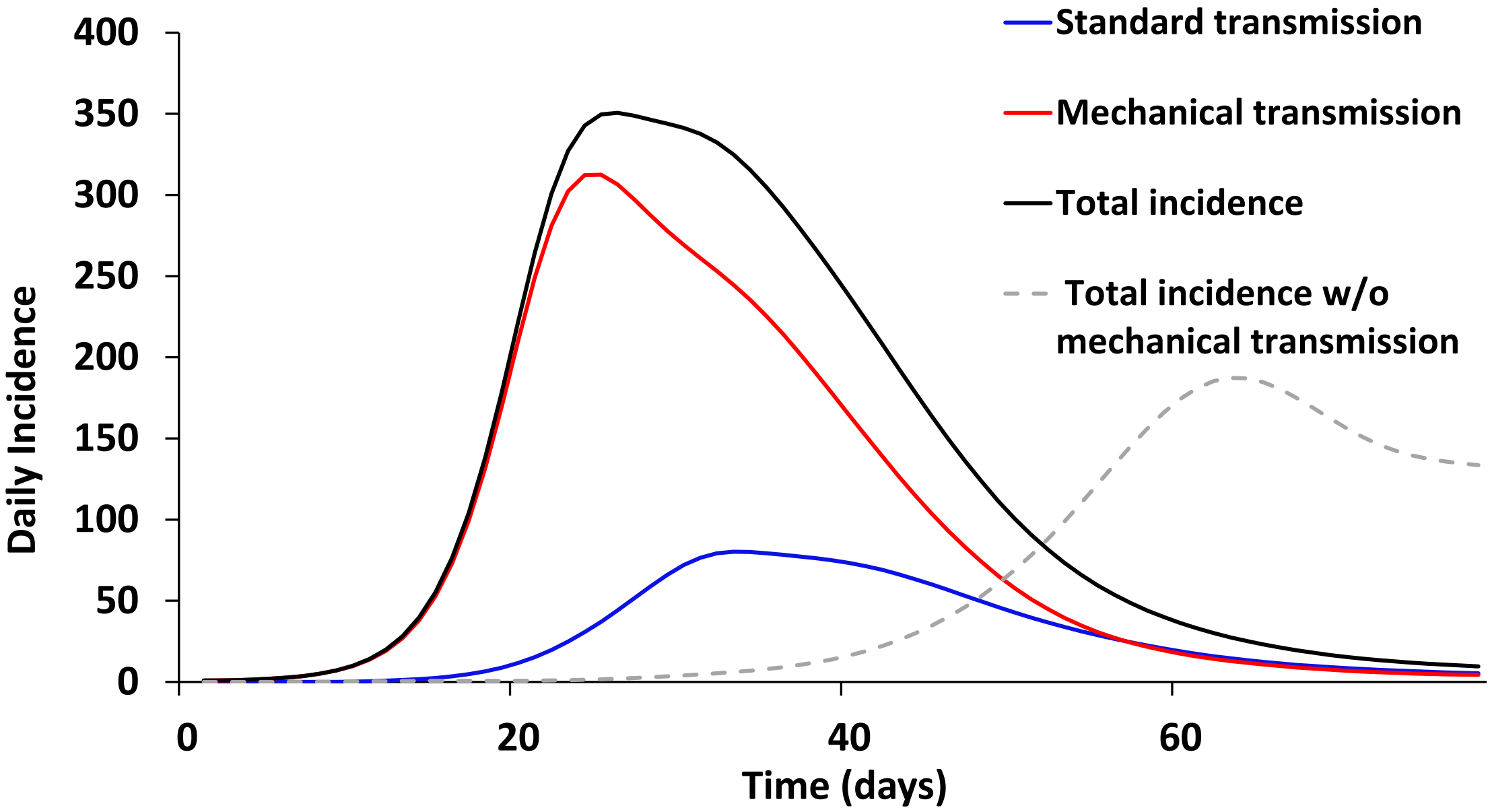

A (p_M=0.1)
Standard transmission
Mechanical transmission
Total incidence
Total incidence w/o mechanical transmission
Daily Incidence
250
200
150
100
50
0
0
20
40
60
80
100
Time (days)

C (p_M=0.4)


Standard transmission
Mechanical transmission
Total incidence
Total incidence w/o mechanical transmission
Daily Incidence
300
250
200
150
100
50
0
0
20
40
60
80
Time (days)


E (p_M=0.8)
Standard transmission
Mechanical transmission
Total incidence
Total incidence w/o mechanical transmission
Daily Incidence
350
300
250
200
150
100
50
0
0
20
40
60
80
Time (days)

F (p_M=1)


Daily Incidence
400
350
300
250
200
150
100
50
0
0
20
40
60
Time (days)
Standard transmission
Mechanical transmission
Total incidence
Total incidence w/o mechanical transmission

Take home messages

DENV2 2015 TW can be immediately mechanically transmitted to naïve mice without the virus undergoing replication within the mosquito

EIP may not be necessary for Dengue transmission

Modelling of mechanical transmission of DENV by *A. aegypti* suggests it could contribute significantly to DENV spread in densely populated regions

This transmission mechanism could also be applicable for the transmission of other *Aedes*-borne diseases (such as Zika),as well as diseases carried by other mosquito species (such as the *Anopheles*-based transmission of malaria)

Wolbachia-Based Mosquito Control:

Progress in Taiwan, Singapore, and Global Perspectives



TAIWAN – TAINAN
研發創新・在地推動



SINGAPORE
科學實證・國家計畫



GLOBAL
國際合作・永續未來



國家蚊媒傳染病防治研究中心
陳俊宏 代理主任



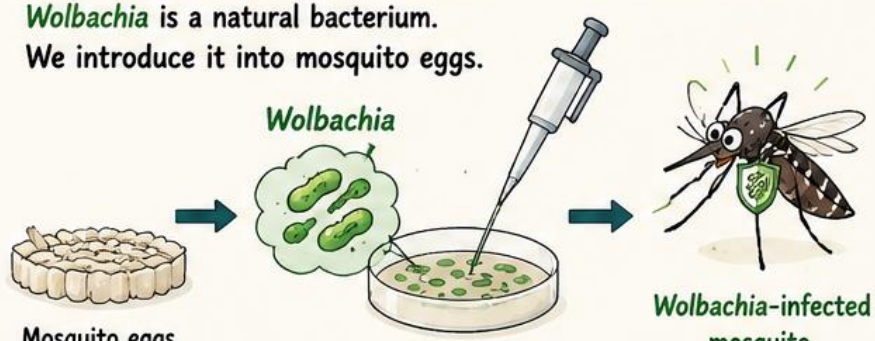
Wolbachia: A VACCINE FOR MOSQUITOES

Immunize mosquitoes, protect humans.



1. VACCINATE MOSQUITOES WITH WOLBACHIA

Wolbachia is a natural bacterium.
We introduce it into mosquito eggs.



2. WOLBACHIA PROTECTS THE MOSQUITO

Blocks viruses



Reduces dengue, Zika and other virus infections

Reproductive advantage



Causes cytoplasmic incompatibility (less viable offspring with wild mosquitoes)

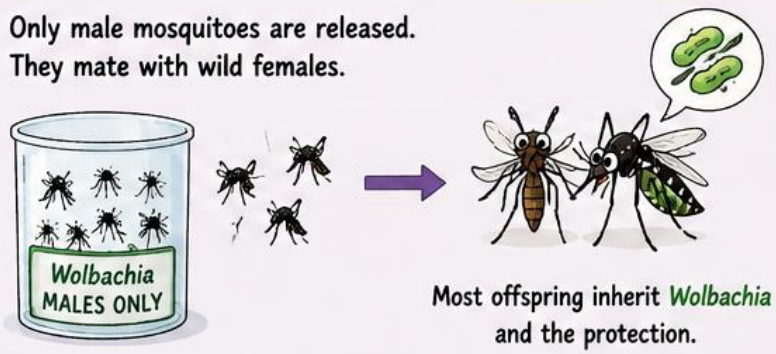
Stable infection



Wolbachia is passed from mother to offspring

3. RELEASE WOLBACHIA MALE MOSQUITOES

Only male mosquitoes are released.
They mate with wild females.



Most offspring inherit *Wolbachia* and the protection.

4. HEALTHIER MOSQUITO POPULATION, LOWER DISEASE RISK FOR HUMANS



Fewer mosquitoes

Lower virus transmission

Lower dengue risk, healthier communities!

WOLBACHIA IS A VACCINE FOR MOSQUITOES



- Natural
- Self-sustaining
- Environmentally friendly

A smart way to fight mosquito-borne diseases!

Southern Taiwan is the major habitat for *Aedes aegypti*, mainly responsible for transmission of Dengue and Zika

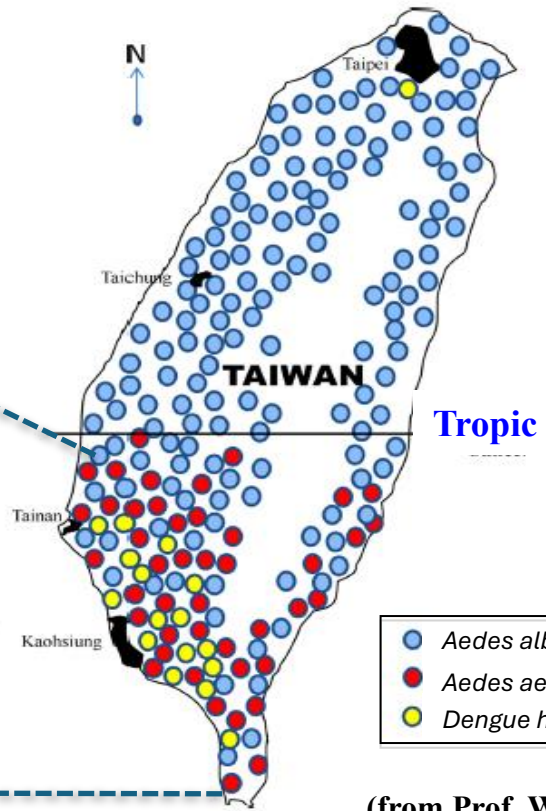
According to the accumulated results obtained from 1998 to 2016, ~96% of 78,022 dengue cases in Taiwan were reported from the three southern cities.

Tainan 26758 (34%)

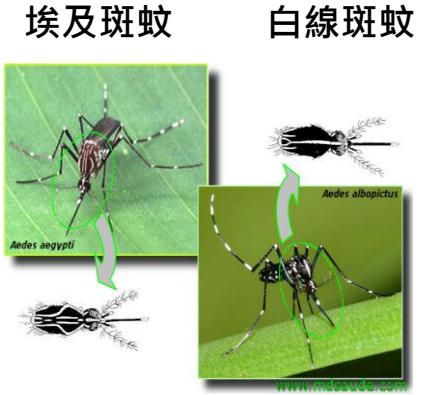
Kaohsiung 45978 (59%)

Pingtung 2167 (3%)

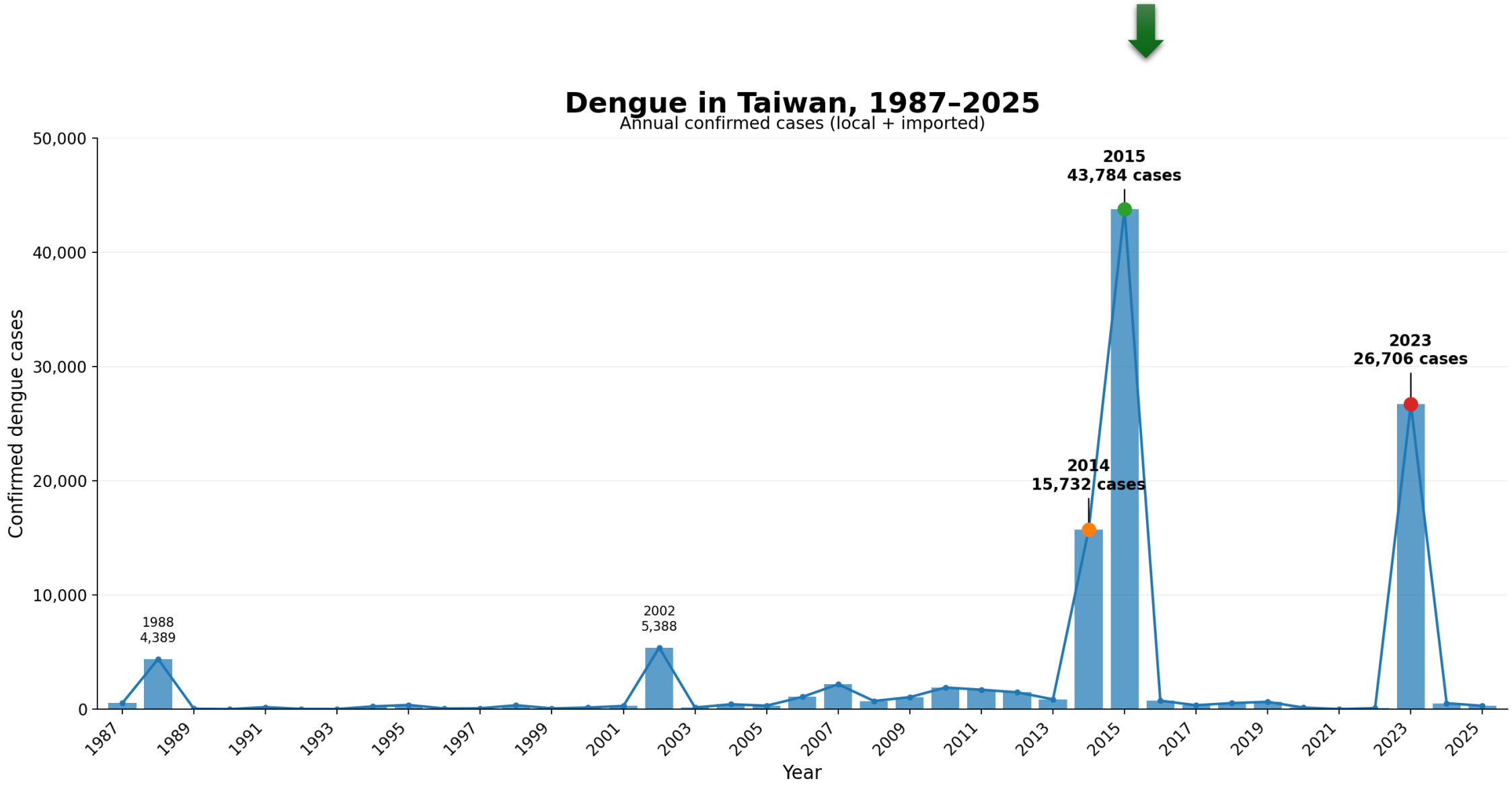
(from Taiwan CDC)



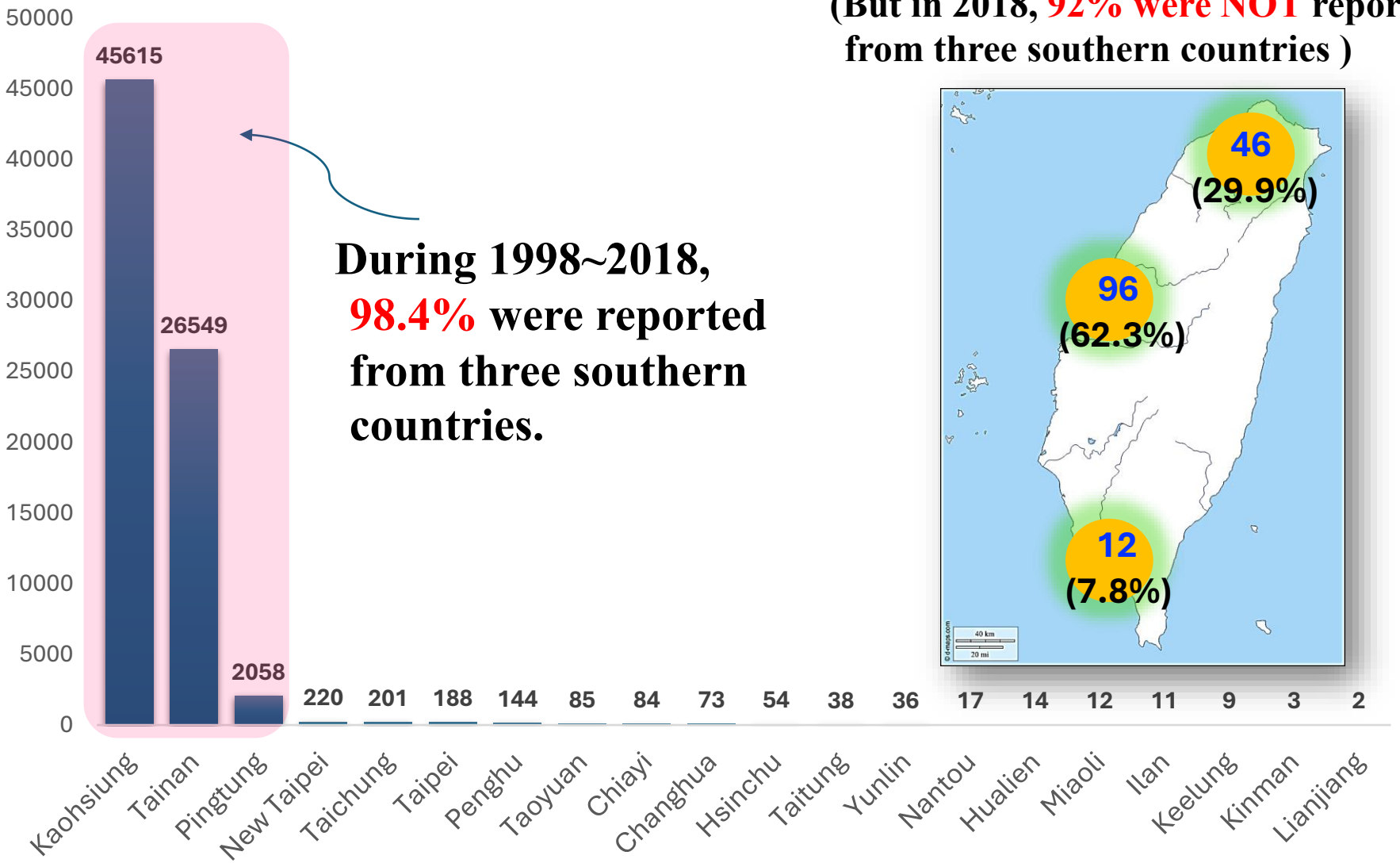
(from Prof. WJ Chen)



Indigenous dengue cases in Taiwan (1987- 2025)



The prevalence of indigenous Dengue cases in Taiwan during 1998~2018



During 1998~2018, **98.4%** were reported from three southern countries.

(But in 2018, **92% were NOT** reported from three southern countries)

Taiwan

The world

1950-1980





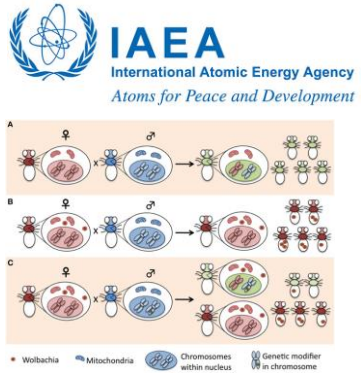


1981-2005









2006-2018







Google

BILL & MELINDA GATES foundation

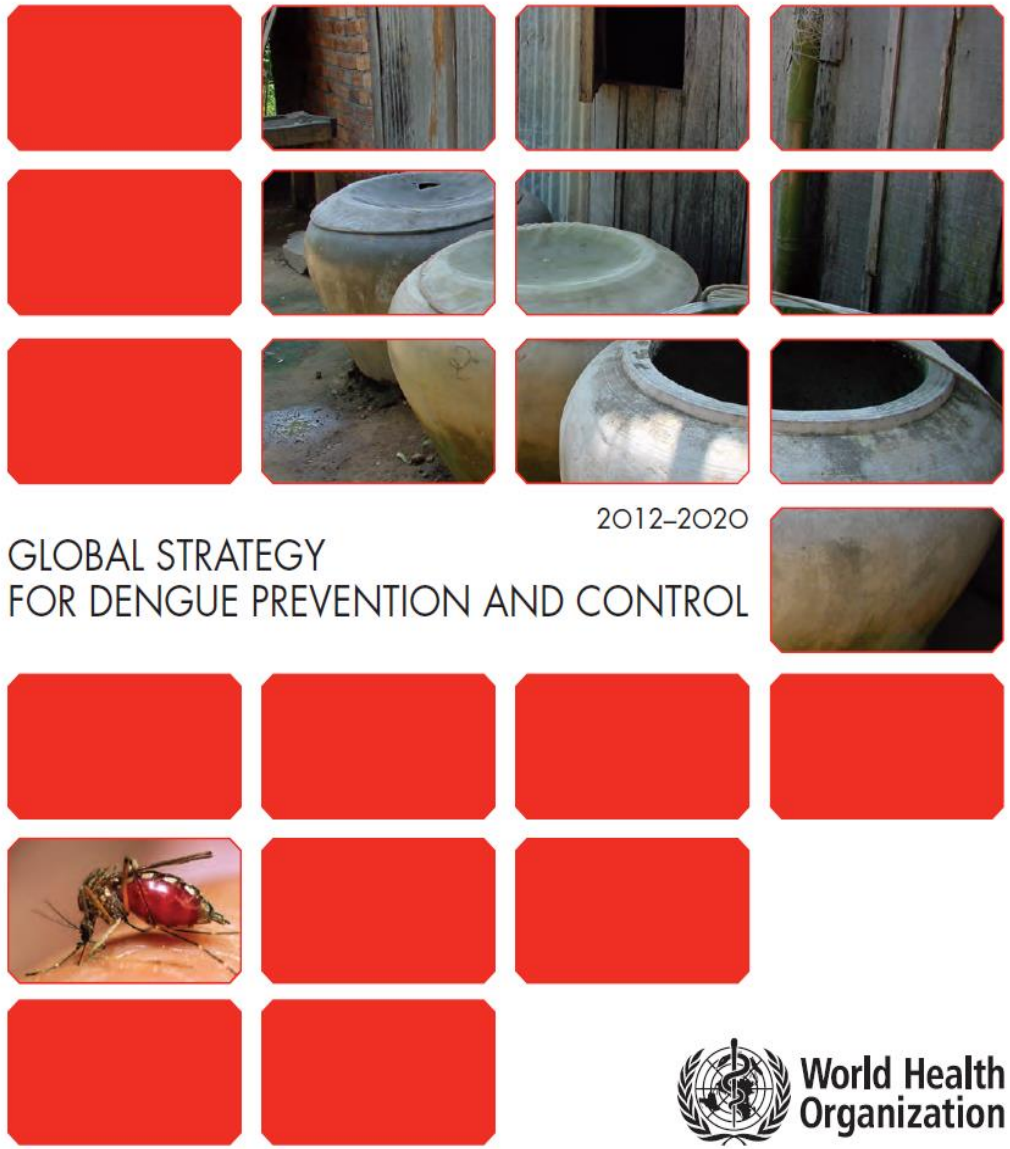
wellcome trust

World Mosquito Program

Eliminate Dengue has changed. So has our name.
Find out why: worldmosquitoprogram.org

PROJECT WOLBACHIA SINGAPORE

新型病媒蚊防治技術-沃爾巴克氏菌



2016 年 3 月 14 日至 15 日，世界衛生組織(WHO)病媒控制諮詢小組審查了五種潛在的病媒控制工具和現有工具，用於應對寨卡病毒爆發，包括：

- 沃爾巴克氏菌 (*Wolbachia*)
- 基因改造蚊子（OX513A）
- 昆蟲不育技術
- 媒介陷阱
- 有吸引力的有毒糖餌。

Wolbachia 防治登革熱：從全球實證到台灣的下一步

六個核心問題｜從科學原理、全球進展到台灣政策選擇

01
什麼是 Wolbachia？
細胞質不相容性（CI）與病媒蚊防治的生物學基礎

02
為什麼需要 Wolbachia？
登革熱持續威脅、殺蟲劑抗藥性與傳統防治的侷限

03
新加坡與全球進展到哪裡？
Replacement vs. Suppression；新加坡隨機試驗保護效益約 71–72%

04
台灣已經做了什麼？
wAlbB-TW 品系、CI / 病毒抑制、大量飼養與半田間 / 田間技術平台

05
台灣需要 Wolbachia 計畫嗎？
應如何納入整合性病媒管理，成為現有防治工具的重要補充？

06
真正的挑戰是什麼？
科學 → 量產 → 法規 → 社會接受 → 成本效益 → 長期監測

全球證據逐漸回答「Wolbachia 有沒有效」；台灣現在需要回答的是：「我們是否準備好使用它？」

Singapore RCT: Lim et al., N Engl J Med, 2026

沃爾巴克氏體(*Wolbachia*)

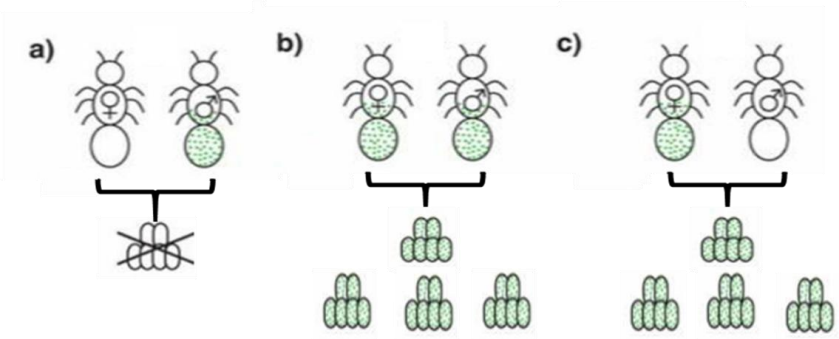
- 天然存在的細菌體
- 通過卵子傳給後代
- 對人類和環境是安全的
- 降低蚊子傳播疾病的能力

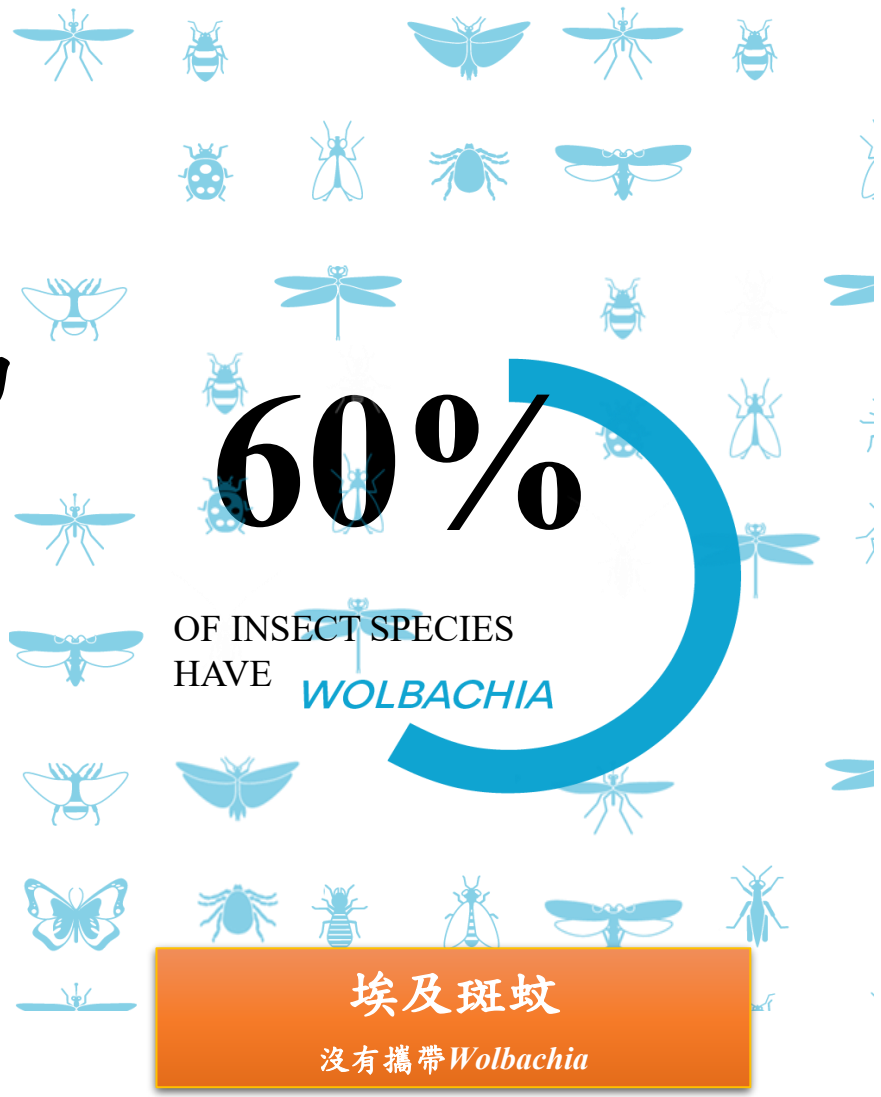
  抗病毒能力

  縮短宿主壽命

  調控宿主生殖機轉

  (Cytoplasmic Incompatibility)



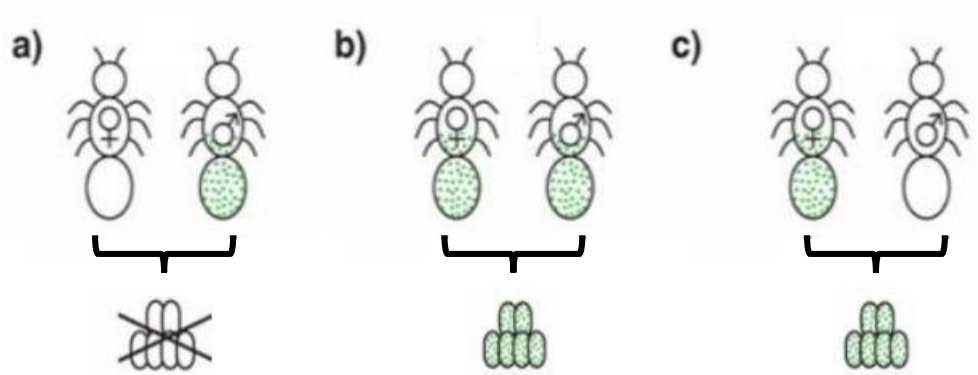


What is *Wolbachia*?

- Naturally occurring bacteria
- Lives inside insect cells
- Transmitted from adult to offspring through the insect's eggs
- Cannot be transmitted to warm-blooded animals
- Safe for humans, animals and the environment

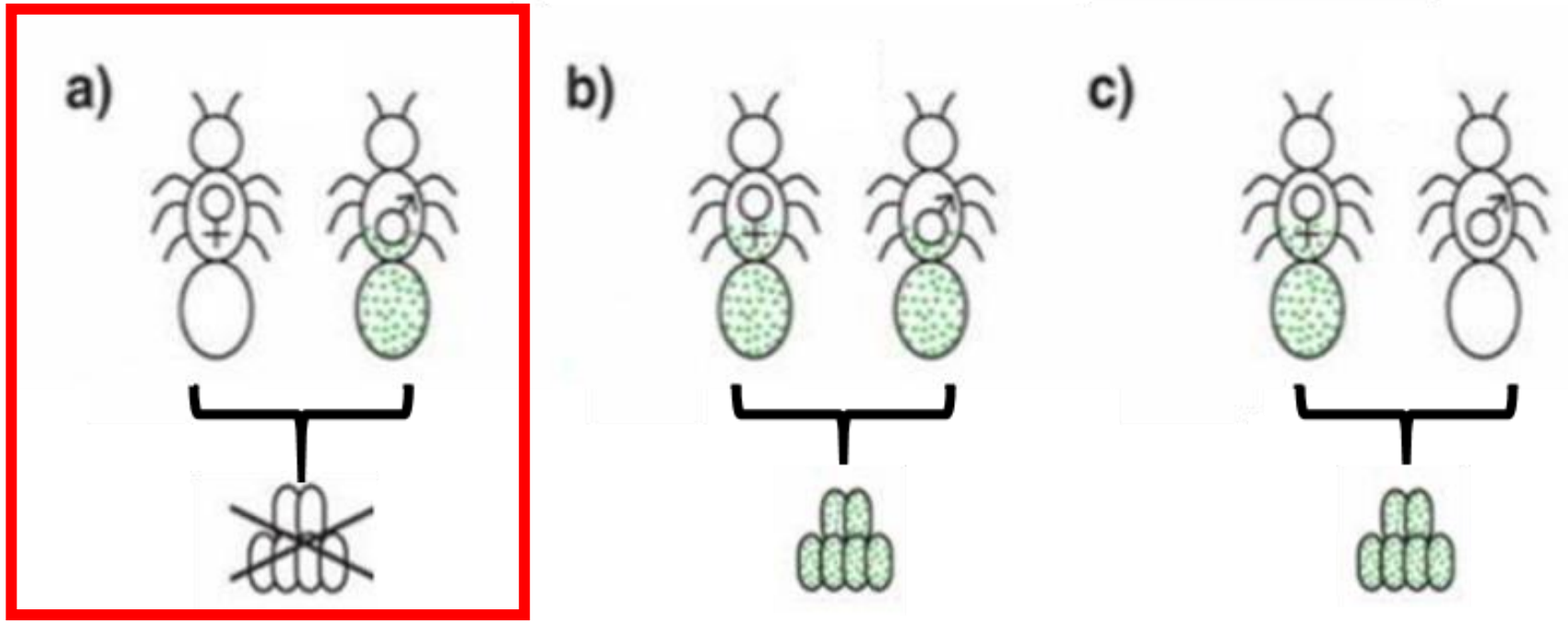


Cytoplasmic Incompatibility



Not genetic modification, not sterile male release, not species eradication
This is a biologically modified replacement strategy for Aedes Aegypti

Cytoplasmic Incompatibility

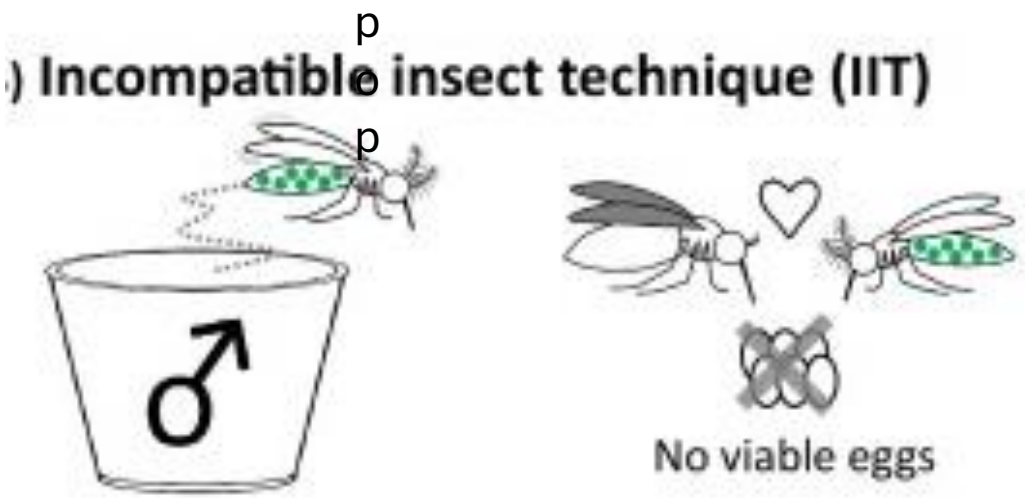


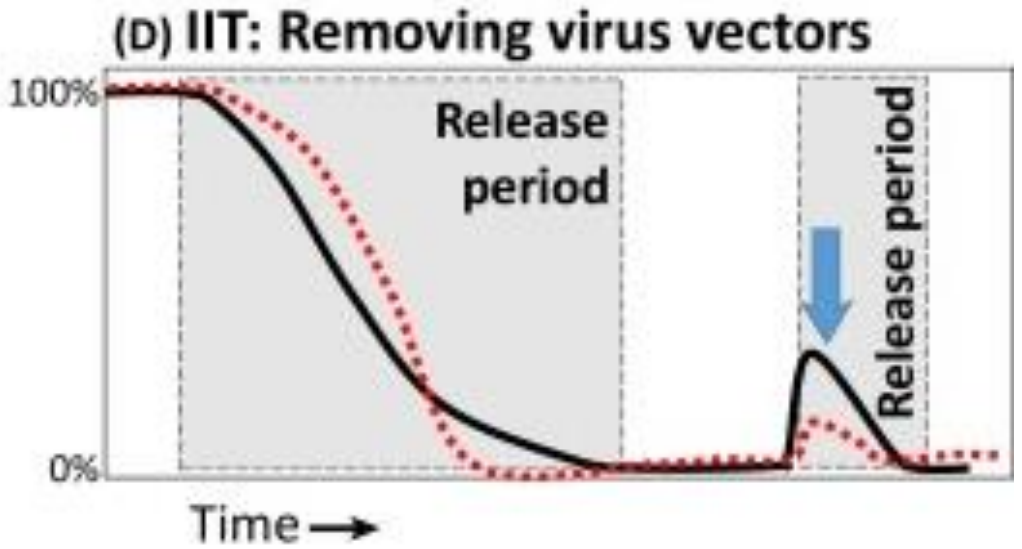
nature microbiology
LETTERS
PUBLISHED: 1 MARCH 2017 | VOLUME: 2 | ARTICLE NUMBER: 17007

A *Wolbachia* deubiquitylating enzyme induces cytoplasmic incompatibility

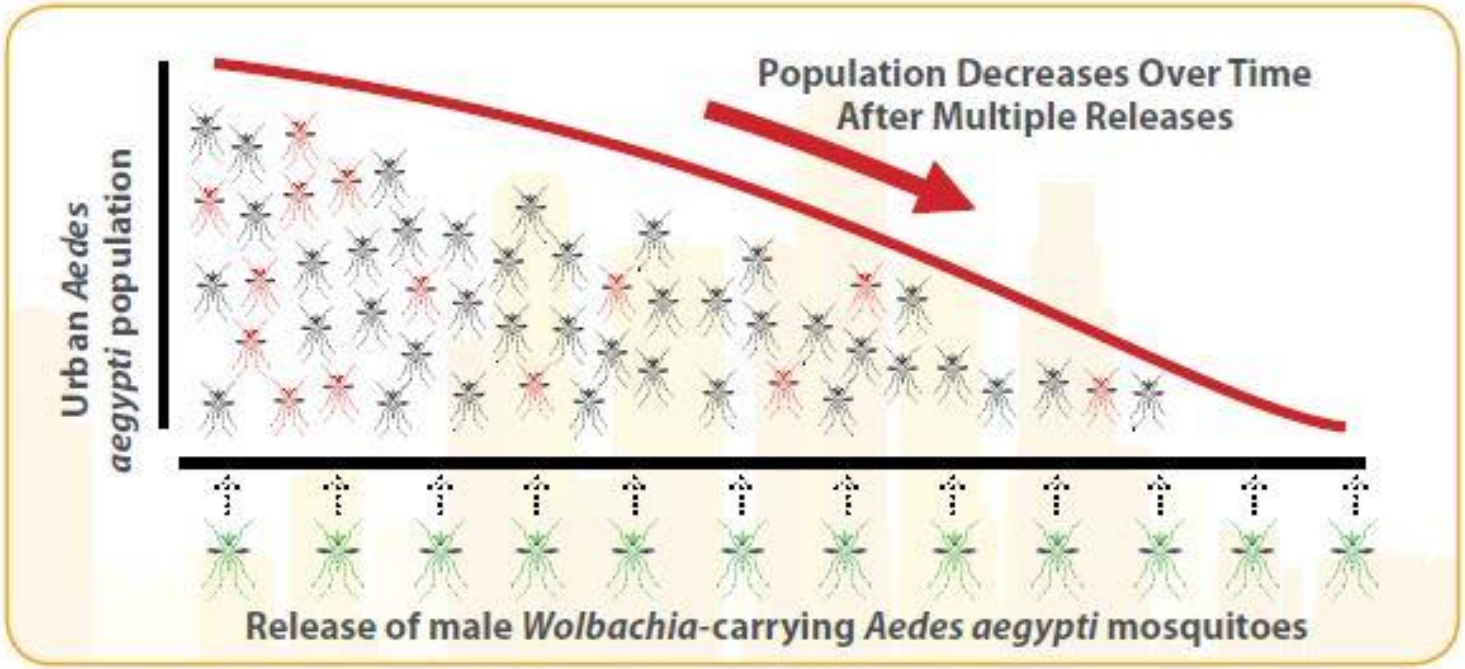
John F. Beckmann[†], Judith A. Ronau[†] and Mark Hochstrasser[1,2]*

Population Suppression





Trends in Parasitology



Wolbachia 品系

Stable *Wolbachia* infections in native and transinfeir reproductive effects (CI – cytoplasmic incompatibility, MT -on), fitness effects, and viral blocking effectiveness where demonstratation unavailable). Effect size is denoted as: high (>90%), mod6), low (<20%) and none (no detectible effects). More details are foun S1

Strain	NATIVE HOST	CI	MT	Viral blockage	TRANSFECTED HOST	CI	MT	Viral blockage
wMel	*Drosophila* [1]	Partial*,[1,2] none[3]	Partial[4]	High[8] moderate[9,10] low[8,9,10,11] none[8]	*Aedes aegypti*[12]	High[12]	High[12]	High[12] moderate[14,15] low[14]
					Ae. albopictus[16]	High[16]	High[16]	High[16,17]
					D. simulans[18]	High[18]	High[18]	Moderate[19,20]
wMelPop	*D. melanogaster*[21]	Partial*,[22] none[21,23]	High[24]	High‡,[10,25]	*Ae. aegypti*[26]	High[26,27]	High[26,27]	High[14,15,30] moderate[30]
					Ae. albopictus[31]	Low[31]	Partial[31]	?
					D. simulans[23]	High*,[23,24,32]	High[23,24,3	?
wAu	*D. simulans*[33]	None[33]	High[33]	High[19,20]	*D. melanogaster*[34]	Low/none[34]	High‡,[34]	High[35]
wMelCS	*D. melanogaster*[36]	Low*,[2] none[37]	?	Moderate[10,25]	*D. simulans*[20]	?	?	High[20]
wInn	*D. innubila*[39]	*Male killing*[39]	High[39]	Low[40]	*D. simulans*[41]	None[41]	High[41]	None[41]
					D. melanogaster[41]	None[41]	Partial[41]	?
wRi	*D. simulans*[42]	Partial*,[42]	High[43] partial[44]	Moderate[19] low[19]	*Ae. albopictus*[46,47]	High[46]	High[46]	?
					D. melanogaster[48]	Low[48]	Partial[48]	?
					D. serrata[49]	High[49]	High[49]	?
					D. yakuba complex[50]	High[50]	High[50]	?
					Laodelphax striatellus[51]	High[51]	Partial[51]	?
wAlbB	*Ae. albopictus*[52]	High[53,54,55]	High[56]	Low[17,58,59]	*Ae. aegypti*[60,61]	High[60] partial[61]	High[60] partial[61]	High[62]
					Ae. polynesiensis[63]	High[63]	High[63]	Moderate[64]
wPip	*Culex pipiens*	High[67]	High[67]	Low[8]	*Ae. albopictus*[69]	High[69]	High[69]	?

取代型

抑制型

02

為什麼需要 Wolbachia？

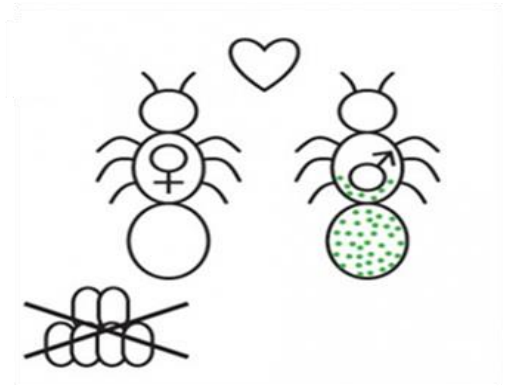
登革熱持續威脅、殺蟲劑抗藥性與傳統防治的侷限

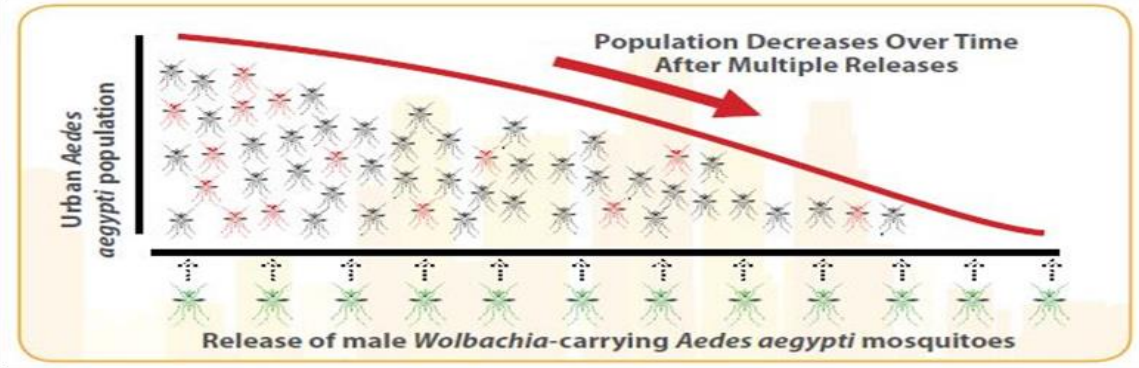
沃爾巴克氏菌防治登革熱試驗國家



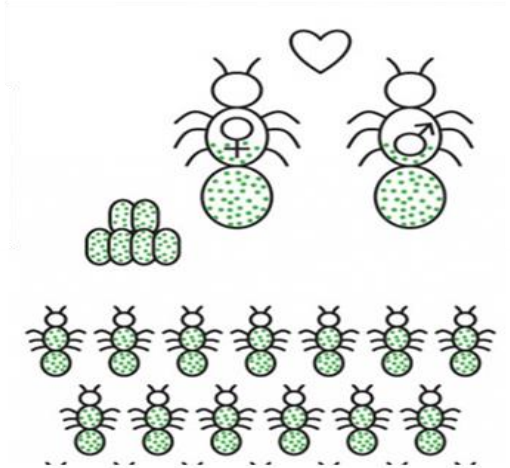
Wolbachia 2 種防治策略

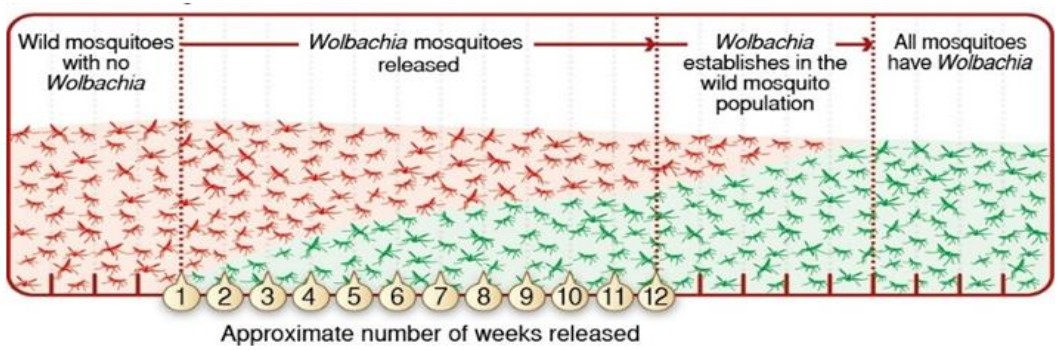
族群抑制：釋放帶*Wolbachia*雄蚊與野生雌蚊交配，無法產生子代。



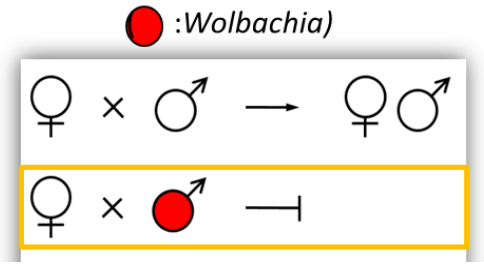


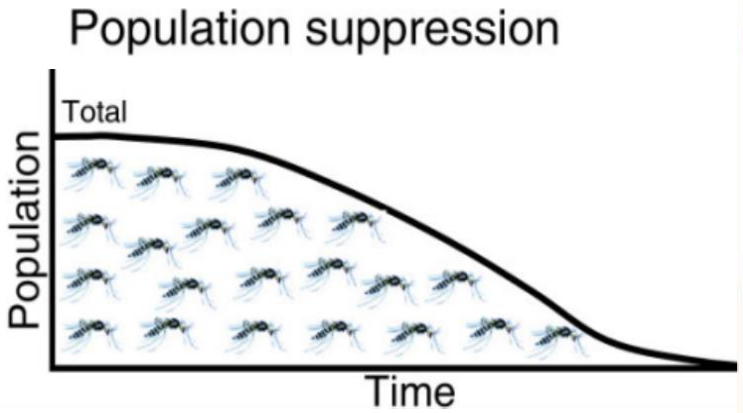
族群取代：釋放帶*Wolbachia*雄雌蚊，逐漸將野生族群替換為帶*Wolbachia*之族群。

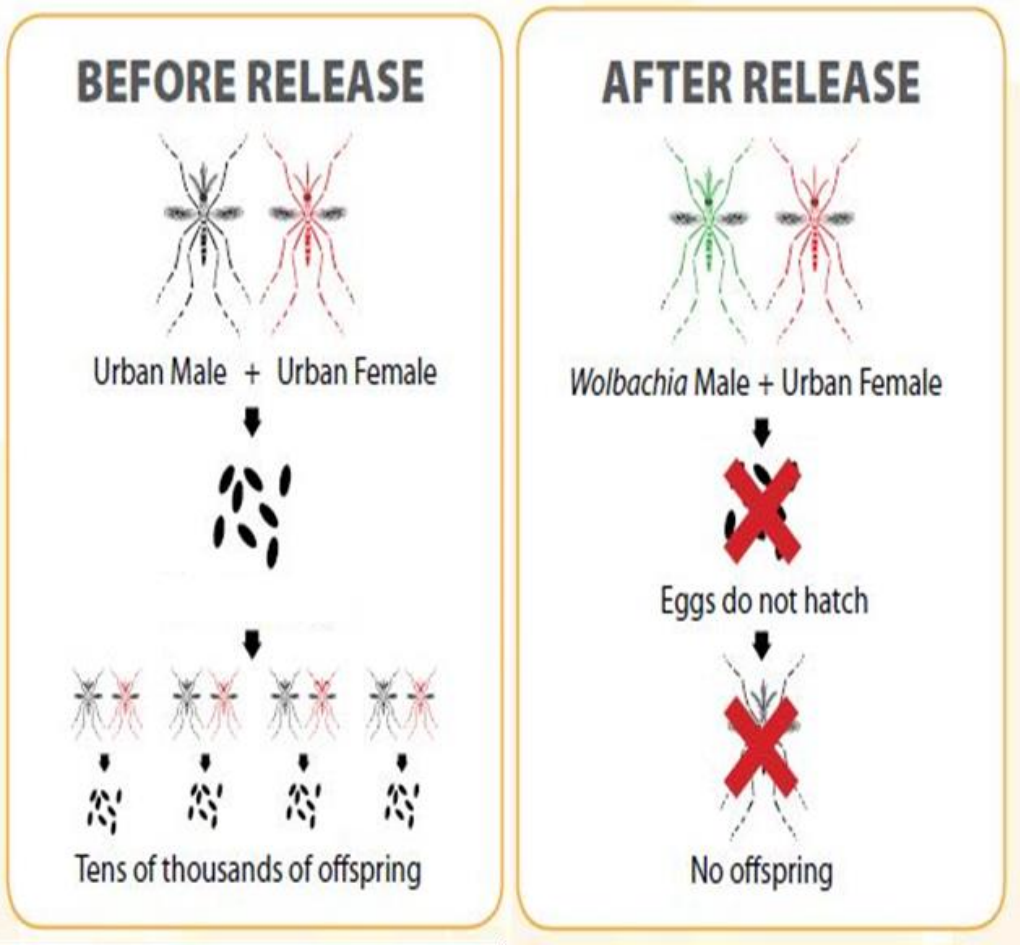


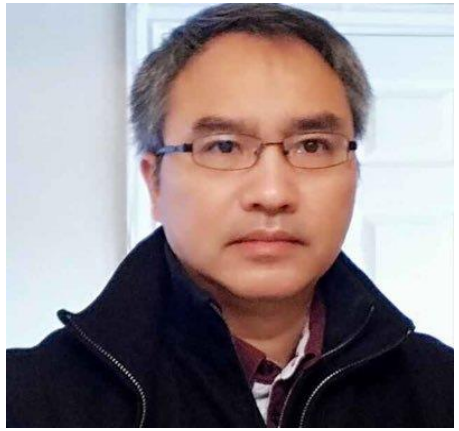


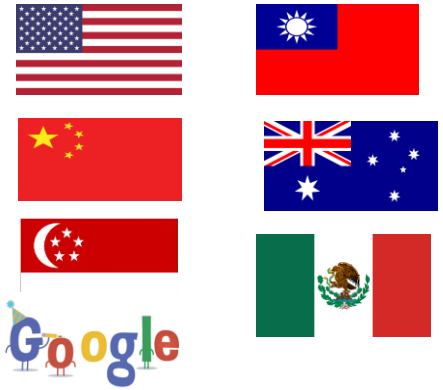
族群抑制











只釋放帶*Wolbachia*雄蚊與野生雌蚊交配，
無法產生子代

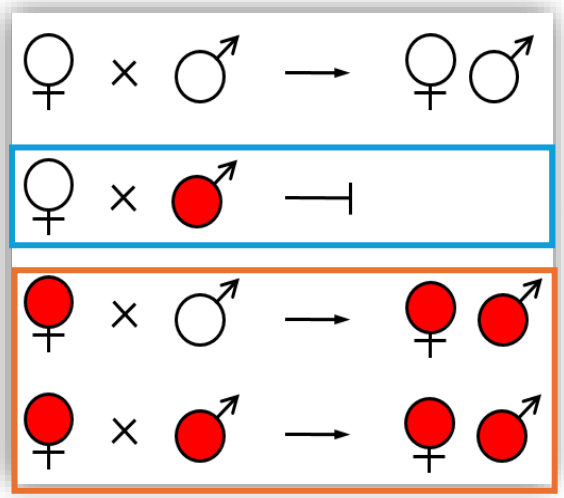
2017 蚊媒中心積極邀請 奚志勇訪台





Wolbachia 2 種防治策略

(● :*Wolbachia*)

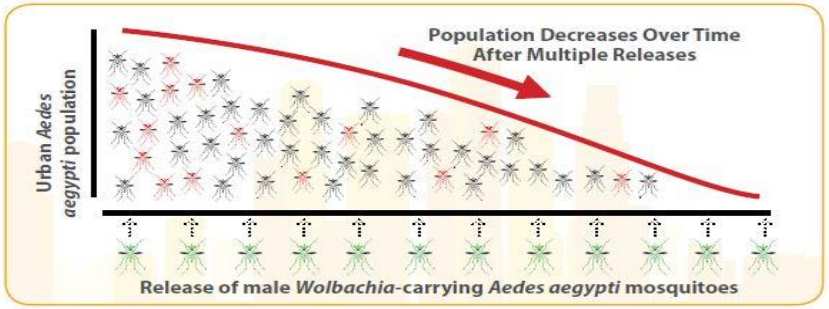


族群抑制

只釋放帶*Wolbachia*雄蚊與野生雌蚊交配，無法產生子代。

優

雄蚊不會叮咬人，釋放後較不會使民眾感受到蚊子增加，也較易政策溝通，成本較高。

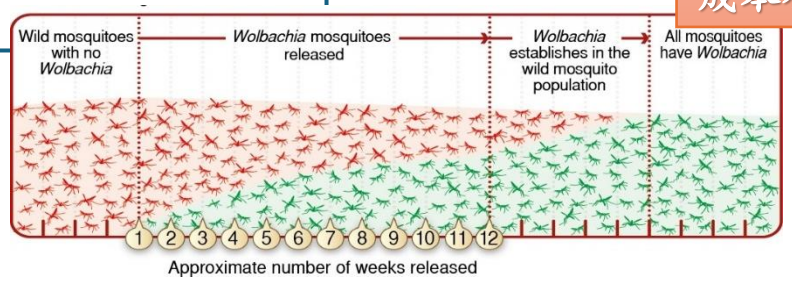


族群取代

帶*Wolbachia*雄蚊、雌蚊皆釋放，逐漸將野生族群替換為帶*Wolbachia*之族群。

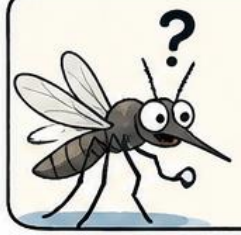
優

製造攜帶 *Wolbachia* 蚊子過程無需雌雄分離，成本相對較低。



Source: 新加坡 *Wolbachia* 計畫民眾衛教手冊

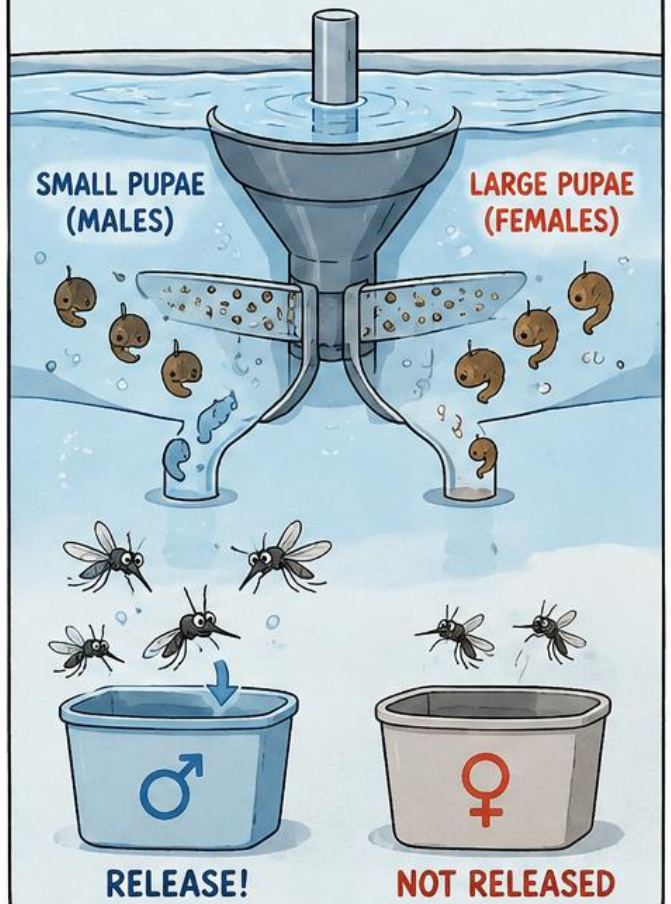
MOSQUITO INDUSTRY FOCUS:
ONE BIG QUESTION – HOW TO SORT MALE MOSQUITOES?





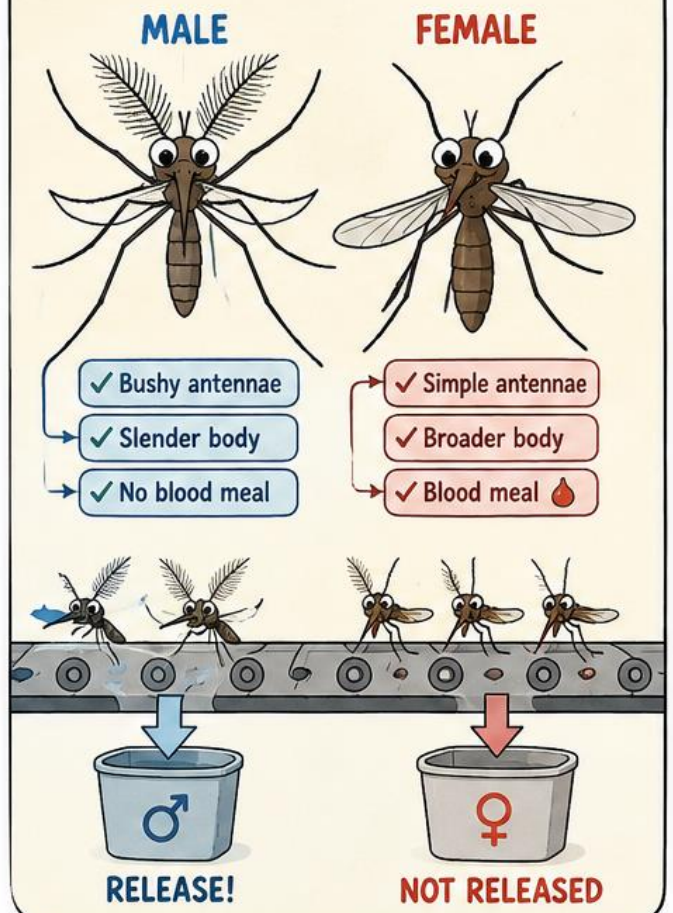
1. PUPAE SIZE SORTING

Sort by size at the pupae stage – males are smaller!



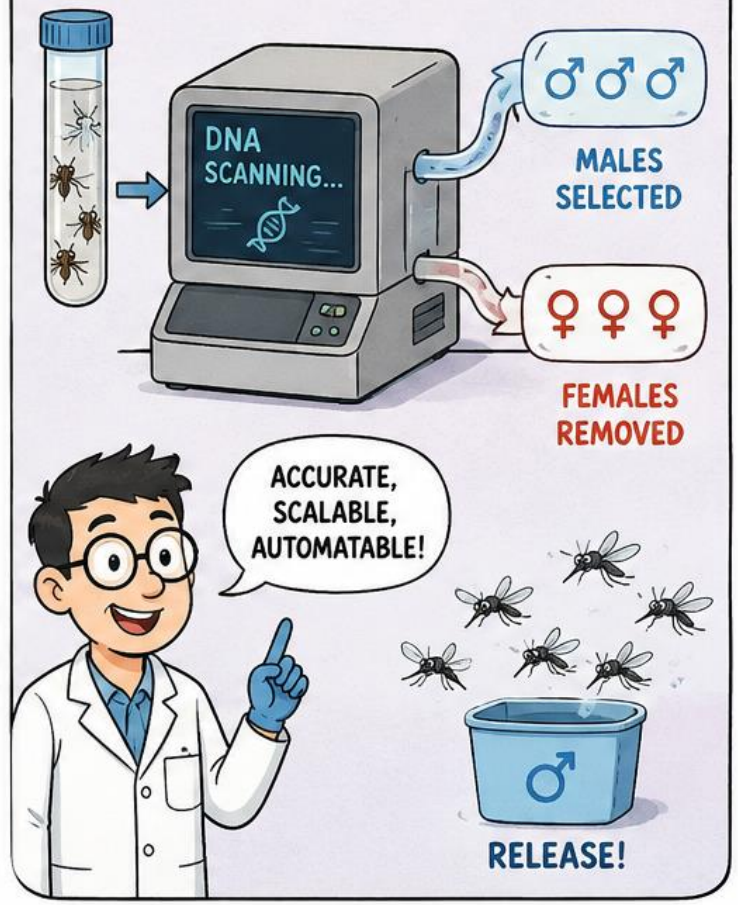
2. ADULT MALE MORPHOLOGY

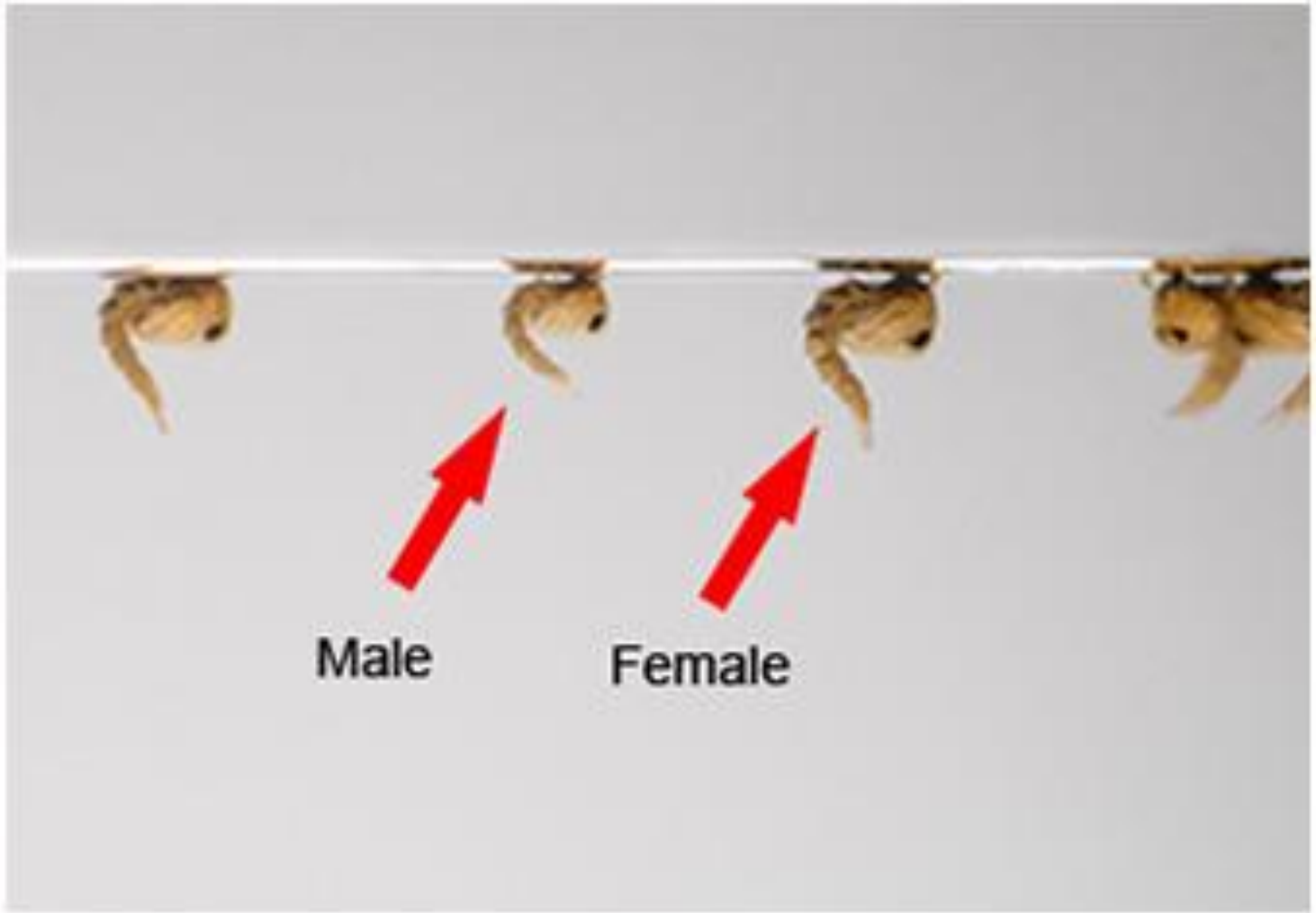
Use unique male features to identify and sort adults!



3. GENETIC (MOLECULAR) SORTING

Use DNA-based markers to distinguish males from females!




Male
Female

03

新加坡與全球進展到哪裡？

Replacement vs. Suppression；從全球實證看防治成效

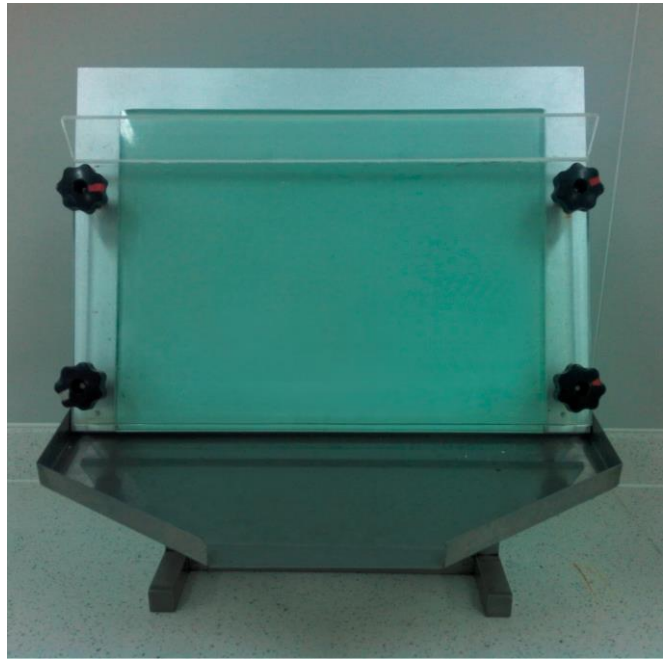
廣州建立第一座蚊子量產工廠 1.0 版







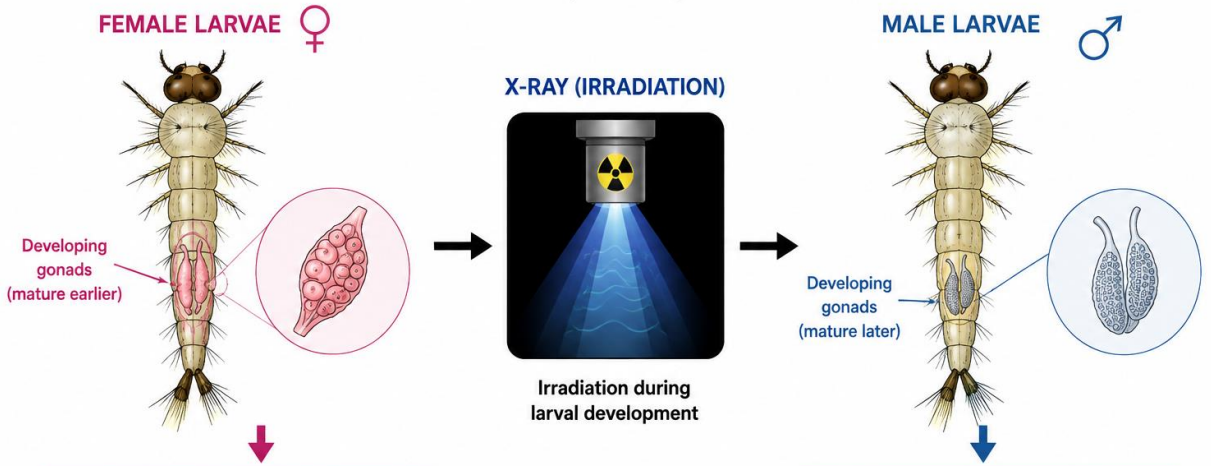
Male and Female separation and 林全的疑問







Aedes Larvae: Female Gonads Mature Earlier Than Male Can Be Disrupted by Irradiation



AFFECTED BY IRRADIATION

X-ray causes damage to early maturing female gonads

↓

Disrupted gonadal development and reduced fertility

Female gonads mature earlier than male gonads and can be disrupted by X-ray irradiation.

NOT AFFECTED

Male gonads mature later and are not affected

↓

Normal gonadal development and fertility



Google 加入賽局，大幅提升技術水準 進入 2.0版



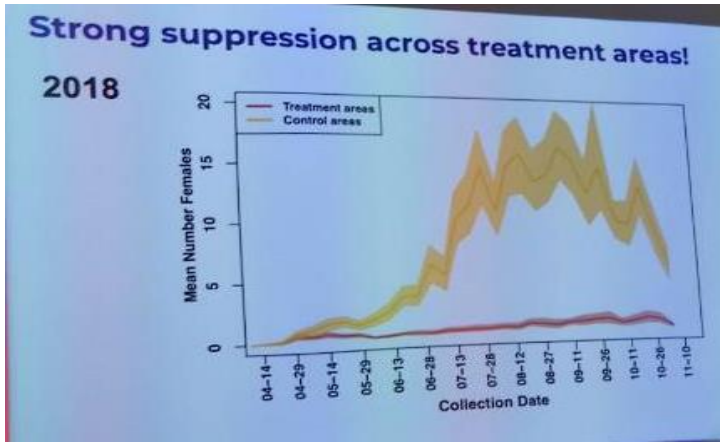




美國-族群抑制計畫

Debug a verily project

- 2017年
- Google Verily
- 加州佛雷斯諾執行
- **95% suppression**





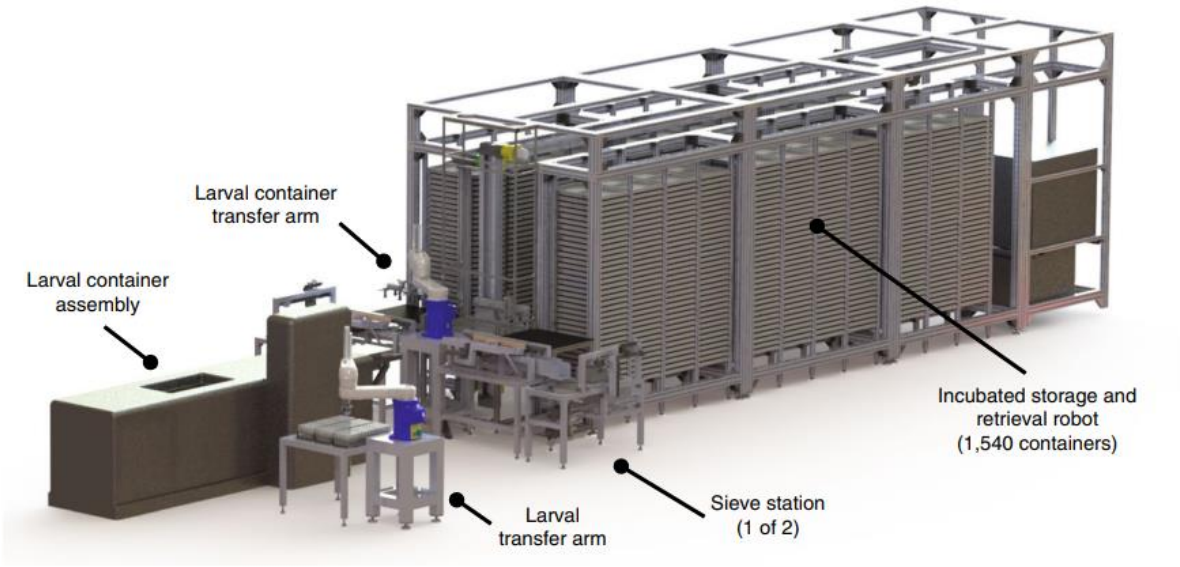


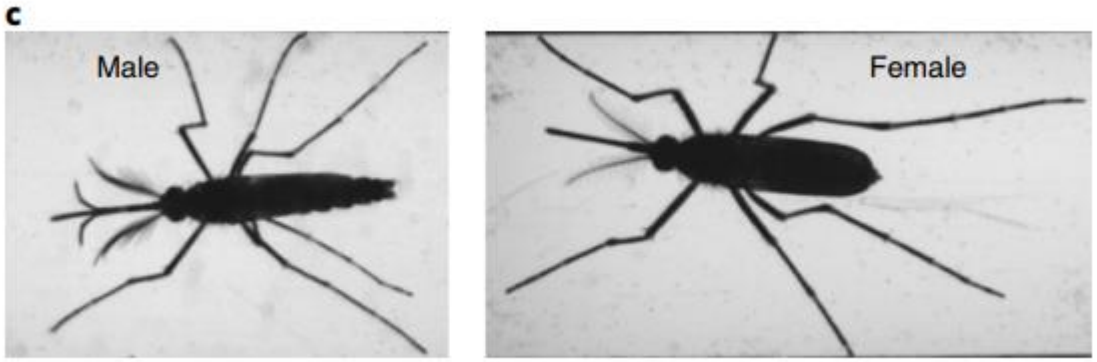
ARTICLES
https://doi.org/10.1038/s41587-020-0471-x
nature biotechnology

Check for updates

Efficient production of male *Wolbachia*-infected *Aedes aegypti* mosquitoes enables large-scale suppression of wild populations

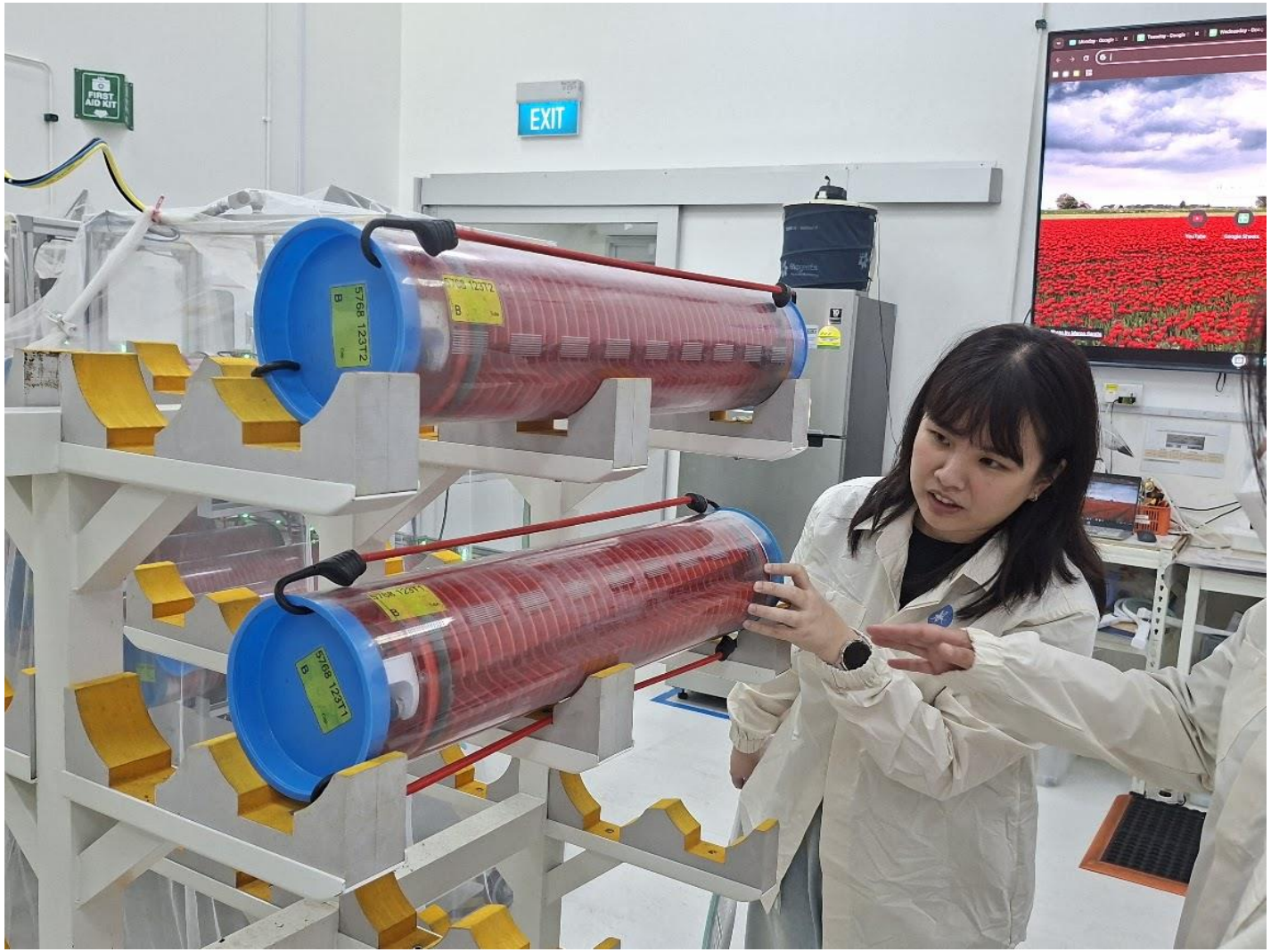
Jacob E. Crawford[1✉], David W. Clarke[1], Victor Criswell[1], Mark Desnoyer[1], Devon Cornel[2], Brittany Deegan[2], Kyle Gong[1], Kaycie C. Hopkins[1], Paul Howell[1], Justin S. Hyde[1], Josh Livni[1], Charlie Behling[1], Renzo Benza[1], Willa Chen[1], Karen L. Dobson[3], Craig Eldershaw[1], Daniel Greeley[1], Yi Han[1], Bridgette Hughes[1], Evdoxia Kakani[1], Joe Karbowski[1], Angus Kitchell[1], Erika Lee[1], Teresa Lin[1], Jianyi Liu[1], Martin Lozano[1], Warren MacDonald[1], James W. Mains[3], Matty Metlitz[1], Sara N. Mitchell[1], David Moore[1], Johanna R. Ohm[1], Kathleen Parkes[1], Alexandra Porshnikoff[1], Chris Robuck[1], Martin Sheridan[1], Robert Sobecki[1], Peter Smith[1], Jessica Stevenson[1], Jordan Sullivan[1], Brian Wasson[1], Allison M. Weakley[1], Mark Wilhelm[1], Joshua Won[1], Ari Yasunaga[1], William C. Chan[1], Jodi Holeman[2], Nigel Snoad[1], Linus Upson[1], Tiantian Zha[1], Stephen L. Dobson[3,4], F. Steven Mulligan[2], Peter Massaro[1] and Bradley J. White[1✉]



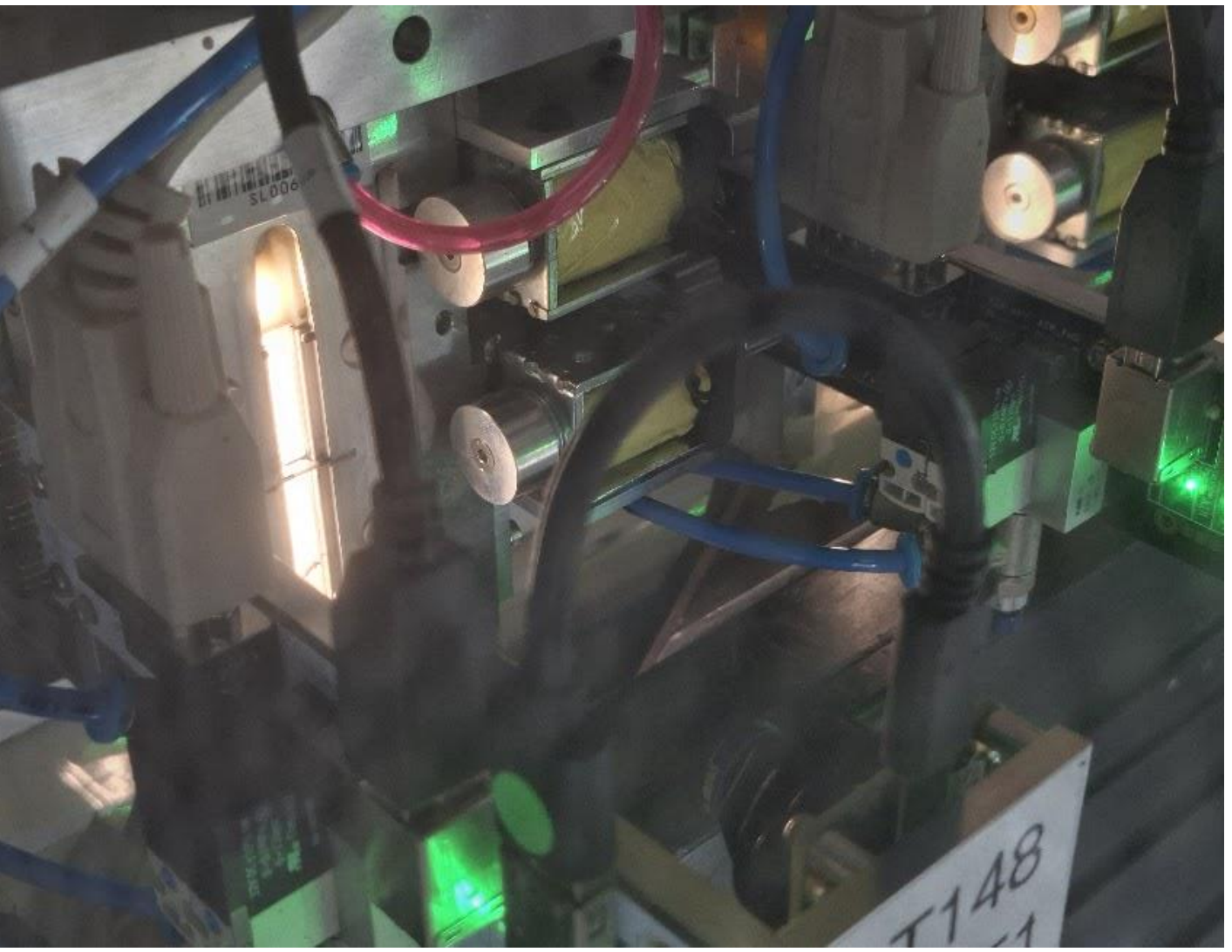


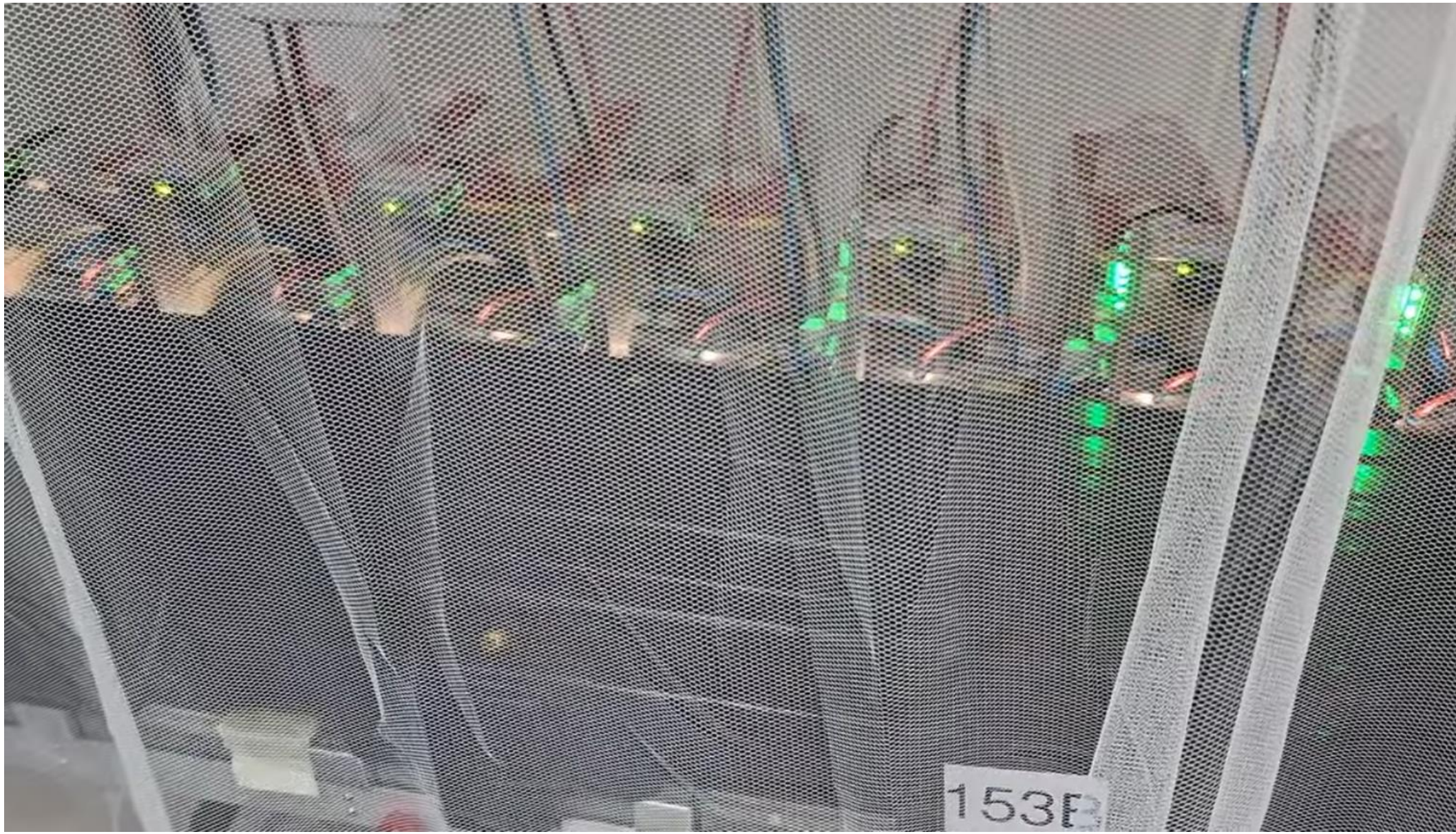



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14977


FIRST AID KIT
EXIT
5768 123T2
5768 123T1




SL006
T148


153B






RAD SOURCE
RAD SOURCE

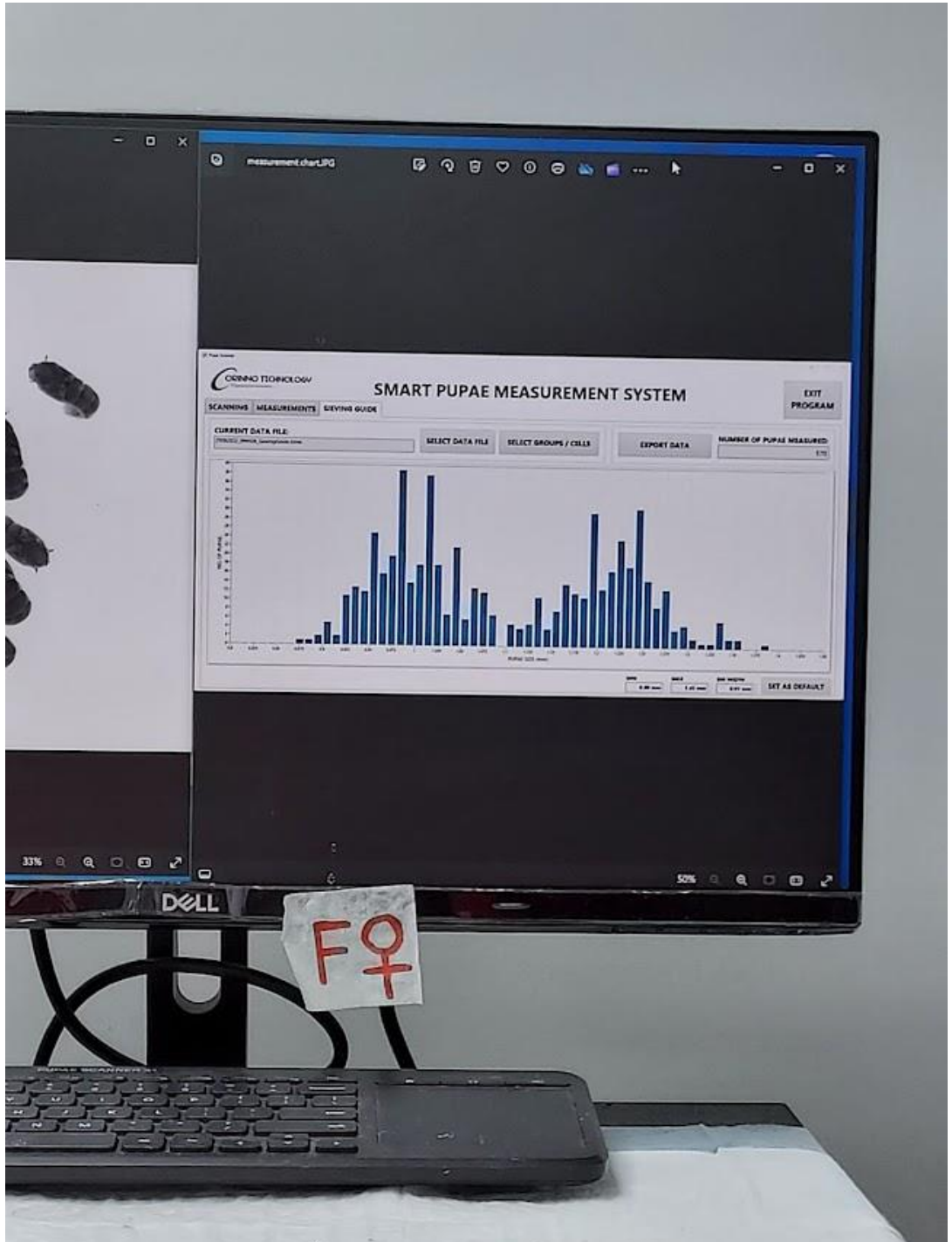


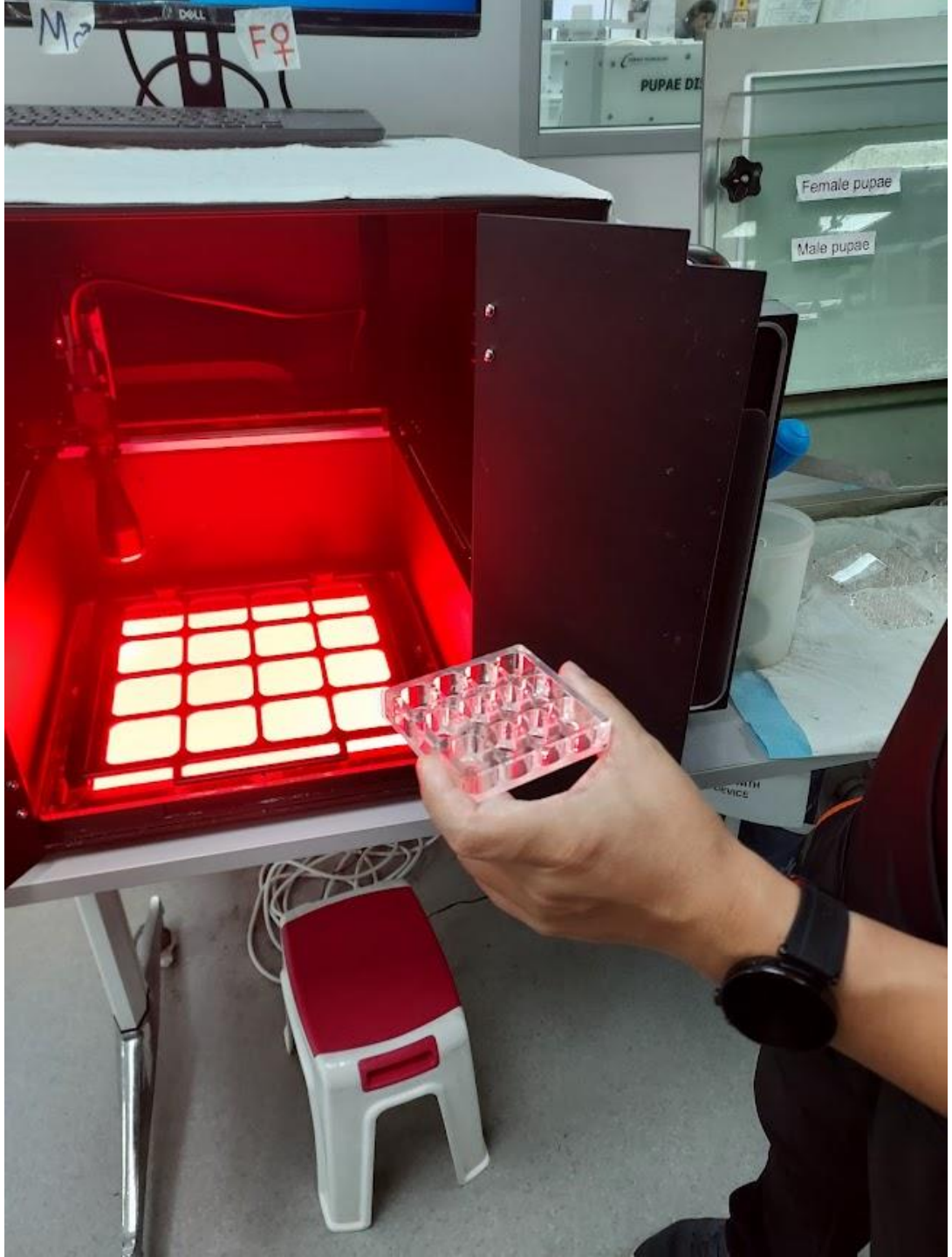







D4


measurement.chartJPG
CORNING TECHNOLOGY
SMART PUPAE MEASUREMENT SYSTEM
EXIT PROGRAM
SCANNING
MEASUREMENTS
SIEVING GUIDE
CURRENT DATA FILE:
SELECT DATA FILE
SELECT GROUPS / CELLS
EXPORT DATA
NUMBER OF PUPAE MEASURED:
SET AS DEFAULT
DELL
F♀


M♂
F♀
PUPAE DI
Female pupae
Male pupae
DELL



SINGAPORE TAKES TWO PARALLEL STRATEGIES
FOR SUSTAINABLE DENGUE CONTROL

1 DEVELOPING GOVERNMENT-BASED TECHNOLOGY

LED BY SINGAPORE'S SCIENTIFIC AGENCIES

RESEARCH & DEVELOPMENT
- Wolbachia research
- Local innovation and IP

SCALE & PRODUCTION
- Mass-rearing of Wolbachia mosquitoes
- Quality & biosecurity

RELEASE & MONITORING
- Field releases in phased approach
- Continuous monitoring and evaluation



OWNED BY SINGAPORE
BUILT FOR LONG-TERM SUSTAINABILITY

COMMON GOAL:
A DENGUE-RESILIENT SINGAPORE



DENGUE CONTROL
FOR A HEALTHIER, SAFER SINGAPORE



Wolbachia mosquitoes
- ✓ Reduce dengue transmission
- ✓ Self-sustaining in the wild
- ✓ Environmentally friendly

2 COLLABORATE WITH GOOGLE VERILY

LEVERAGING GLOBAL INNOVATION & DATA SCIENCE

verily  Google

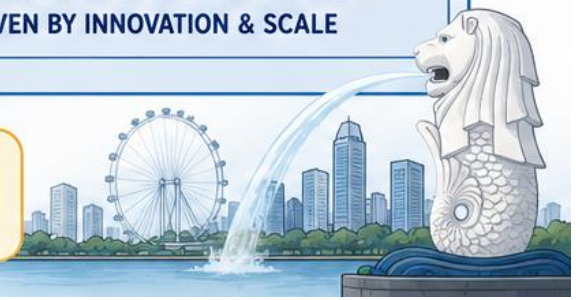
verily
WOLBACHIA PROJECT



STRAIN DEVELOPMENT	EFFICACY TESTING	DATA & AI	PROGRAM OPTIMIZATION
Advanced Wolbachia strains	Lab & field evaluation	Data science & modeling for decision support	Improve impact, efficiency & scalability

POWERED BY GLOBAL PARTNERSHIP
DRIVEN BY INNOVATION & SCALE





TWO PARALLEL PATHS, ONE SHARED MISSION:
SCIENCE. INNOVATION. PARTNERSHIP. FOR A DENGUE-RESILIENT SINGAPORE.

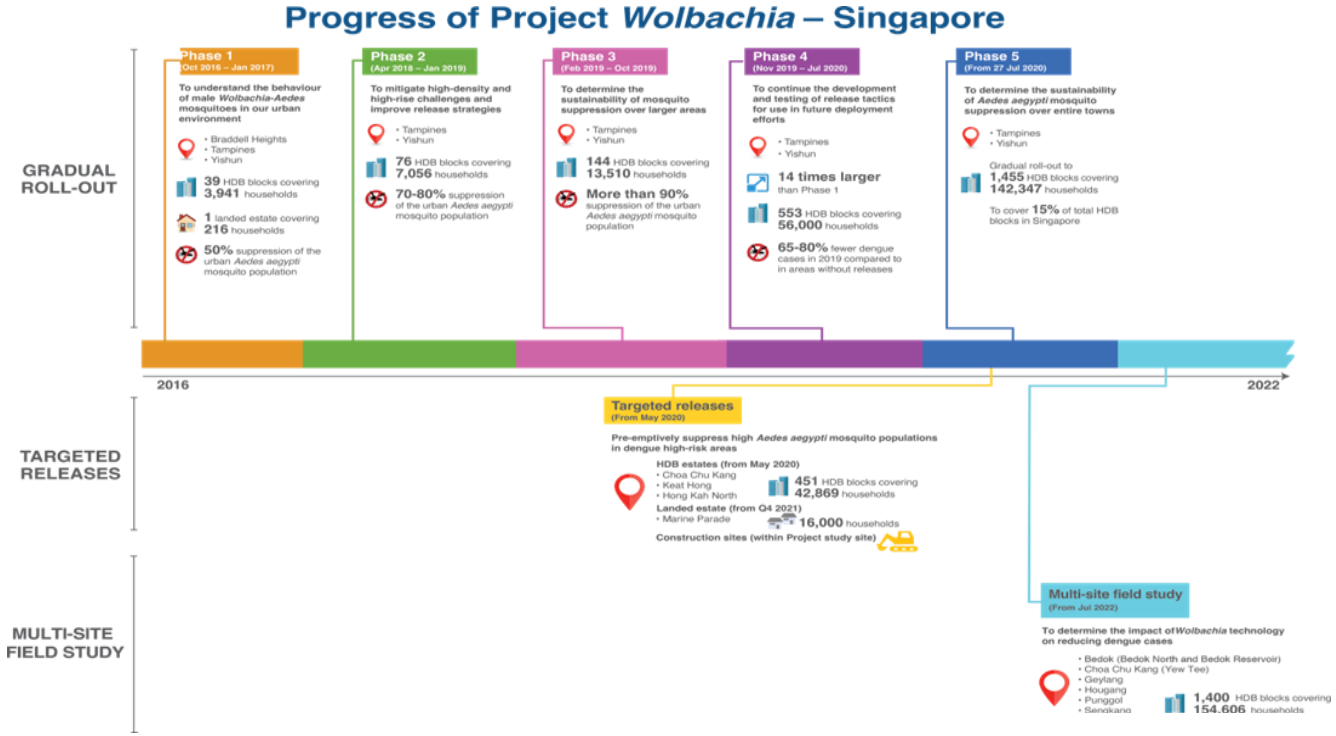
新加坡自 2016-2019 自行開發量產機器及技術







新加坡--族群抑制計畫 2019-2026





Phase 5 Field Study

2 towns: Tampines, Yishun

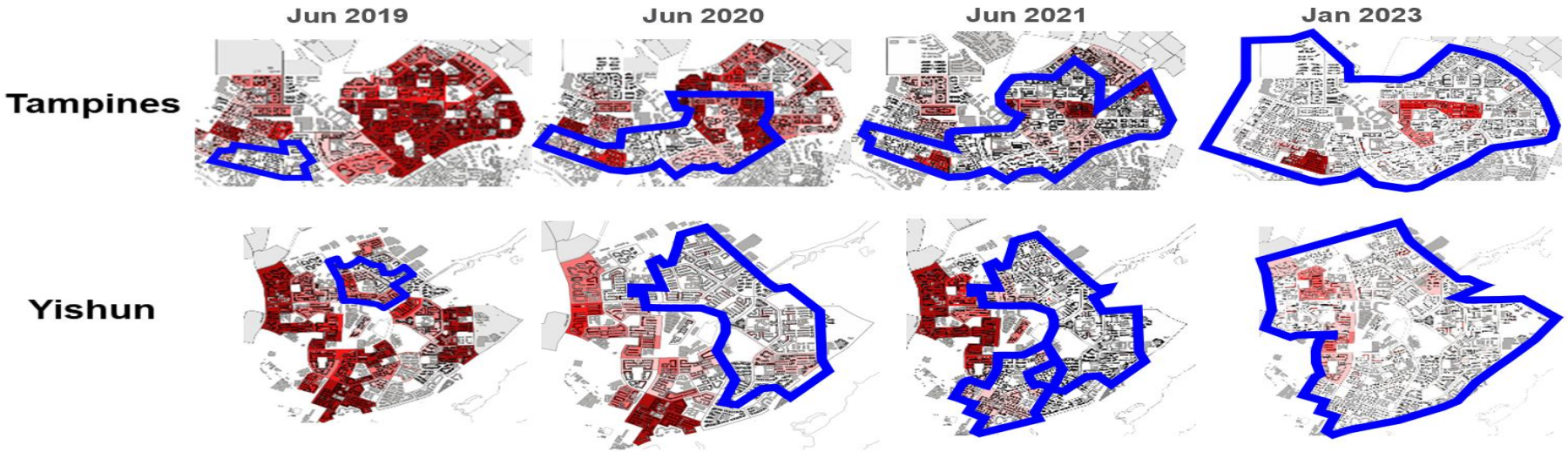
1,455 HDB Blocks covering 142,347 households

To determine if suppression across whole towns can be sustained with fewer male *Wolbachia-Aedes* mosquitoes

From Jul 2020

Phase 5 releases covers whole of Tampines and Yishun to maintain mosquito suppression with smaller or less frequent releases. The ability to maintain suppression with reduced release numbers is important to allow *Wolbachia-Aedes* technology to be sustainably deployed on a larger scale in the future. These studies will expand NEA's repertoire of potential deployment strategies, and inform future deployment plans for *Wolbachia-Aedes* technology





Google Debug Singapore: Automated Production Line 蚊子生產革命

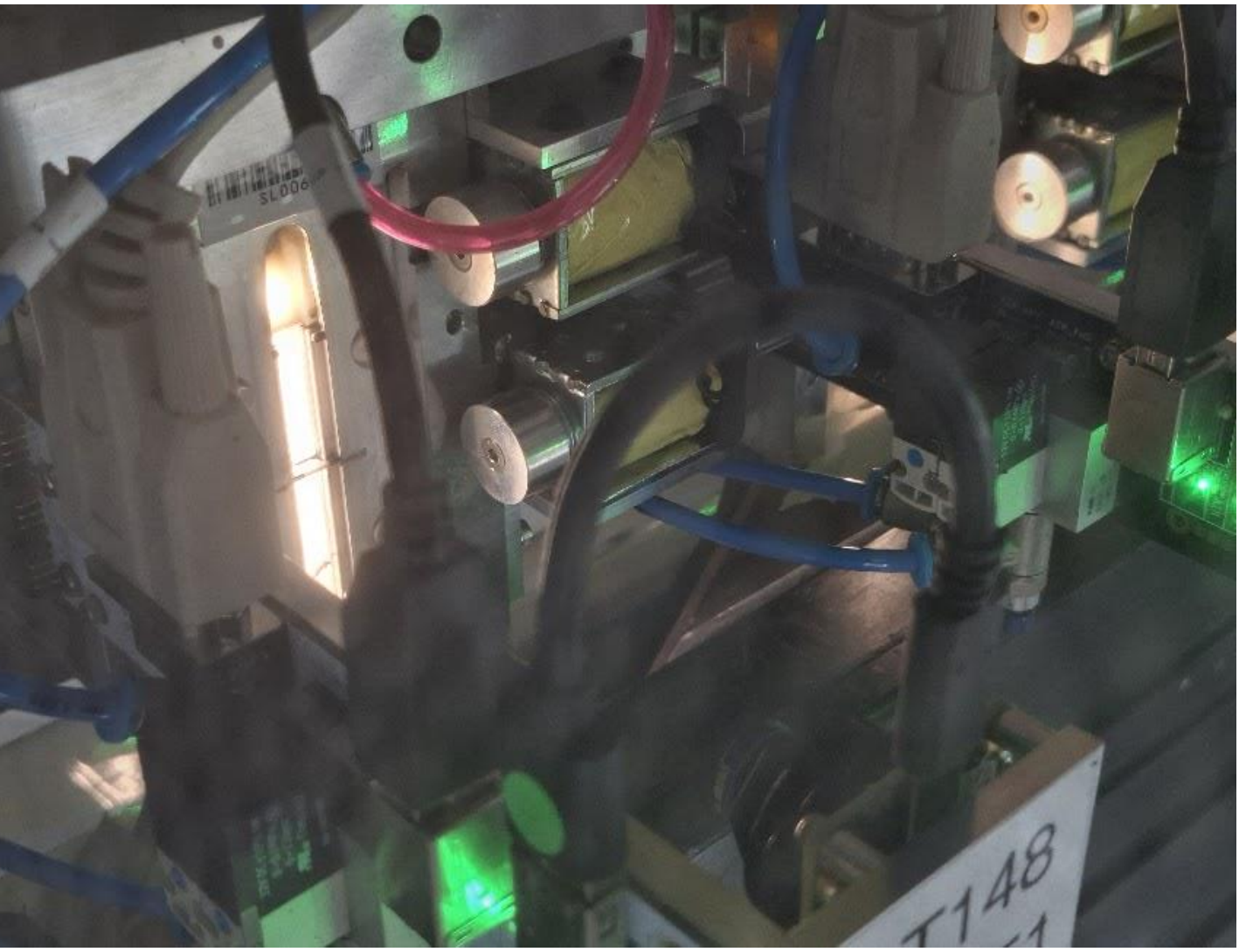


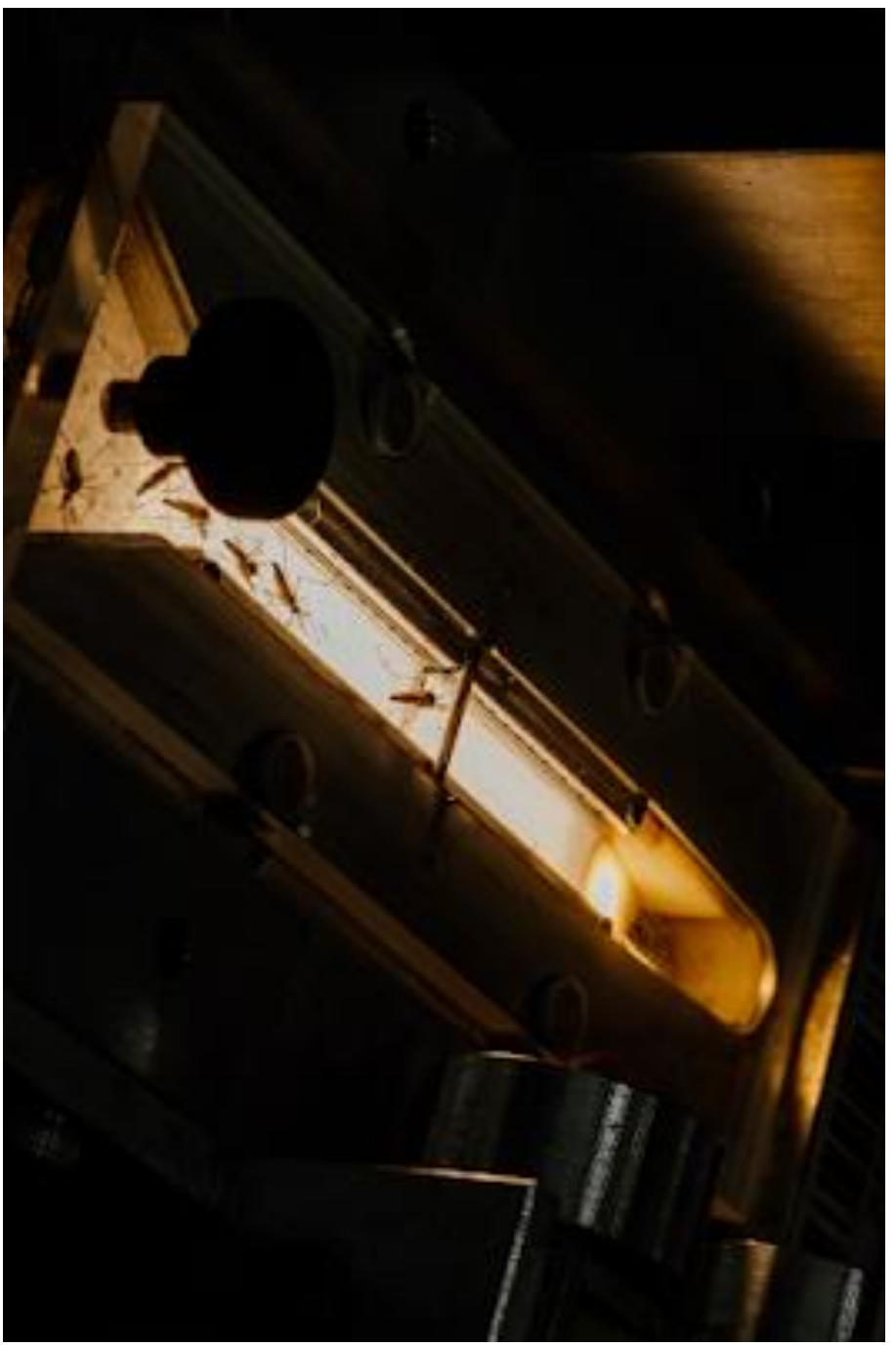



RAD SOURCE
RAD SOURCE






T148



Releasing Trays



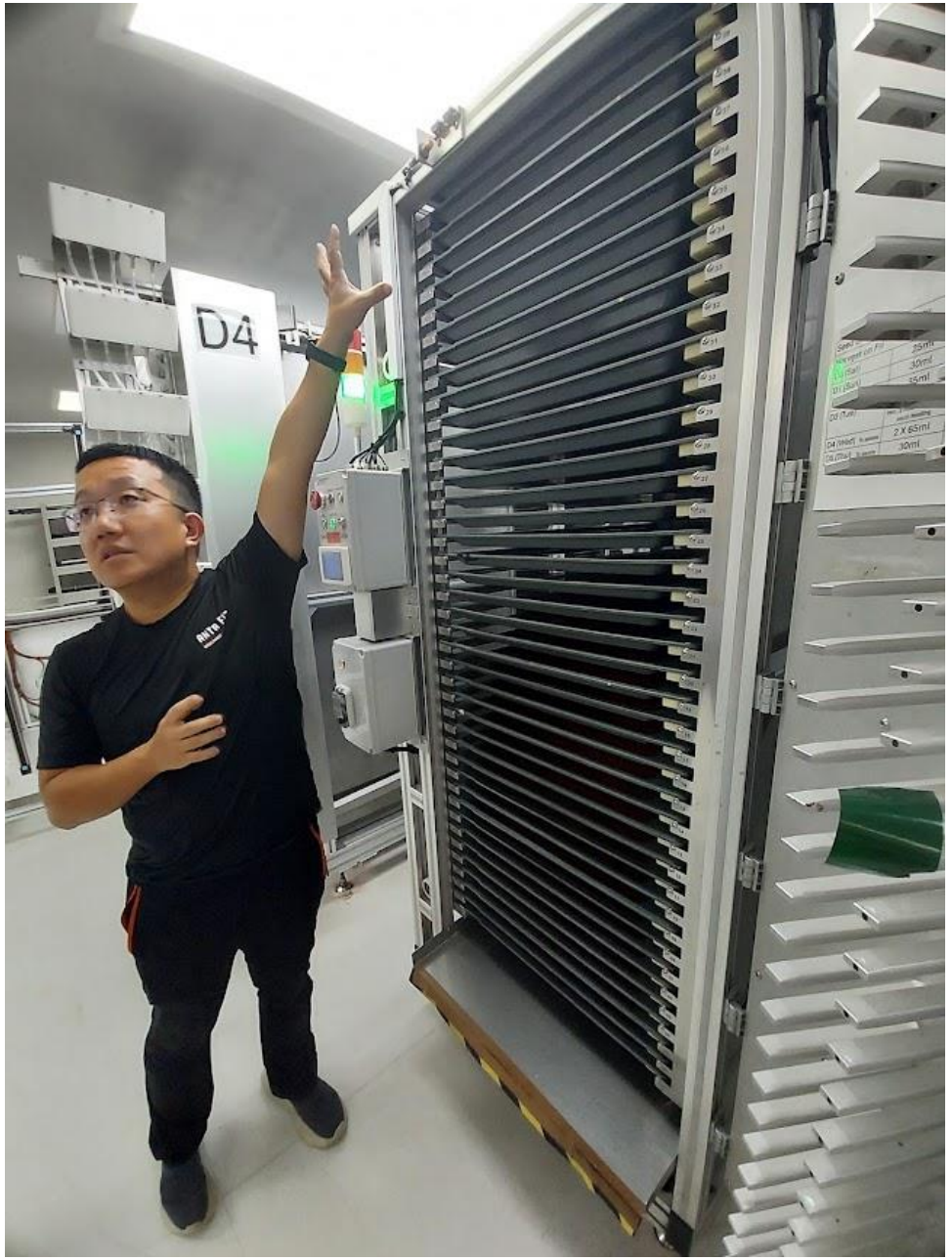


Singapore Mosquito Facility Visit: From Research to Deployment 公衛國力的展示

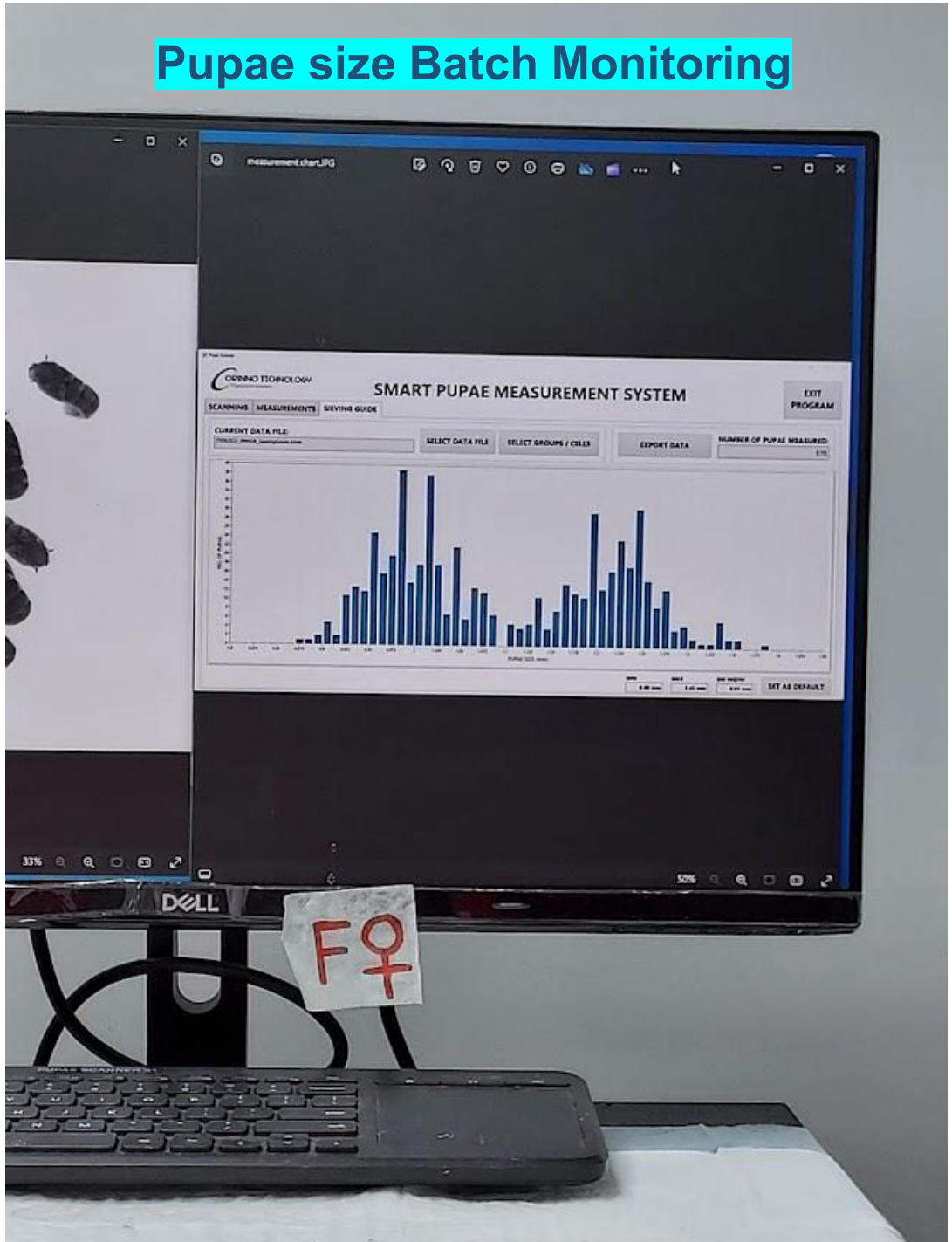


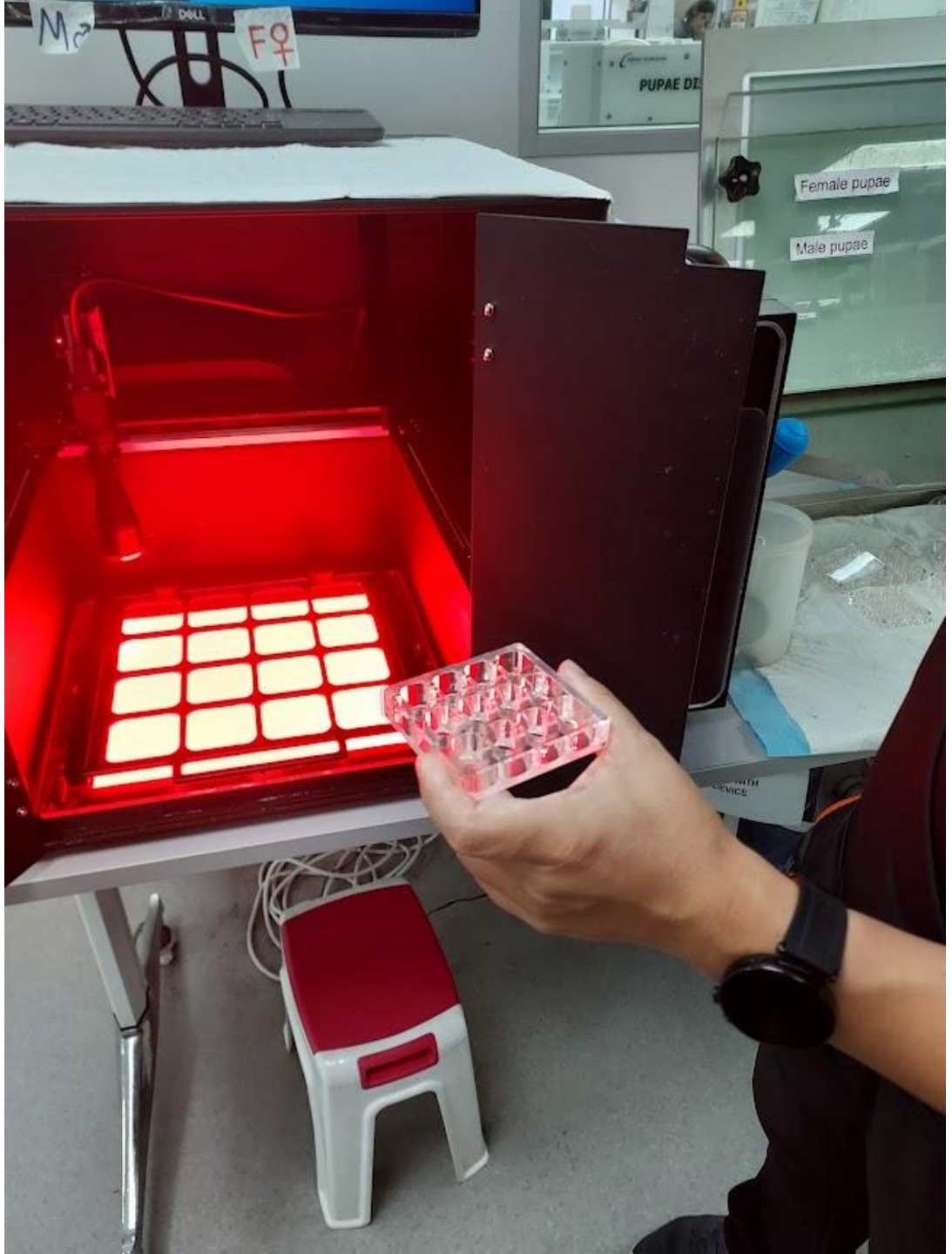
High-Density Tray rearing system



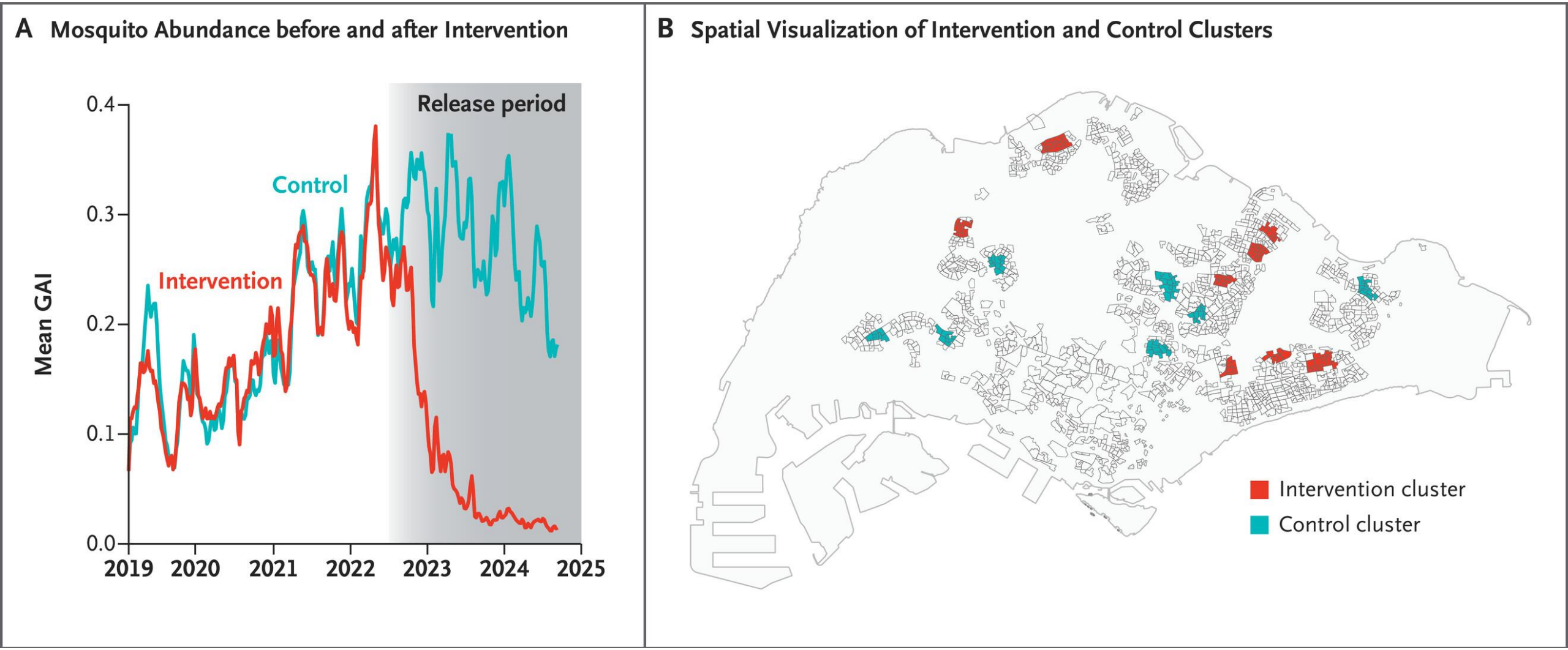


Pupae size Batch Monitoring





Wolbachia Male Release Suppresses Aedes aegypti



新加坡證明：放 Wolbachia 雄蚊，可以同時「減蚊」與「減少登革熱」

The NEW ENGLAND JOURNAL of MEDICINE

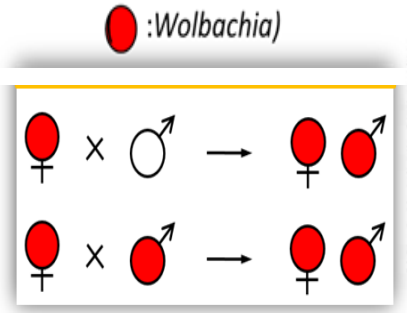
ORIGINAL ARTICLE

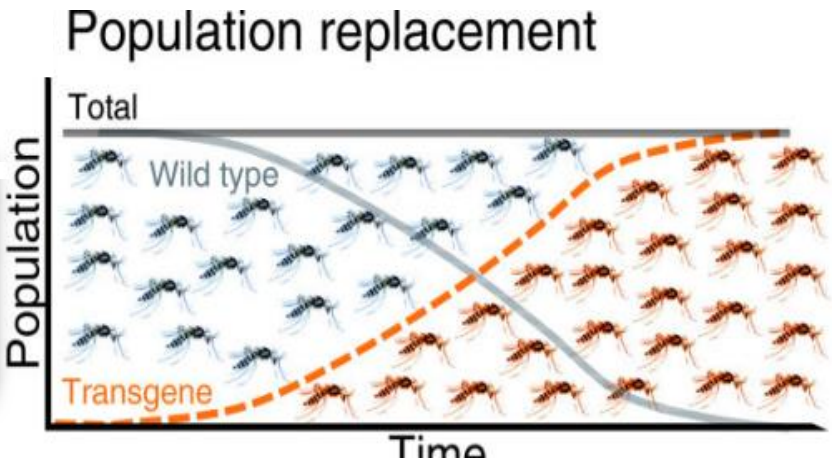
Dengue Suppression by Male Wolbachia-Infected Mosquitoes

J.T. Lim,[1] C.-S. Chong,[2] C.-C. Chang,[2,3] D. Mailepessov,[2] B. Dickens,[4] Y.L. Lai,[2] L. Deng,[2] C. Lee,[2] L.Y. Tan,[2] G. Chain,[2] M.F. Zulkifli,[2] J.W.K. Liew,[2] K. Vasquez,[2] M.L. Chau,[2] Y. Ng,[2] V. Lee,[4,5] J.C.C. Wong,[2] S. Sim,[2] C.H. Tan,[2] and L.C. Ng,[2,4,6] for the Project Wolbachia–Singapore Consortium*

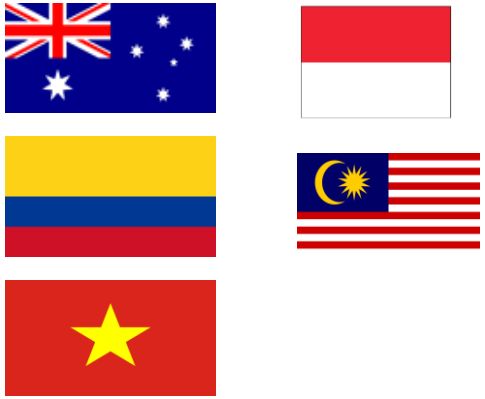
- 野生埃及斑蚊大幅下降
- 登革熱陽性率：21% → 6%
-
- 登革熱保護效力達 71–72%
-
- 不同年齡與性別皆有效
-
- 效果可持續至 24 個月

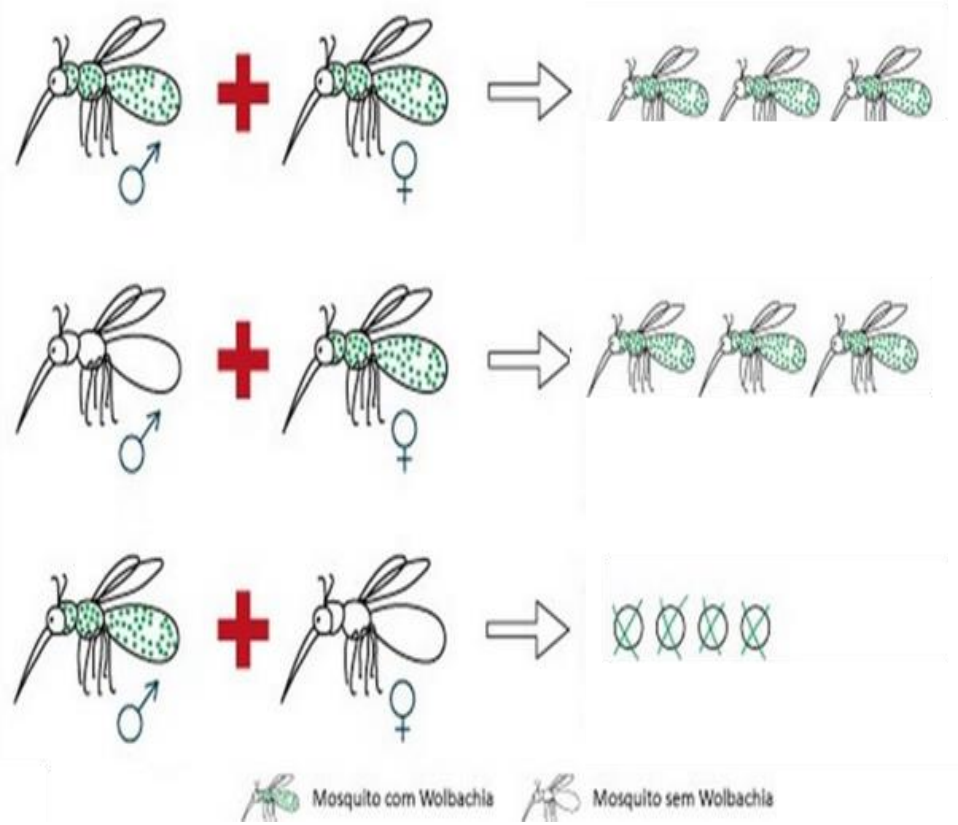
族群取代 便宜的方案





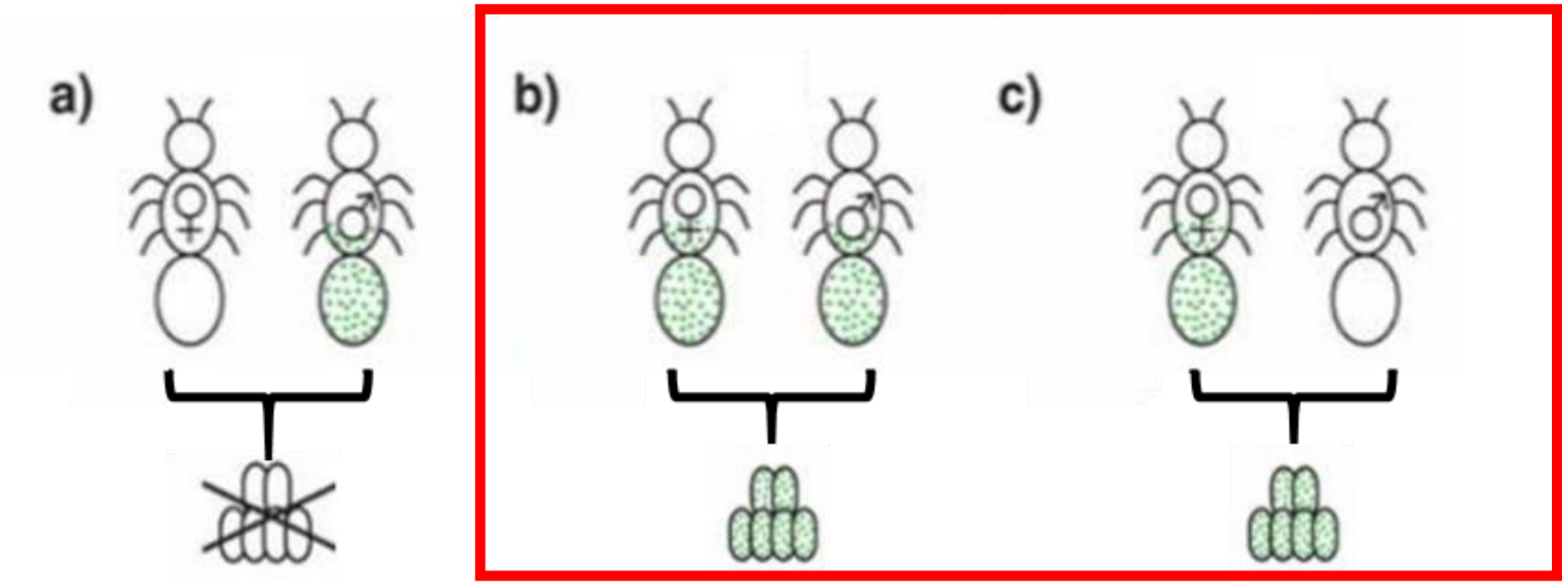






帶*Wolbachia*雄蚊、雌蚊皆釋放，
逐漸將野生族群替換為帶*Wolbachia*之族群

Cytoplasmic Incompatibility

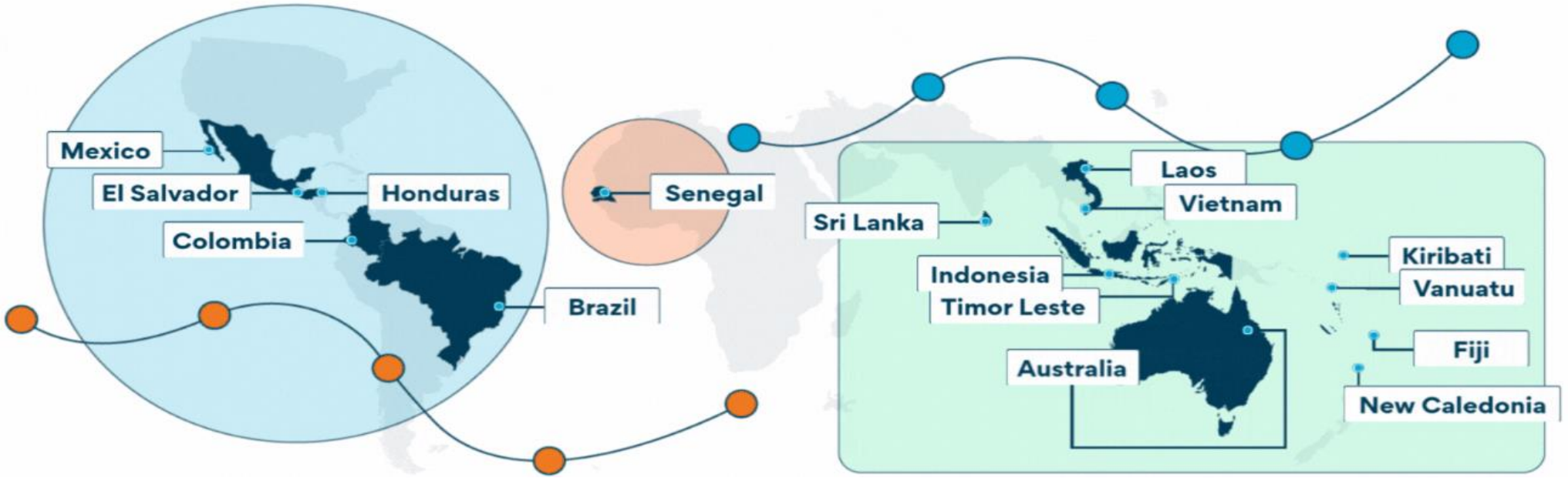


世界蚊子計畫

World Mosquito Program (WMP)

◆國際性的非政府組織 (NGO)，致力於發展和實施 *Wolbachia*，將其作為減少斑蚊傳播疾病的有效方法

◆從小型試點項目到數百萬人在城市範圍內的部署 *Wolbachia*蚊。



取代型門檻相對低，施放雄雌蚊（蛋紙）







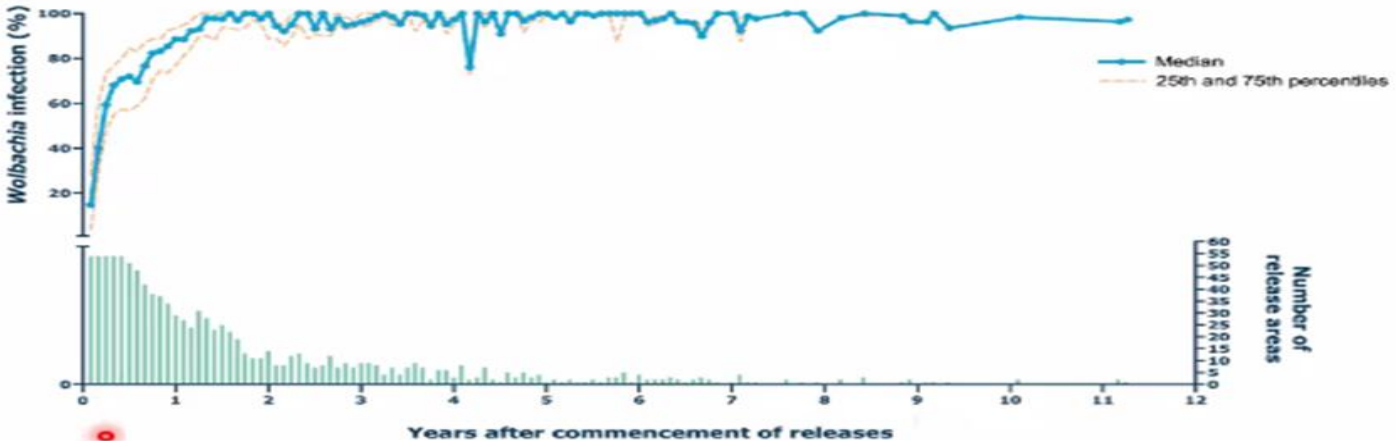


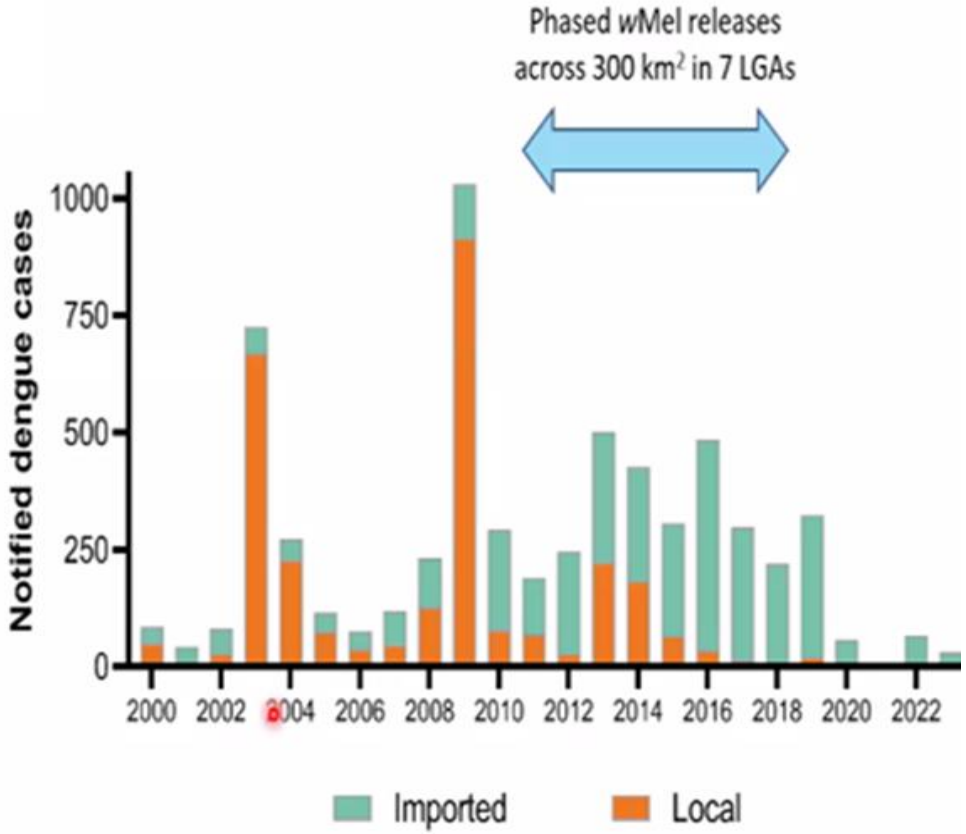
澳洲—族群取代計畫

Deployment at scale | Australia

2011 – 2017: **325,000 people** (300 km²)

- Mix of **adult & egg** releases
- *Wolbachia* sustained at high level **>10 years** post-release

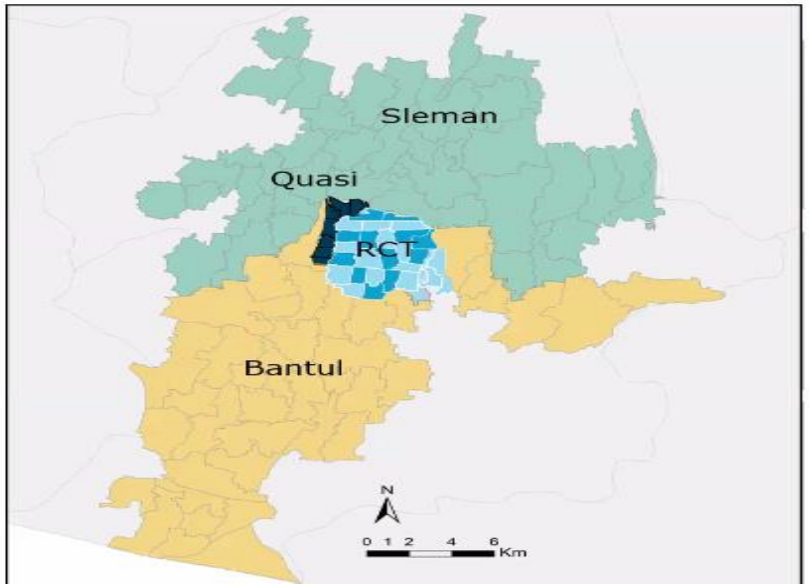




印尼—日惹市族群取代計畫

2014–2022: **1.8 million people** (540 km²)

2014	Peri-urban pilot releases
2016	Quasi-experimental deployment
2017-2020	AWED cluster randomised trial
2020-2021	Fill-in of Yogya untreated areas
2021-2022	Area-wide deployment in Yogyakarta Special Region





Efficacy of Wolbachia-infected mosquito deployments for the control of dengue. Utarini et. al., 2021 Efficacy of Wolbachia-Infected Mosquito Deployments for the Control of Dengue. *N Engl J Med*. Jun 10;384(23):2177-2186.

*95% confidence interval: 65% to 85%
**95% confidence interval: 66% to 94%

接受高雄市韓市長指派帶領高雄市規劃取代型 Wolbachia 防治計畫





高雄市官員參訪印尼日惹 Wolbachia 計畫



釋放感染沃爾巴克氏體（Wolbachia）之病媒蚊對登革熱防治的效力

The NEW ENGLAND JOURNAL of MEDICINE

ESTABLISHED IN 1812 JUNE 10, 2021 VOL. 384 NO. 23

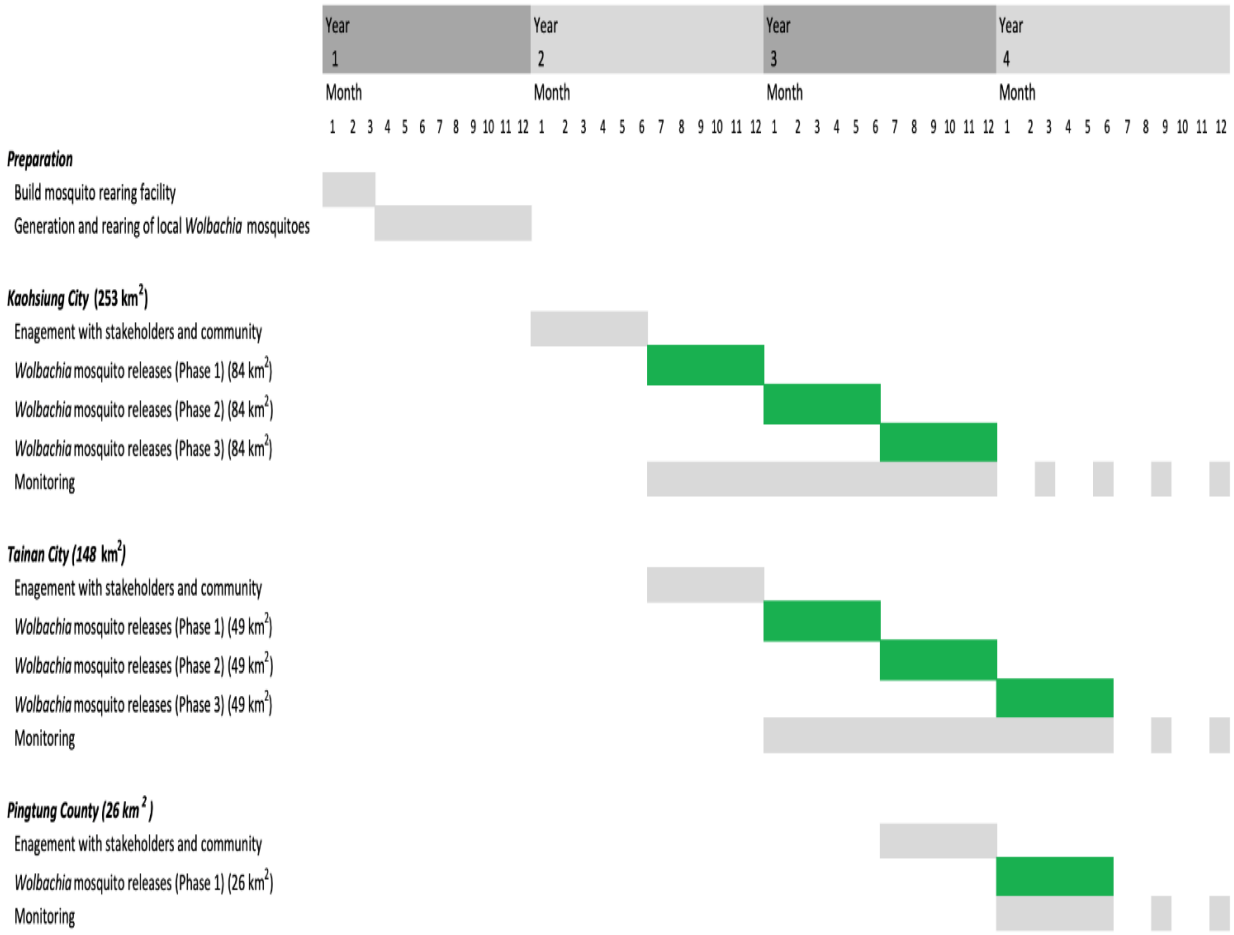
Efficacy of Wolbachia-Infected Mosquito Deployments for the Control of Dengue

A. Utarini, C. Indriani, R.A. Ahmad, W. Tantowijoyo, E. Arguni, M.R. Ansari, E. Supriyati, D.S. Wardana, Y. Meitika, I. Ernesia, I. Nurhayati, E. Prabowo, B. Andari, B.R. Green, L. Hodgson, Z. Cutcher, E. Rancès, P.A. Ryan, S.L. O'Neill, S.M. Dufault, S.K. Tanamas, N.P. Jewell, K.L. Anders, and C.P. Simmons, for the AWED Study Group*

- Wolbachia 成功穩定建立於蚊群 — wMel 盛行率中位數 95.8%
- 登革熱病例降低 77% — 保護效力 77.1%
- 四種登革病毒血清型皆有效
- 登革熱住院降低 86% — 保護效力 86.2%
- Wolbachia 建立越高，保護效果越好

Wolbachia 防治計畫在台灣施行規劃距離最近的一次…

「世界蚊子計畫 (WMP)」之評估，高雄都會區，施放的規劃 ：





04

台灣已經做了什麼？

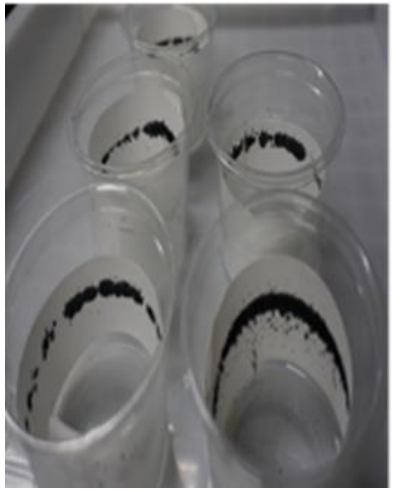
wAlbB-TW 品系、CI / 病毒抑制、大量飼養與半田間技術平台

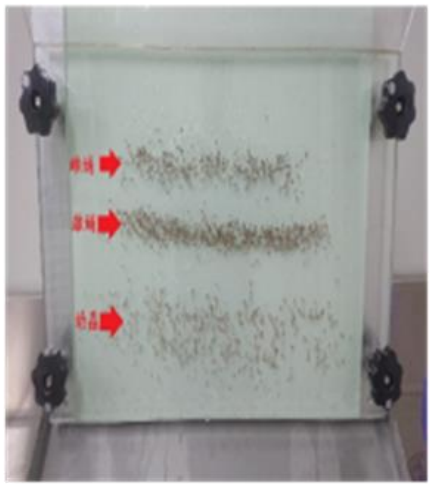
台灣 *Wolbachia*防治技術平台建置

Egg	Larvae	pupae separation	Adult	Exp. area





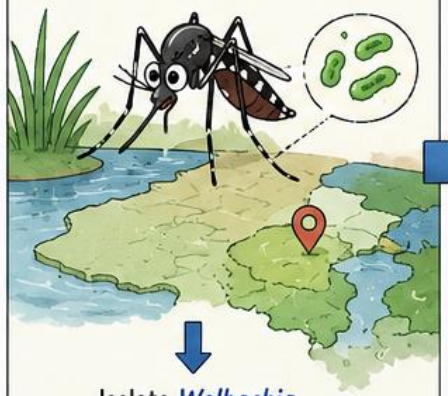




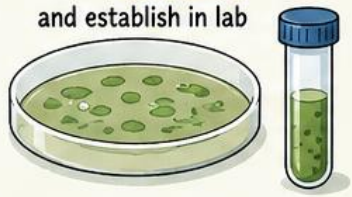
Building Local *Wolbachia* for Local *Aedes*

Use local *Wolbachia* strain + local *Aedes* genetic background = local adaptation, better stability

1. Collect local *Wolbachia* from field mosquitoes



Isolate *Wolbachia* and establish in lab



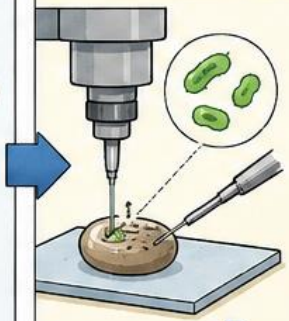
2. Establish antibiotic-cured local *Aedes* line





Remove native *Wolbachia* with antibiotic treatment and recover to obtain *Wolbachia*-free line

3. Microinject local *Wolbachia* into local *Aedes* embryos



Inject local *Wolbachia* suspension into embryos



Rear injected survivors to adults

4. Establish and stabilize local *Wolbachia* line

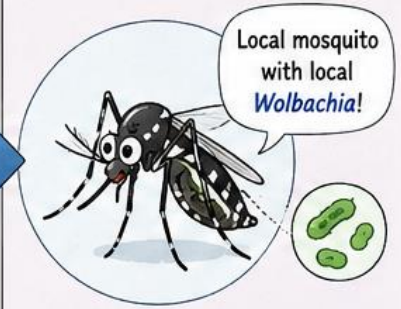


Screen for *Wolbachia* infection, density and maternal transmission



Backcross with local *Aedes* to stabilize *Wolbachia* in the local genetic background

5. Local *Wolbachia* strain is ready!



- High maternal transmission
- Appropriate density
- Stable in local environment
- Strong pathogen blocking and population suppression

RESULT: LOCAL WOLBACHIA STRAIN

Derived from local *Wolbachia* + local *Aedes*

Local *Wolbachia*



Adapted to local environment and hosts

+

Local *Aedes* genetic background



Local adaptation and fitness

=

Local *Wolbachia* strain in local *Aedes*

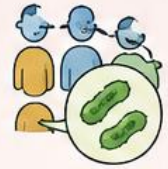


Better stability, sustainability and field performance

WHY LOCAL IS BETTER



Better adaptation to local climate and ecology



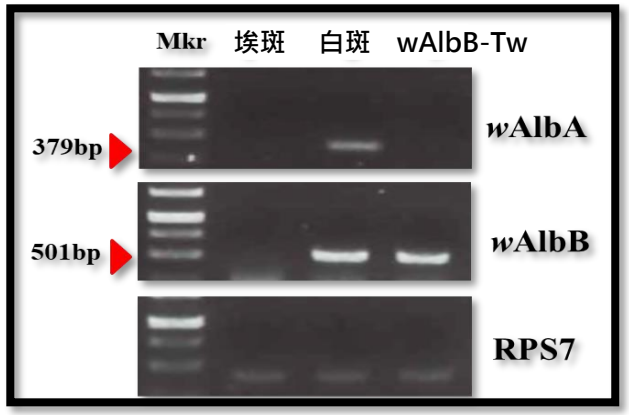
Higher stability and persistence in the field

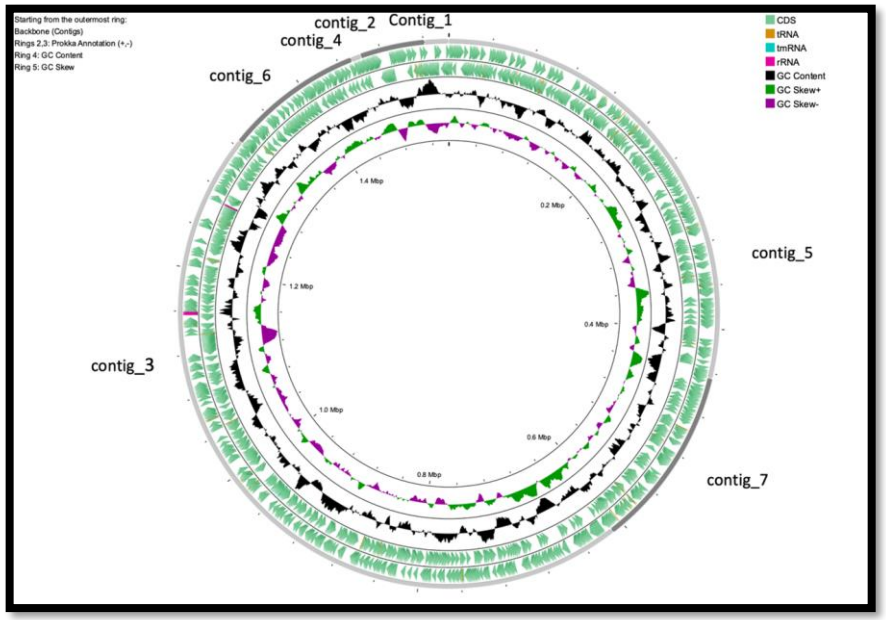


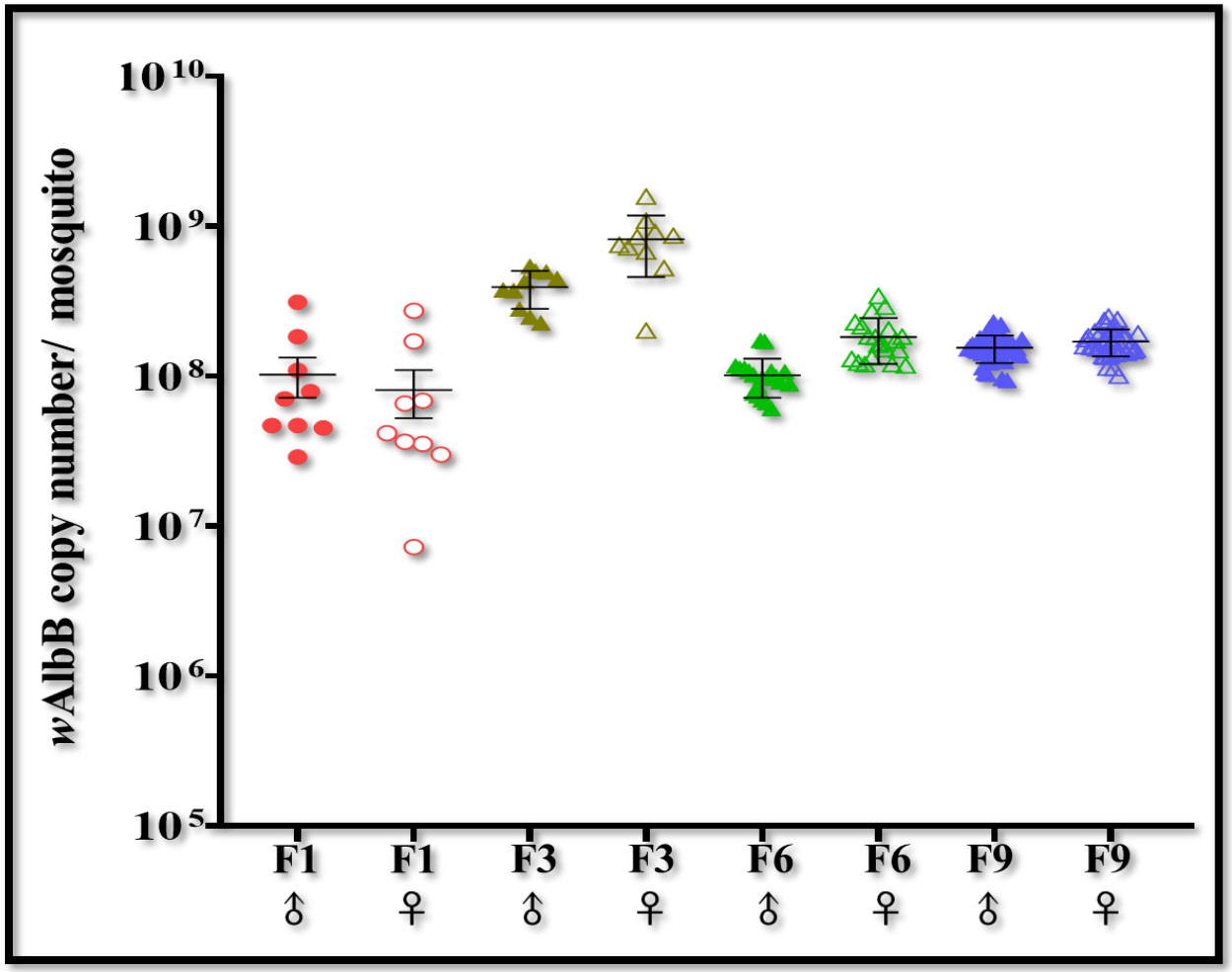
Stronger, durable control of dengue and other diseases

*w*AlbB-TW埃及斑蚊感染株

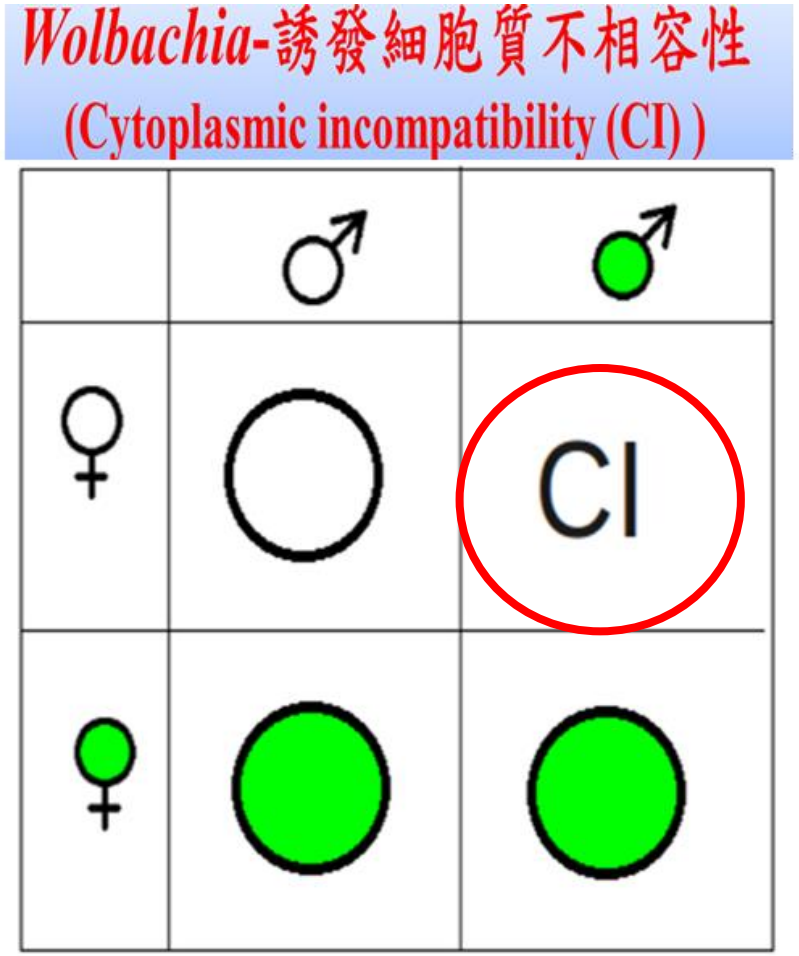
定性與定量分析

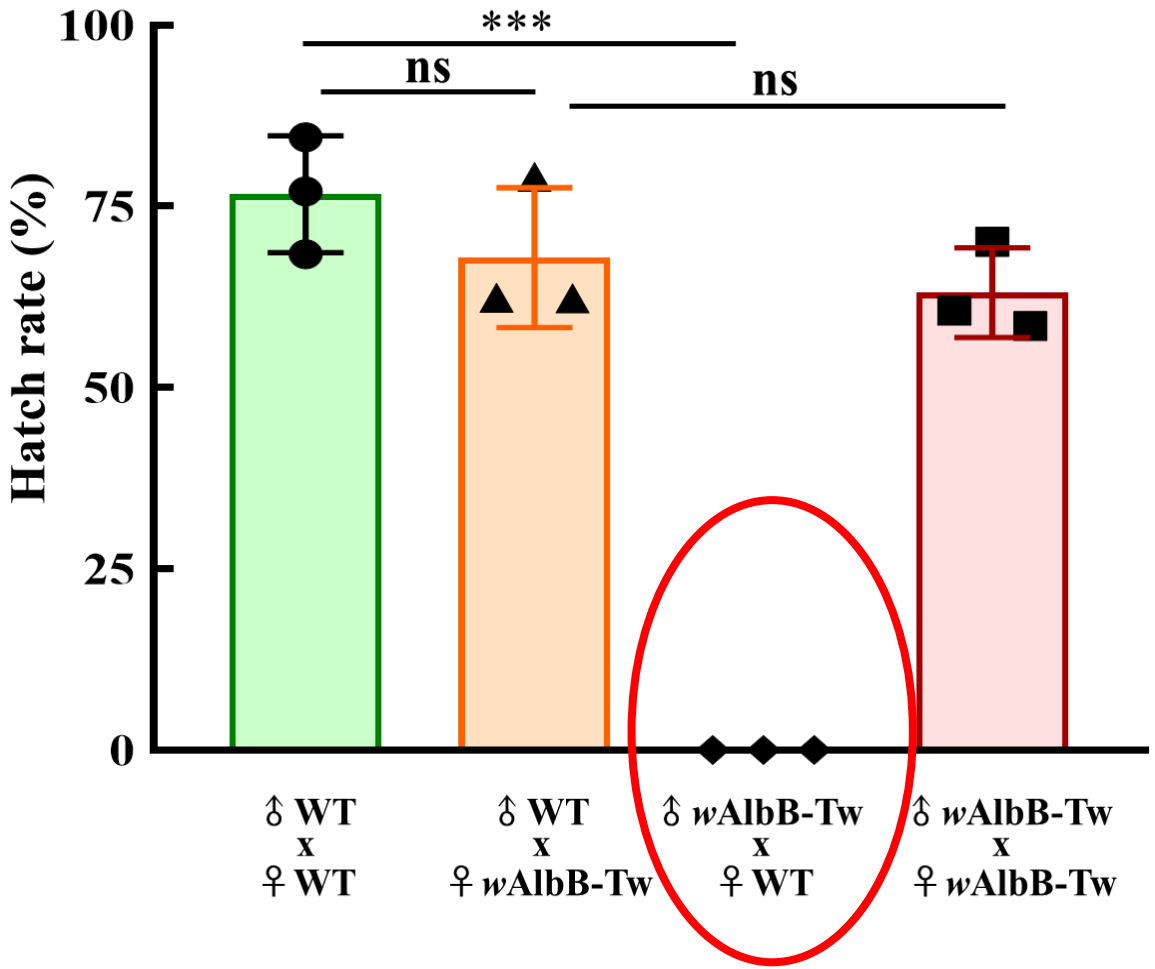




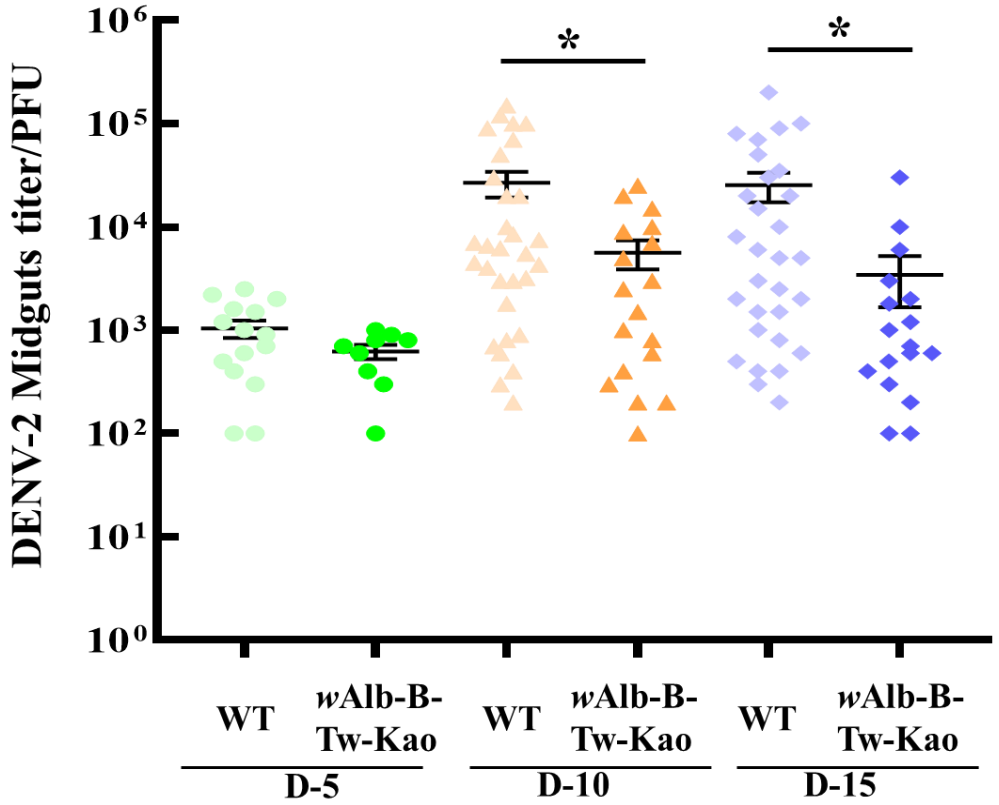


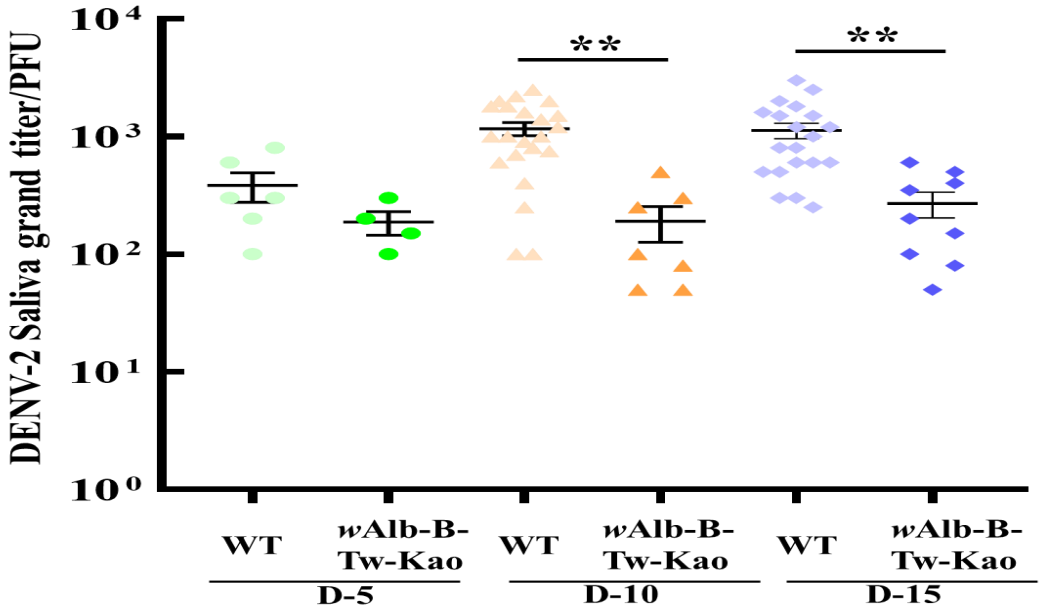
*w*AlbB-Tw 感染株細胞質不相容性(CI)

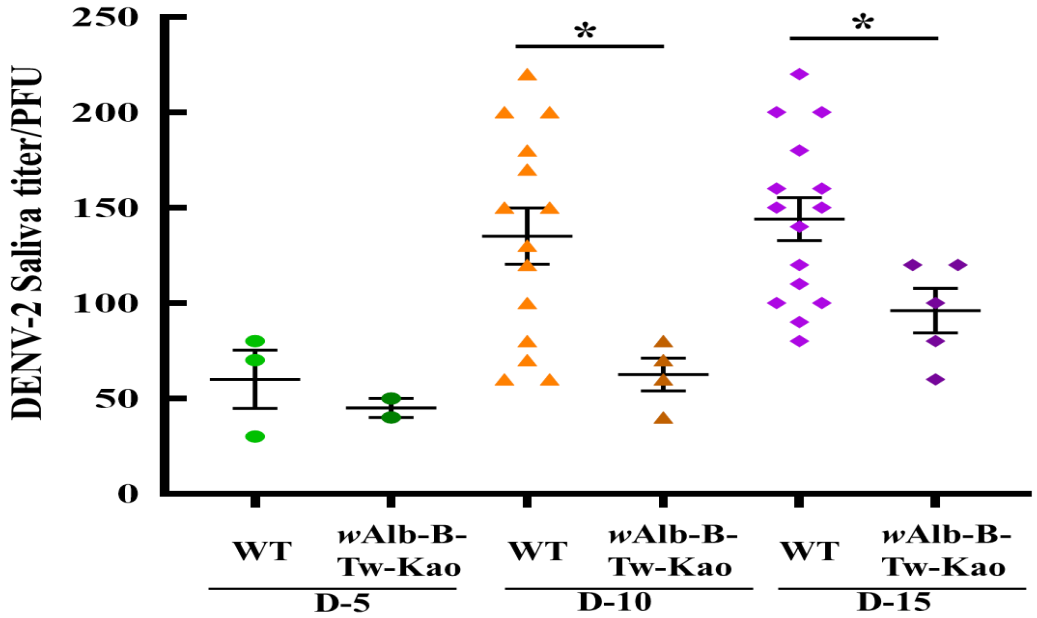




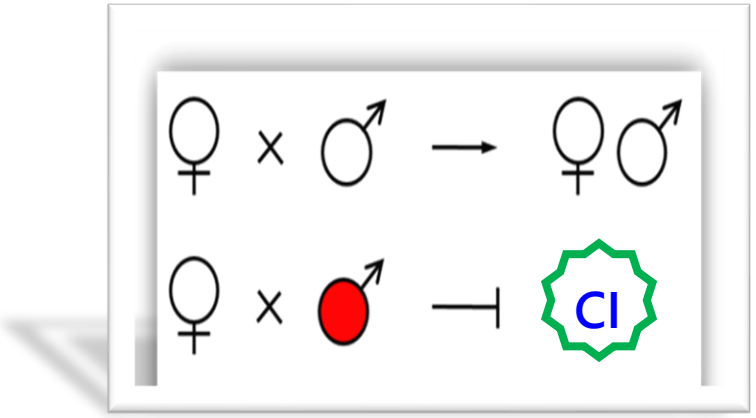
*w*AlbB-Tw 感染株抑制登革熱

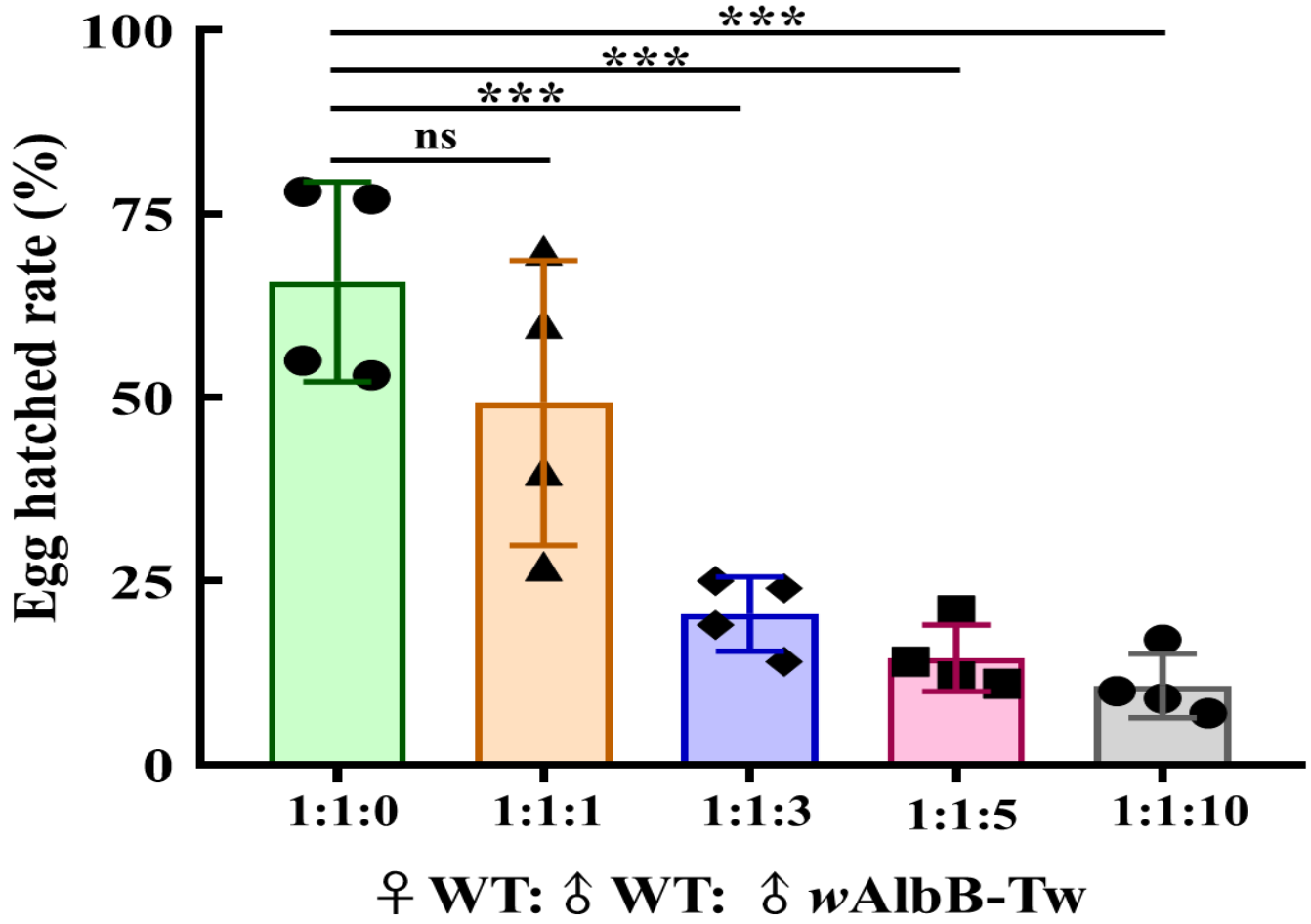




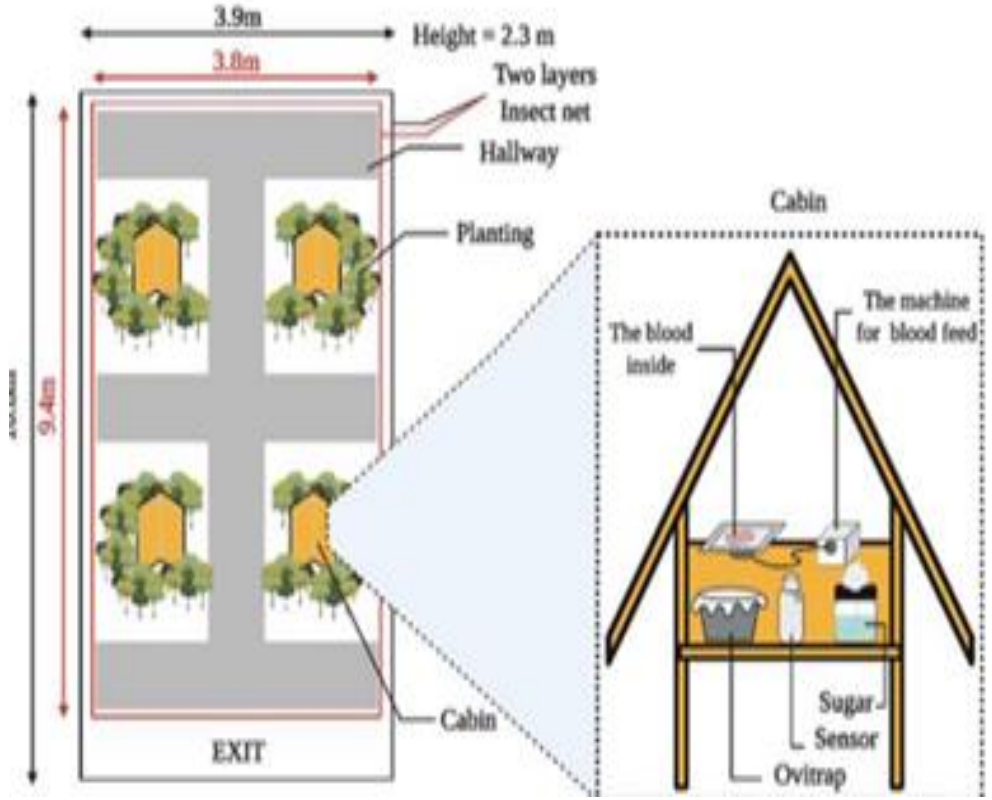


*w*AlbB-Tw 感染雄蚊不同釋放比例對蚊群密度的抑制效應





田間測試場域建立


3.9m
Height = 2.3 m
3.8m
Two layers
Insect net
Hallway
Planting
9.4m
Cabin
EXIT
Cabin
The blood inside
The machine for blood feed
Sugar
Sensor
Ovitrap











05

台灣需要 Wolbachia 計畫嗎？

如何納入整合性病媒管理，成為現有防治工具的重要補充？

台灣需要 Wolbachia 計畫嗎？

1. **誰是決策者？— Who decides?→ 需要一個明確的中央主責單位（Lead Agency）。**
2. **誰出錢？— Who pays?**
   真正進入實際防治後，經費不能只靠**研究計畫**。需要的是多年期的公共衛生預算，包括：**蚊子大量生產、品質控制、運輸、釋放、監測、人力與成效評估**。
   → 關鍵問題應從「研究經費有多少？」改成：**政府願意為每年降低多少登革熱病例，投入多少長期防治經費？**
3. **誰來執行？— Who operates?**
   研究機構可以建立技術與提供科學證據，但不是全國蚊媒防治的執行單位。比較合理的架構是：
   研究機構 → 技術與品質引進與改進
   中央政府 → 政策、法規、經費
   地方政府 → 現場釋放與社區溝通
   第三方／專家 → 獨立成效評估
4. **錢花下去，有沒有比現在更有效？— Is it worth the money?** 要比較：
   Wolbachia 成本 vs. 傳統噴藥＋清除孳生源＋疫情爆發醫療成本＋社會經濟損失。
5. **誰對最後結果負責？— Who owns the outcome?病媒蚊下降多少？登革熱病例下降多少？每避免一個病例的成本是多少？能否持續 5–10 年？**
   → 因此台灣需要的不只是 Wolbachia Project，而是建立一個 National Wolbachia Program（**國家級 Wolbachia 防治計畫**）。

從「研究計畫」走向「國家公共衛生政策」，才是台灣 Wolbachia 下一階段真正的挑戰。

06

真正的挑戰是什麼？

科學 → 量產 → 法規 → 社會接受 → 成本效益 → 長期監測

台灣 Wolbachia 計畫的挑戰

1. **法規核准與政策支持**
   - 如何從「研究試驗」正式跨入**戶外釋放與常規公共衛生防治**
   - 需要衛福部、疾管署、環境部及地方政府建立明確的審查與**經費**來源。**四年一次選舉政策無法連續**。
2. **決定使用何種方式，抑制型或取代型？ 境內建廠生廠或境外輸入**
3. **建立大規模、穩定且低成本的生產系統**
   - 從實驗室養蚊走向**每週數百萬隻雄蚊**
   - 關鍵包括自動化飼養、精準雌雄分離、品質控制，X 光機的建置。
4. **如果噴藥方式政府已經滿意，如何與現行噴藥防治整合**
   - 台灣疫情發生時仍高度依賴化學殺蟲劑

5. **社區接受度與風險溝通**
   - 最大問題可能不是「Wolbachia 安不安全」，而是民眾會問：**「為什麼政府要在我家附近放蚊子？」**
   - 必須清楚解釋「只釋放不吸血的雄蚊」、雌蚊污染率、監測方式，以及居民可以預期看到什麼。台灣研究也指出，雄蚊 suppression strategy 在社區接受度上具有優勢。
6. **證明真實世界的公共衛生成效與永續性**
   - 等待新加坡及其他國家**革熱病例真的下降**
   - 需要長期評估成本效益、蚊群回復、外來蚊遷入與是否需要持續釋放。

THANK YOU!

從科學證據，到台灣的下一步

Wolbachia is ready. Are we ready?

科學已經證明可行，下一步是我們的選擇。



Chun-Hong Chen, Ph.D.

National Health Research Institutes, Taiwan